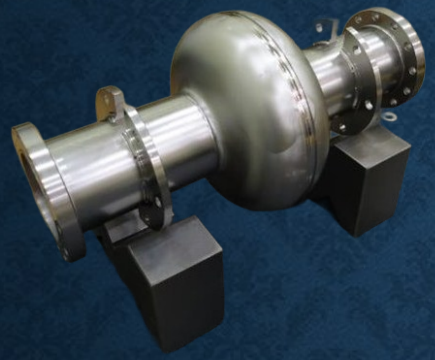
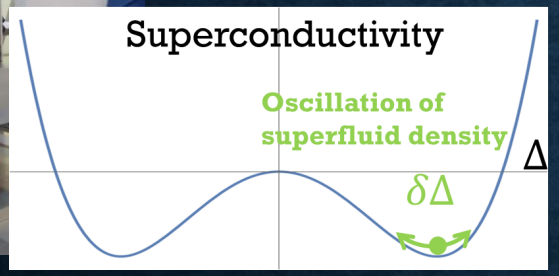
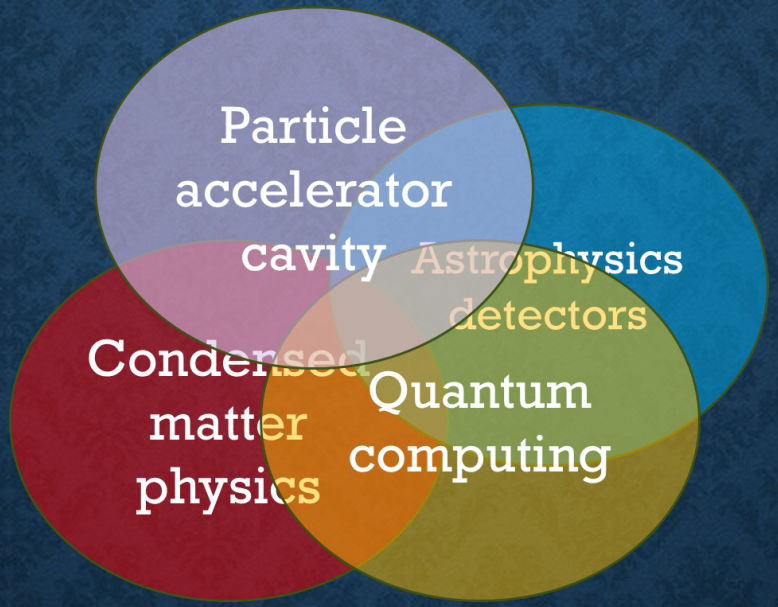
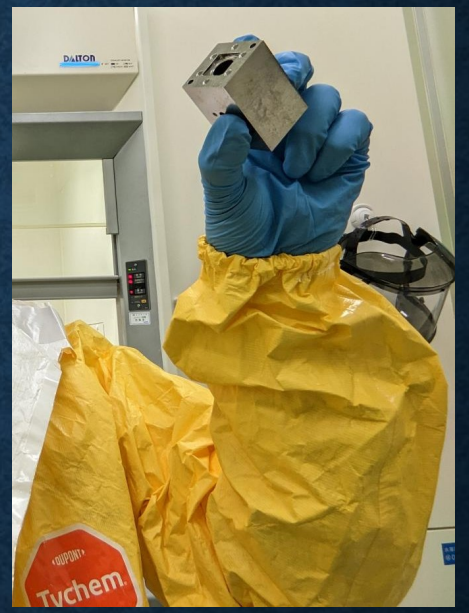
Superconducting Cavities and Their Physics:
From Particle Accelerators to Quantum Technologies









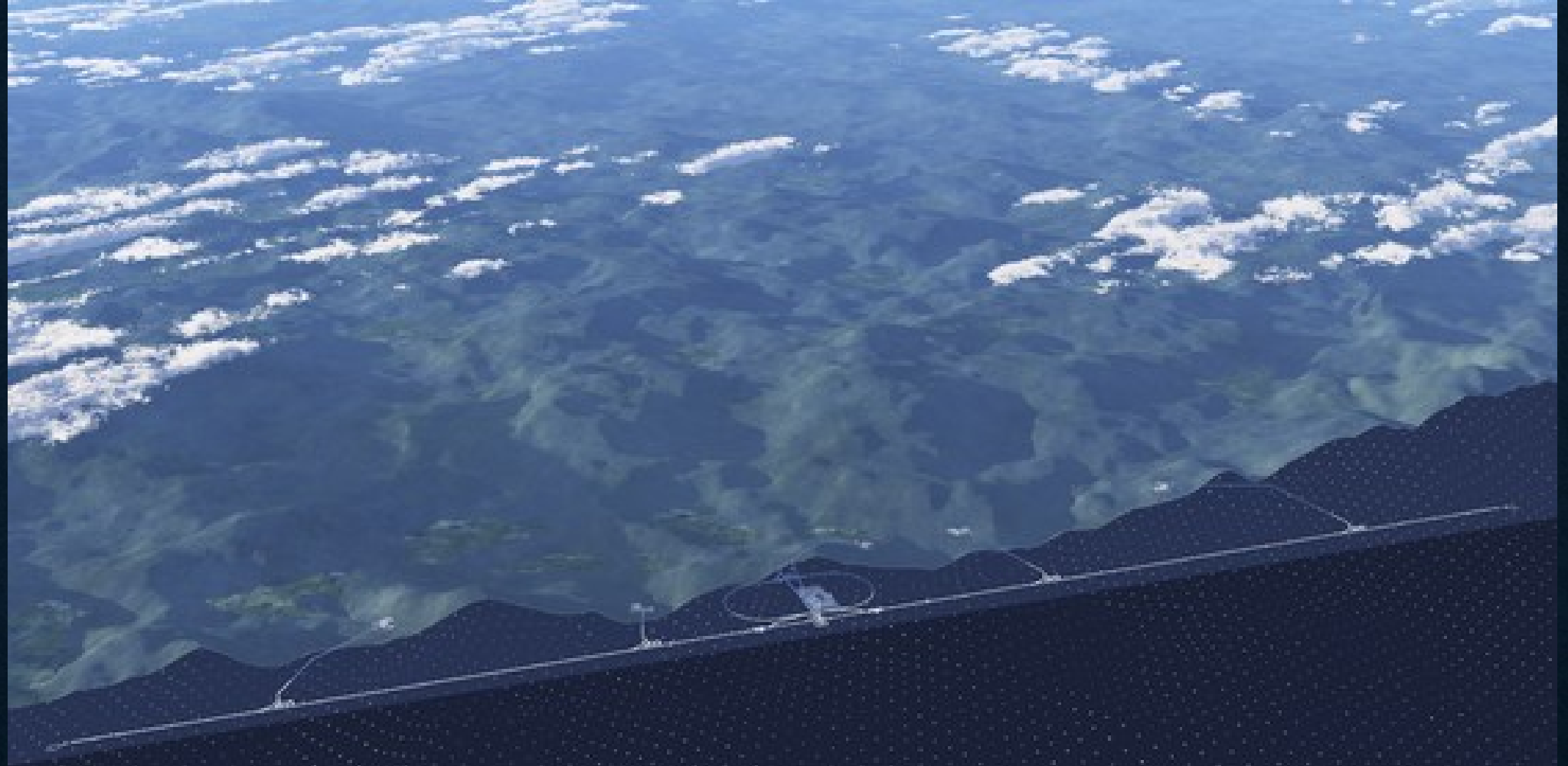




High energy Accelerator Research Organization (KEK)
Innovation Center for Applied Superconducting Accelerators (iCASA)

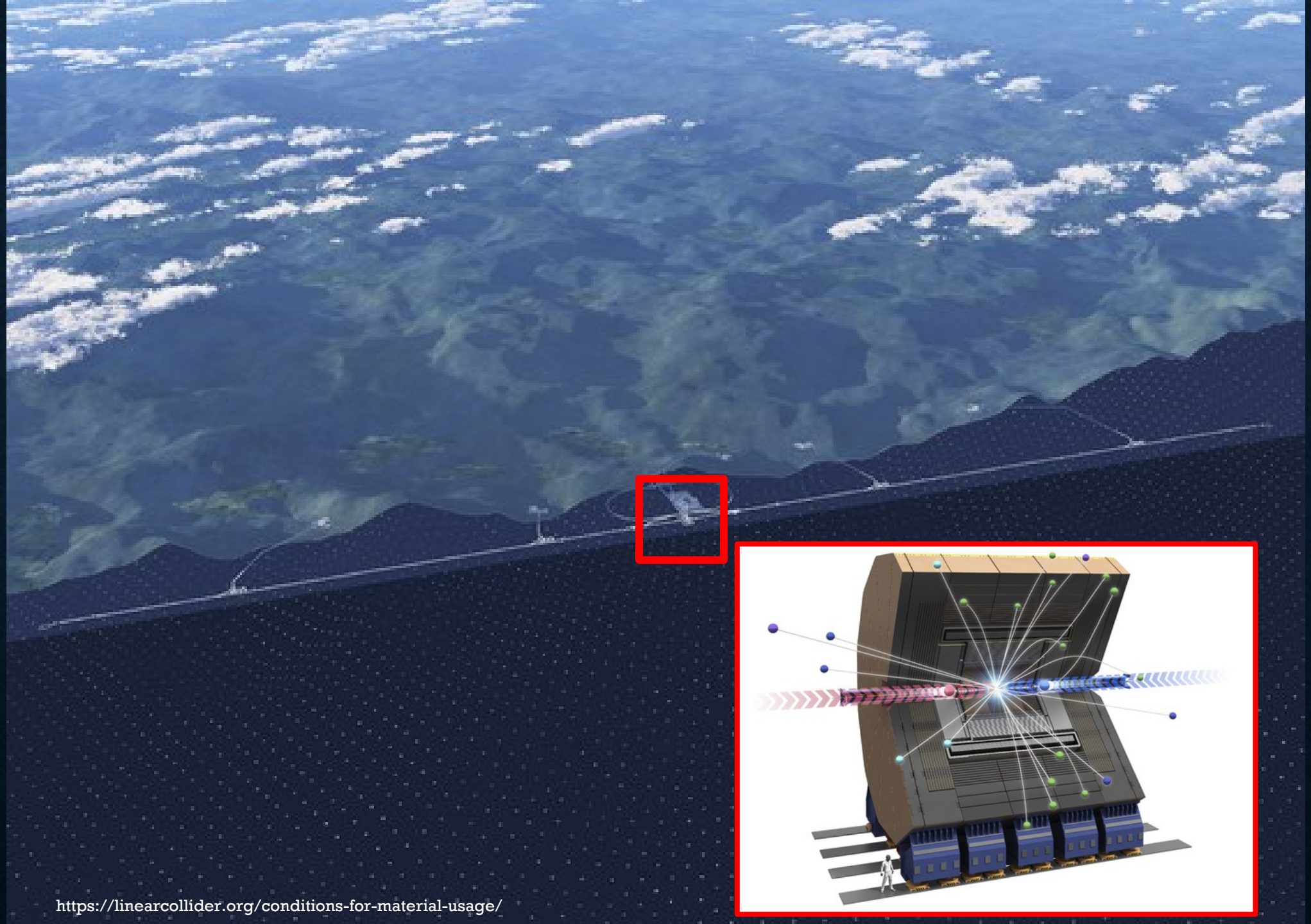
Takayuki Kubo

https://kbtkphys.github.io/
https://scholar.google.com/citations?user=MN4RQIsAAAAJ

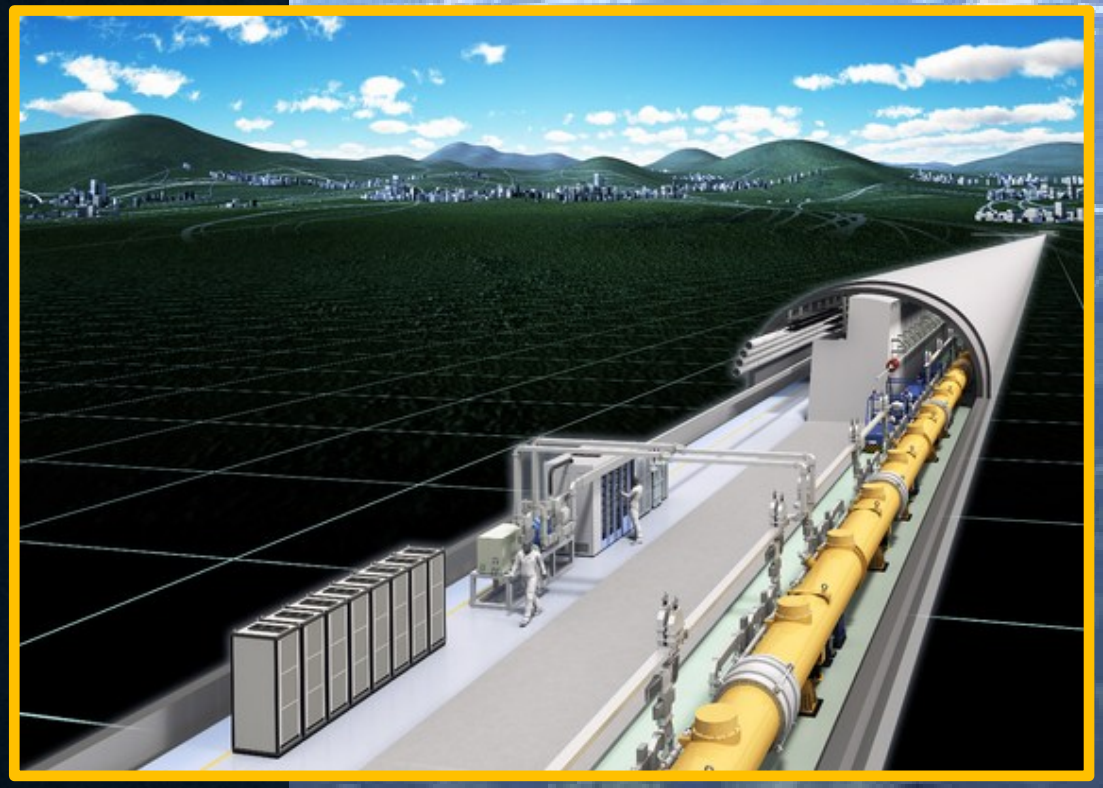


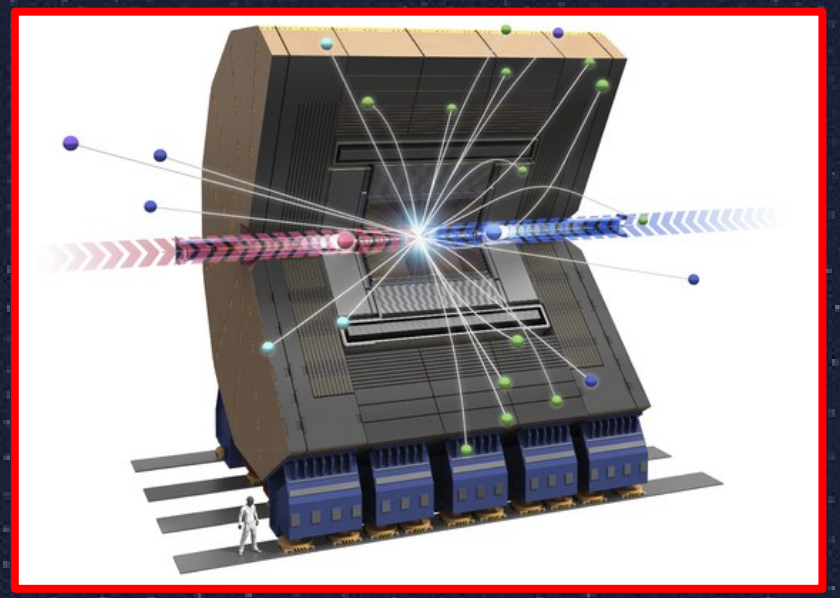
Here we use the International Linear Collider (ILC), an electron–positron Higgs factory, **as an example**, but the European X-ray Free-Electron Laser (European XFEL) in Germany and the Linac Coherent Light Source II (LCLS-II) at SLAC in California are also based on similar superconducting RF linac technology.

https://linearcollider.org/conditions-for-material-usage/

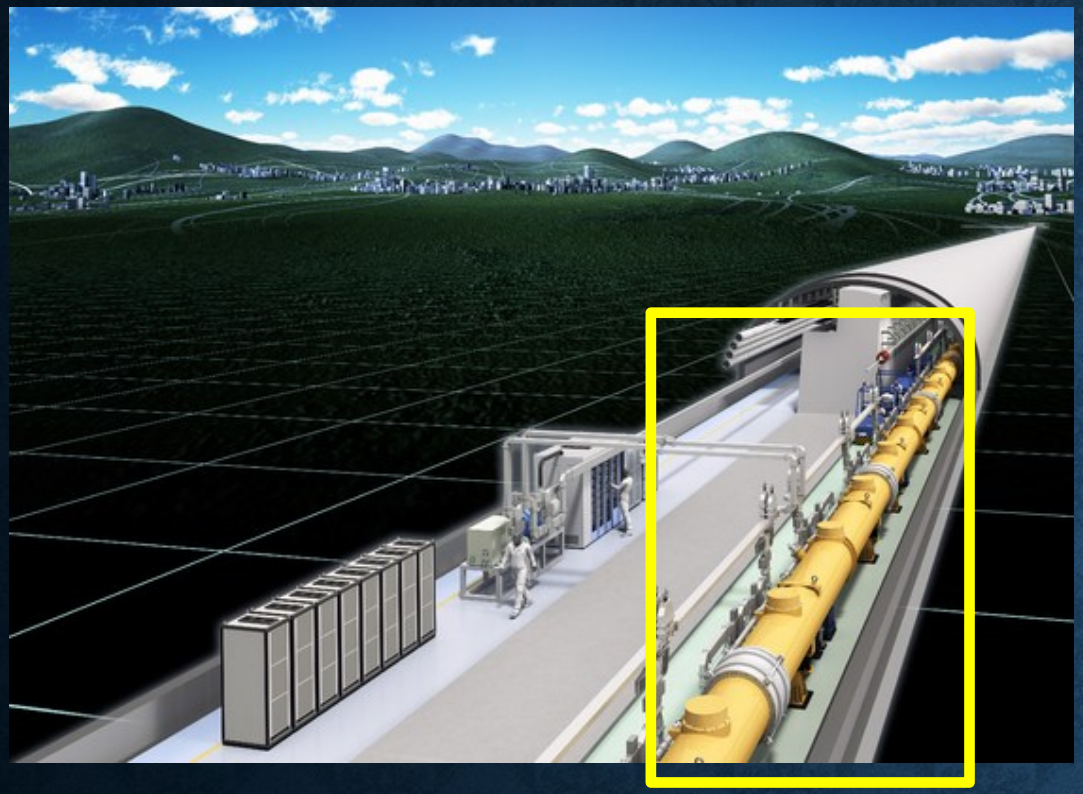


https://linearcollider.org/conditions-for-material-usage/





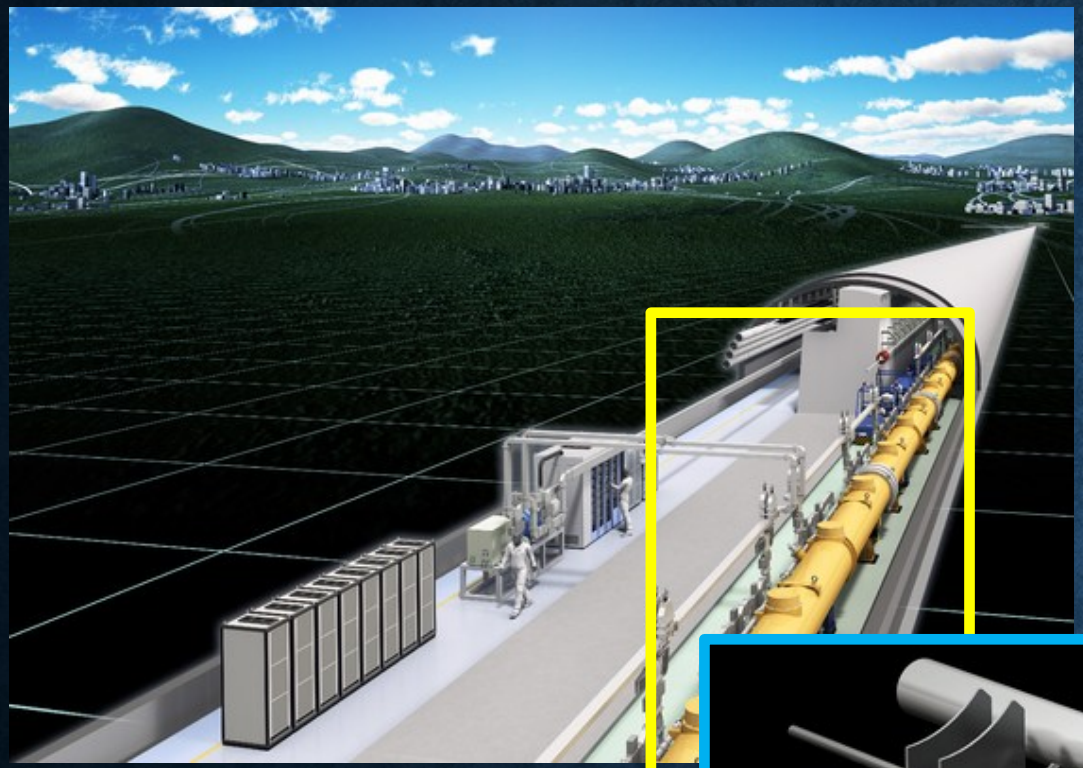
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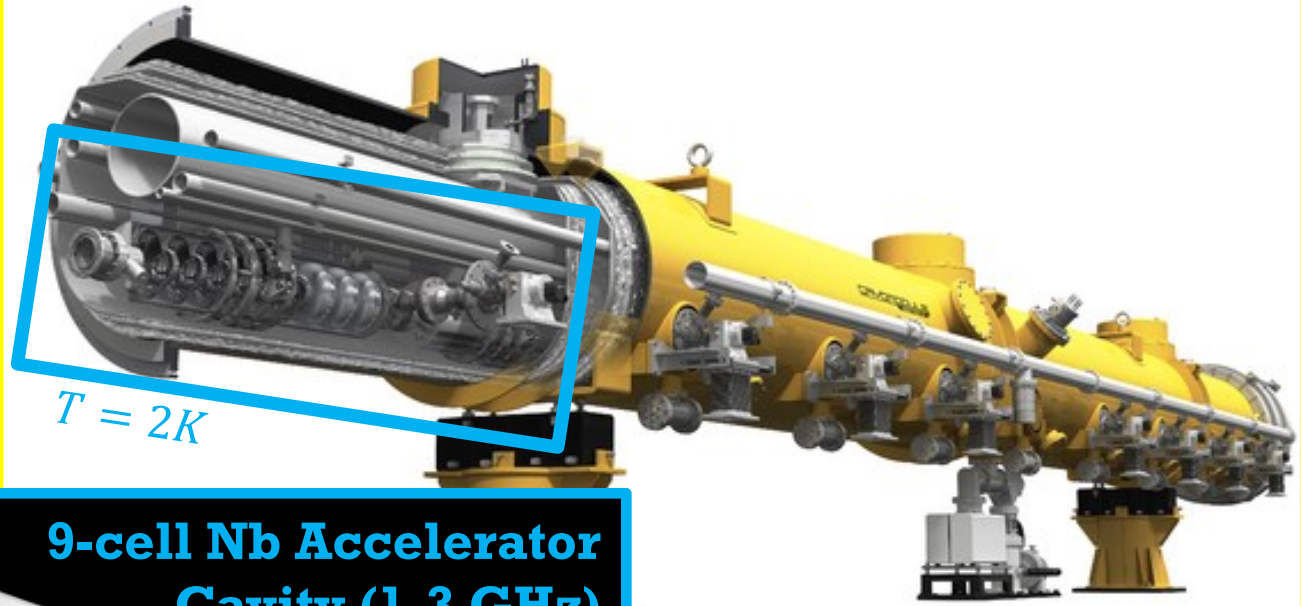
Cryomodule



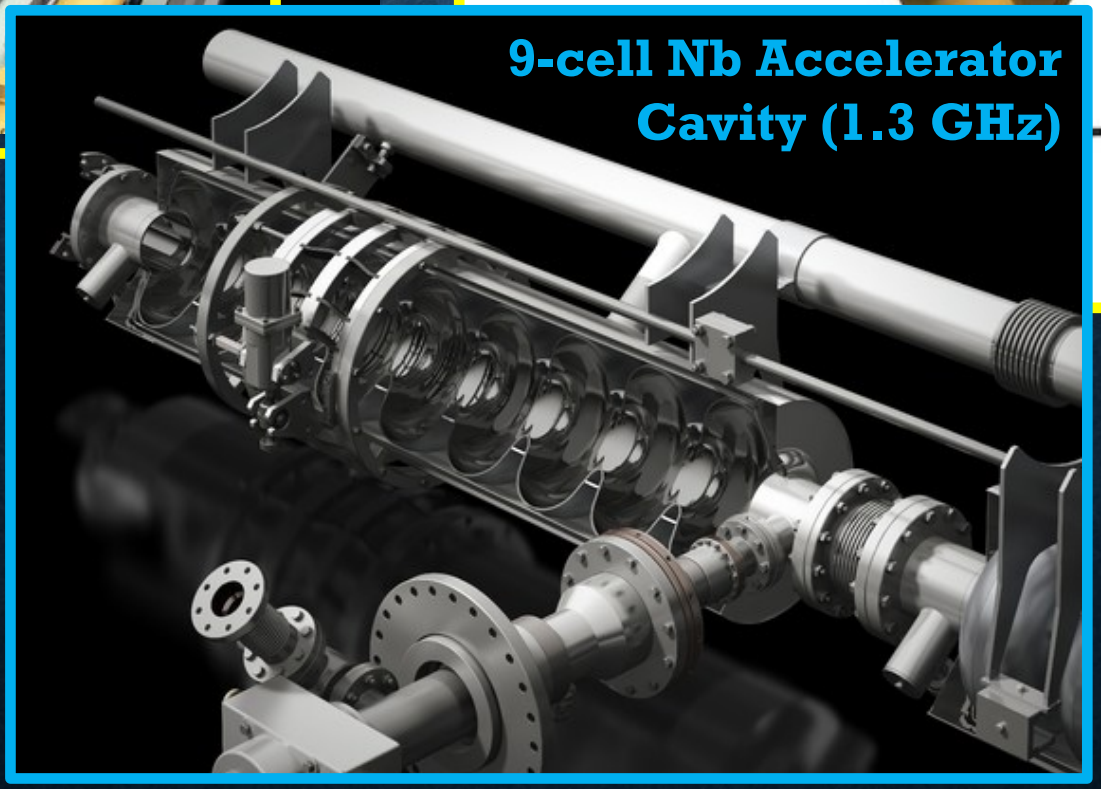
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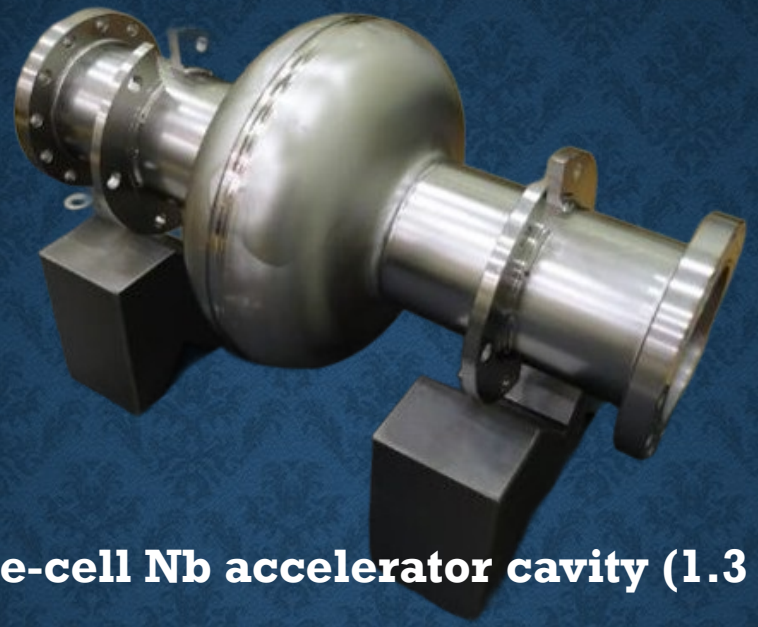
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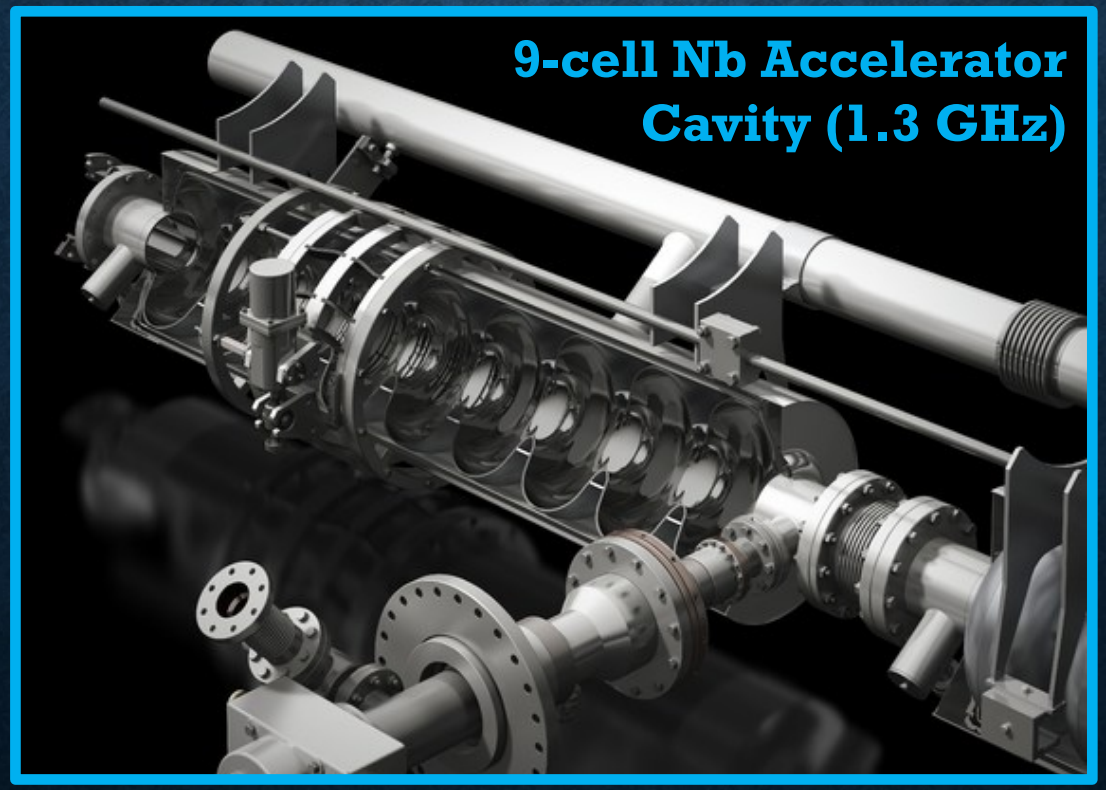
9-cell Nb Accelerator Cavity (1.3 GHz)

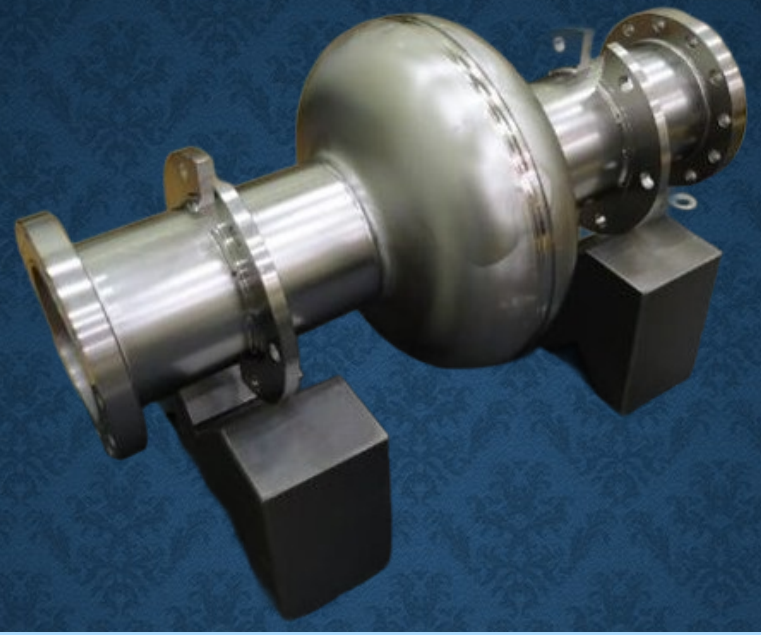


https://linearcollider.org/conditions-for-material-usage/

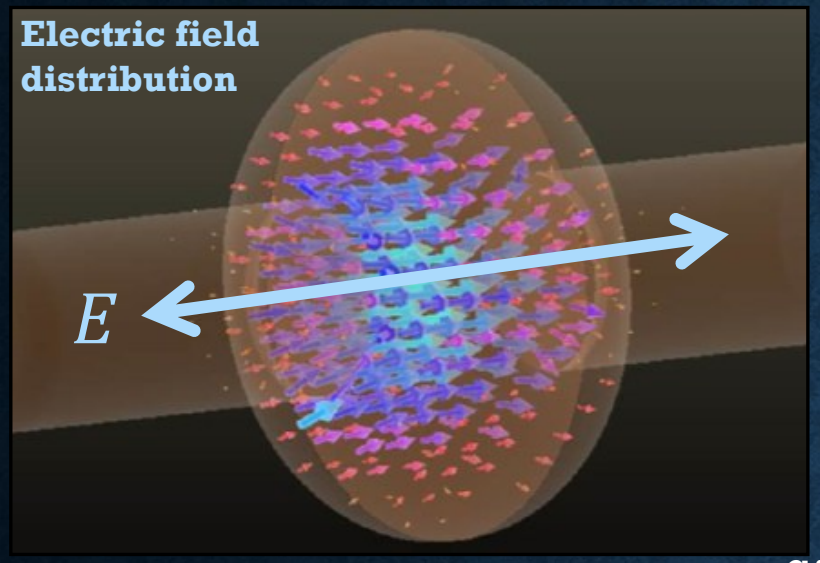


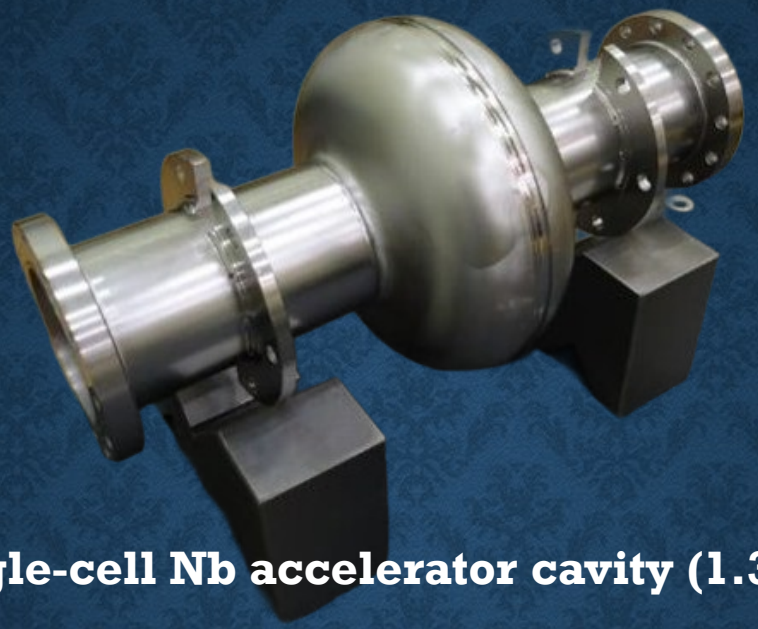
Single-cell Nb accelerator cavity (1.3 GHz)



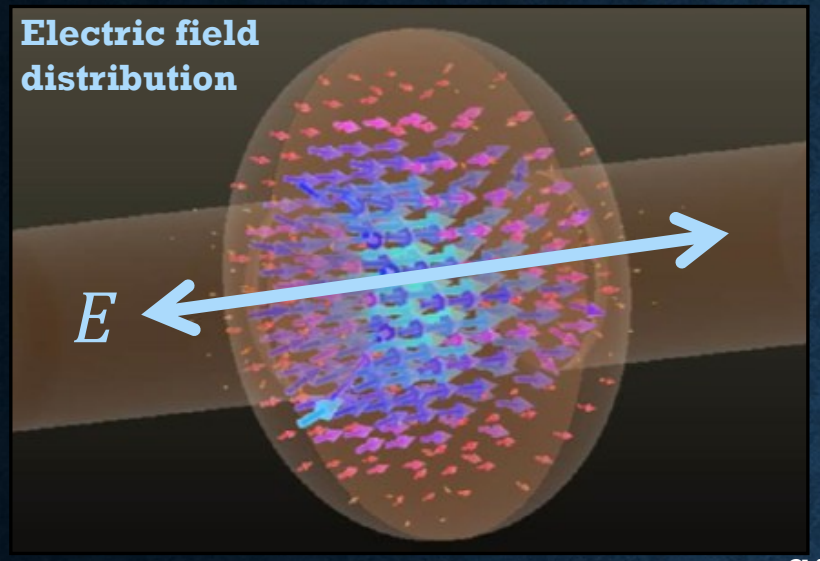


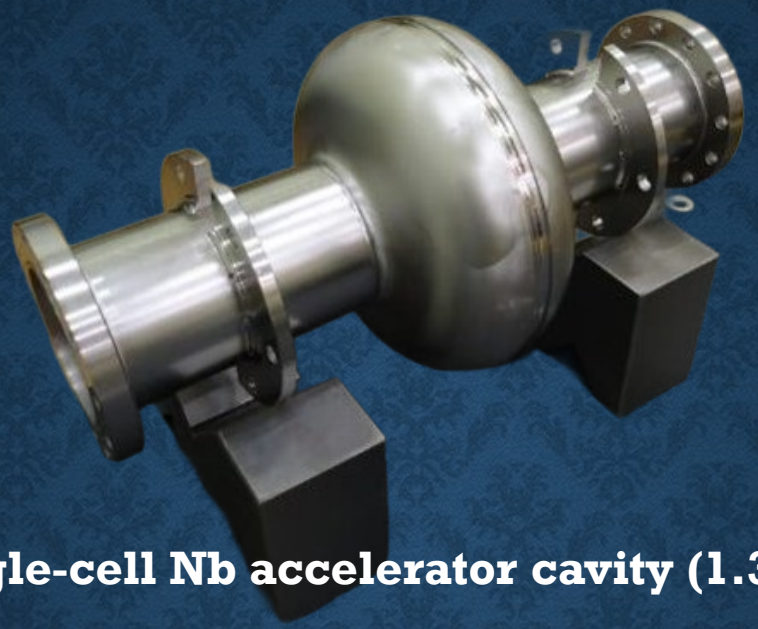
Overview

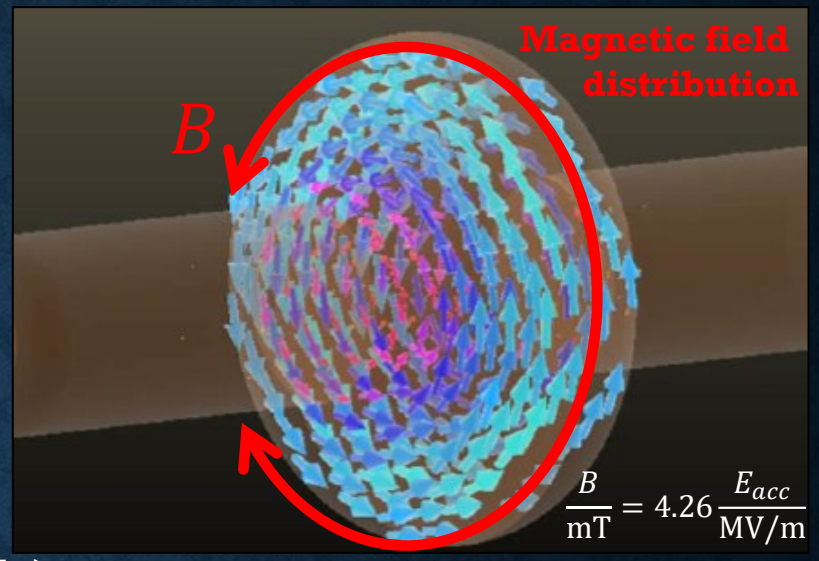




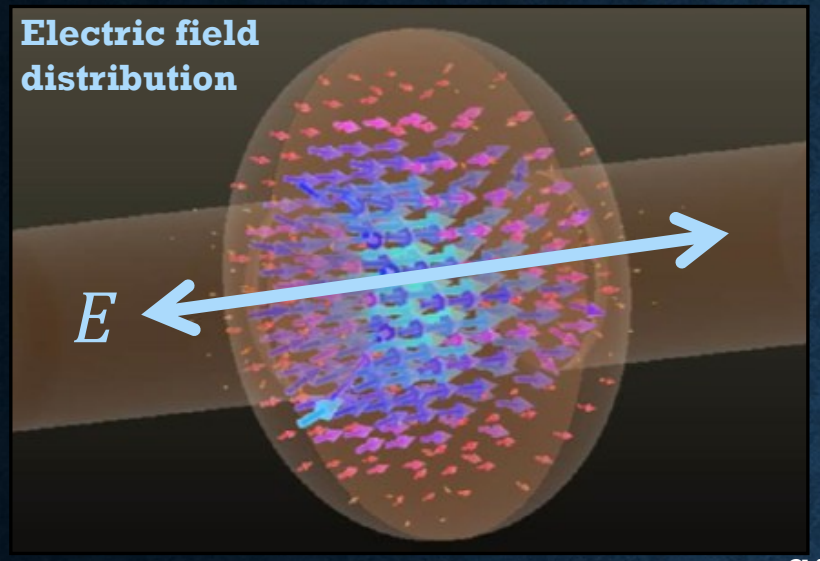
Single-cell Nb accelerator cavity (1.3 GHz)

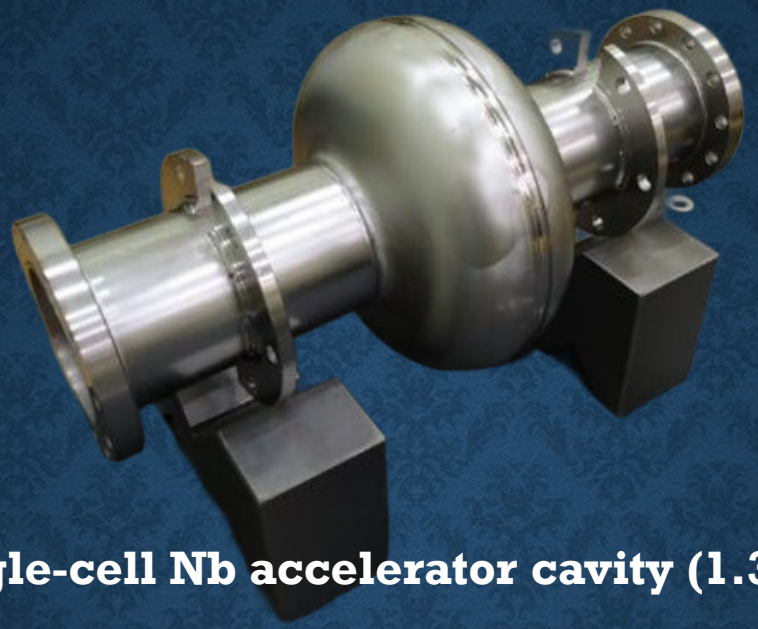


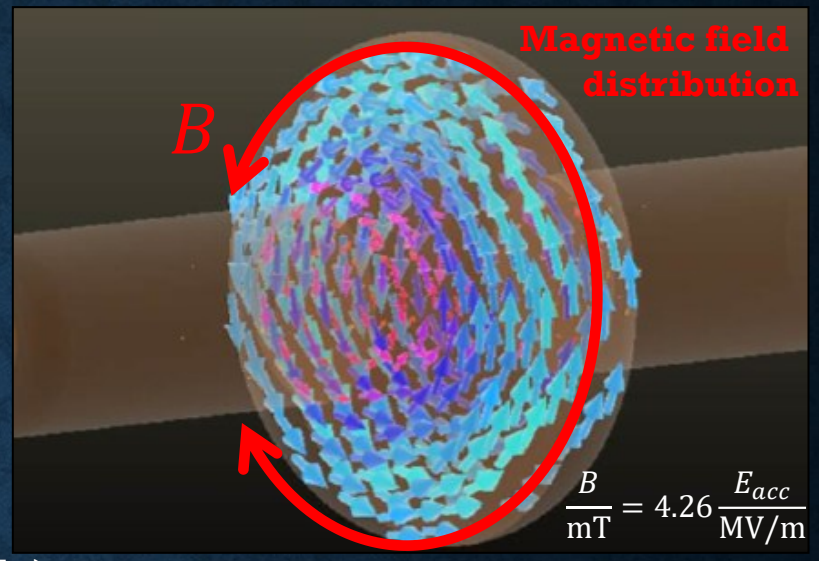




Single-cell Nb accelerator cavity (1.3 GHz)





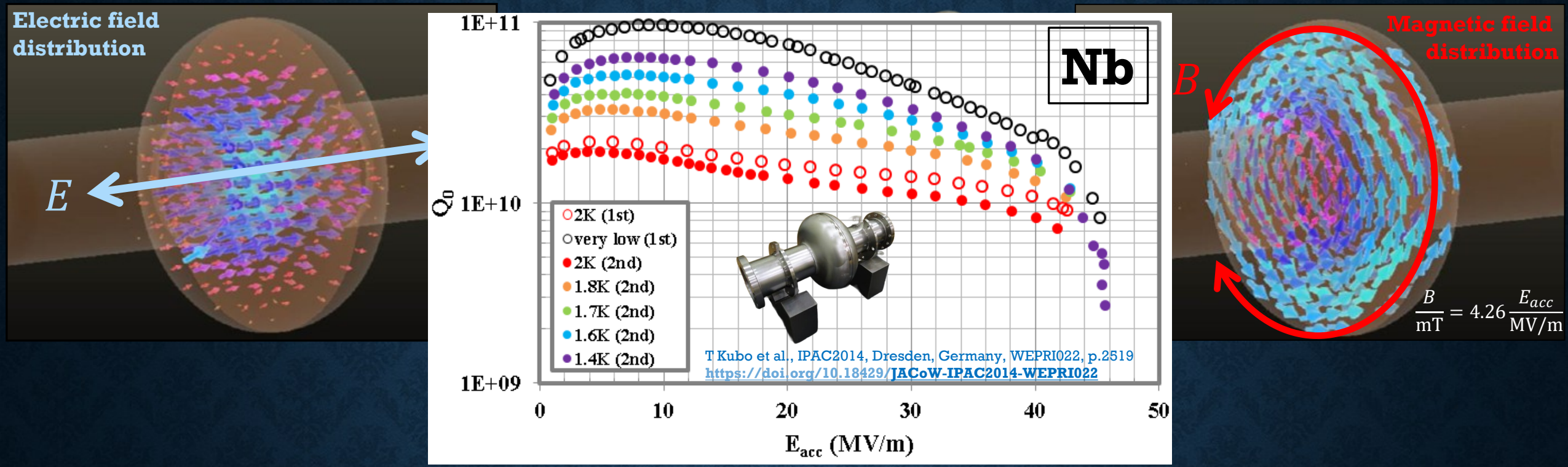


Single-cell Nb accelerator cavity (1.3 GHz)

There are two key performance metrics

(1) Maximum field (stability limit of the vortex-free state)

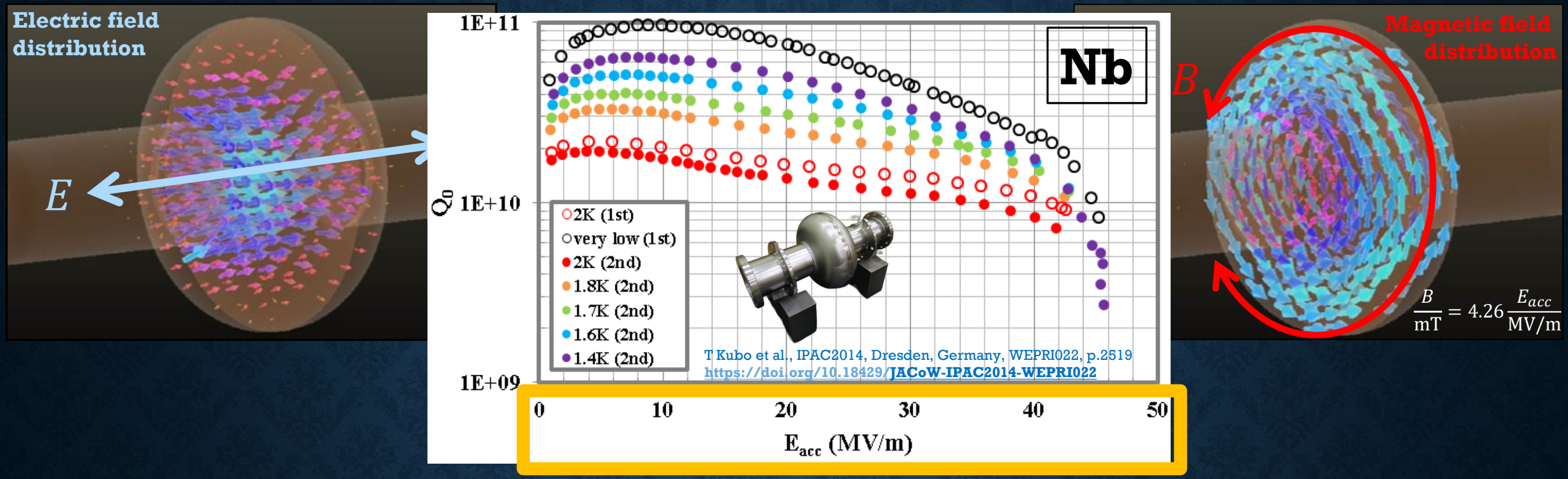
(2) Quality factor Q_0 (smallness of dissipation)



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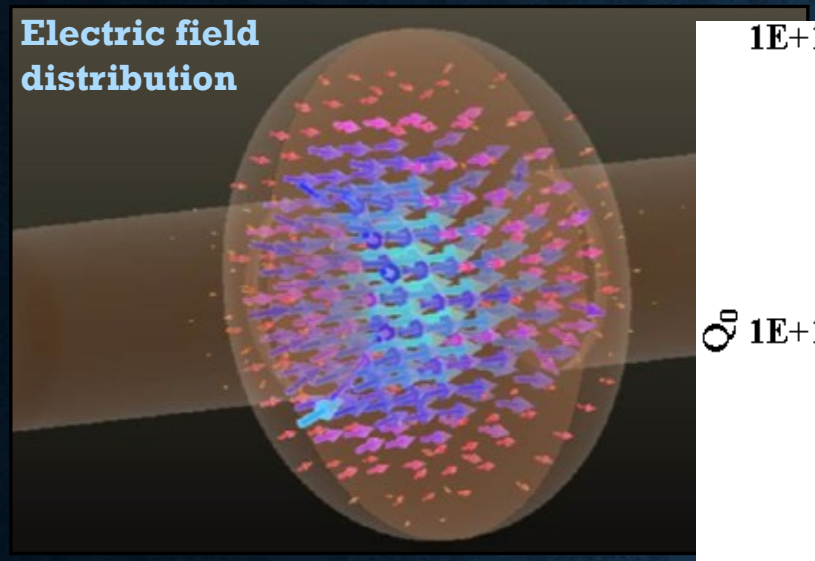
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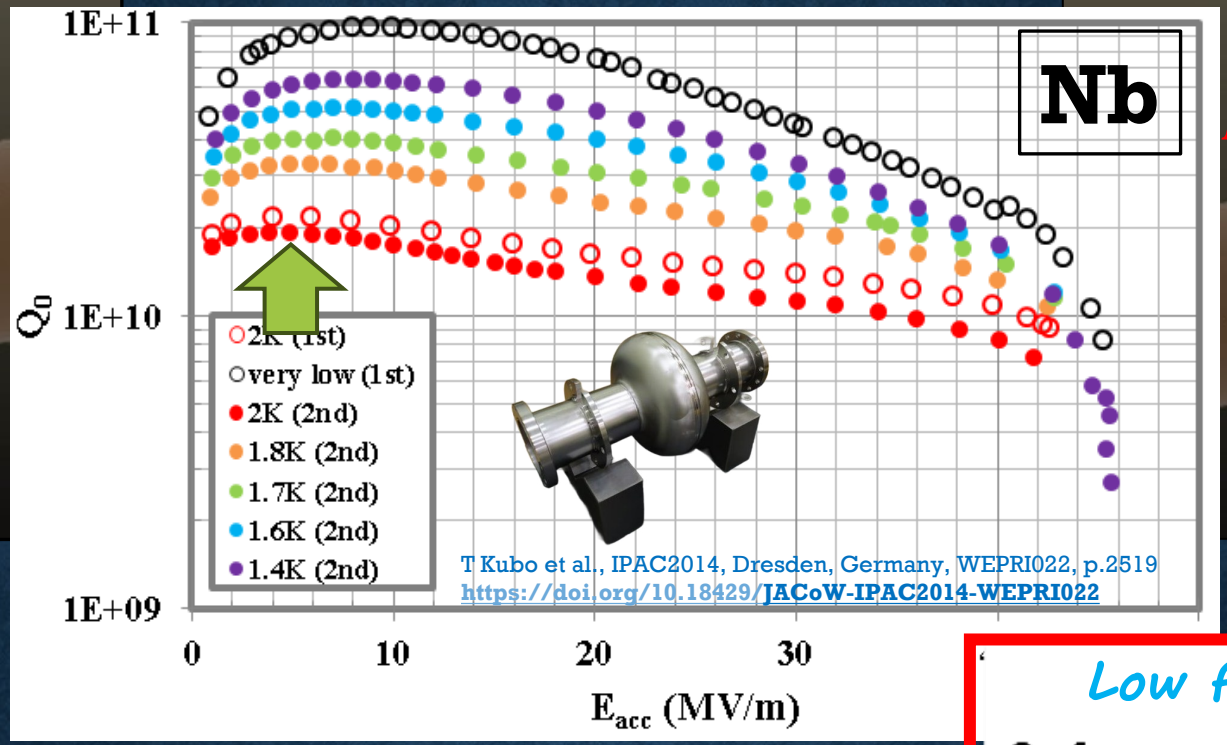


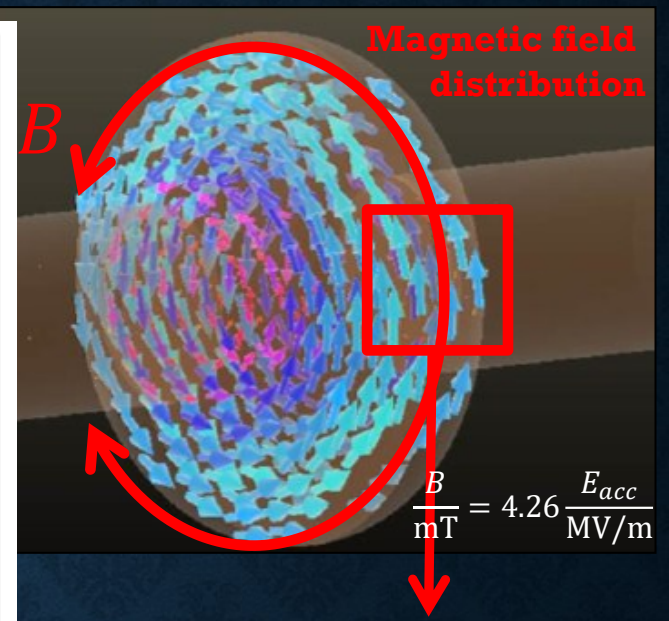
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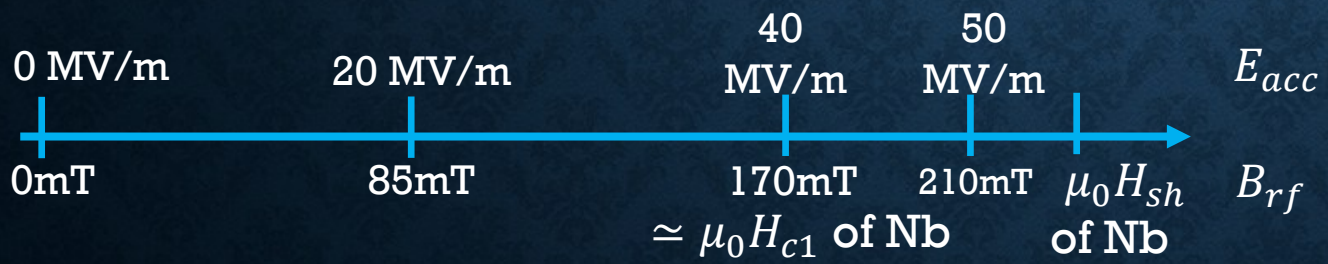
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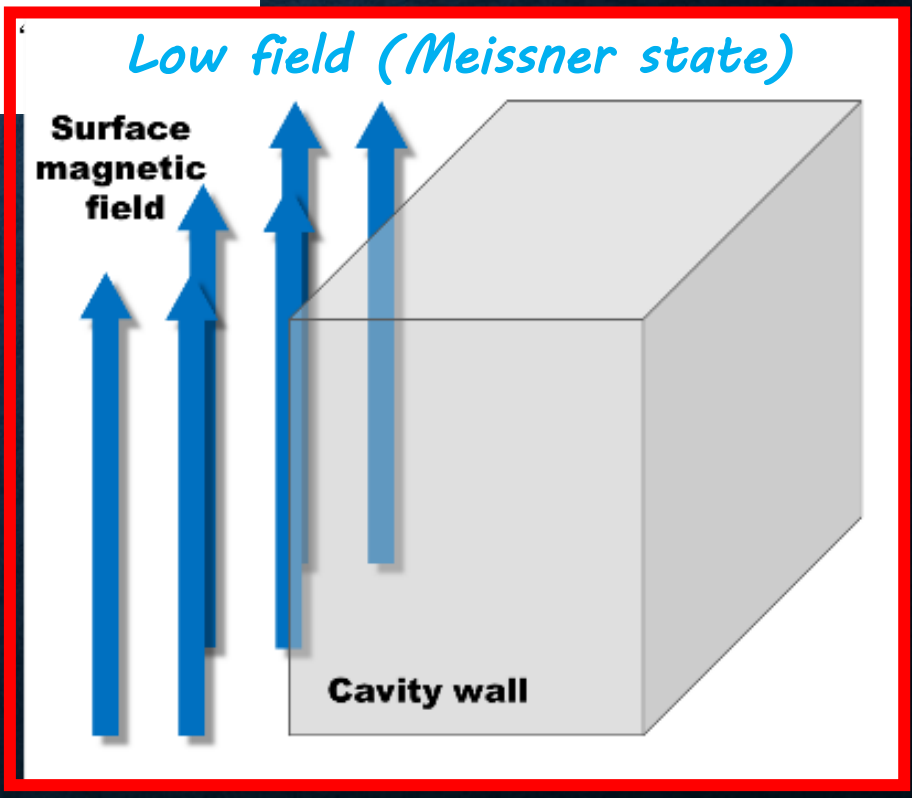


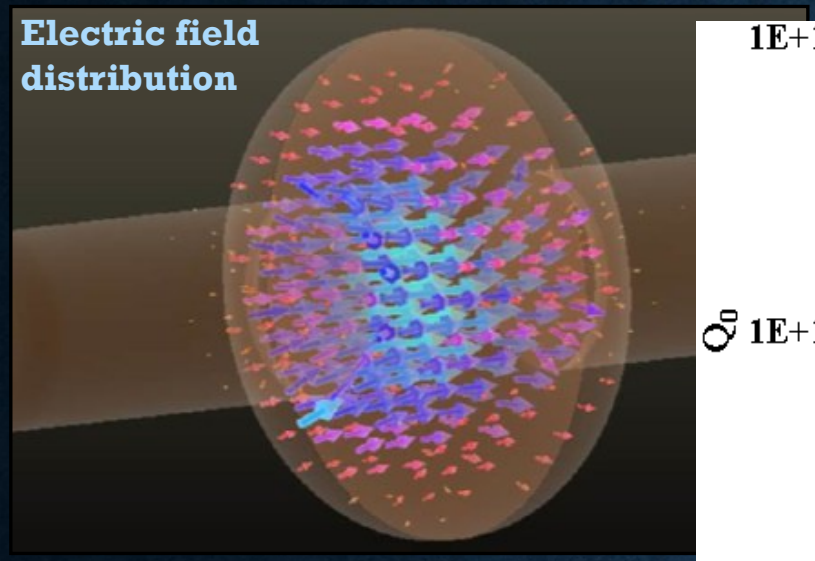


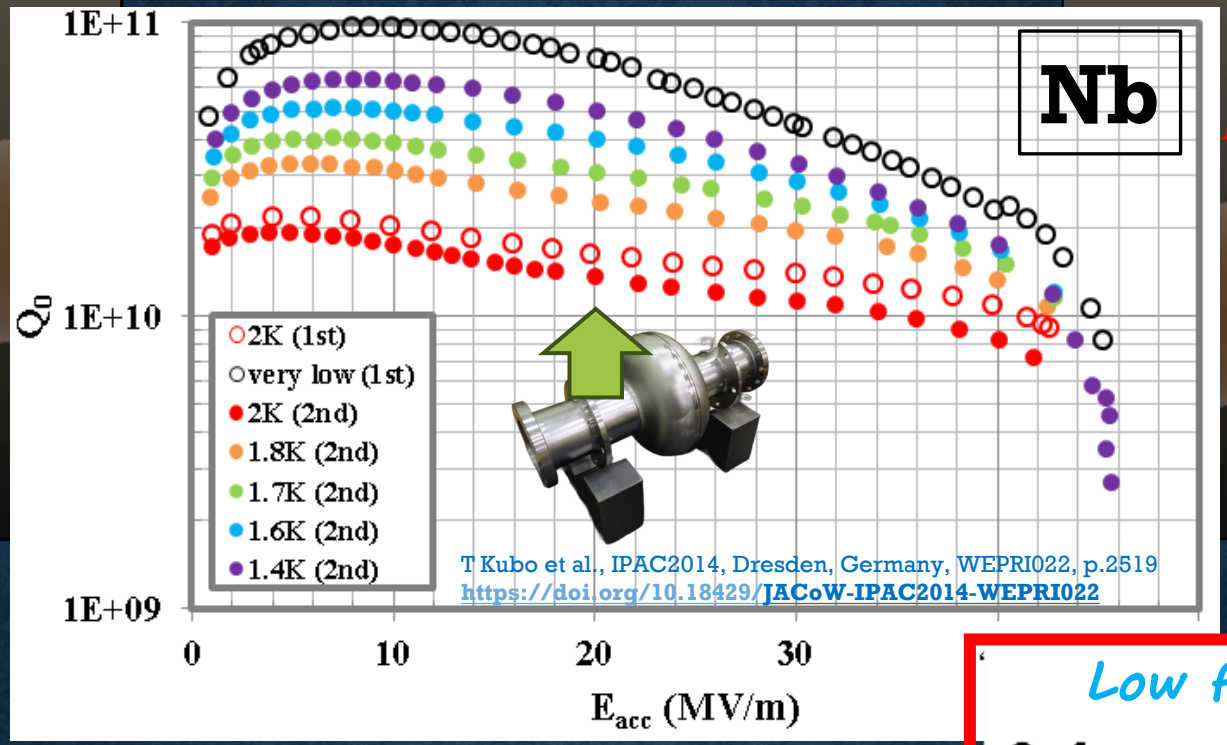
Note: Cavities are operated under the **Meissner state**

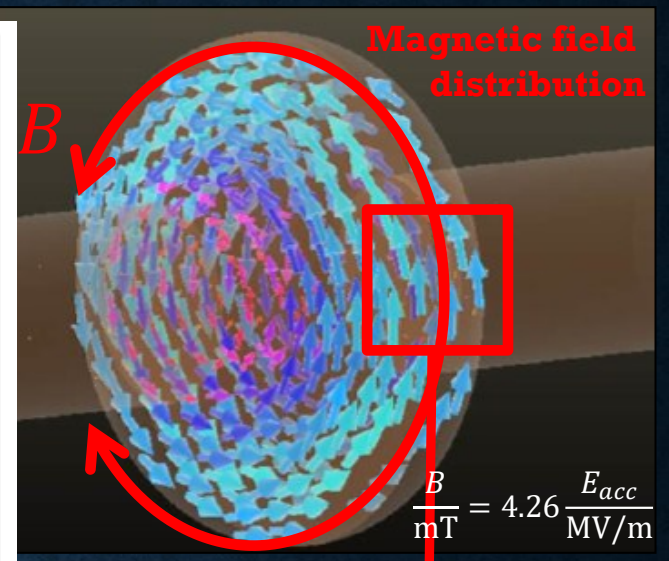


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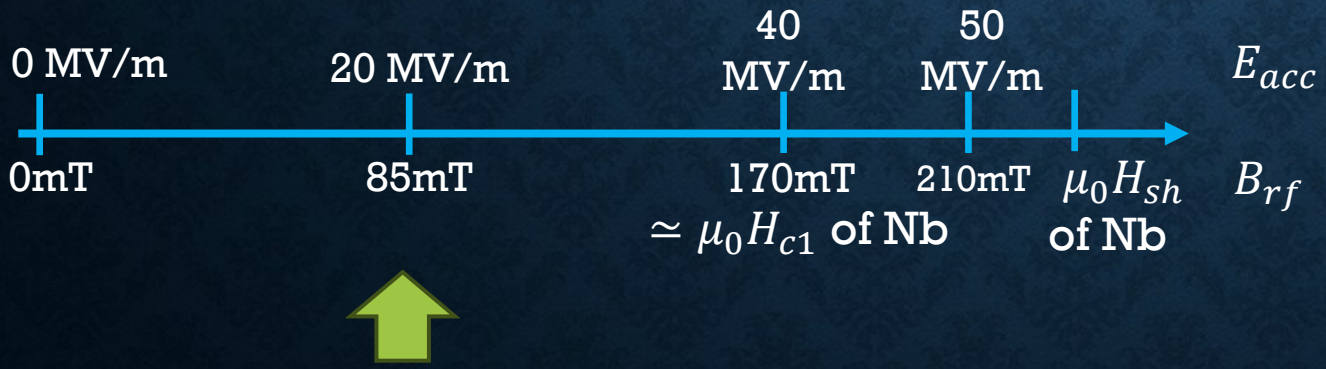




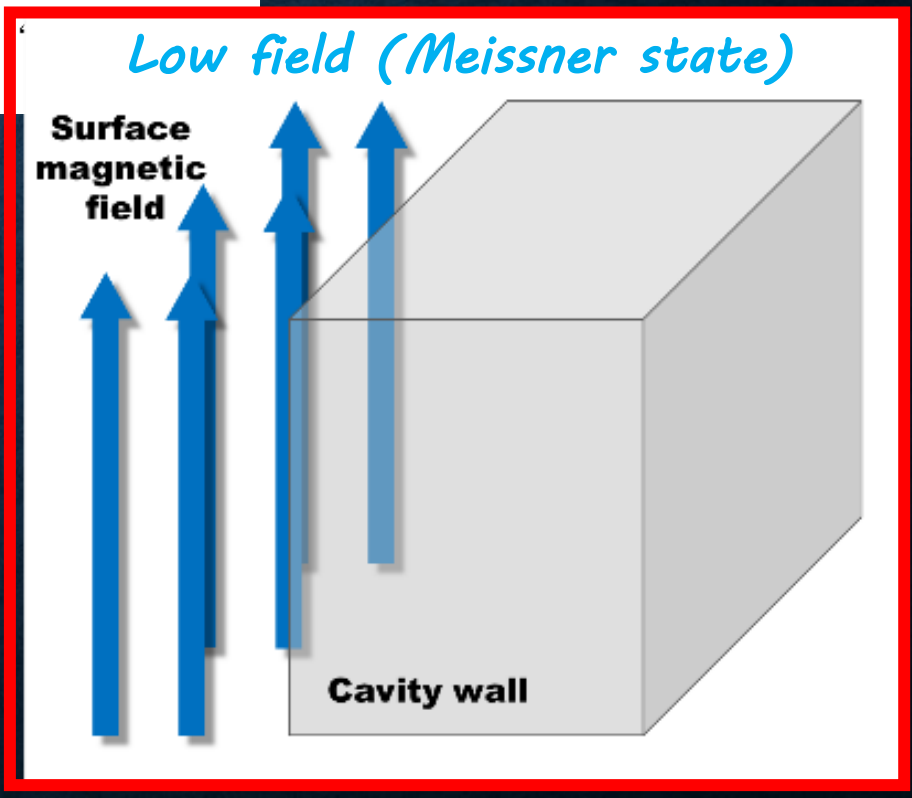


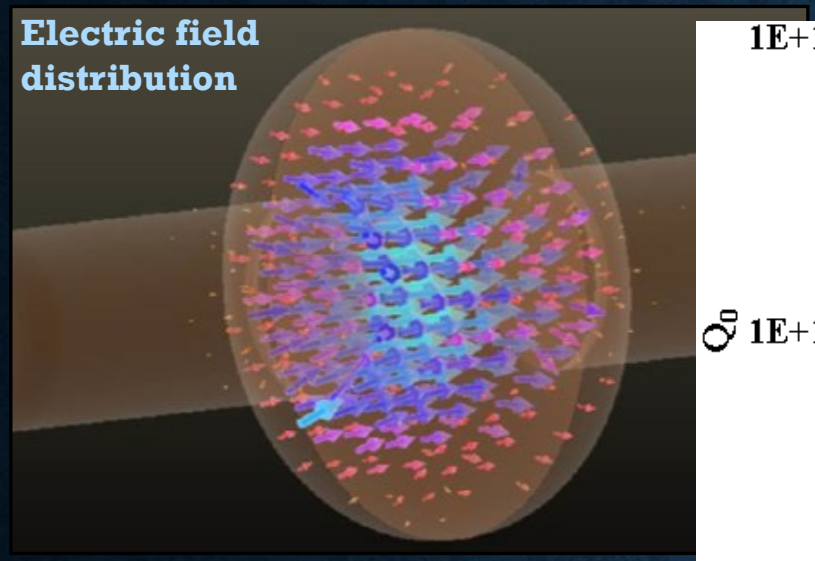


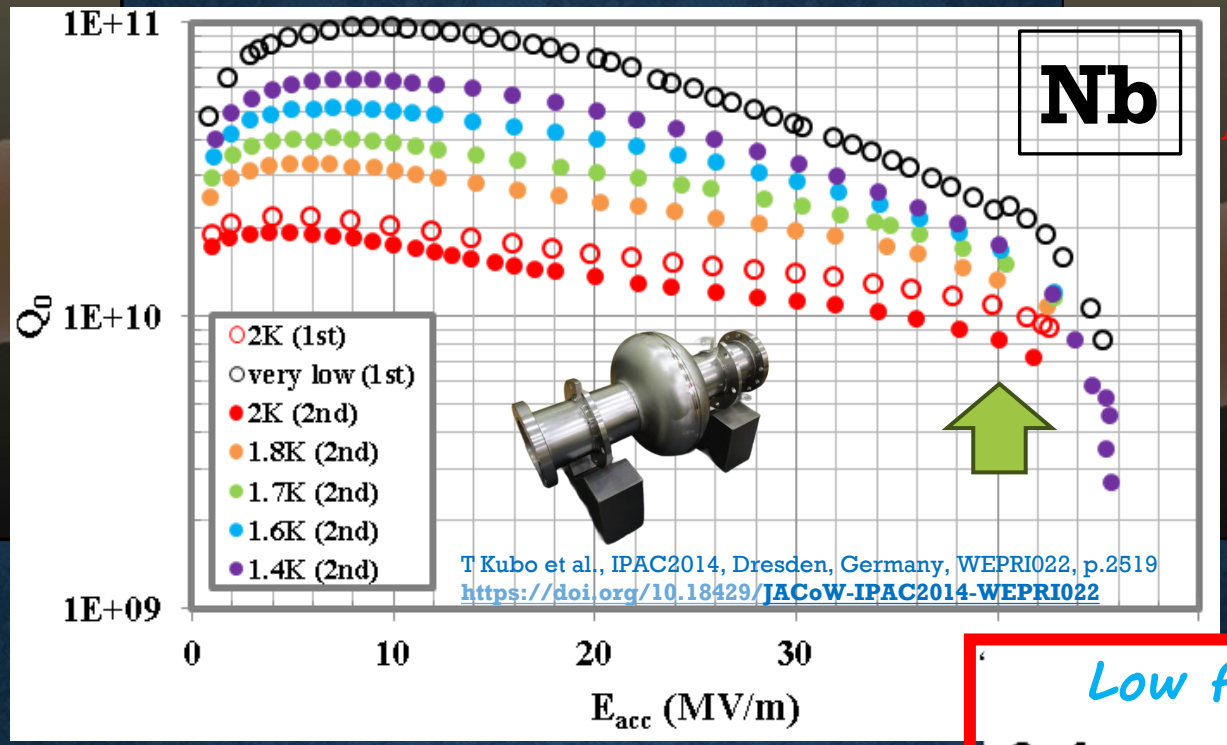
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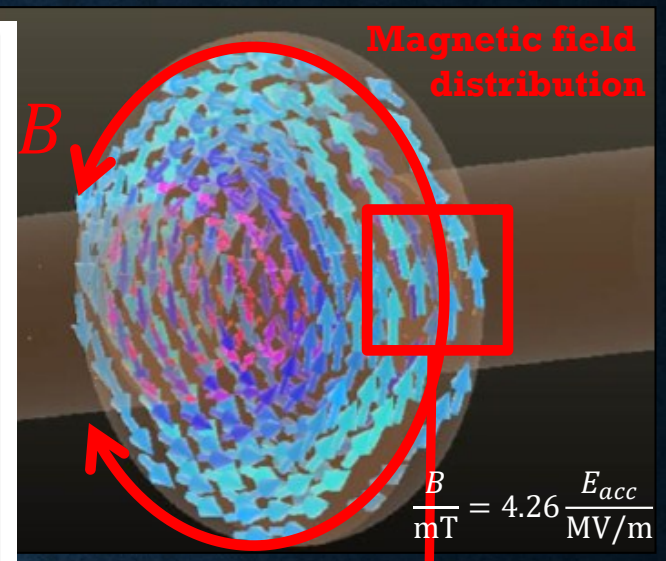


(⌐■_■) good

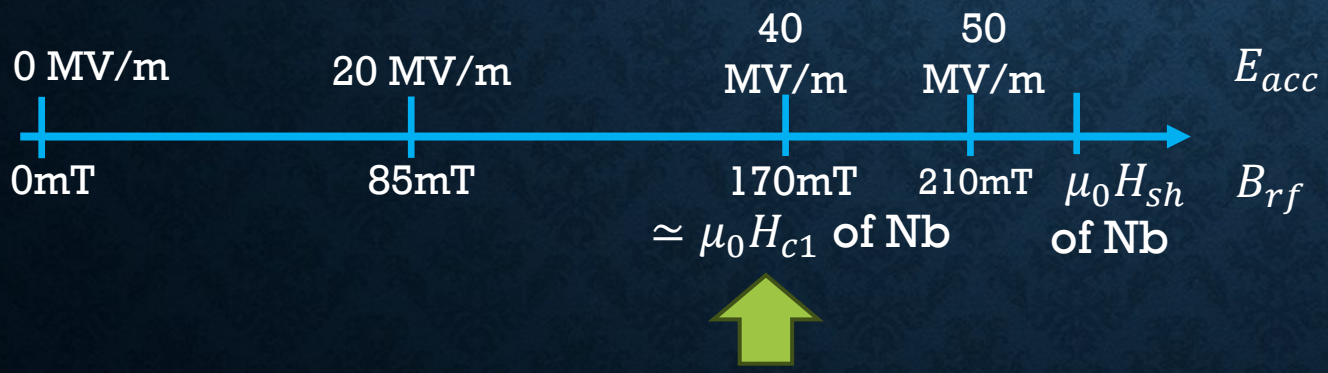




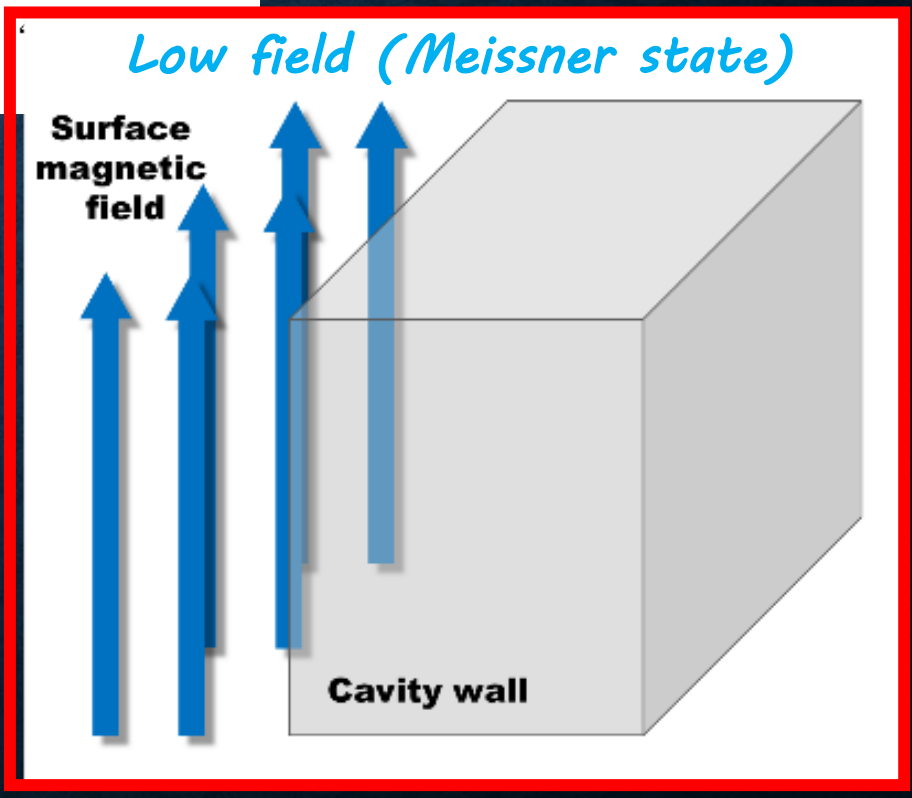


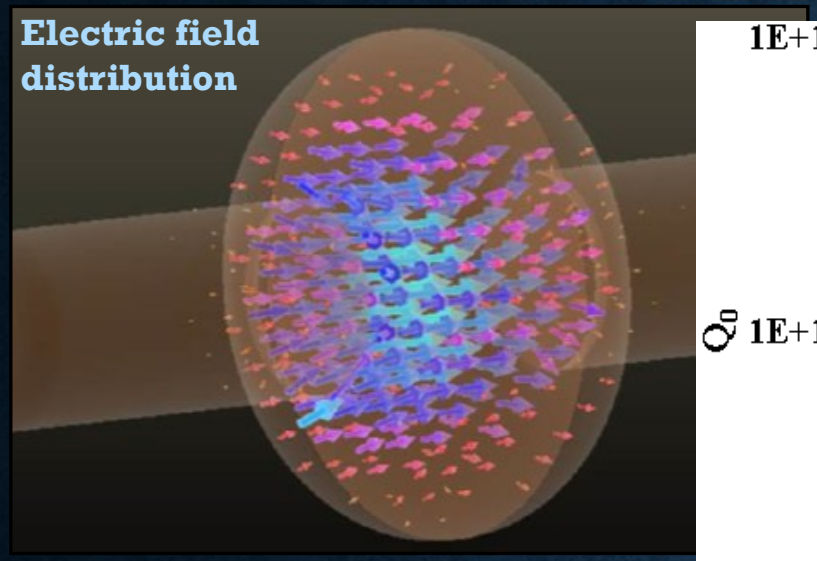


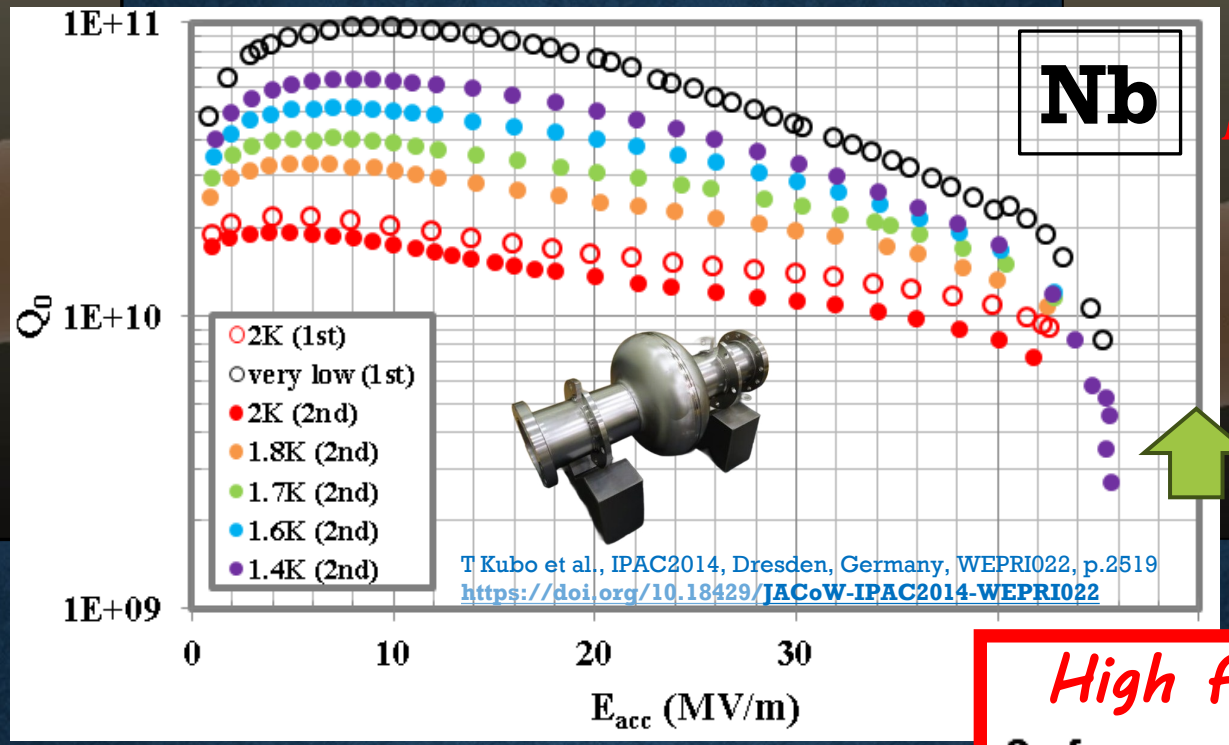
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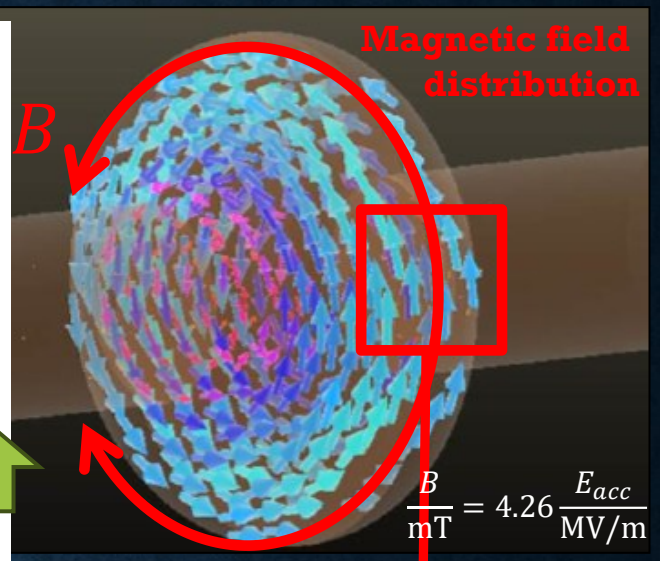
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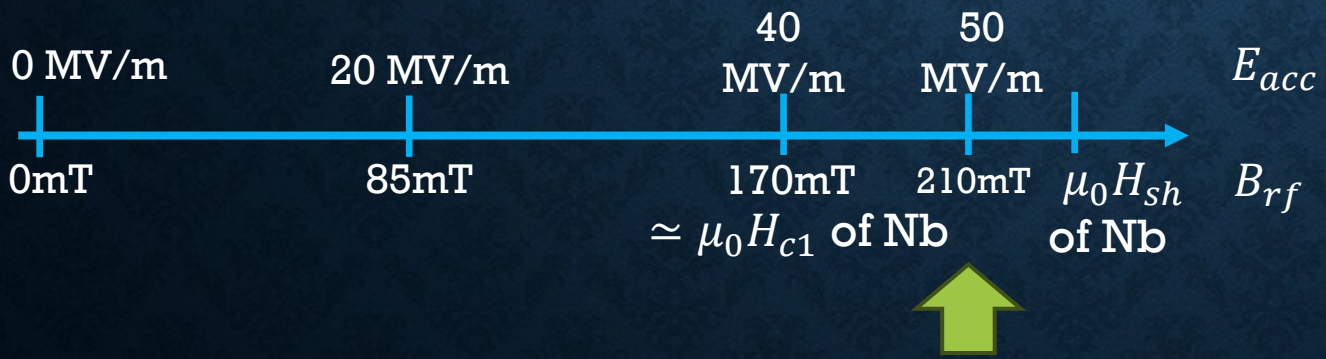




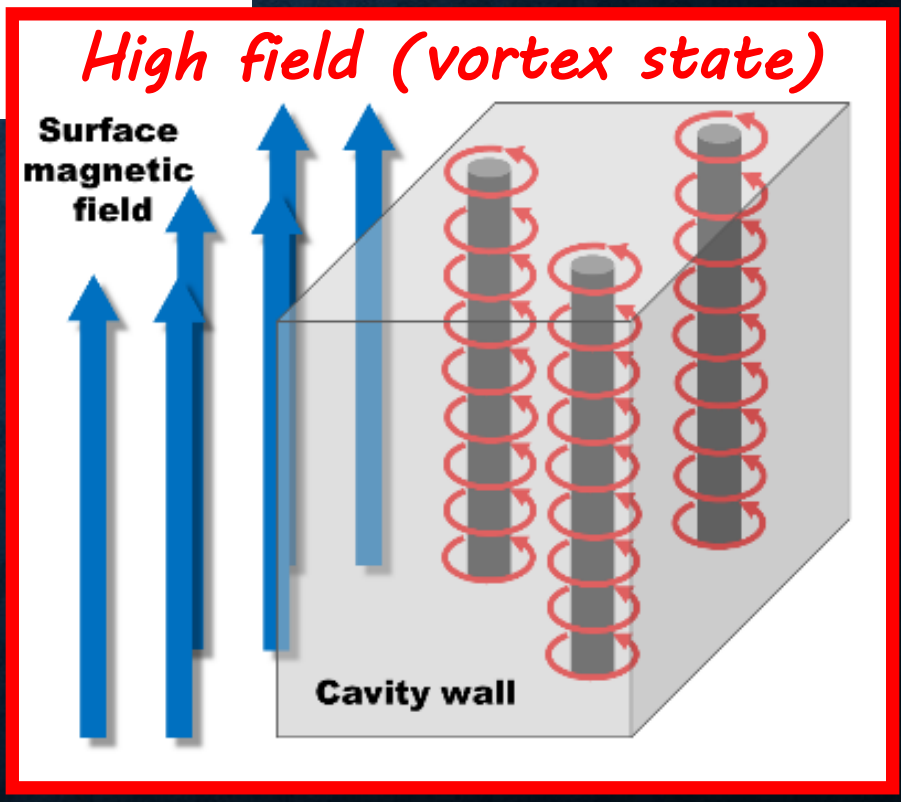
T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022



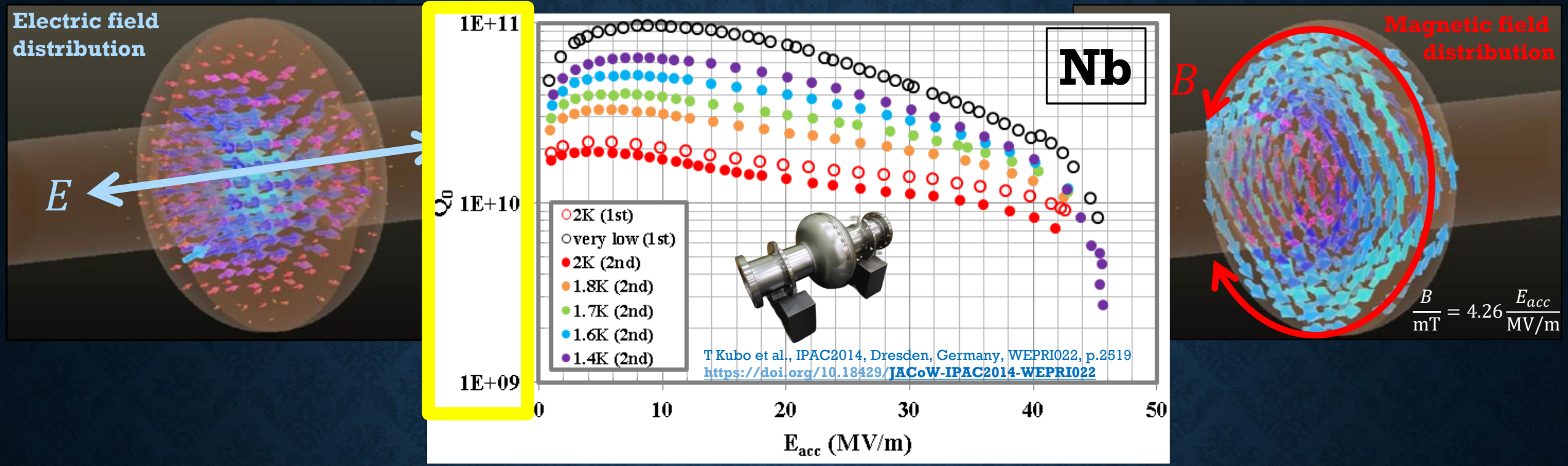
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＼(^o^)／オワタ
I'm doomed



UHM (Nov. 5, 2026)



There are two key performance metrics

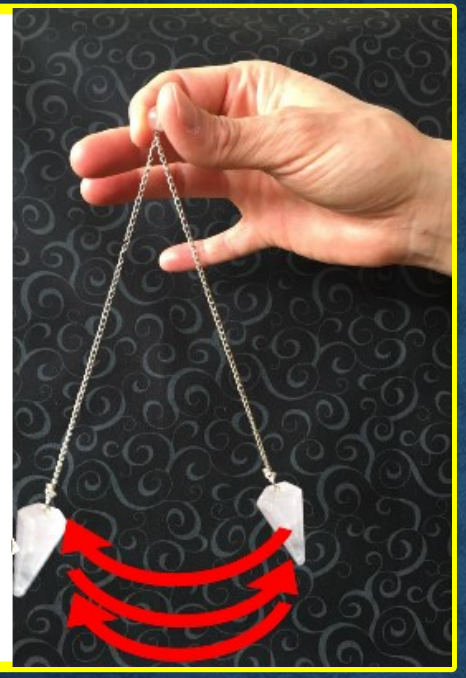
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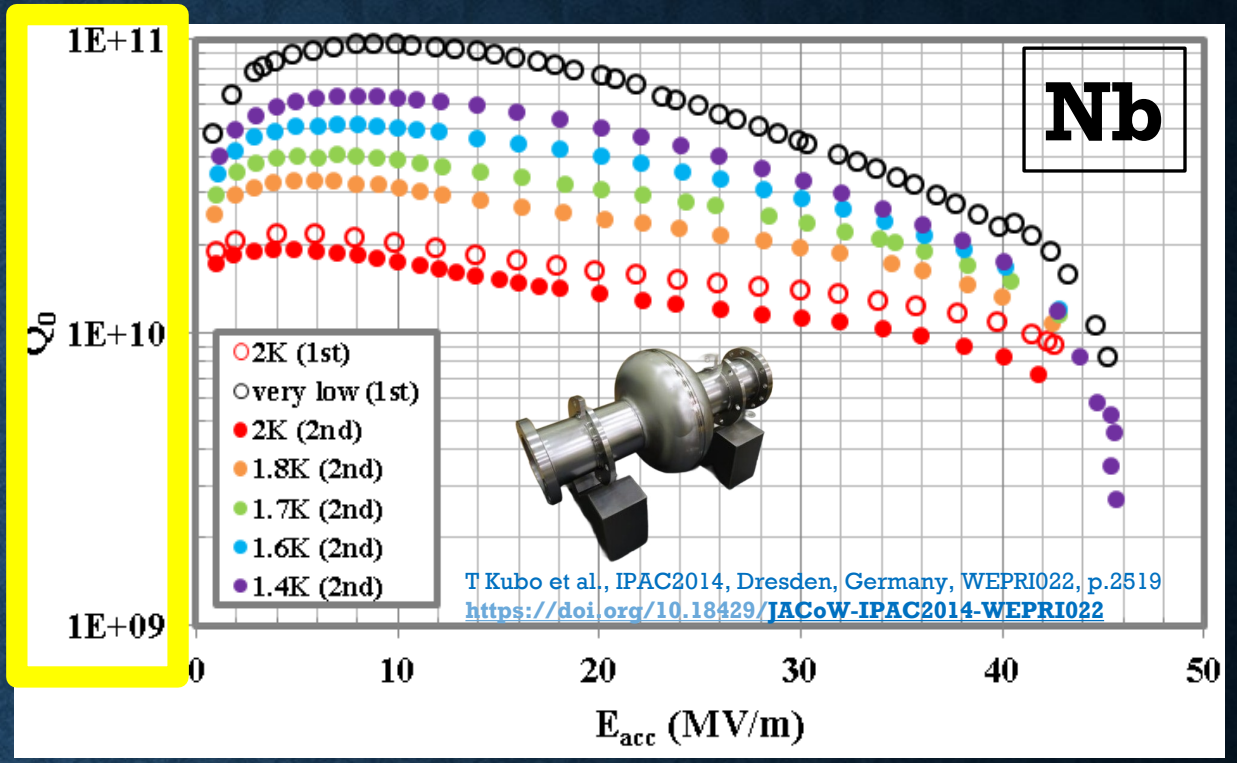
(2) Quality factor Q_0 (smallness of dissipation)

$$Q_0 = \frac{\omega U}{P} \sim \frac{U}{P_1} \propto \frac{1}{R_s}$$

P_1 : dissipation per RF cycle

Q_0 : how many cycles it takes for the stored energy to completely dissipate





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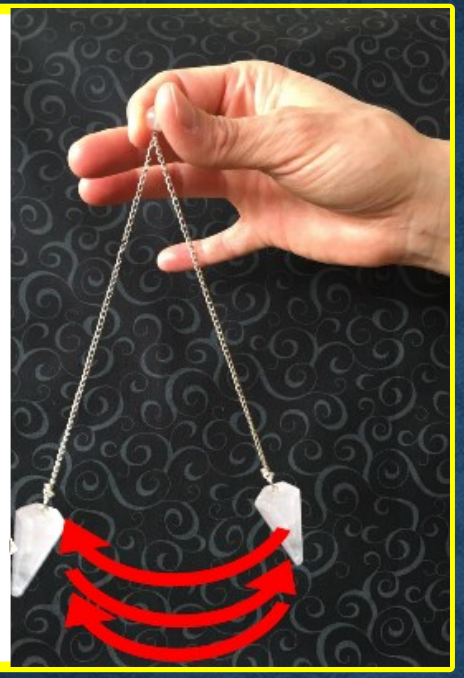
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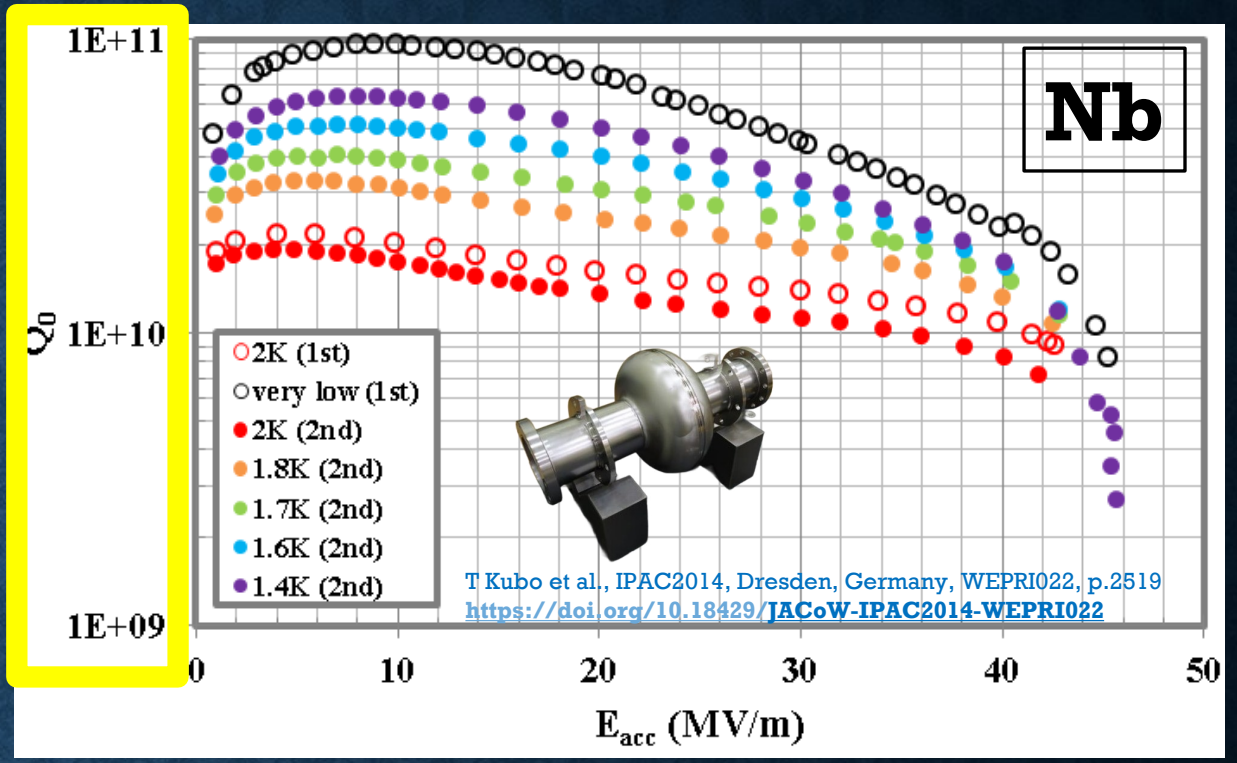
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What are the causes of dissipation?





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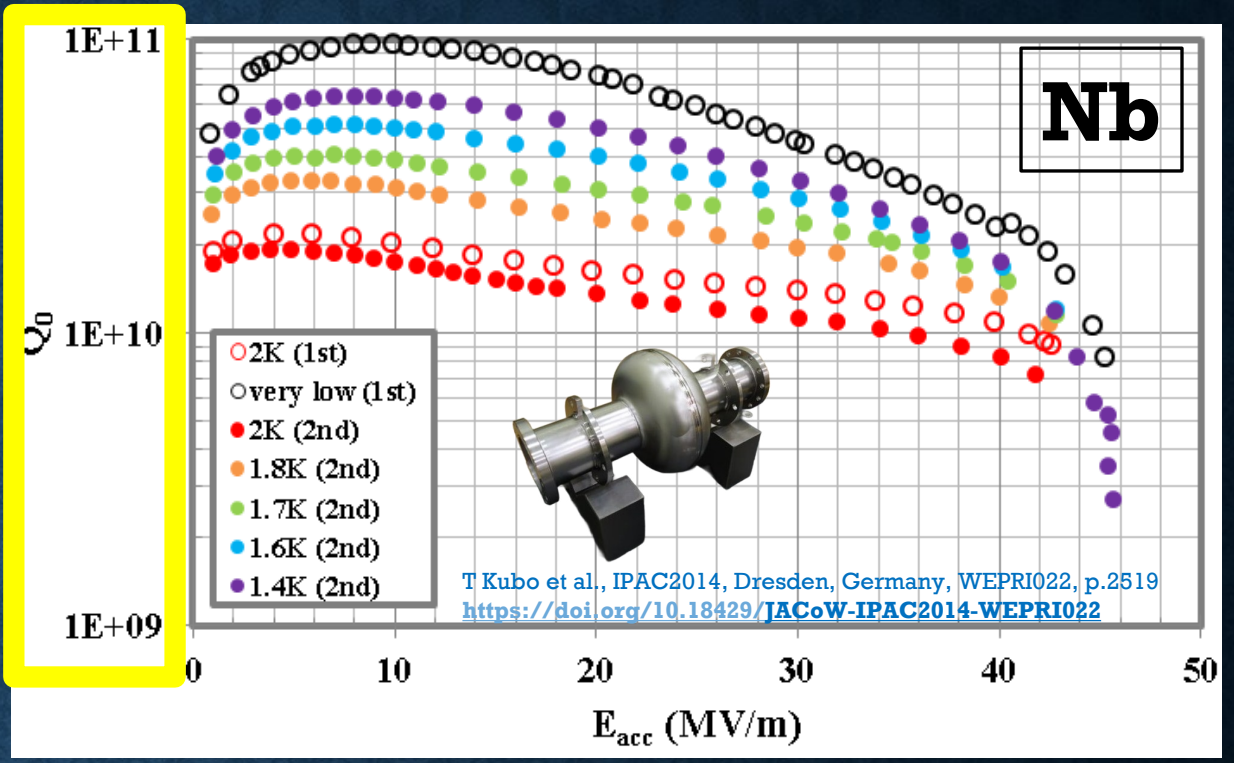
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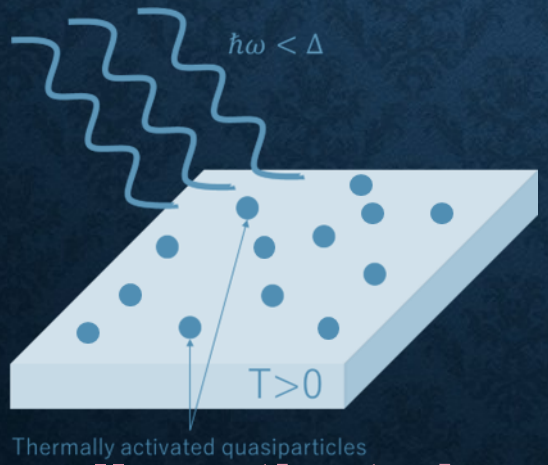




T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

There are two different temperature regimes

At a higher T ($T/T_c \gtrsim 0.2$)



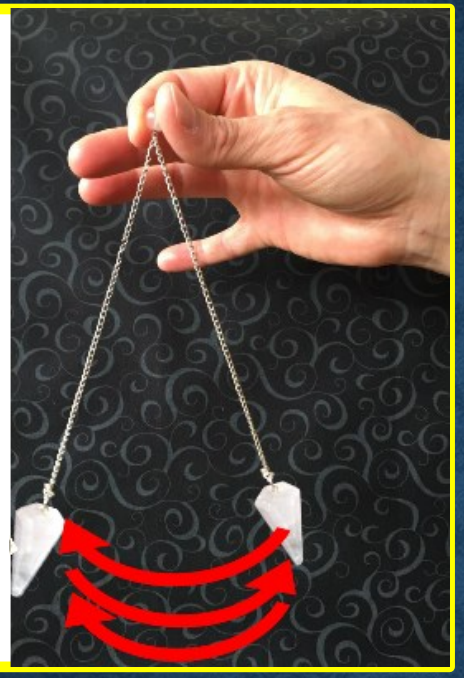
thermally activated quasiparticles

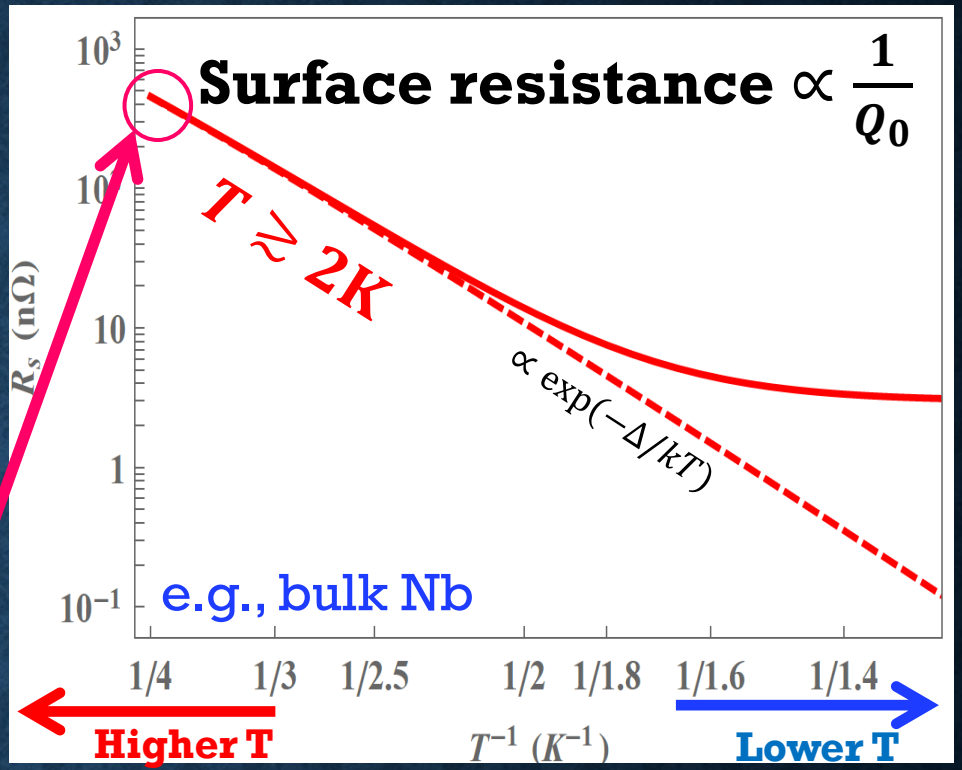
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P_1 : dissipation per RF cycle

Q_0 : how many cycles it takes for the stored energy to completely dissipate

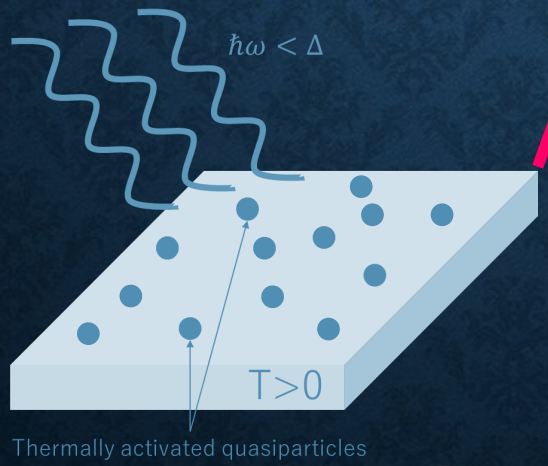
What are the causes of dissipation?





There are two different temperature regimes

At a higher T ($T/T_c \gtrsim 0.2$)



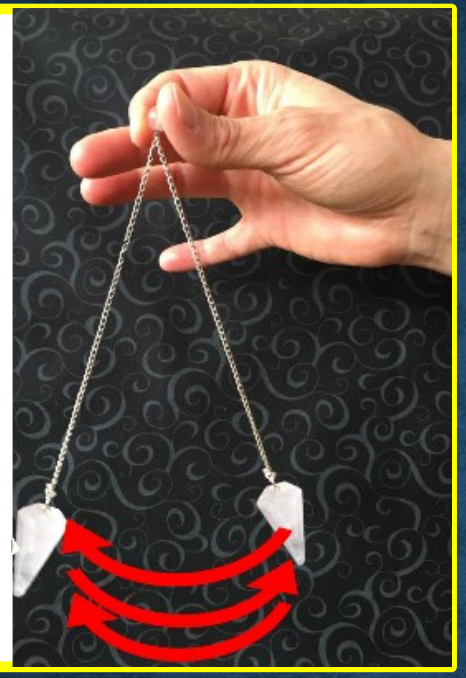
The contribution from **thermally activated quasiparticles** decreases exponentially [$\propto \exp(-\Delta/kT)$].

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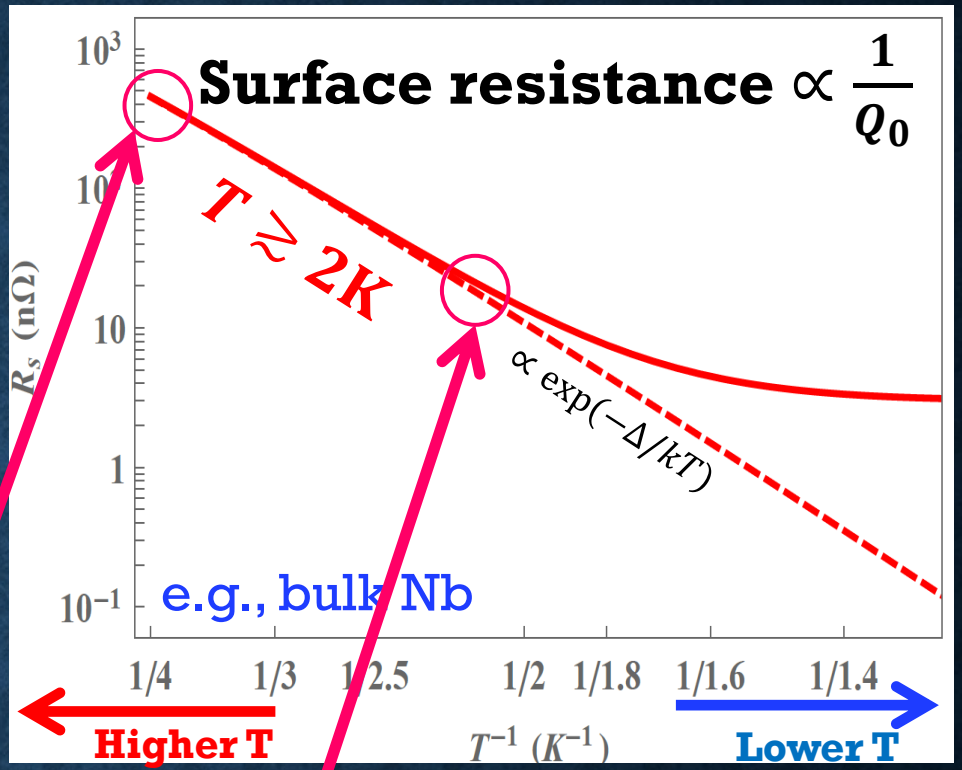
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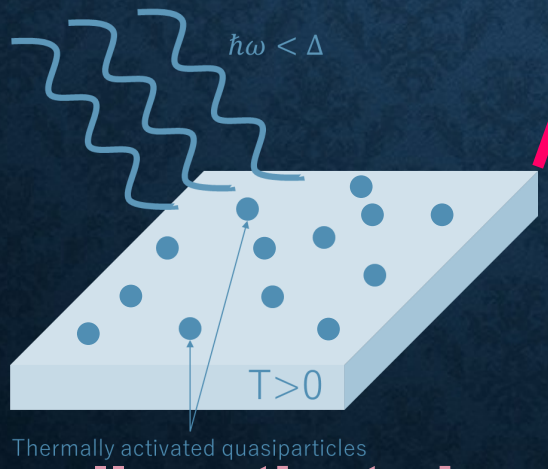
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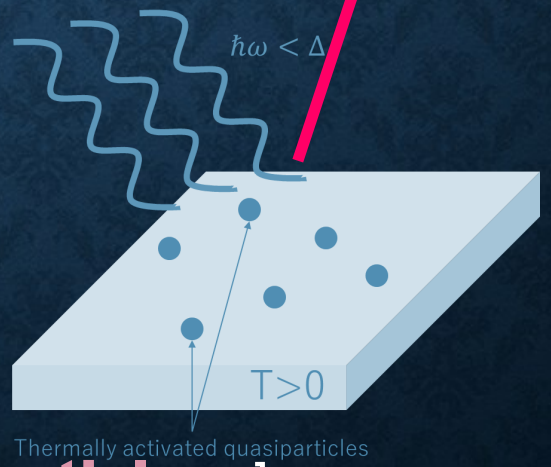
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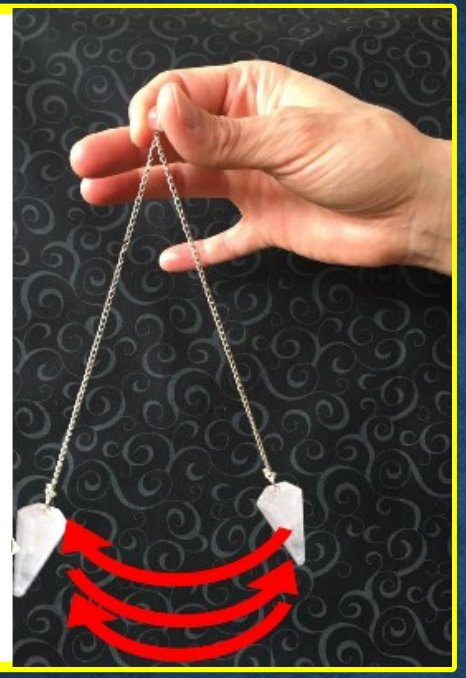
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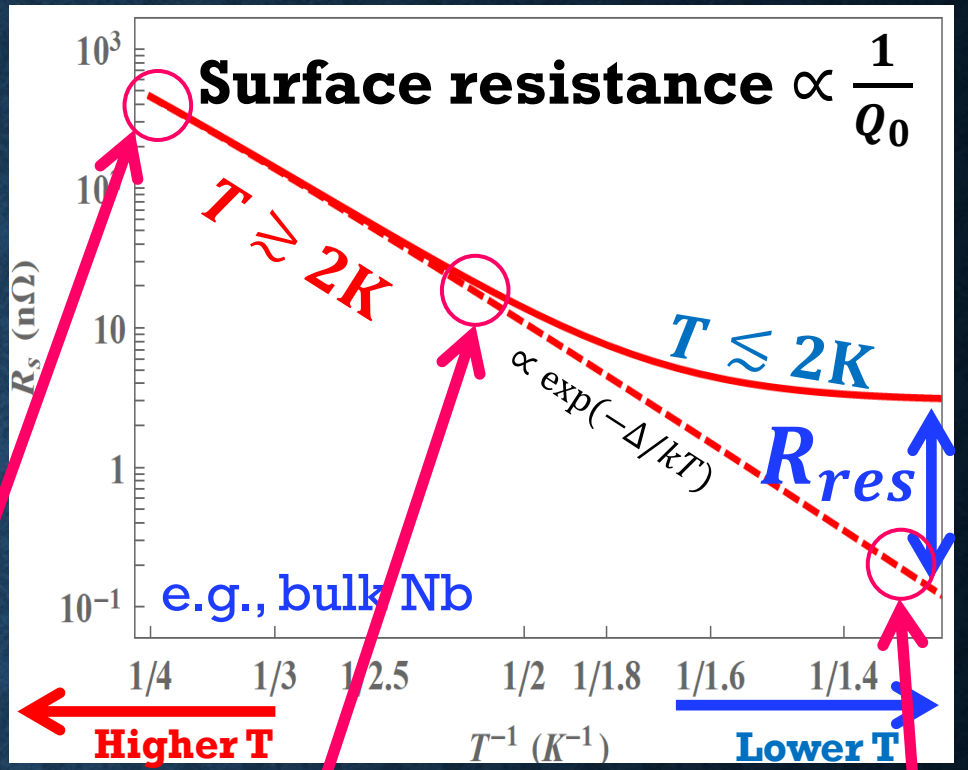
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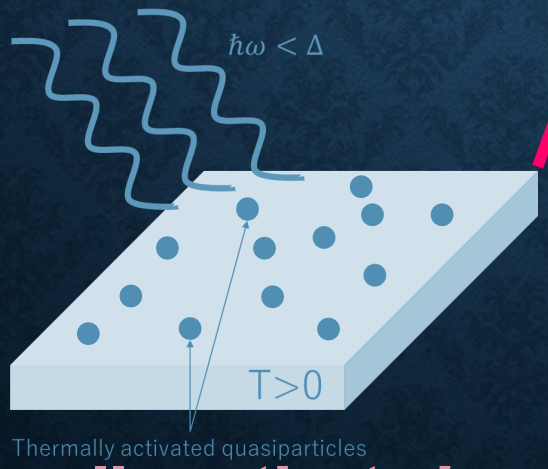
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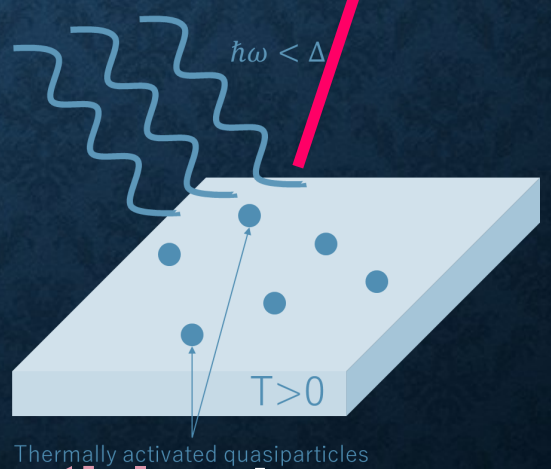
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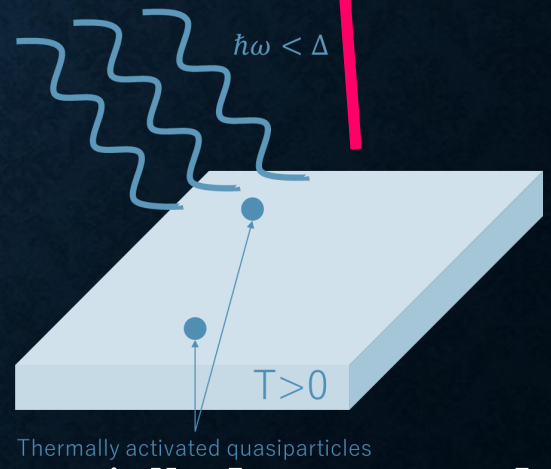


There are two different temperature regimes









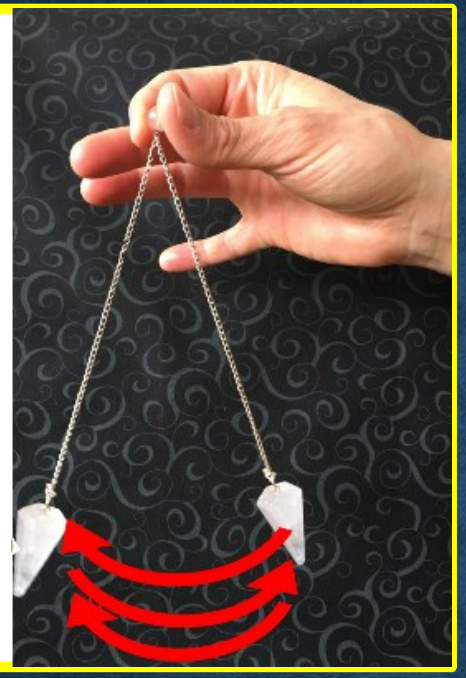
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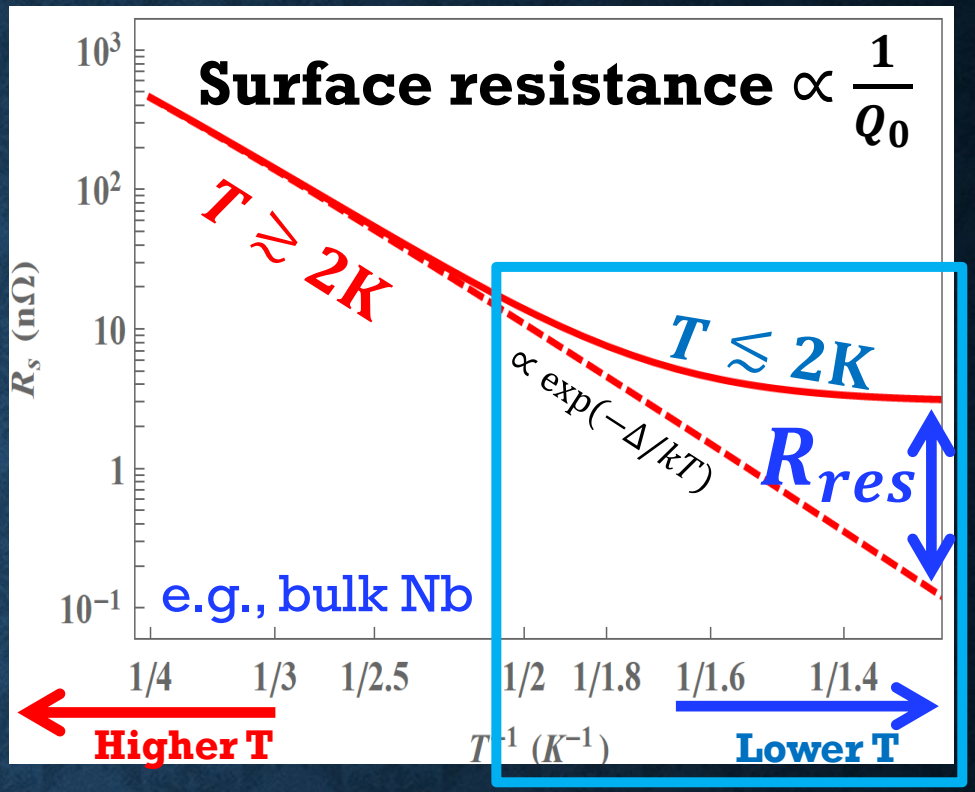
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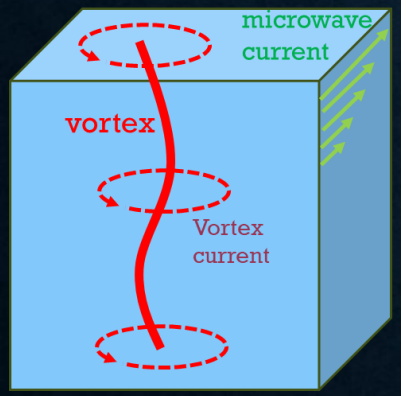
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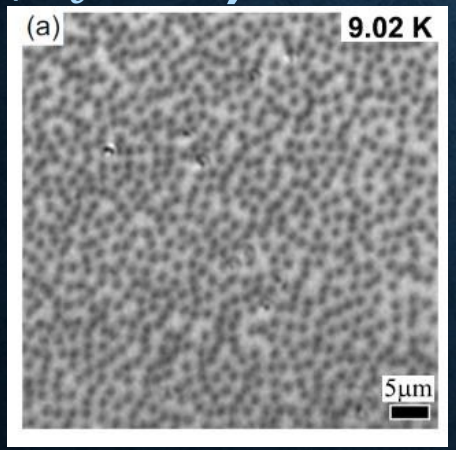




There are two different temperature regimes

At a lower T ($T/T_c \lesssim 0.2$)





Trapped ambient vortex

Microwave-driven vortex dynamics cause dissipation, which is only weakly dependent on temperature, leading to an elevated surface resistance: R_{res}

Figures adapted from Ooi et al., PRB **111**, 094519 (2025)
© American Physical Society.

Videos of vortex dynamics in **cavity-grade Nb (RRR>300)** can be found in Ooi et al., PRB **111**, 094519 (2025)

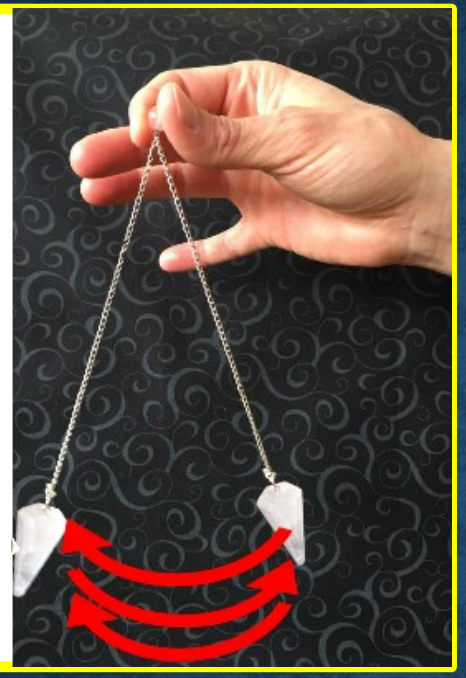
See also,
A. Gurevich and G. Ciovati, PRB **87**, 054502 (2013)
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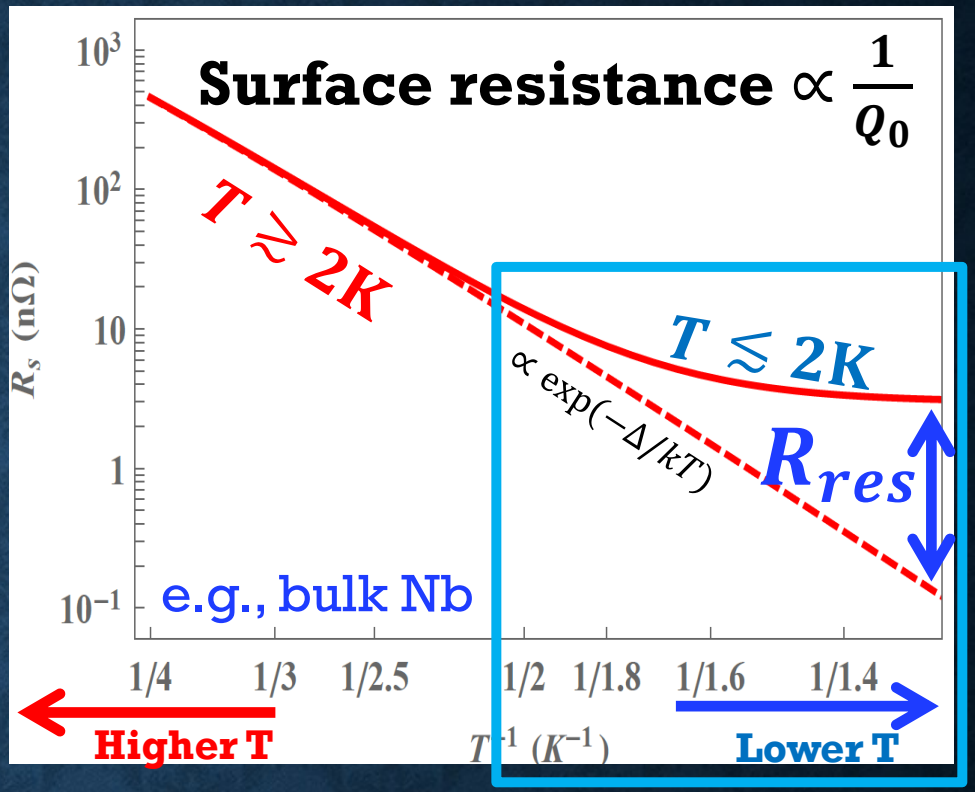
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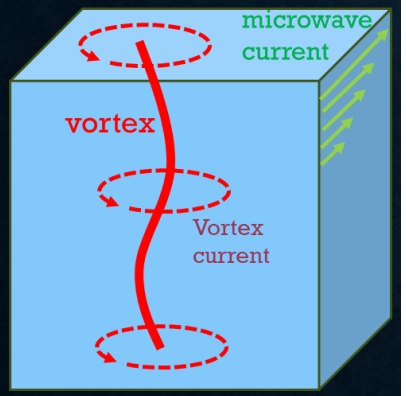
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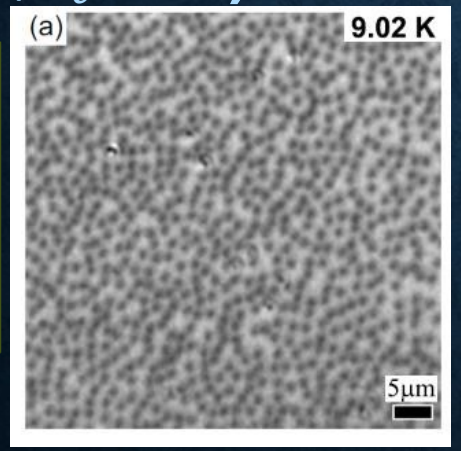




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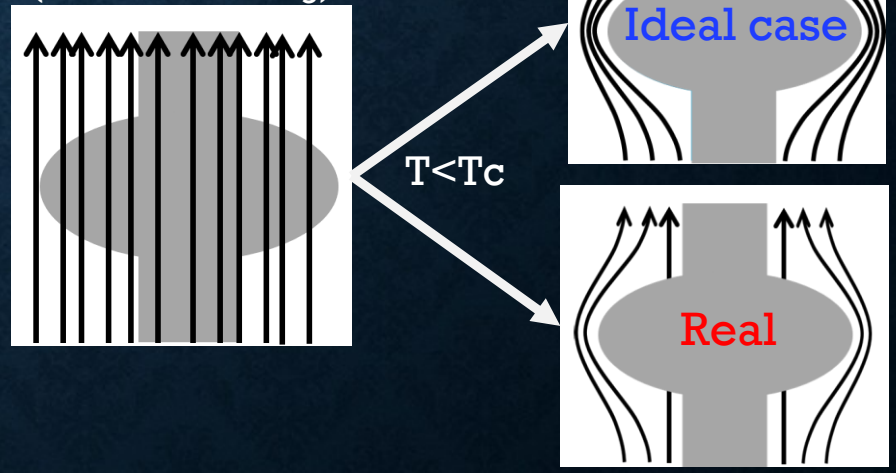


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Trapped ambient vortex

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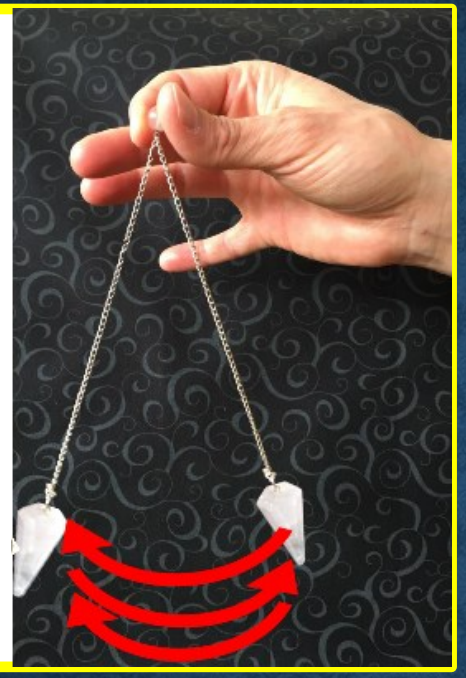
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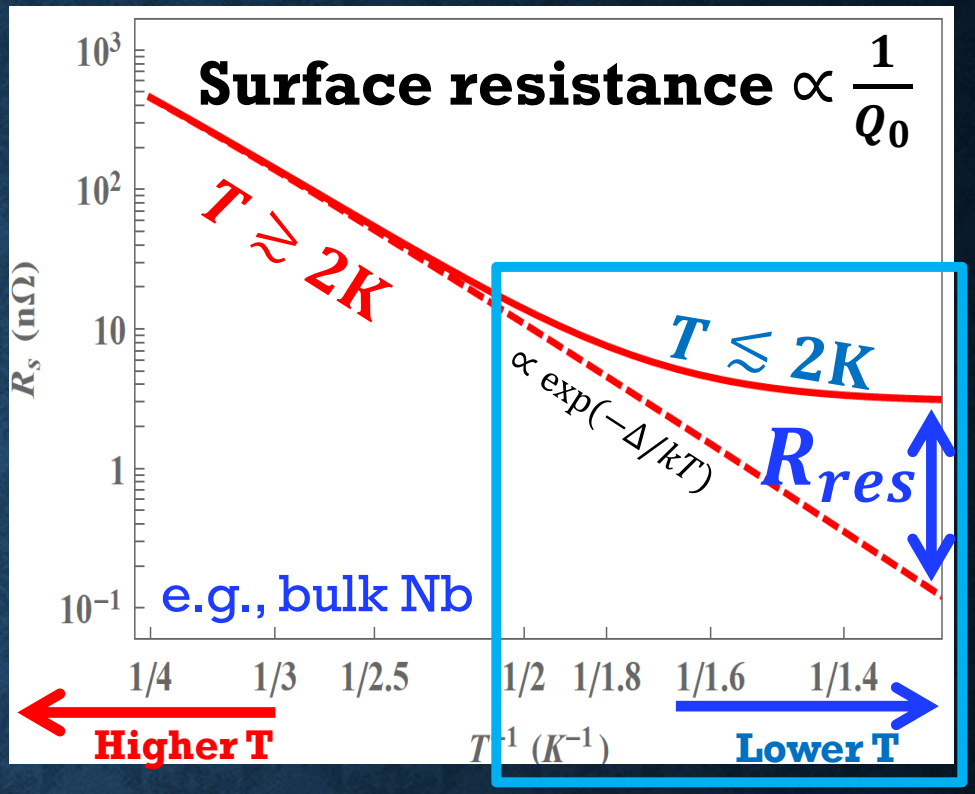
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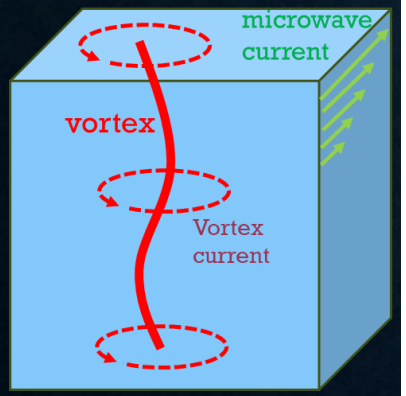
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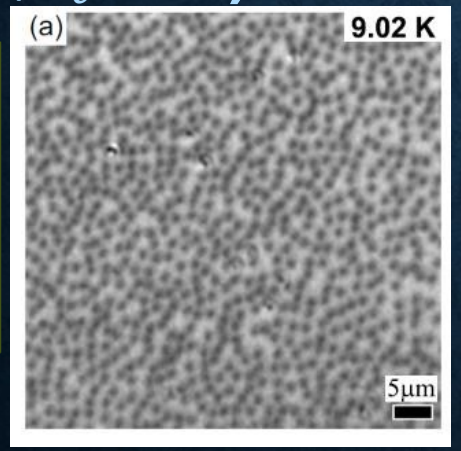




There are two different temperature regimes

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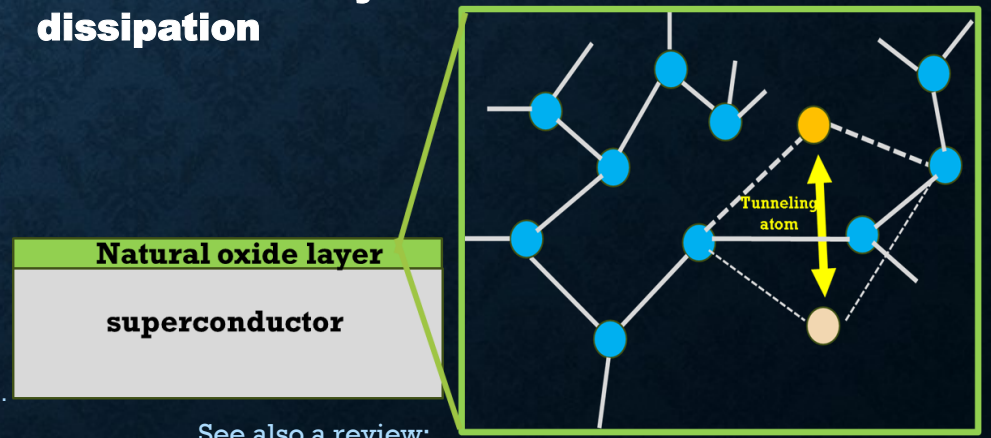




Trapped ambient vortex

Microwave-driven vortex dynamics cause dissipation, which is only weakly dependent on temperature, leading to an elevated surface resistance: R_{res}

Two-level system (TLS) defects on the natural oxide layer also contributes to the dissipation



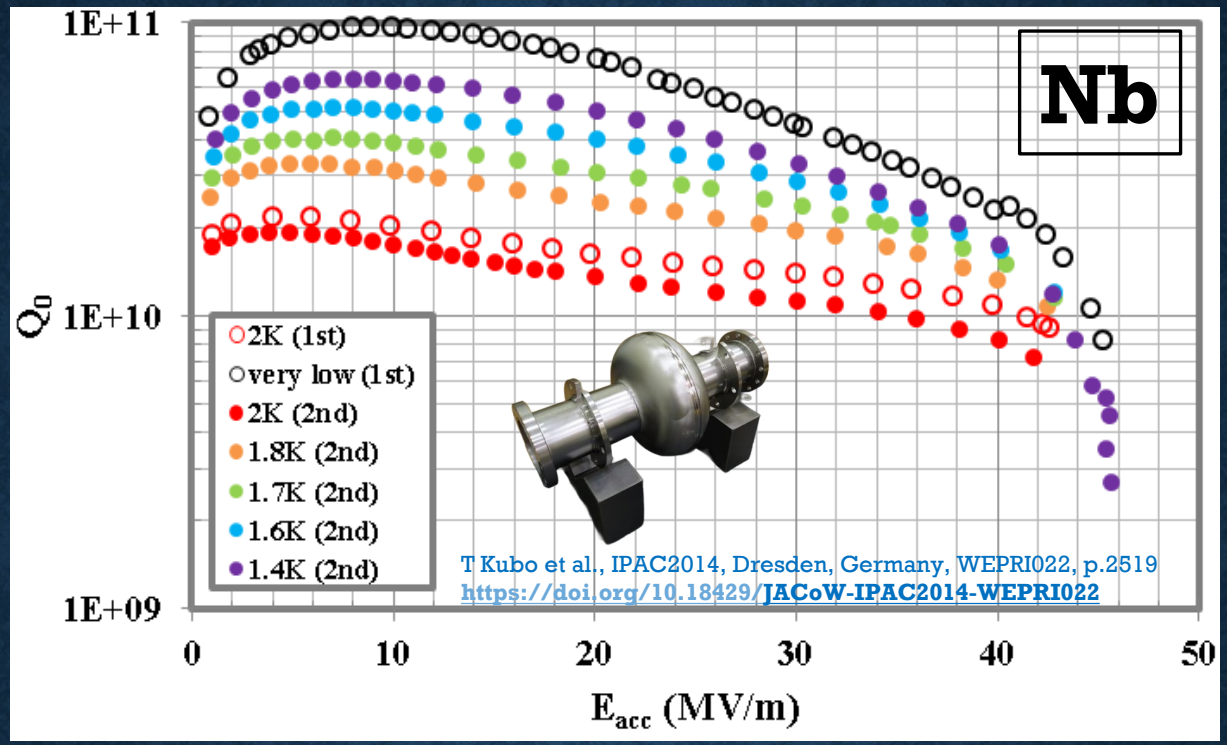
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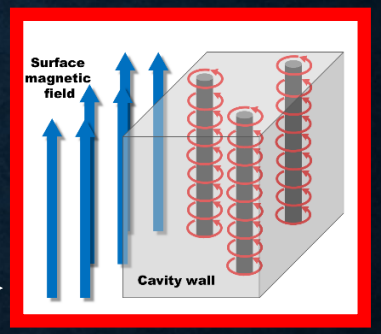
See also a review:
Müller et al Rep. Prog. Phys. **82**, 124501 (2019)

Summary of this part

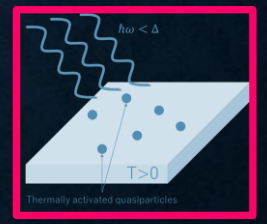


There are two key performance metrics

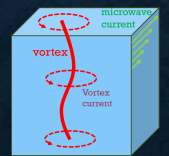
(1) Maximum field (stability of the vortex-free state)

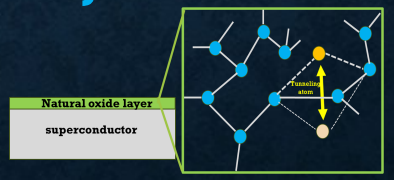


(2) Quality factor Q_0 (smallness of dissipation)



- **Quasiparticles (at $T \gtrsim 2$ K for Nb)**
- **Trapped vortex, TLS etc. (at $T \lesssim 2$ K for Nb)**





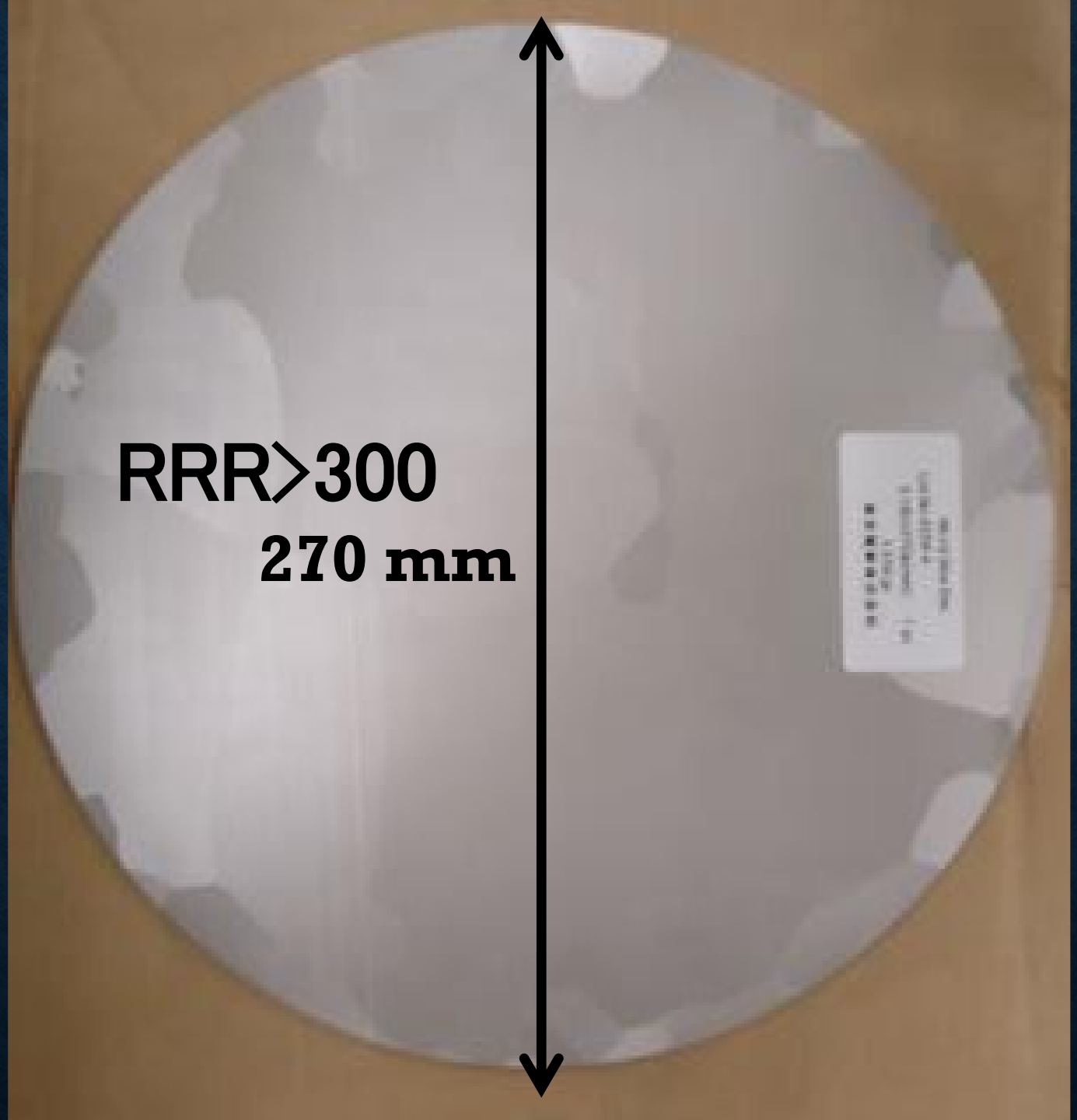


Nb cavity technology for accelerators:

Classical ILC recipe as a representative example

ILC recipe

- Materials
- Fabrication
- Electropolishing(1) (100-200μm)
- Annealing (750-900°C, 2-3hours)
- Electropolishing(2) (several μm)
- High pressure rinsing
- Assembling
- Low-temperature baking (100-140°C, 48hours)
- Test cavity



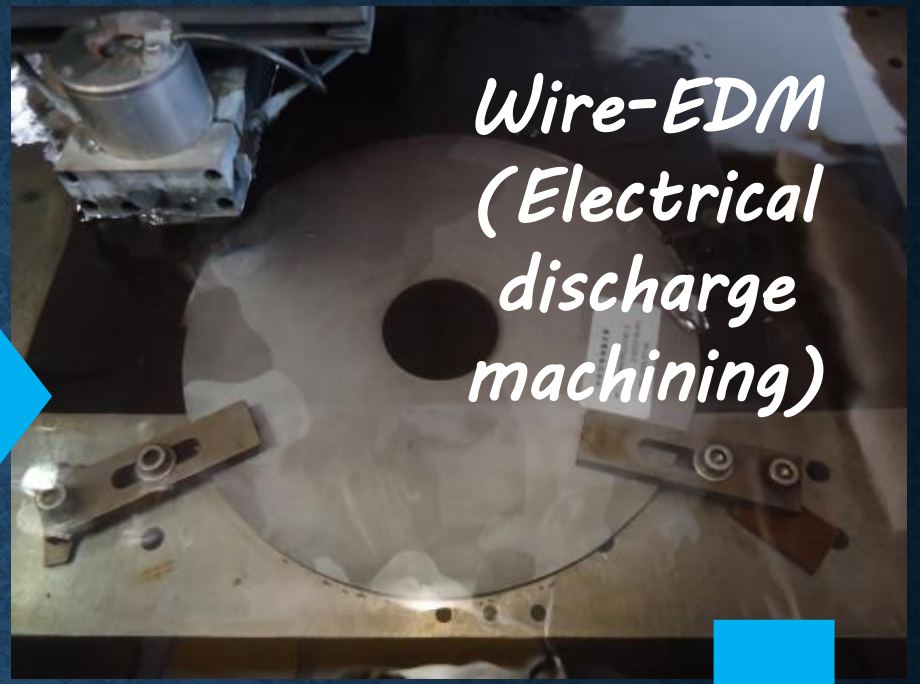
High purity Nb

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Let's start fabrication.





Wire-EDM (Electrical discharge machining)





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Electron beam welding

Cavity material quality is maintained

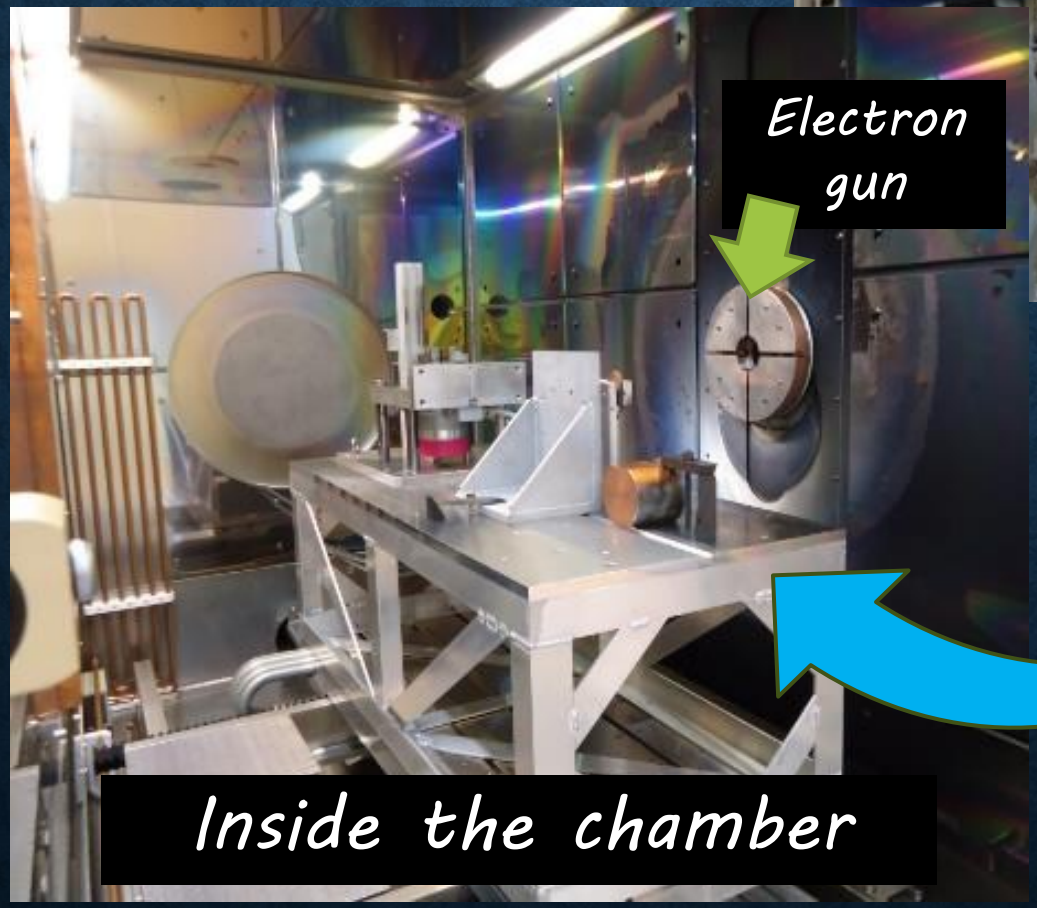


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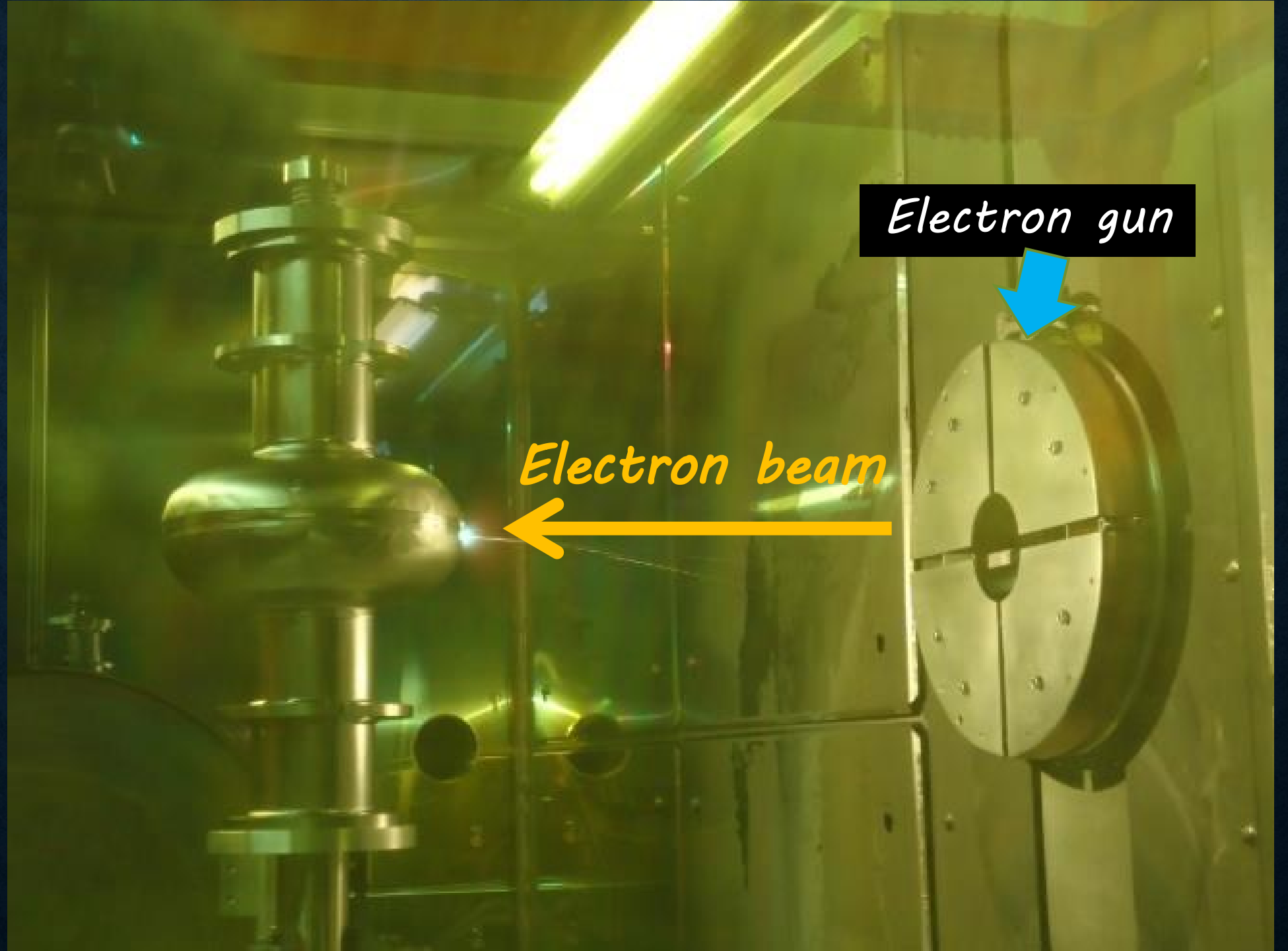






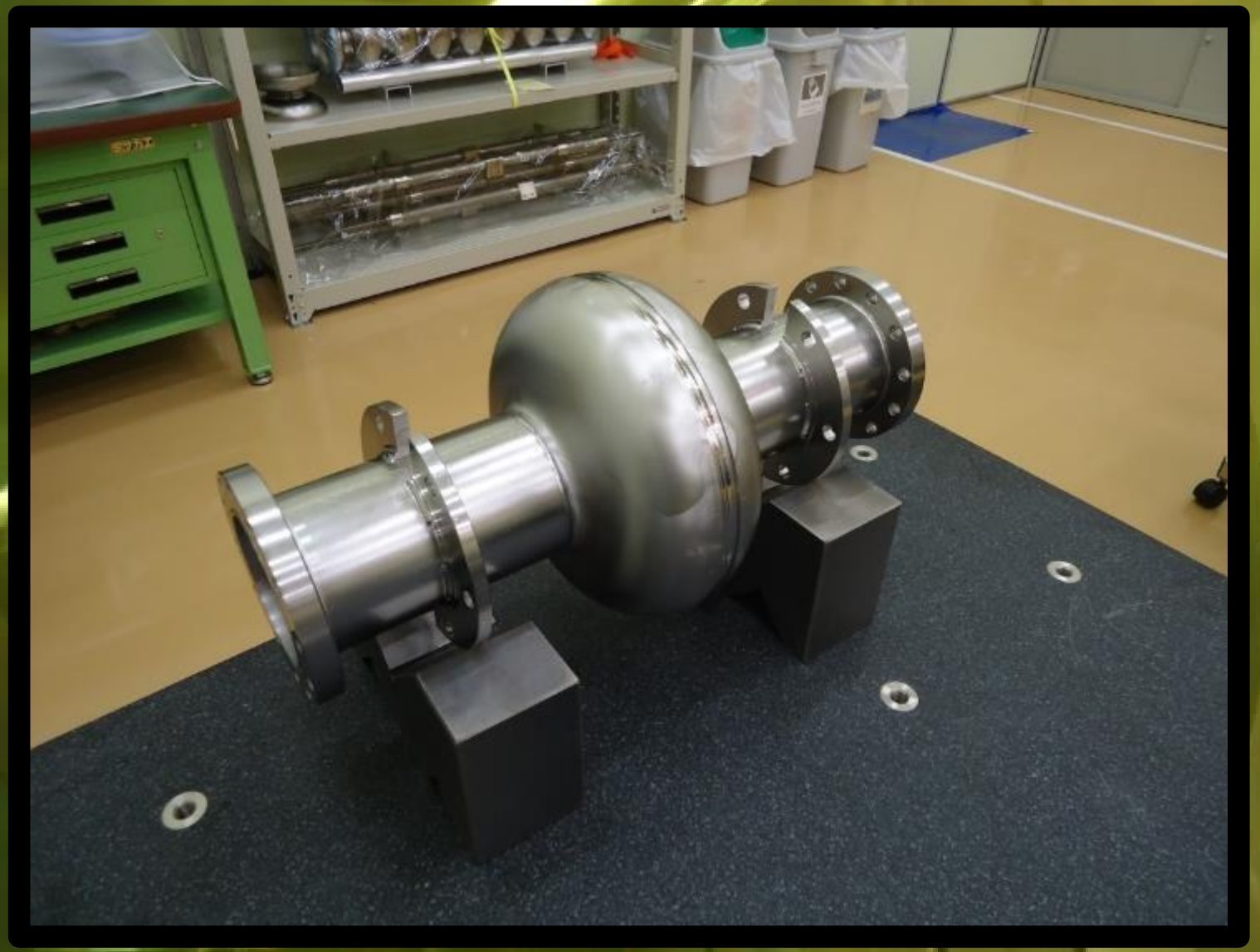
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- Test cavity

Electropolishing (EP)

- The electrolyte is a mixture of (9:1) sulfuric acid and hydrofluoric acid (HF).
- A DC voltage is applied with Nb as the anode and aluminum as the cathode.
- Niobium is oxidized to form Nb_2O_5.
- This oxide layer is dissolved and polished away by HF present in the electrolyte.











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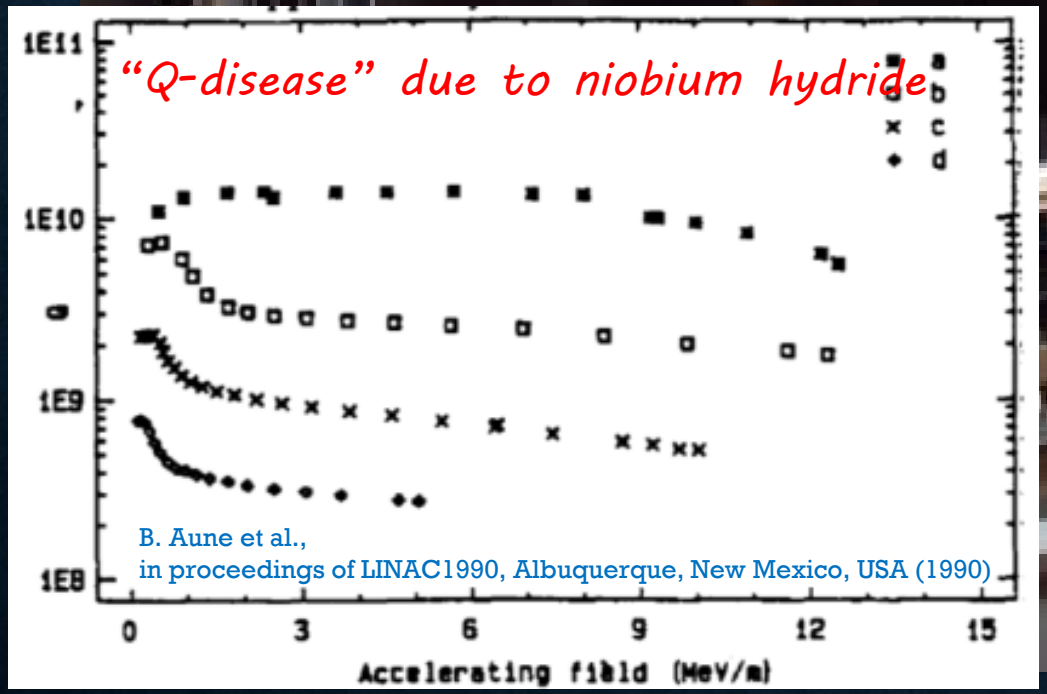


Annealing

- To degas hydrogen — prevents hydrogen Q-disease (requires >700 ℃)

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- **Annealing (750-900℃, 2-3hours)**
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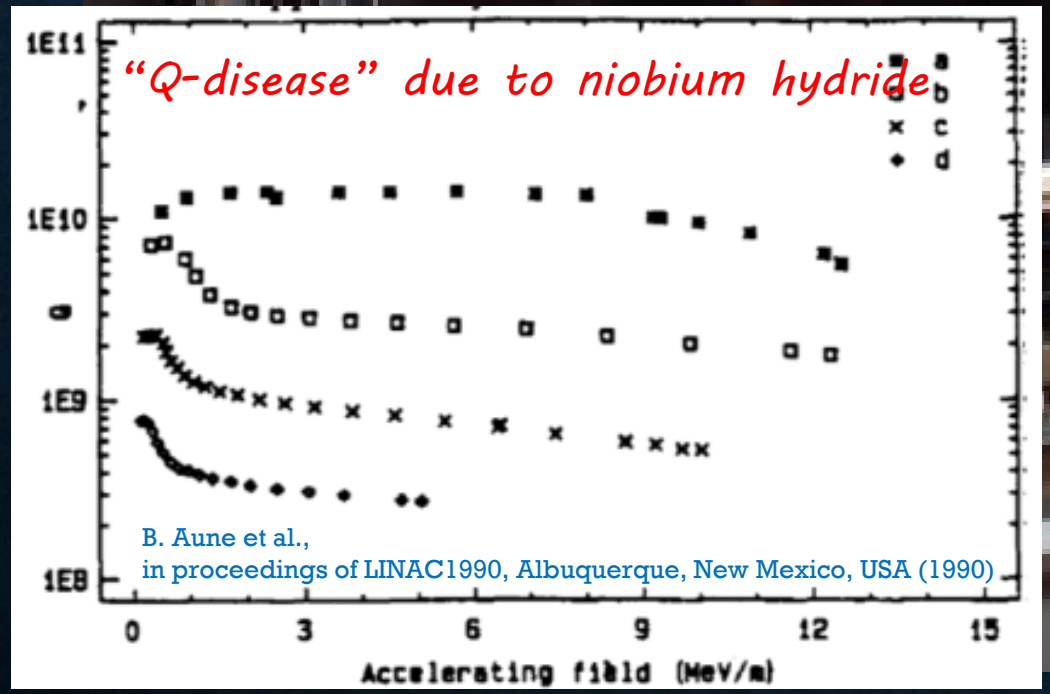


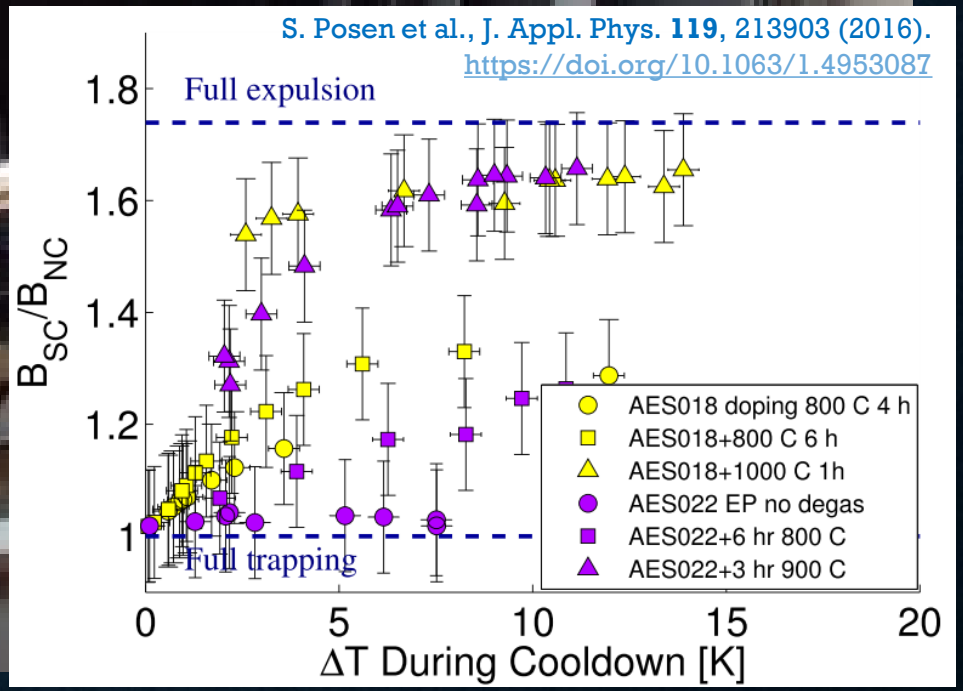
Annealing

- To degas hydrogen — prevents hydrogen Q-disease (requires >700 ℃)
- To reduce pinning centers — minimizes trapped flux (requires ~900 ℃)

ILC recipe

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- Fabrication
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- **Annealing (750-900℃, 2-3hours)**
- Electropolishing(2) (several μm)
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- Assembling
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- Test cavity





A second EP step, removing only a few microns.

ILC recipe

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- Low-temperature baking (100-140°C, 48hours)
- Test cavity

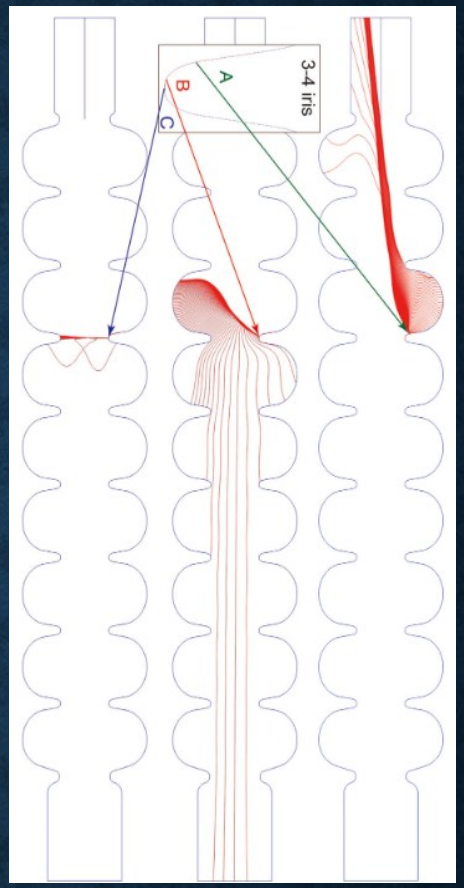


ILC recipe

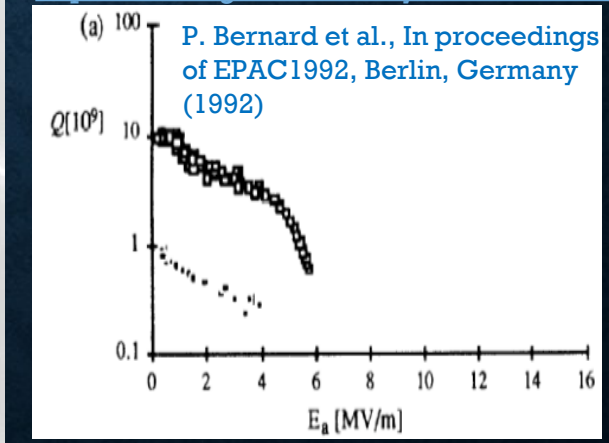
- Materials
- Fabrication
- Electropolishing(1) (100-200μm)
- Annealing (750-900°C, 2-3hours)
- Electropolishing(2) (several μm)
- High pressure rinsing
- Assembling
- Low-temperature baking (100-140°C, 48hours)
- Test cavity



Courtesy of Mathieu Omet (KEK)



H. Sakai et al.,
Phys. Rev. Accel. Beams **22**,022002 (2019)
https://doi.org/10.1103/PhysRevAccelBeams.22.022002



P. Bernard et al., In proceedings of EPAC1992, Berlin, Germany (1992)

ILC recipe

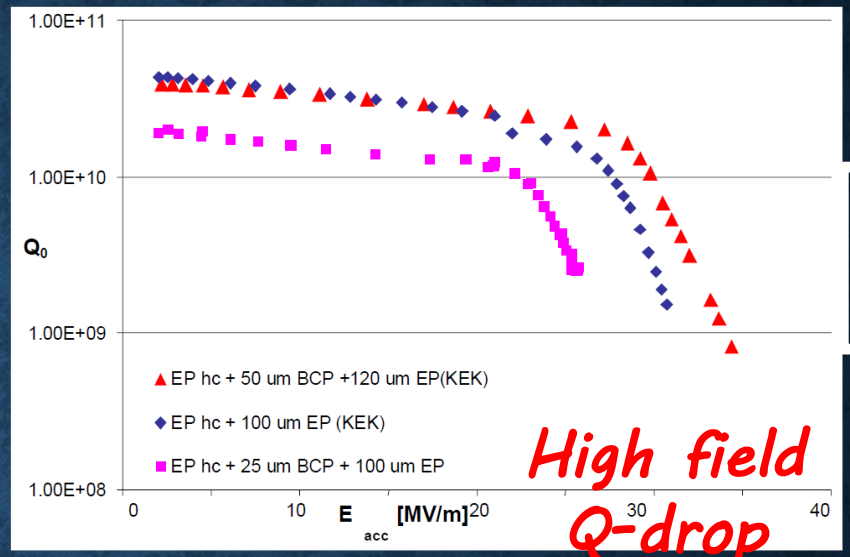
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- Electropolishing(2) (several μm)
- High pressure rinsing
- Assembling
- **Low-temperature baking (100-140°C, 48hours)**
- Test cavity

120°C—48h bake



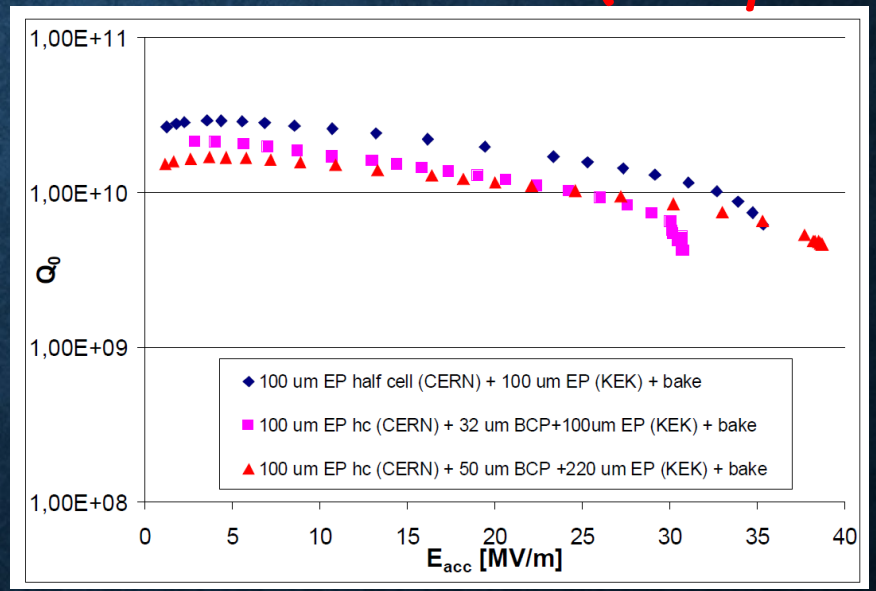
Courtesy of Mathieu Omet (KEK)

L. Lilje et al., in Proceedings of SRF1999, Santa Fe, New Mexico, USA (1999), p. 74, TUA001.



Without baking

High field Q-drop



With baking

Spatial variation of the mean free path on the scale of the penetration depth suppresses the surface current, thereby increasing the maximum achievable field.

T. Kubo, Supercond. Sci. and Technol. **30**, 023001 (2017); Supercond Sci. and Technol. **34**, 045006 (2021)
V. Ngampruetikorn and J. A. Sauls, Phys. Rev. Research **1**, 012015(R) (2019)

ILC recipe

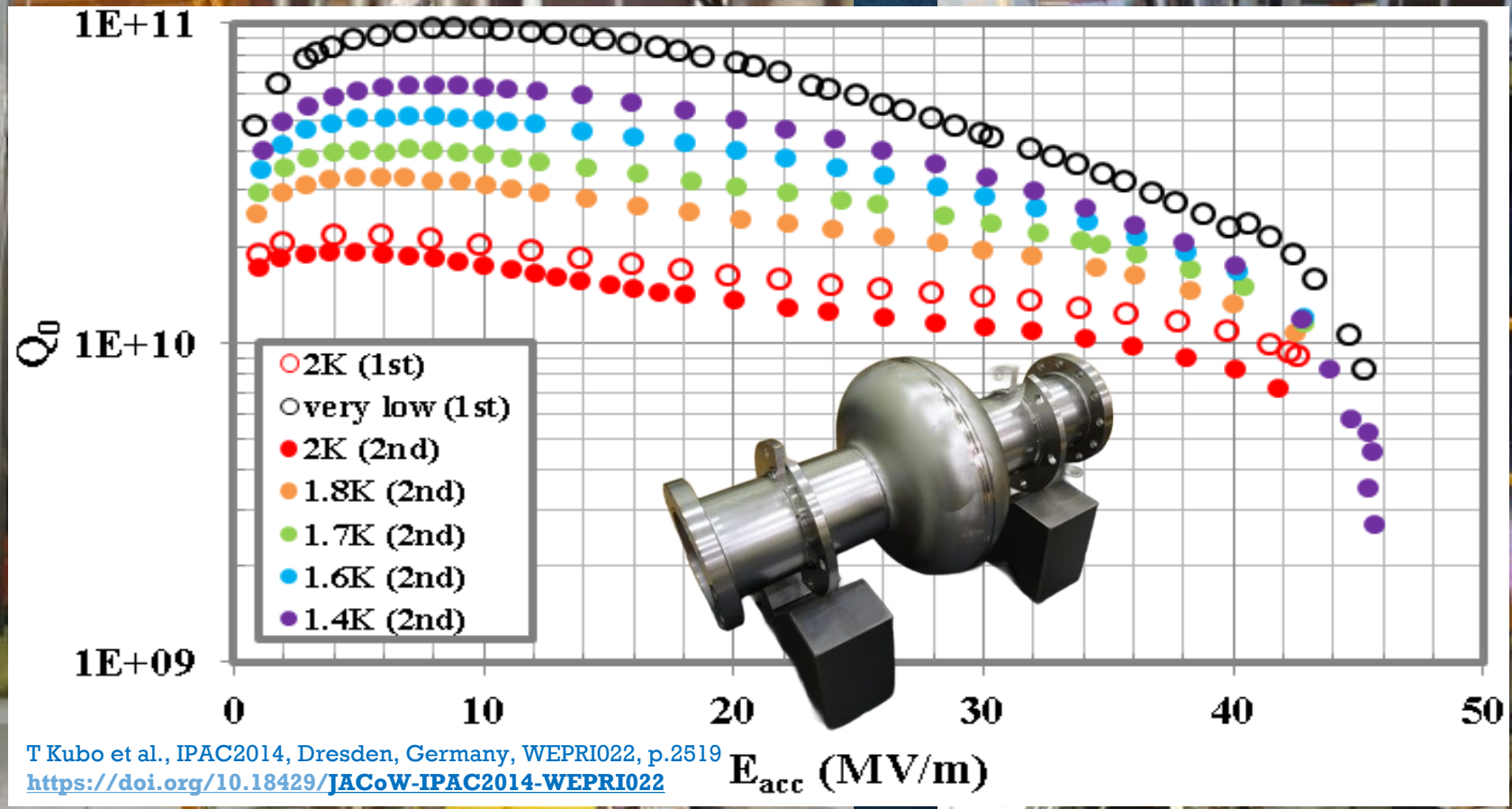
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- Assembling
- Low-temperature baking (100-140℃, 48hours)
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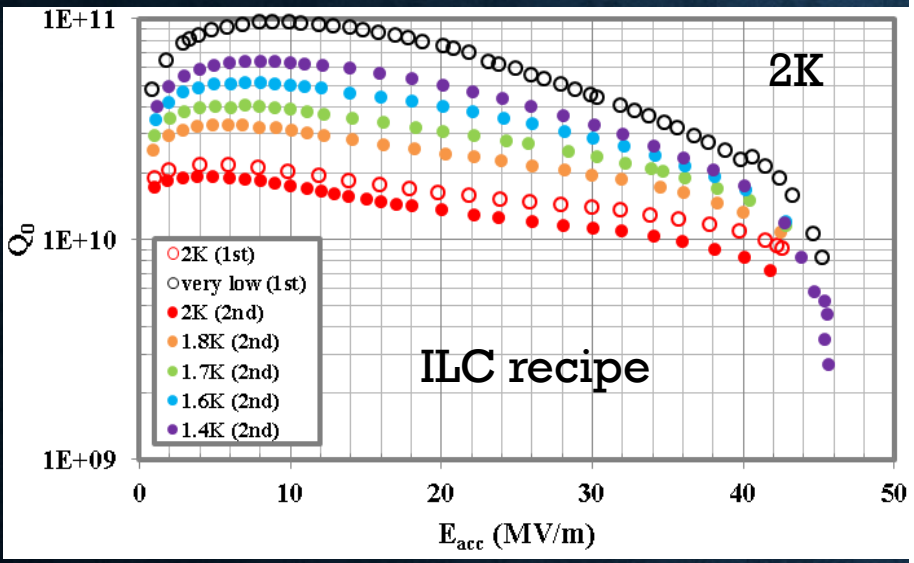
ILC recipe

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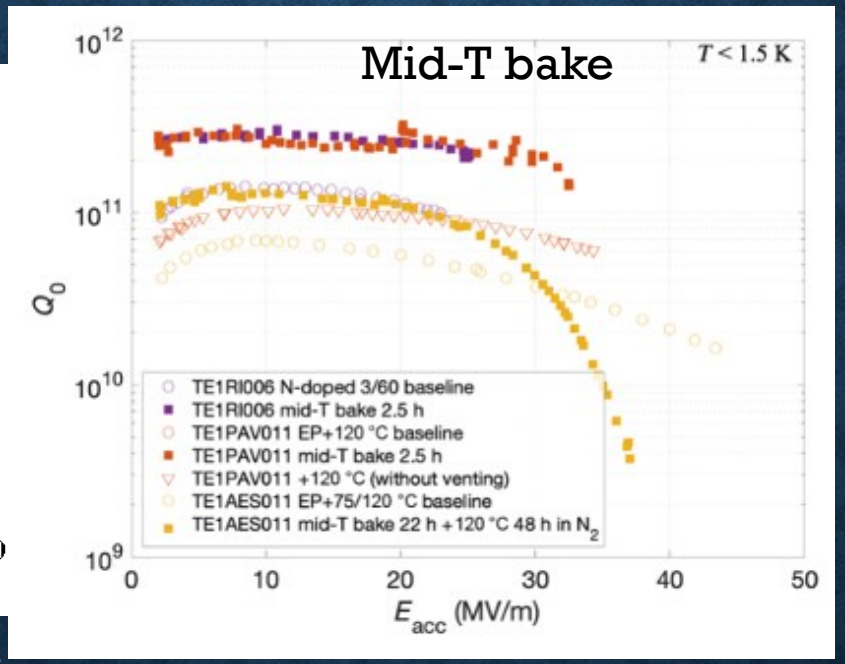


T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

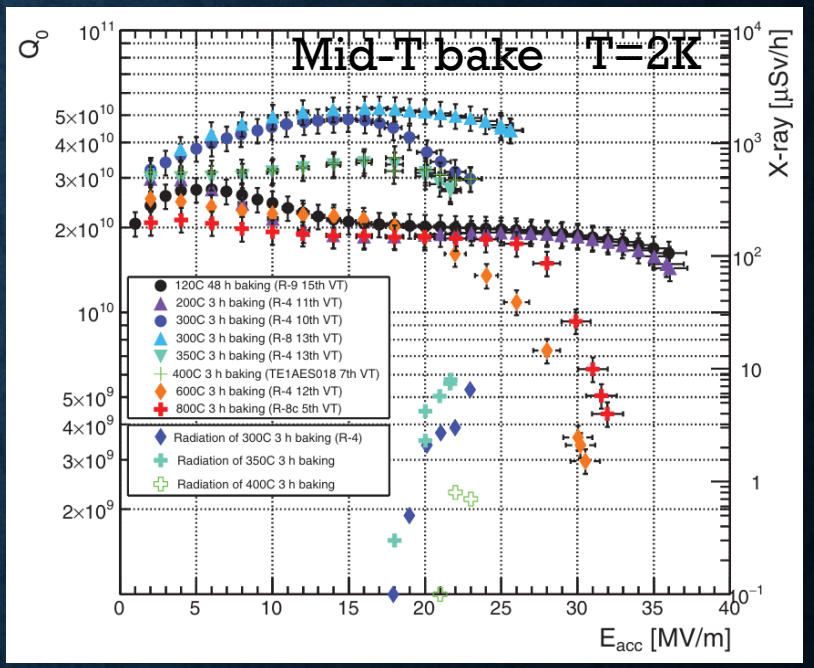
Many modern recipes exist: N-doping, N-infusion, two-step bake, mid-T bakes, etc. **They are all variants of the classical recipe**, with some steps replaced or modified.



T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

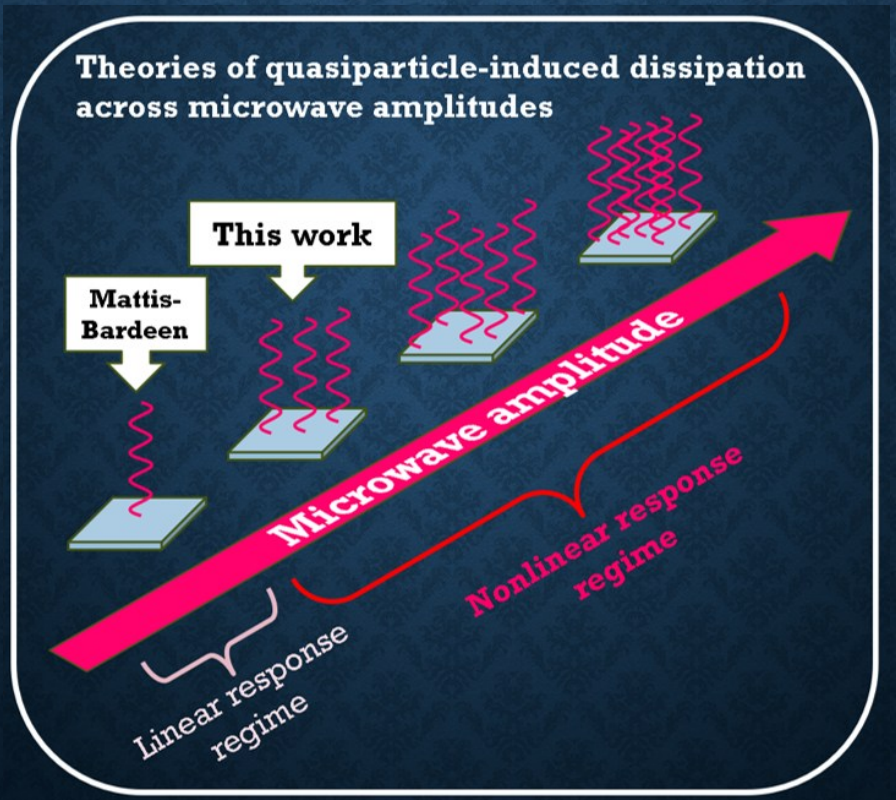


S. Posen et al., Ultralow Surface Resistance via Vacuum Heat Treatment of Superconducting Radio-Frequency Cavities,
Phys. Rev. Applied **13**, 014024 (2020)
https://doi.org/10.1103/PhysRevApplied.13.014024



H. Ito et al., Influence of furnace baking on Q–E behavior of superconducting accelerating cavities,
Prog. Theor. Exp. Phys. **2021**, 071G01 (2021)
https://doi.org/10.1093/ptep/ptab056

Understanding more about physics of Q factor



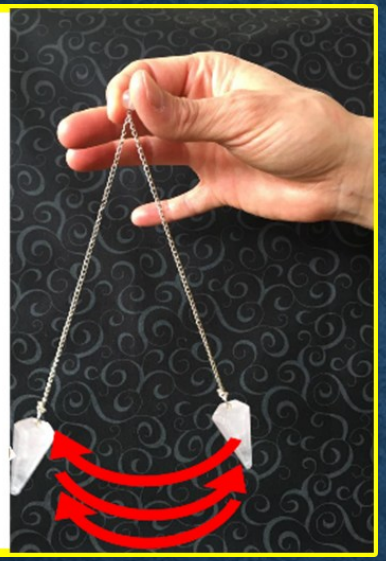
In the accelerator regime (T~2K), the quality factor is largely determined by quasiparticle dissipation.

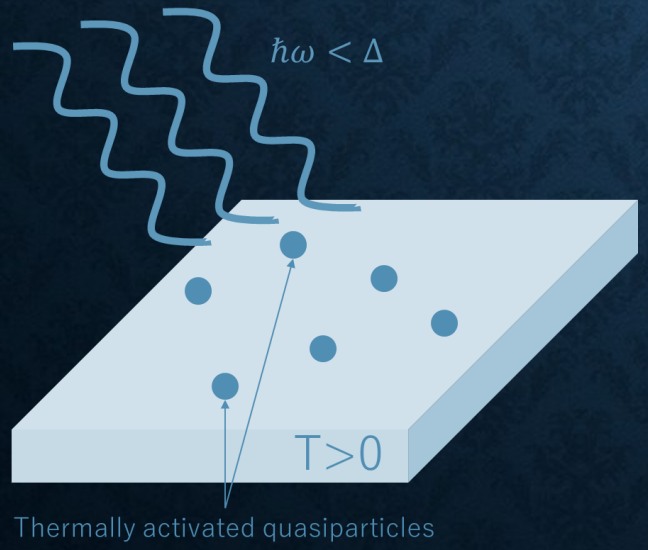
$$Q_0 = \frac{\omega U}{P} \sim \frac{U}{P_1} \propto \frac{1}{R_s}$$

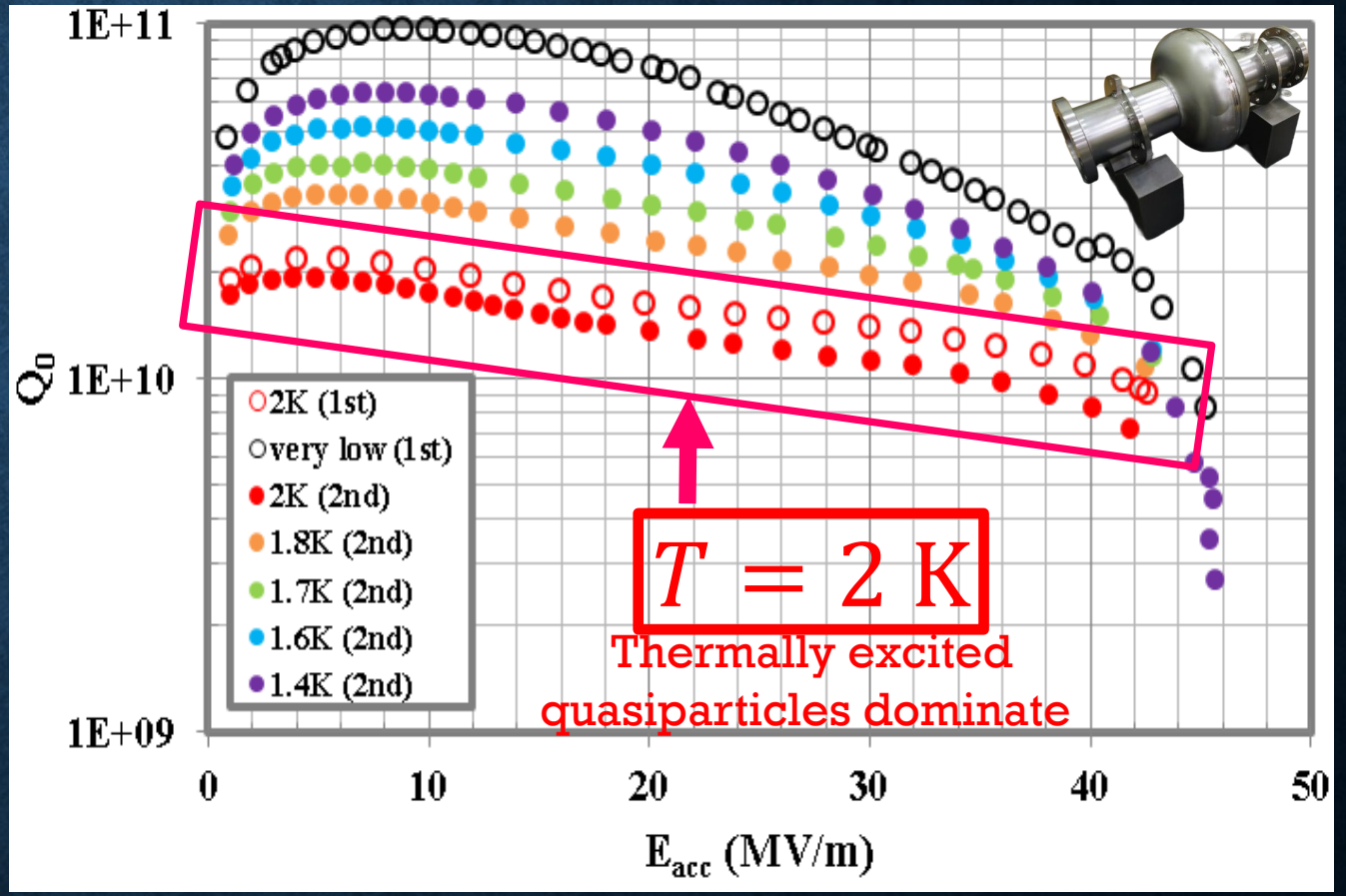
P_1 : dissipation per RF cycle

Q_0 : how many cycles it takes for the stored energy to completely dissipate

What are the causes of dissipation?







T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

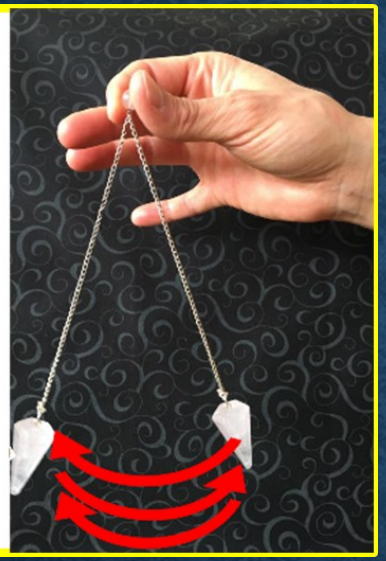
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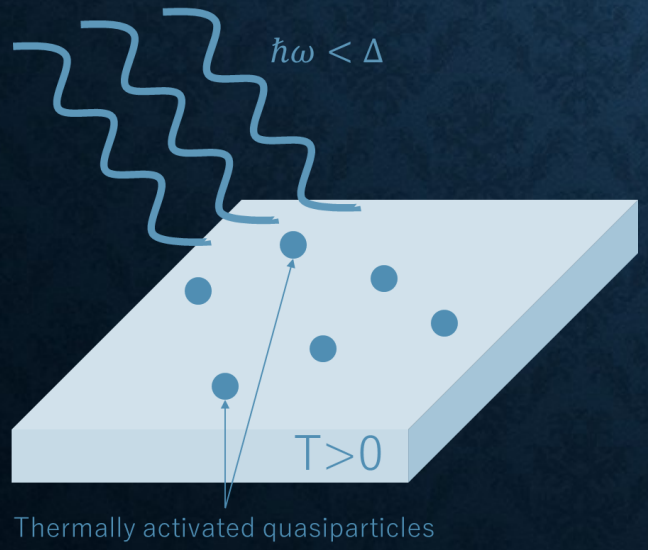
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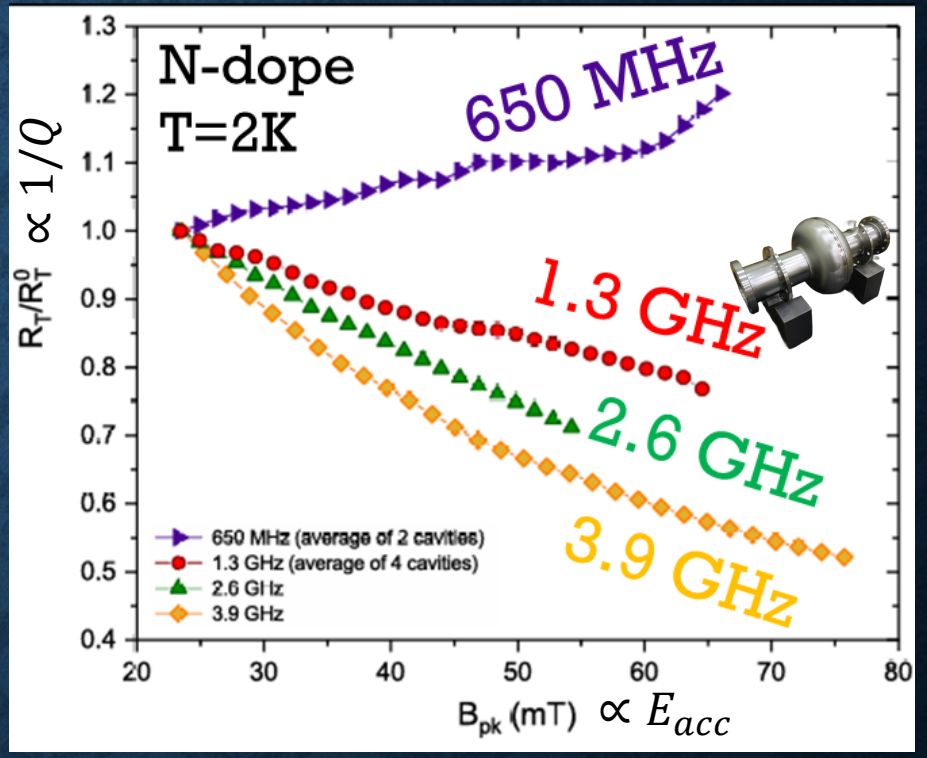
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What are the causes of dissipation?

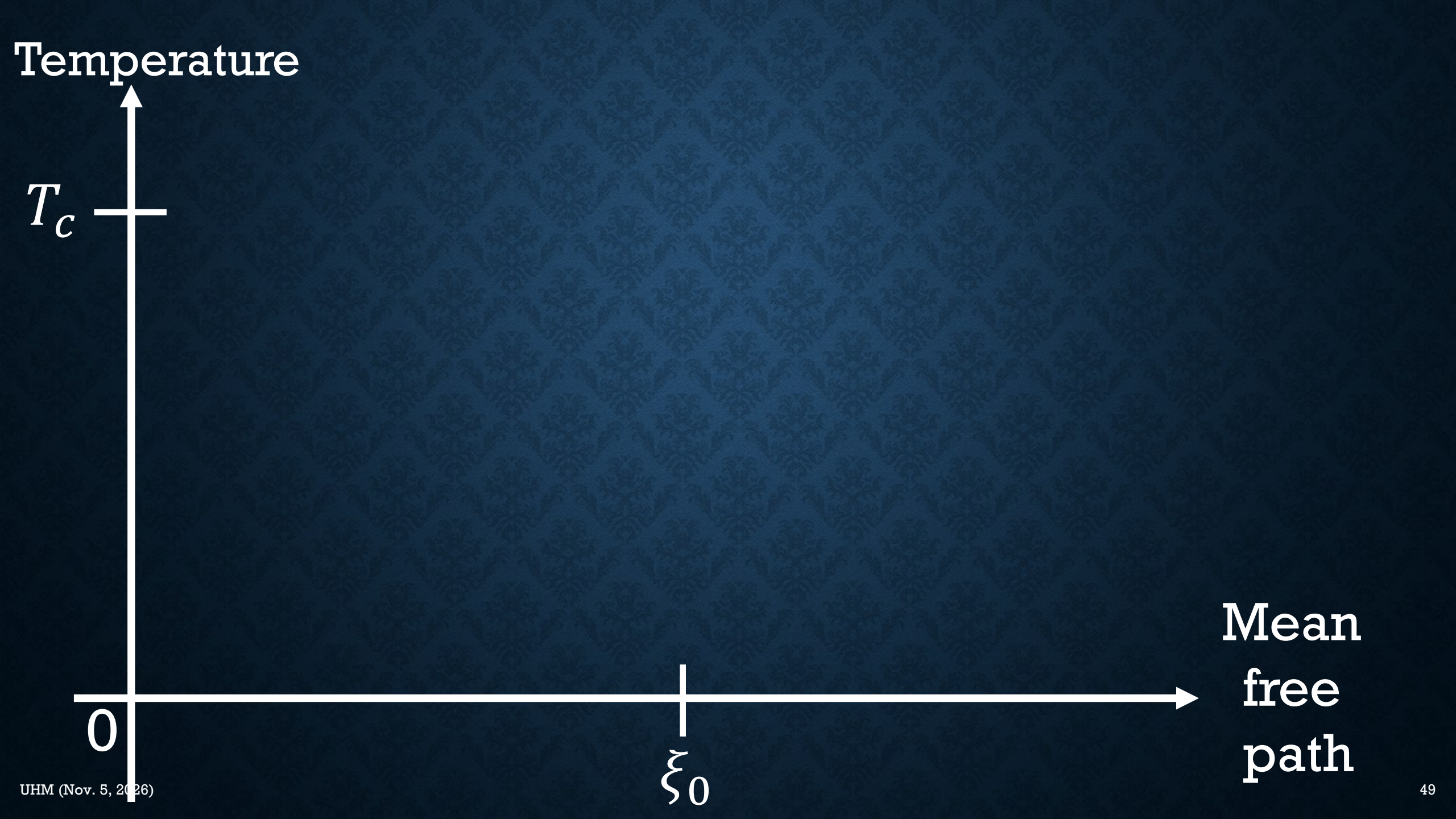


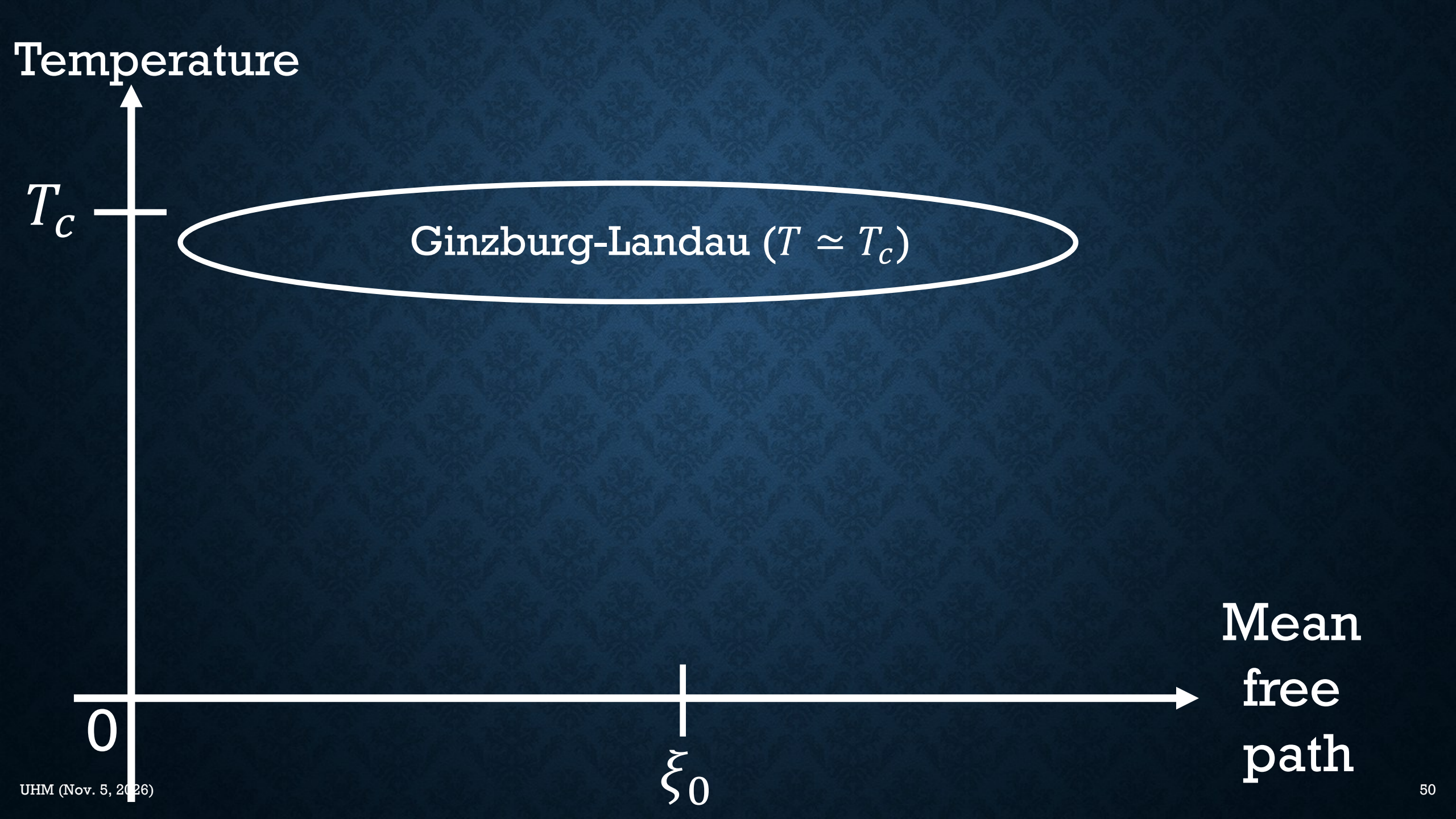


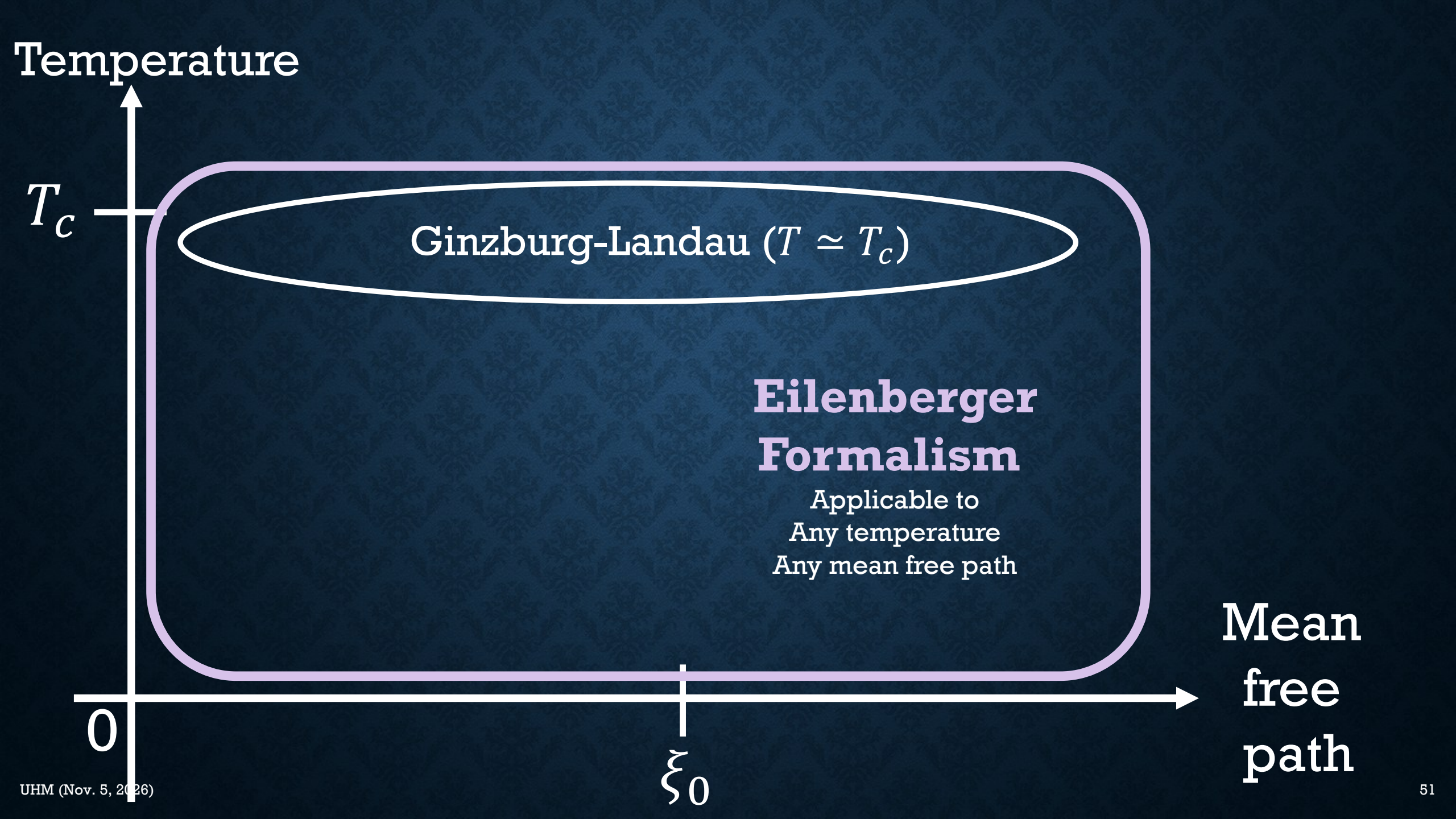


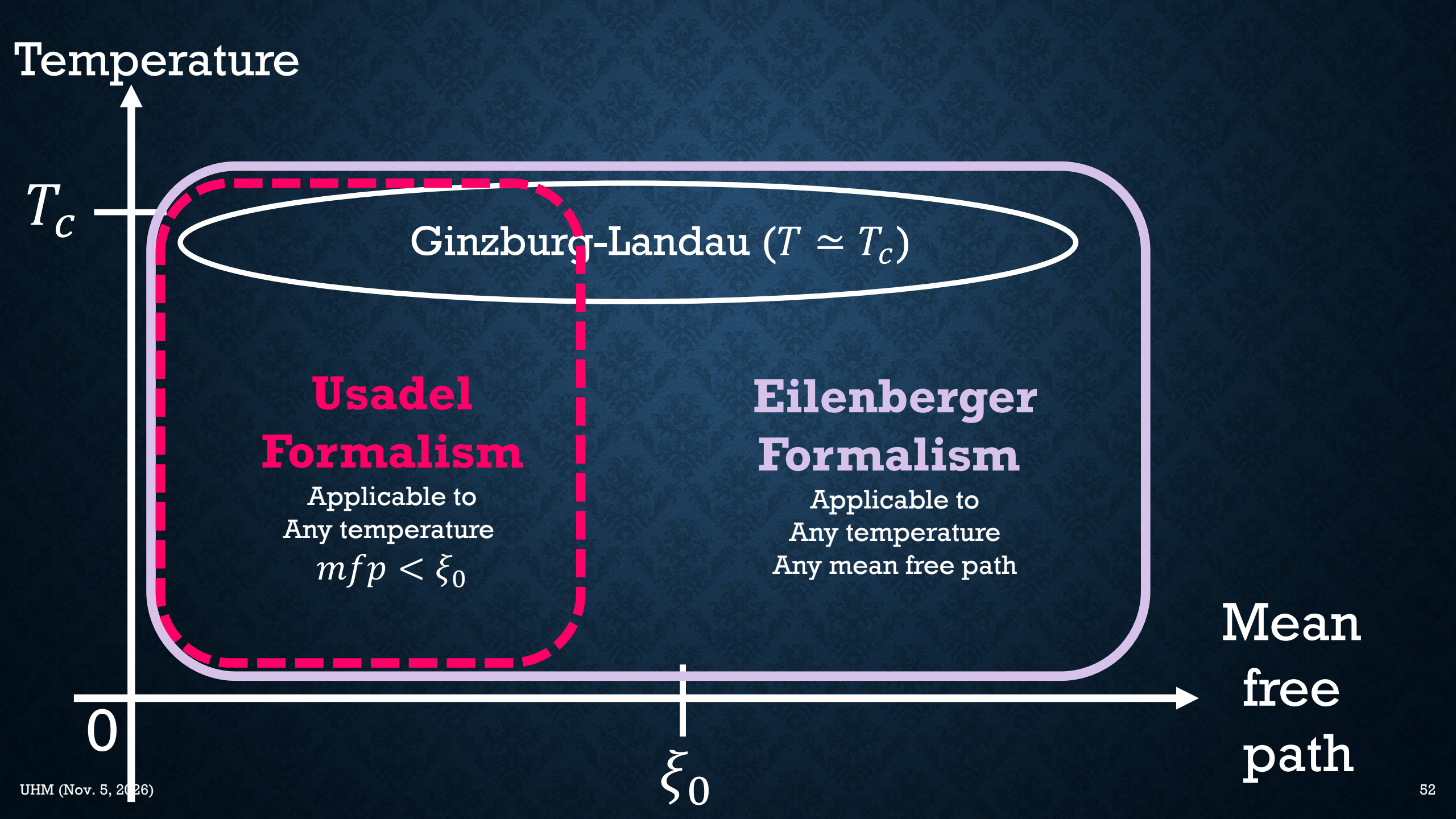
M. Martinello et al., Phys. Rev. Lett. **121**, 224801 (2018)
DOI: https://doi.org/10.1103/PhysRevLett.121.224801

Actually, this is a very challenging problem

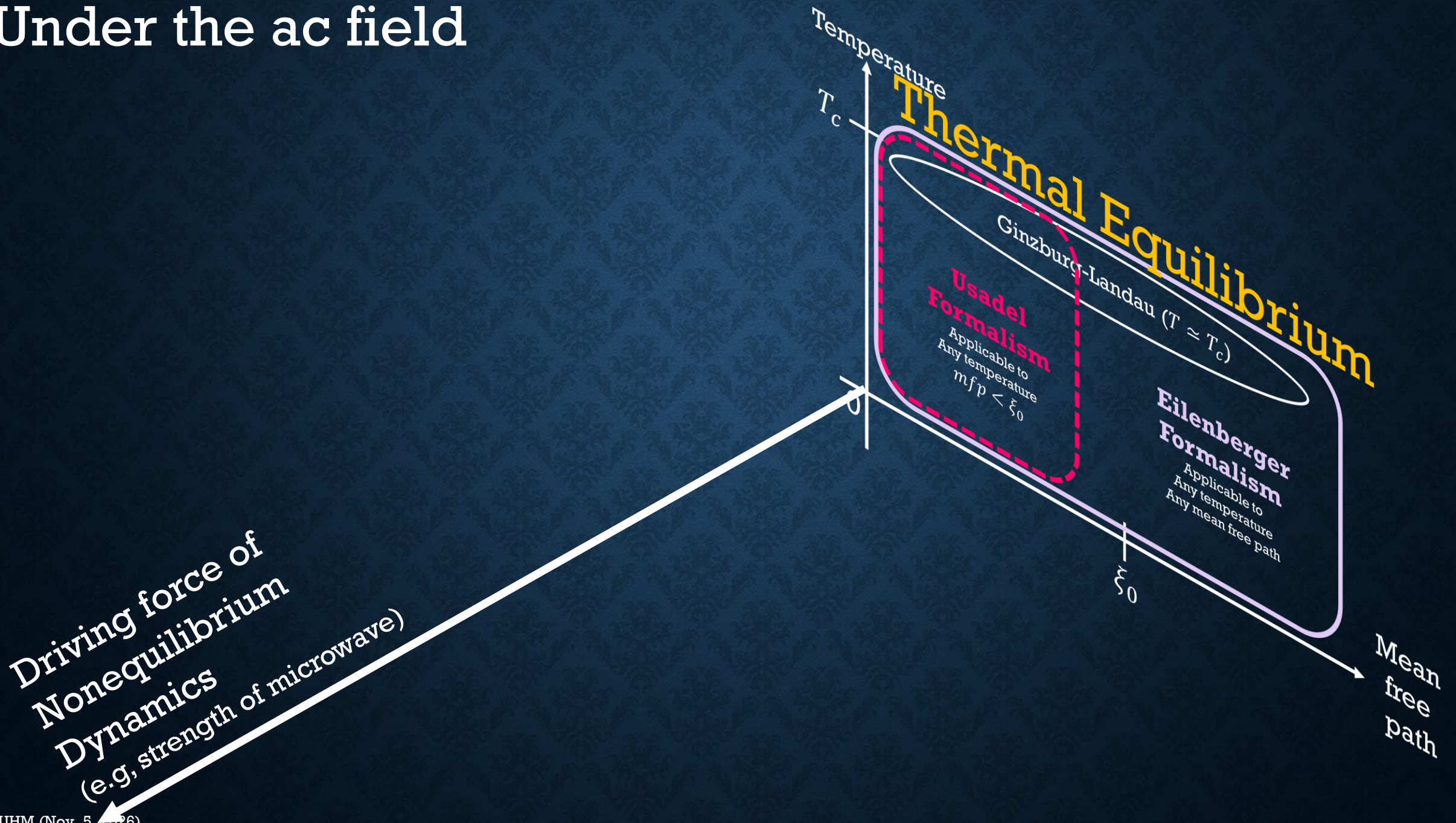

Temperature
T_c
0
ξ_0
Mean free path


Temperature
T_c
Ginzburg-Landau ($T \simeq T_c$)
0
ξ_0
Mean free path

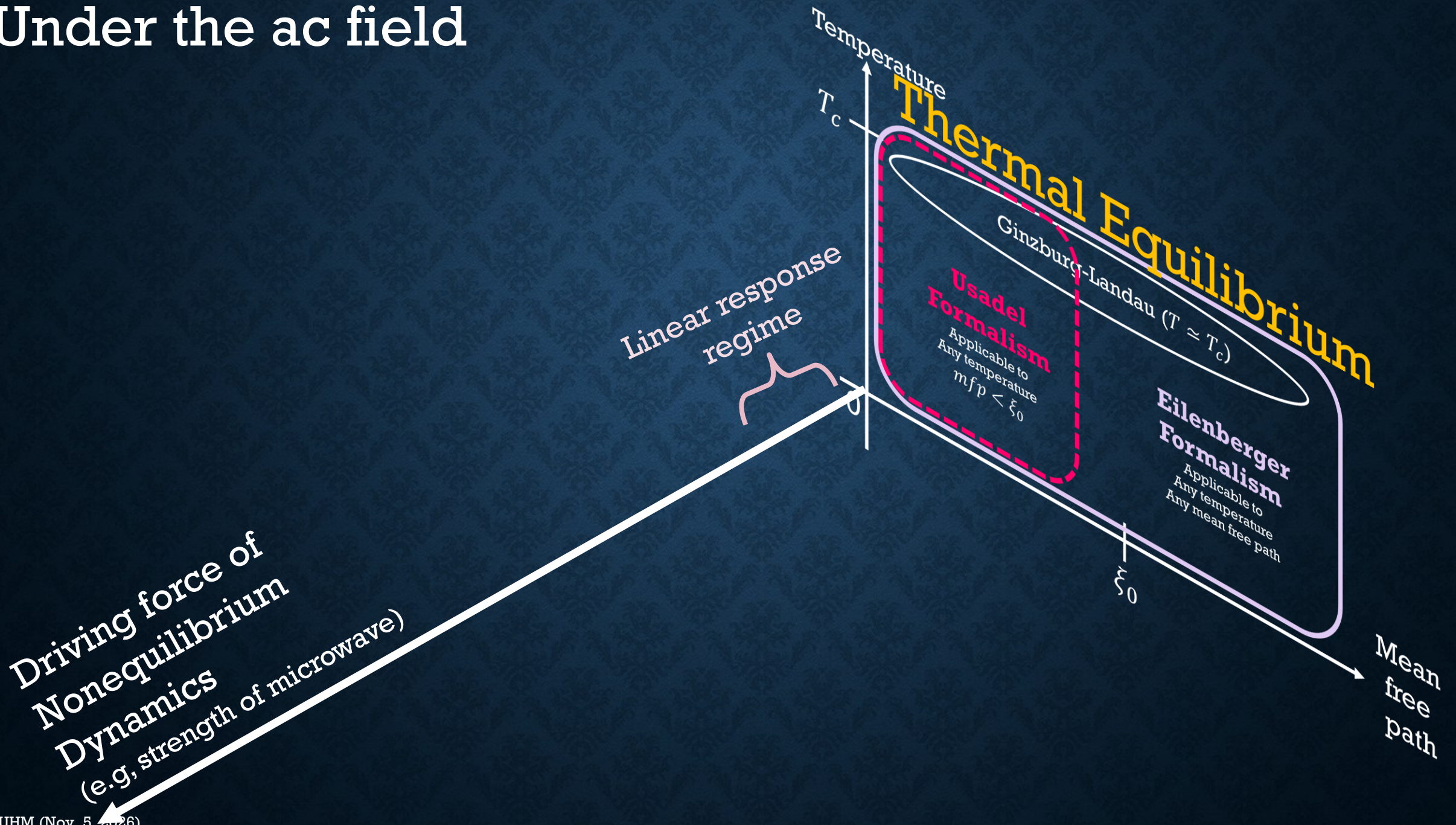

Temperature
T_c
Ginzburg-Landau ($T \simeq T_c$)
Eilenberger Formalism
Applicable to
Any temperature
Any mean free path
0
ξ_0
Mean free path


Temperature
T_c
Ginzburg-Landau ($T \simeq T_c$)
Usadel Formalism
Applicable to
Any temperature
$mfp < \xi_0$
Eilenberger Formalism
Applicable to
Any temperature
Any mean free path
0
ξ_0
Mean free path

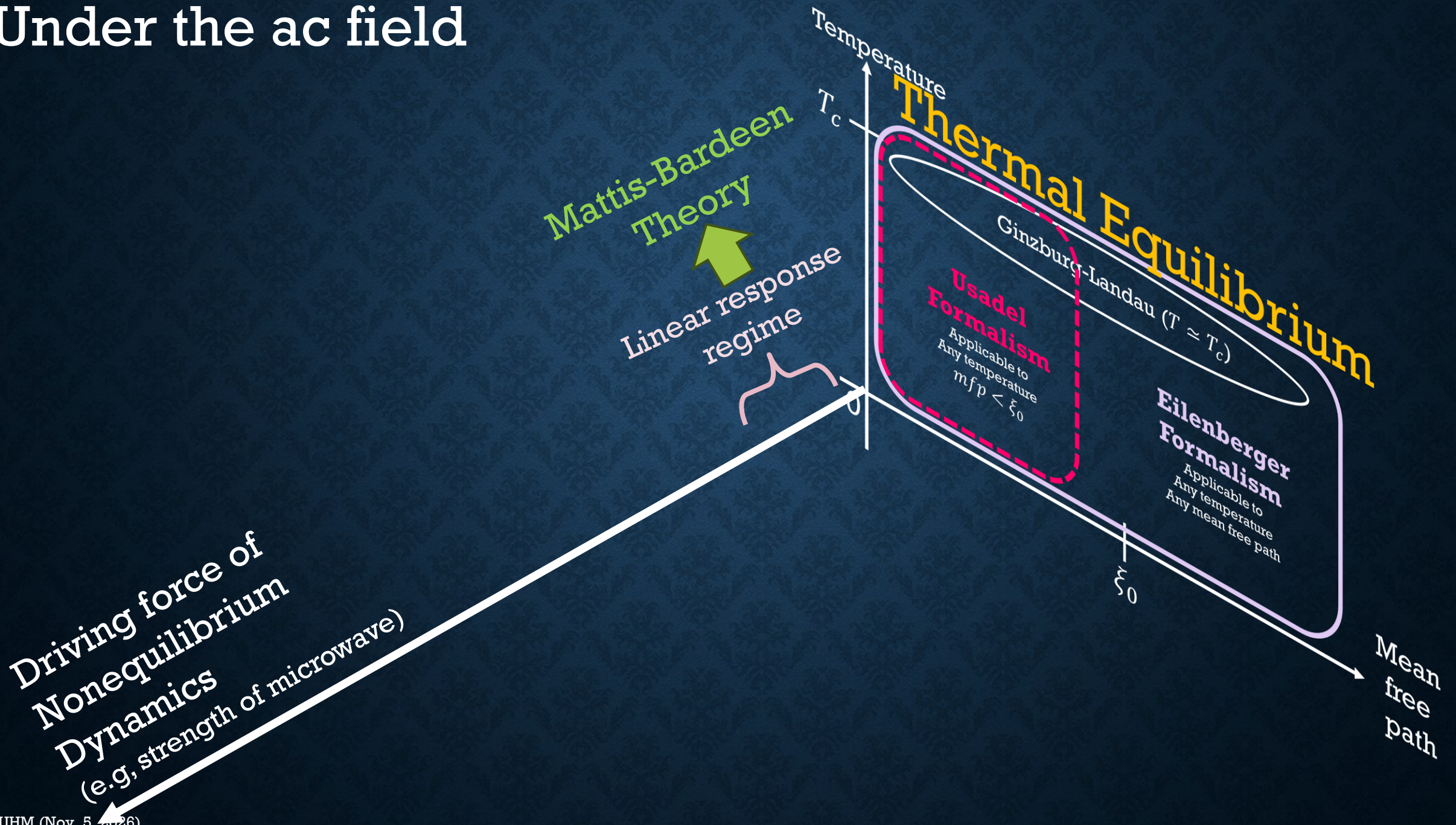
Under the ac field



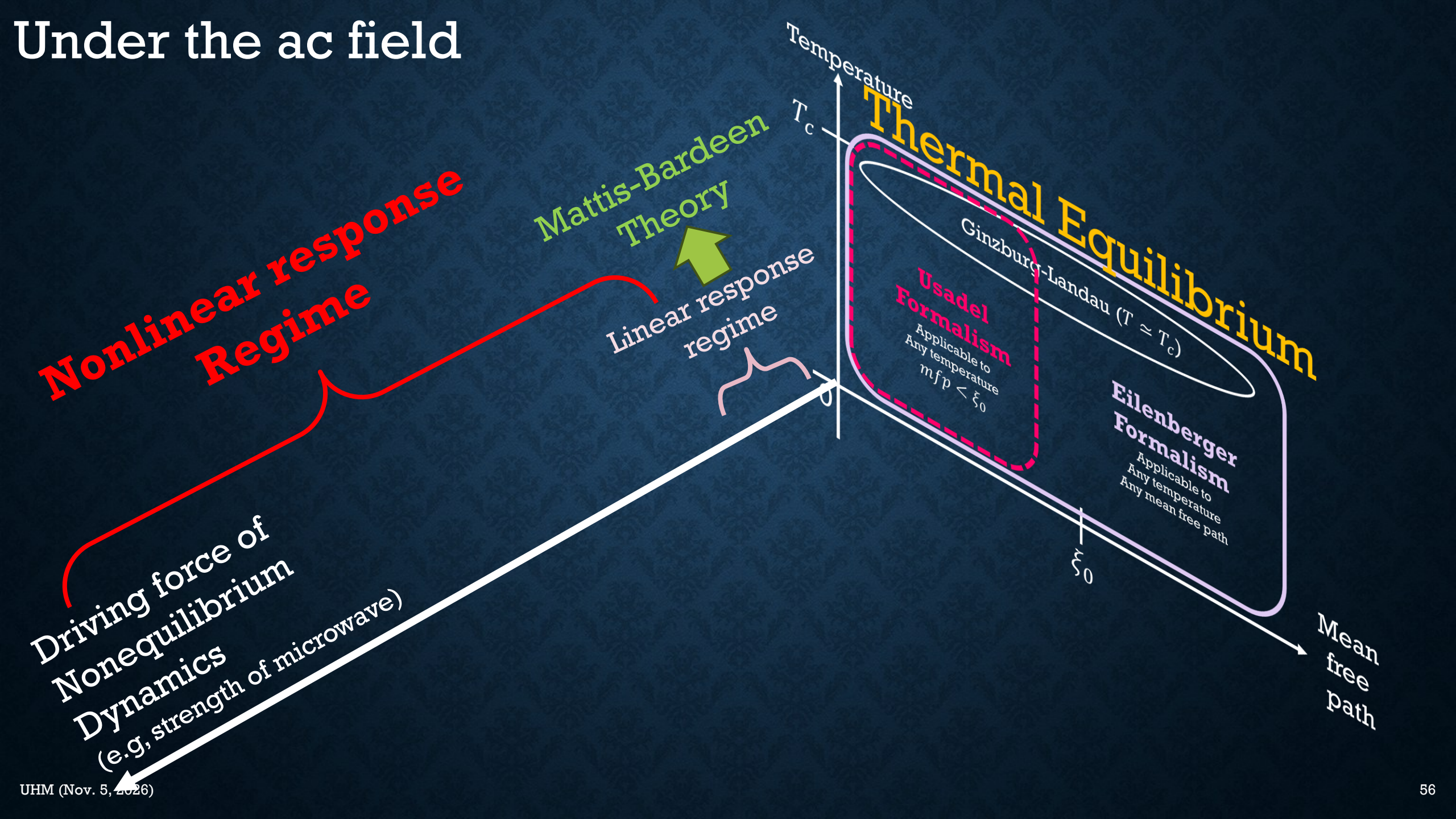
Under the ac field



Under the ac field



Under the ac field



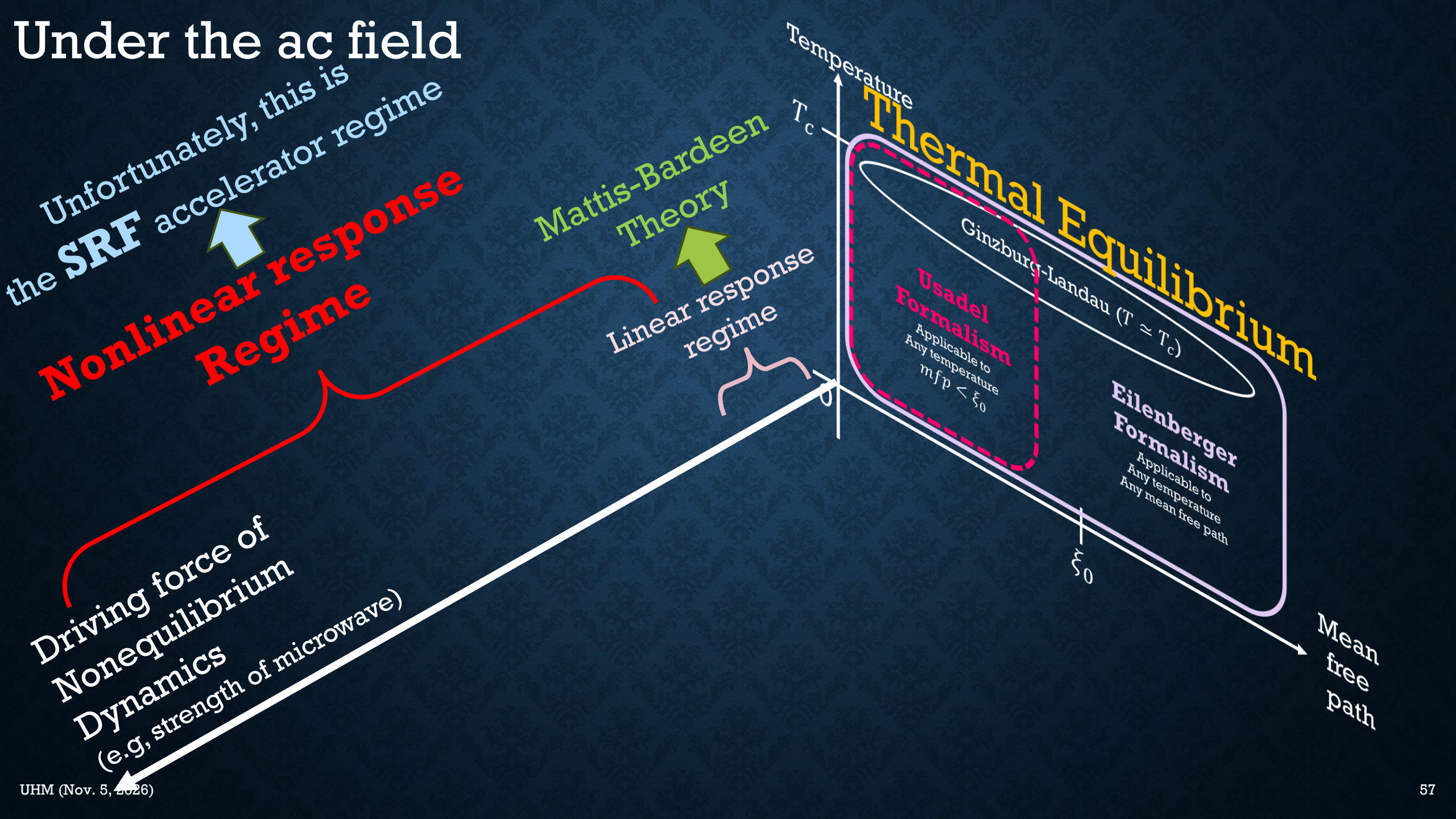
Under the ac field

Unfortunately, this is the **SRF** accelerator regime

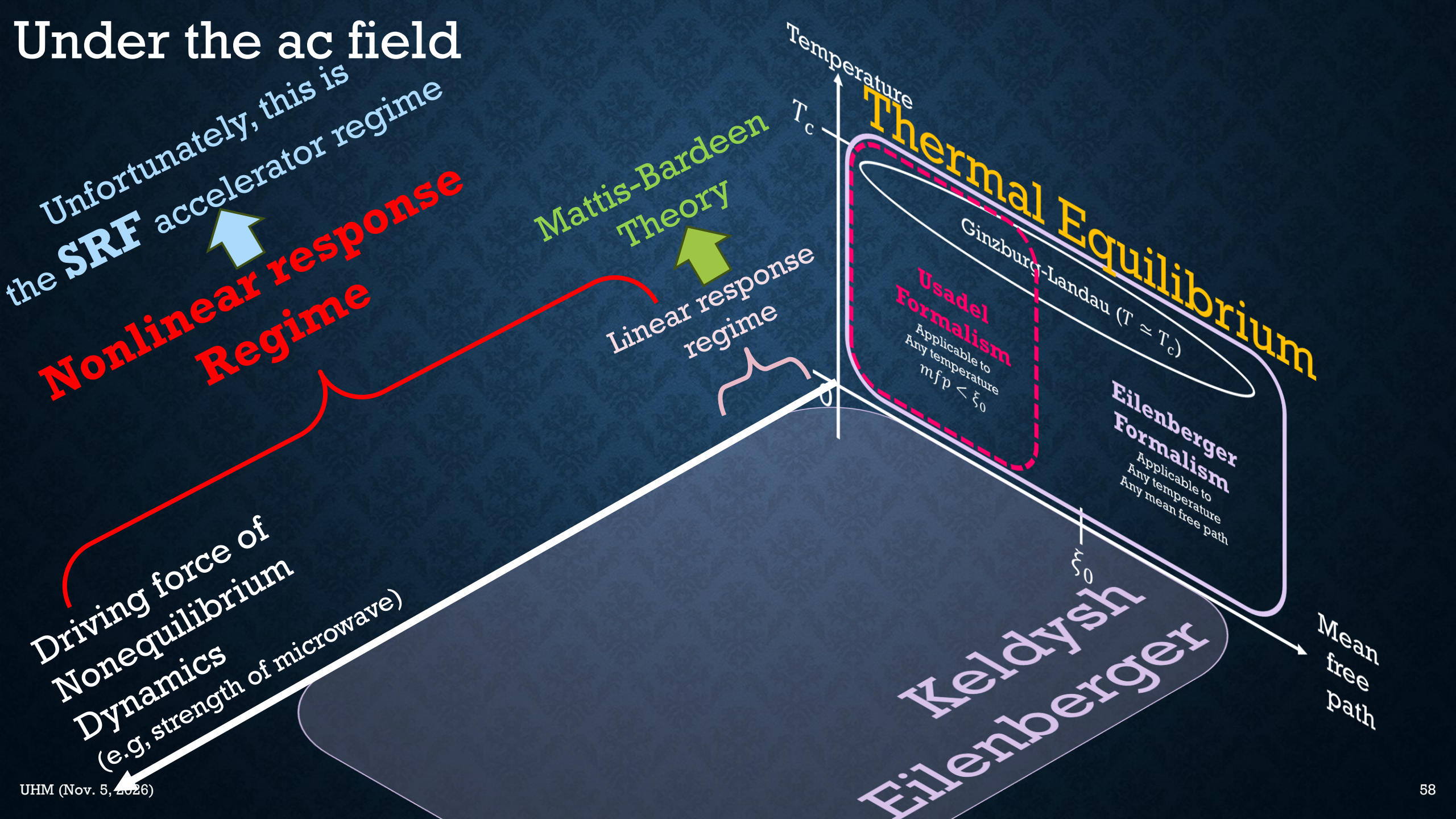
Nonlinear response Regime

Mattis-Bardeen Theory

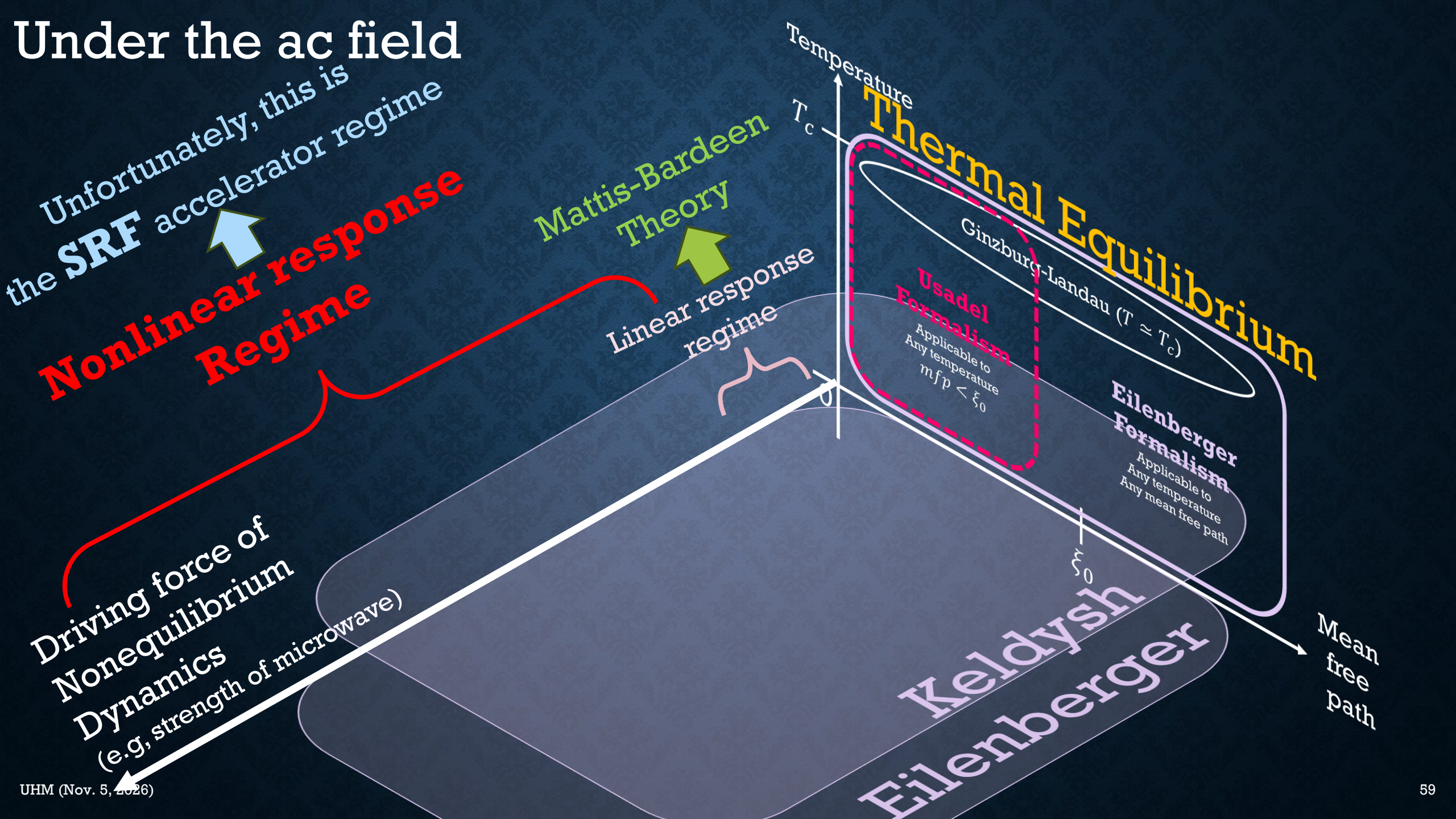
Linear response regime



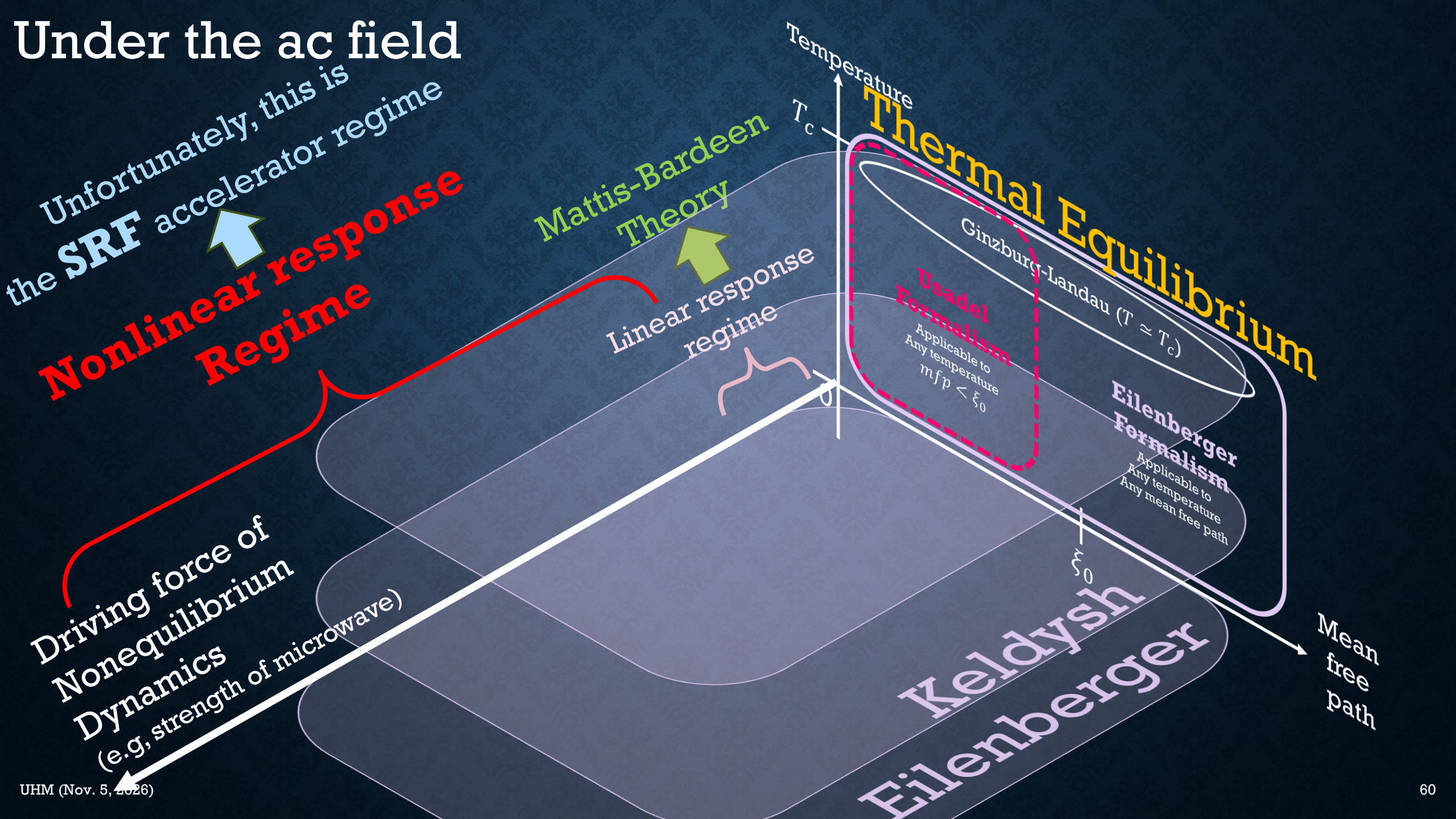
Under the ac field



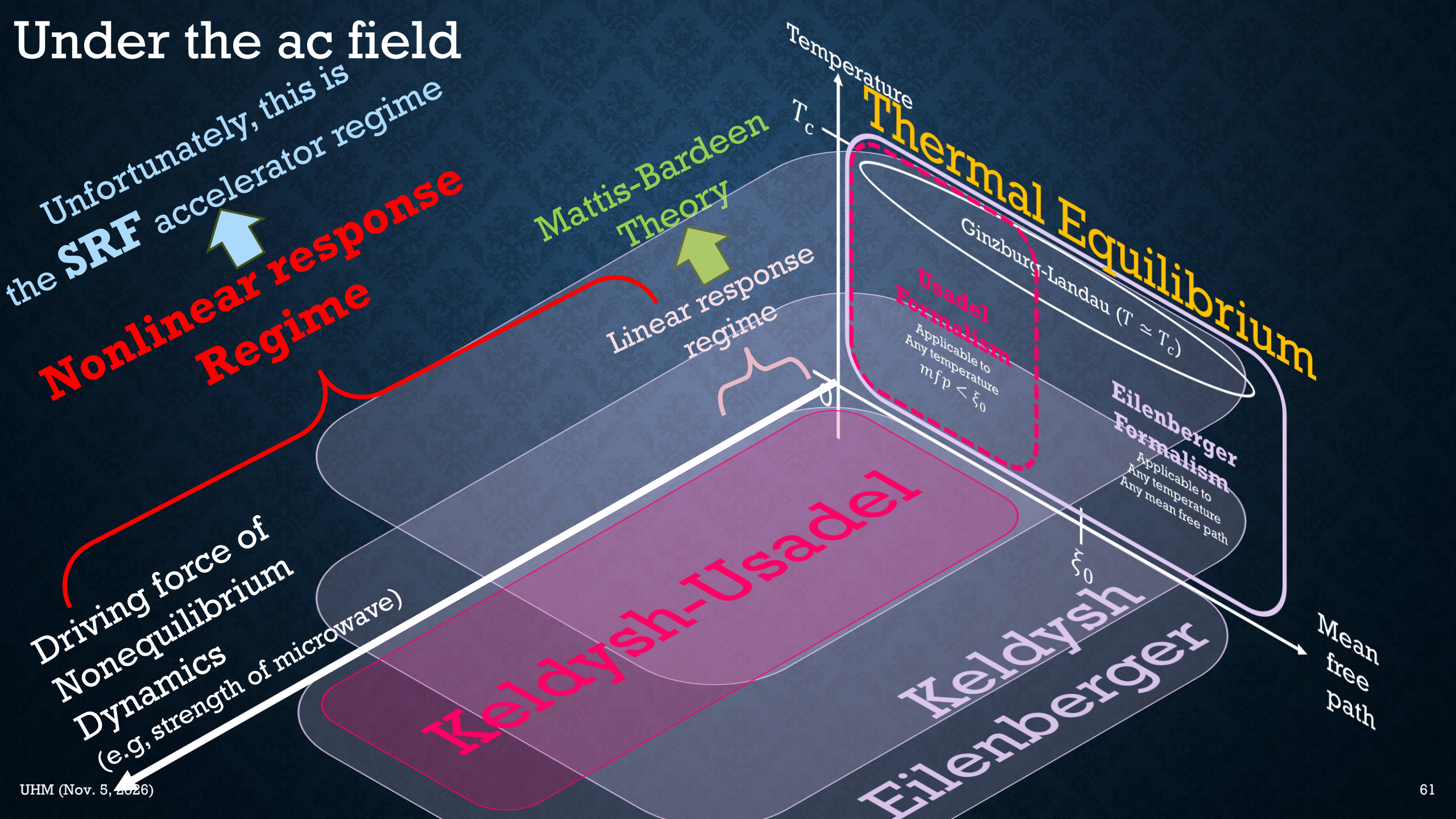
Under the ac field



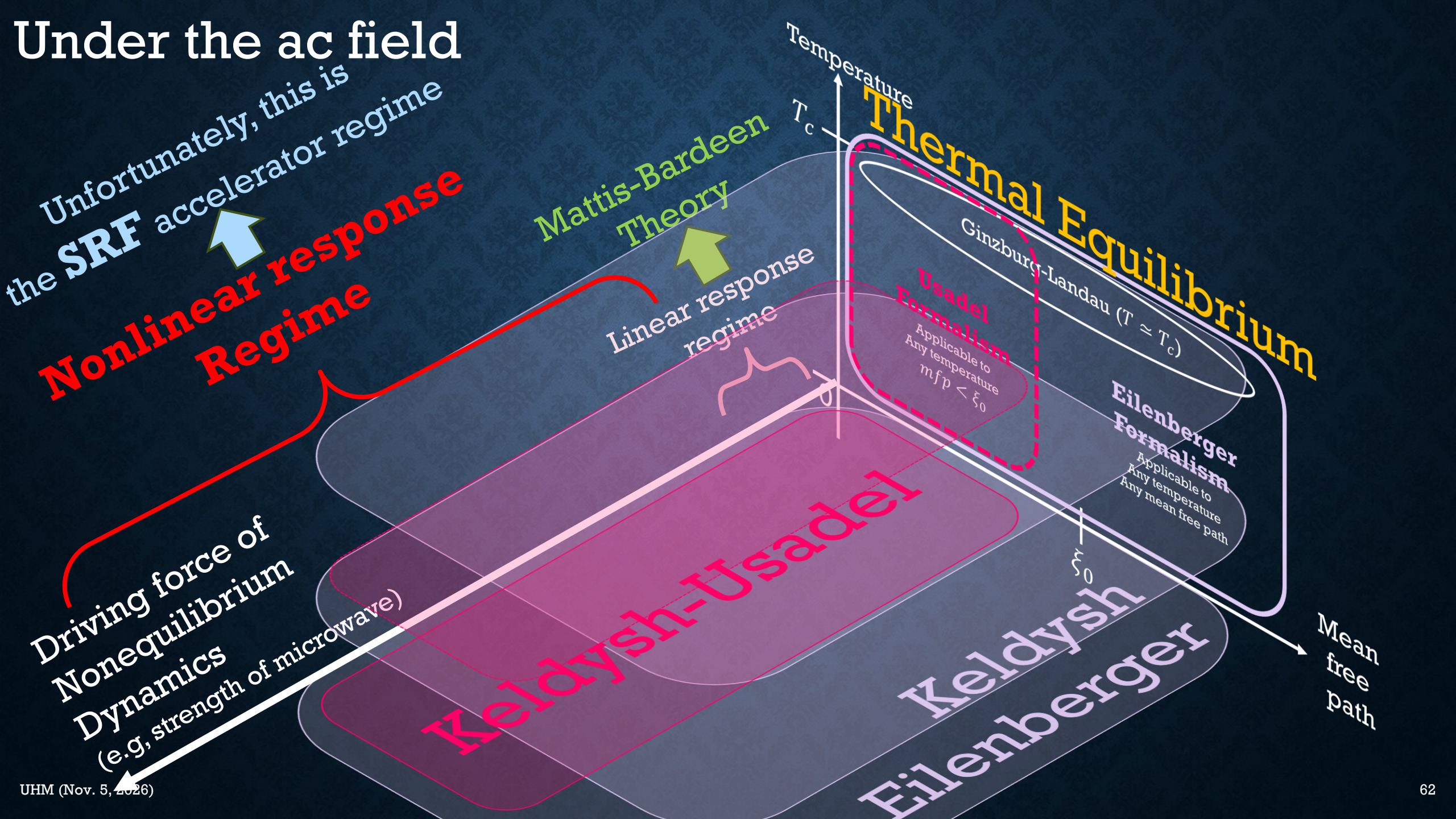
Under the ac field



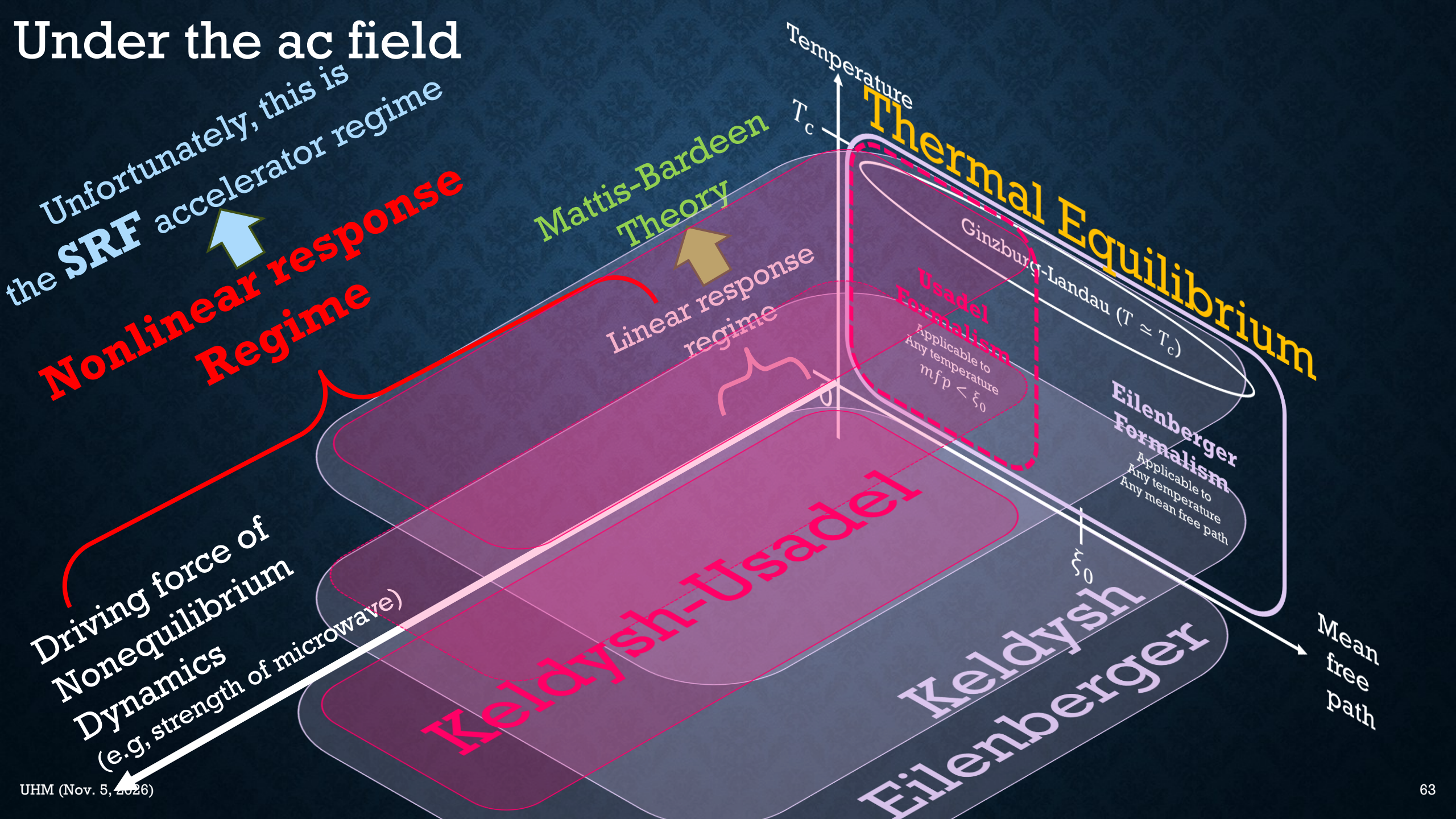
Under the ac field



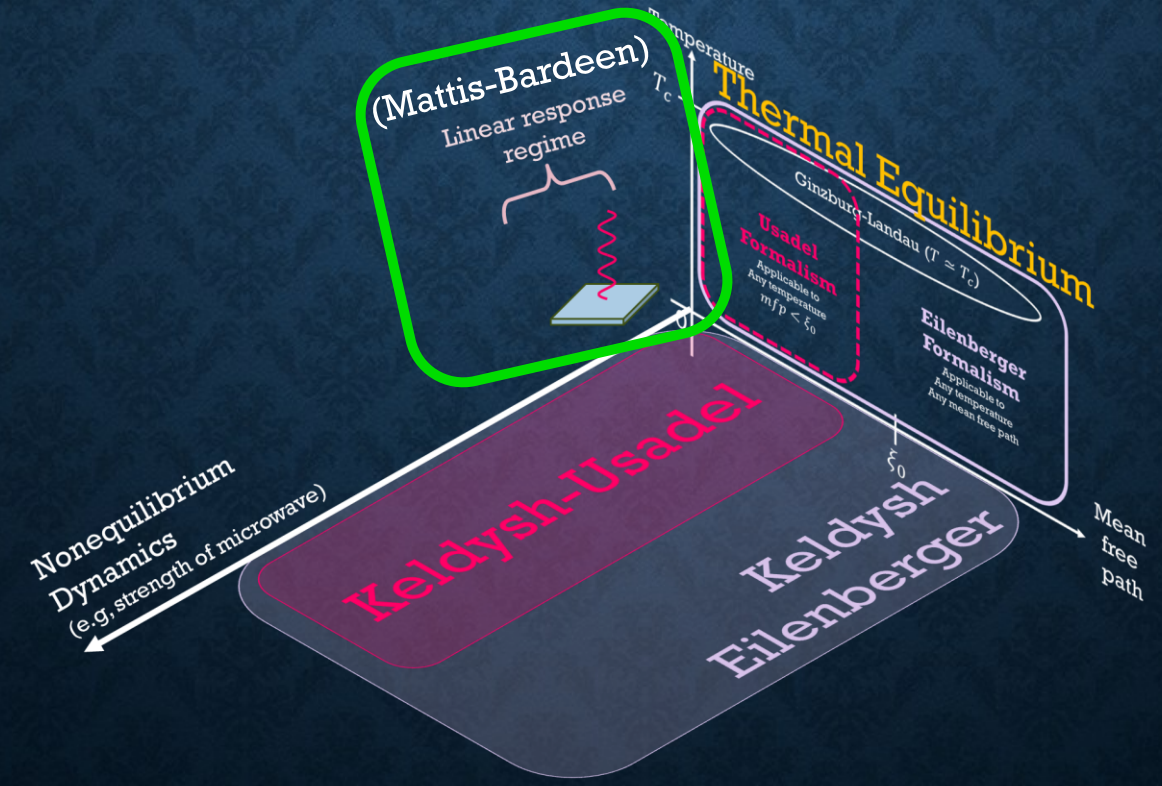
Under the ac field



Under the ac field



Let us start from the linear response



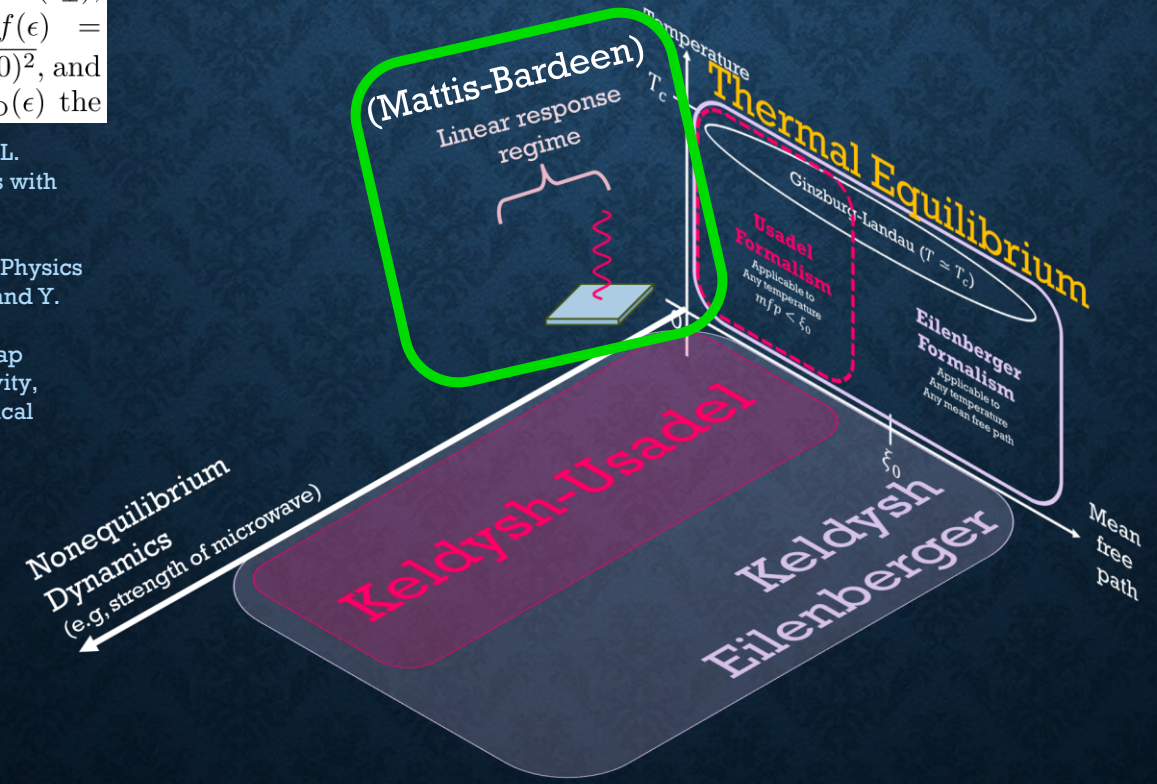
Let us start from the linear response

$$\sigma(\ell, T, \omega) = \sigma_1 + i\sigma_2$$
$$= \frac{-3i\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} d\epsilon \left[\kappa \mathcal{T}_- + \kappa^* \mathcal{T}_+ + \kappa^a (\mathcal{T}_+ - \mathcal{T}_-)\right],$$
$$\kappa = \frac{1}{3} \frac{g_+ g_- + f_+ f_- - 1}{d_+ + d_-},$$
$$\kappa^a = \frac{1}{3} \frac{-g_+ g_-^* - f_+ f_-^* - 1}{d_+ - d_-^*}.$$

$\tau = \ell/v_f$ is the electron relaxation time. We define $\epsilon_\pm := \epsilon \pm \hbar\omega/2$ and set $g_\pm := g(\epsilon_\pm)$, $f_\pm := f(\epsilon_\pm)$, $d_\pm := d(\epsilon_\pm)$, and $\mathcal{T}_\pm := \mathcal{T}(\epsilon_\pm)$, where $g(\epsilon) = -i(\epsilon + i0)/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $f(\epsilon) = -i\Delta/\sqrt{\Delta^2 - (\epsilon + i0)^2}$, $d(\epsilon) = i\gamma + i\sqrt{\Delta^2 - (\epsilon + i0)^2}$, and $\mathcal{T}(\epsilon) := \tanh(\epsilon/2k_BT) = 1 - 2f_{\rm FD}(\epsilon)$, with $f_{\rm FD}(\epsilon)$ the

- W. Zimmermann, E. Brandt, M. Bauer, E. Seider, and L. Genzel, Optical conductivity of BCS superconductors with arbitrary purity, Physica C **183**, 99 (1991).
- D. Rainer and J. A. Sauls, Strong coupling theory of superconductivity, in Superconductivity: From Basic Physics to the Latest Developments, edited by P. N. Butcher and Y. Lu (World Scientic, Singapore, 1995), p. 45.
- T. Kubo, Effects of Nonmagnetic Impurities and Subgap States on the Kinetic Inductance, Complex Conductivity, Quality Factor, and Depairing Current Density, Physical Review Applied **17**, 014018 (2022)

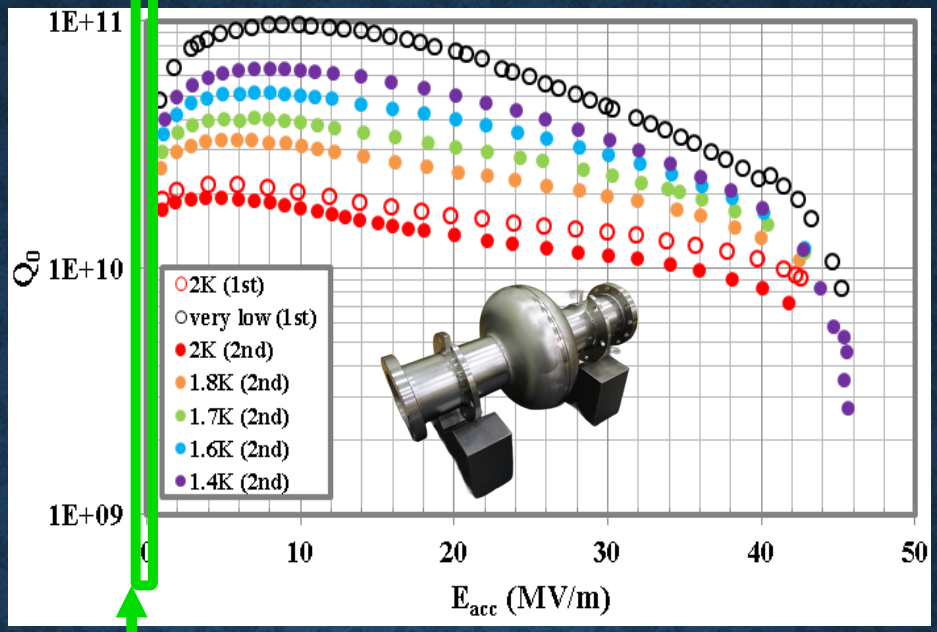
The theory was established decades ago.
We just need to plug in parameters such as temperature.



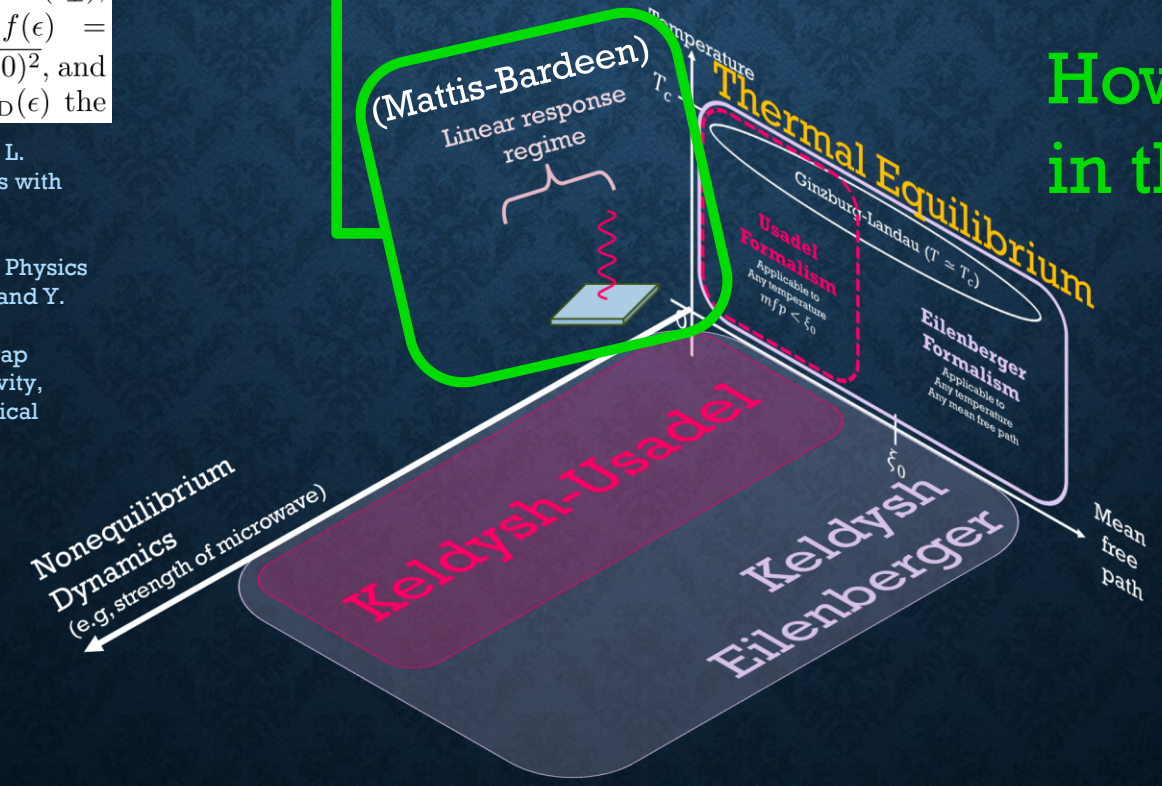
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- D. Rainer and J. A. Sauls, Strong coupling theory of superconductivity, in Superconductivity: From Basic Physics to the Latest Developments, edited by P. N. Butcher and Y. Lu (World Scientic, Singapore, 1995), p. 45.
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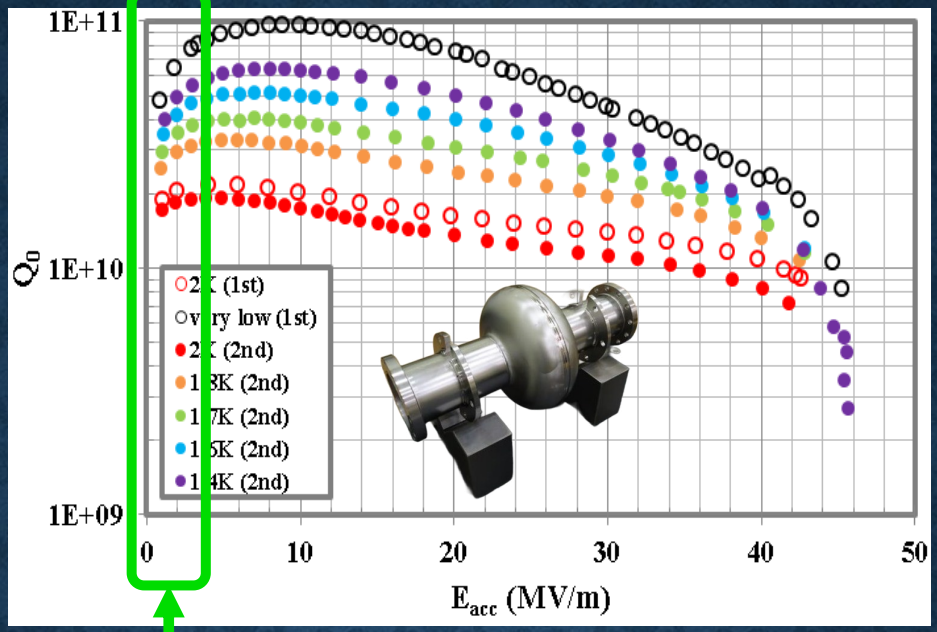


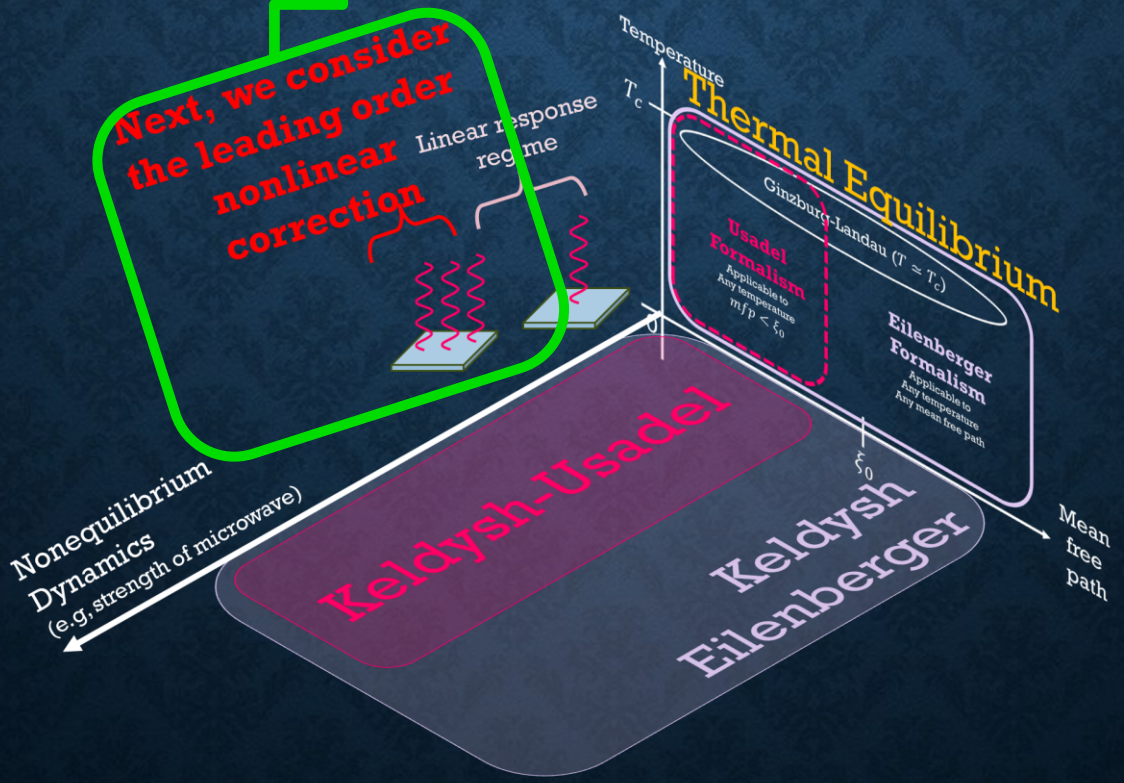
T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022



However, it is only valid in the limit $E_{acc} \to 0$

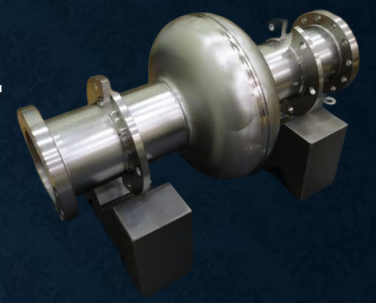
The leading-order nonlinear correction allows us to calculate the field dependence in the weak-field regime.





Keldysh–Usadel Equation for SC under a strong ac field

T. Kubo,
Phys. Rev. Applied
24, 064061 (2025)
https://doi.org/10.1103/8fh8-nknj



$$
\begin{aligned}
&-i\frac{s}{8}\sum_{\eta=\pm\omega_{\rm ac}}\sum_{\eta'=\pm\omega_{\rm ac}}2\pi\delta(\omega+\eta+\eta')\times\\
&\Big[\hat{\tau}_3\hat{g}_e^r\Big(\epsilon-\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^r\Big(\epsilon+\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\\
&-\hat{g}_e^r\Big(\epsilon-\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^r\Big(\epsilon+\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\Big]\\
=&\Big(\epsilon+\frac{\hbar\omega}{2}\Big)\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)-\Big(\epsilon-\frac{\hbar\omega}{2}\Big)\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\\
&+\delta\hat{\Delta}(\omega)\hat{g}_e^r\Big(\epsilon-\frac{\hbar\omega}{2}\Big)-\hat{g}_e^r\Big(\epsilon+\frac{\hbar\omega}{2}\Big)\delta\hat{\Delta}(\omega)\\
&+[\hat{\Delta}_e,\delta\hat{g}^r(\epsilon,\omega)].
\end{aligned}
\tag{11}
$$

$$
\delta\Delta(\omega)=-\frac{\mathcal{G}}{8}\int d\epsilon\,\mathrm{Tr}[(-i\tau_2)\delta\hat{g}^K(\epsilon,\omega)]. \tag{15}
$$

$$
\begin{aligned}
&-i\frac{s}{8}\sum_{\eta=\pm\omega_{\rm ac}}\sum_{\eta'=\pm\omega_{\rm ac}}2\pi\delta(\omega+\eta+\eta')\times\\
&\Big[\hat{\tau}_3\hat{g}_e^R\Big(\epsilon-\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^K\Big(\epsilon+\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\\
&-\hat{g}_e^R\Big(\epsilon-\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^K\Big(\epsilon+\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\\
&+\hat{\tau}_3\hat{g}_e^K\Big(\epsilon-\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^A\Big(\epsilon+\frac{\hbar\eta}{2}+\frac{\hbar\eta'}{2}\Big)\\
&-\hat{g}_e^K\Big(\epsilon-\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\hat{g}_e^A\Big(\epsilon+\frac{\hbar\eta}{2}-\frac{\hbar\eta'}{2}\Big)\hat{\tau}_3\Big]\\
=&\Big(\epsilon+\frac{\hbar\omega}{2}\Big)\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)-\Big(\epsilon-\frac{\hbar\omega}{2}\Big)\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\\
&+\delta\hat{\Delta}(\omega)\hat{g}_e^K\Big(\epsilon-\frac{\hbar\omega}{2}\Big)-\hat{g}_e^K\Big(\epsilon+\frac{\hbar\omega}{2}\Big)\delta\hat{\Delta}(\omega)\\
&+[\hat{\Delta}_e,\delta\hat{g}^K(\epsilon,\omega)].
\end{aligned}
\tag{13}
$$

We solve these equations to obtain the **nonequilibrium corrections** ($\delta\hat{g}^{R,A,K}$ and $\delta\Delta$).

To obtain the current response, we substitute the solutions ($\delta\hat{g}^{R,A,K}$, $\delta\Delta$) into

$$
\begin{aligned}
\mathbf{J}(t)&=\mathbf{J}^{(1)}(t)+\mathbf{J}^{(3)}(t)\\
&=\frac{\sigma_n}{8e}\int\Big\{\mathbf{S}^{(1)}(\epsilon,t)+\mathbf{S}^{(3)}(\epsilon,t)\Big\}d\epsilon,
\end{aligned}
\tag{49}
$$

$$
\mathbf{S}^{(1)}(\epsilon,t)=\mathrm{Tr}\big[\hat{\tau}_3\big\{\hat{g}_e^R\circ(\hat{\partial}\circ\hat{g}_e^K)+\hat{g}_e^K\circ(\hat{\partial}\circ\hat{g}_e^A)\big\}\big] \tag{50}
$$

$$
\begin{aligned}
\mathbf{S}^{(3)}(\epsilon,t)=\mathrm{Tr}\big[\hat{\tau}_3\big\{&\hat{g}_e^R\circ(\hat{\partial}\circ\delta\hat{g}^K)+\hat{g}_e^K\circ(\hat{\partial}\circ\delta\hat{g}^A)\\
&+\delta\hat{g}^R\circ(\hat{\partial}\circ\hat{g}_e^K)+\delta\hat{g}^K\circ(\hat{\partial}\circ\hat{g}_e^A)\big\}\big].
\end{aligned}
\tag{51}
$$

← Linear response

← **Non**linear correction of $\mathcal{O}(A^3)$

Schematically, the results can be illustrated as

Linear response

Nonlinear corrections

Current $J \sim \boxed{Agg} + \boxed{Ag\delta g}$ $\delta g \sim A^2 + \delta\Delta$ and $\delta\Delta \sim \Psi A^2$

Schematically, the results can be illustrated as

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Nonlinear corrections

Current $J \sim \boxed{Agg} + \boxed{Ag\delta g}$ $\delta g \sim A^2 + \delta\Delta$ and $\delta\Delta \sim \Psi A^2$

$$\sim A(A^2 + \delta\Delta)$$

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$\sim A(A^2 + \delta\Delta)$

Time averaged $\delta\Delta$: **Eliashberg effect**

Time dependent $\delta\Delta$: **Higgs mode**

Note: This theory automatically includes the **Eliashberg effect** (microwave-induced gap enhancement) as well as the **Higgs-mode** contribution.

Eliashberg effect? Higgs mode?

Eliashberg effect? Higgs mode?

The variation of order parameter $\delta\Delta$ does not appear in standard linear-response theories such as Mattis–Bardeen theory because $\delta\Delta$ is an $\mathcal{O}(A^2)$ effect, but it does emerge in nonlinear response.

$$\delta\Delta \propto A_0^2 \cos^2 \omega t$$

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$$\delta\Delta \propto A_0^2 \cos^2 \omega t \ \propto 1 + \cos 2\omega t$$

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Time-averaged nonequilibrium gap variation

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Time-averaged nonequilibrium gap variation

Time dependent nonequilibrium gap variation

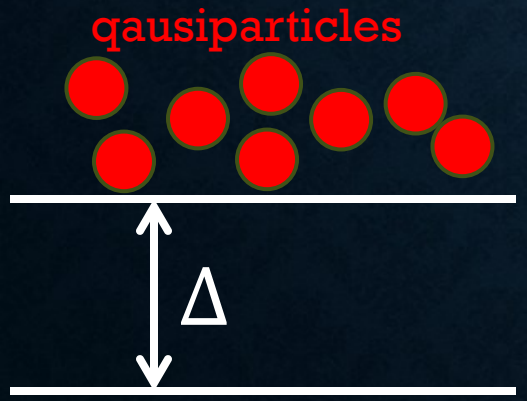
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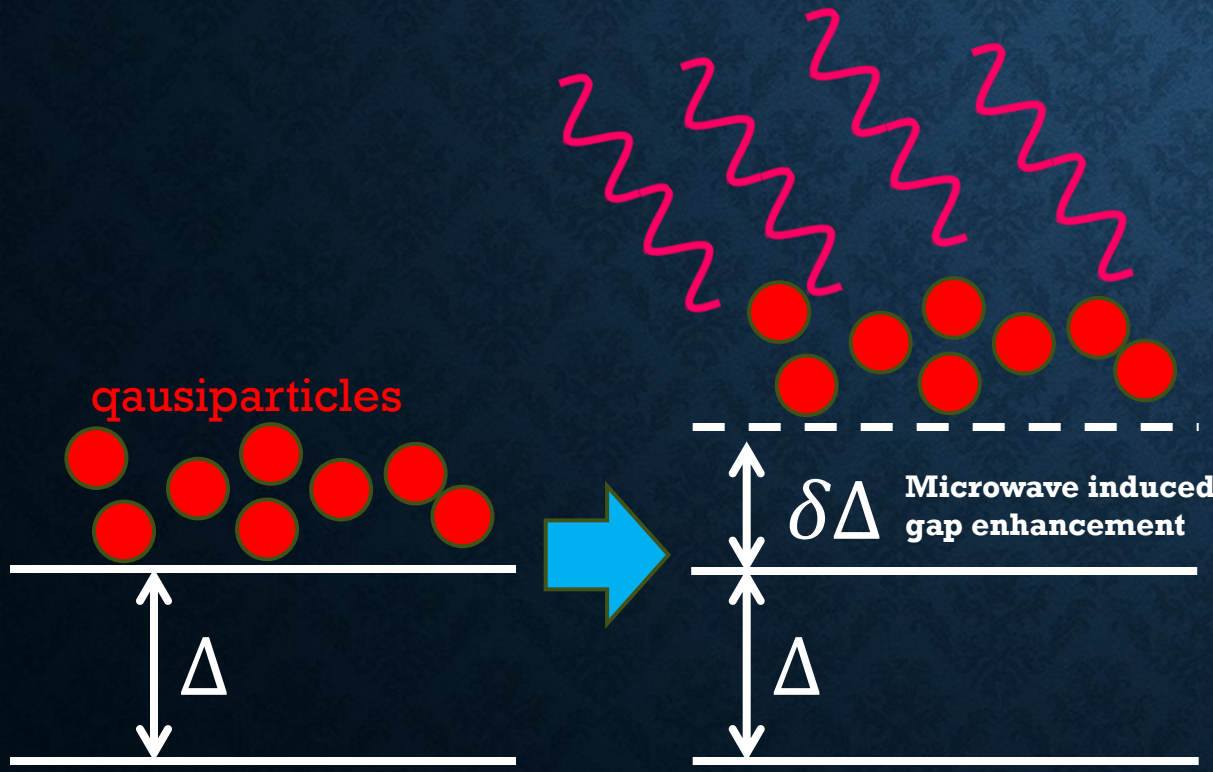
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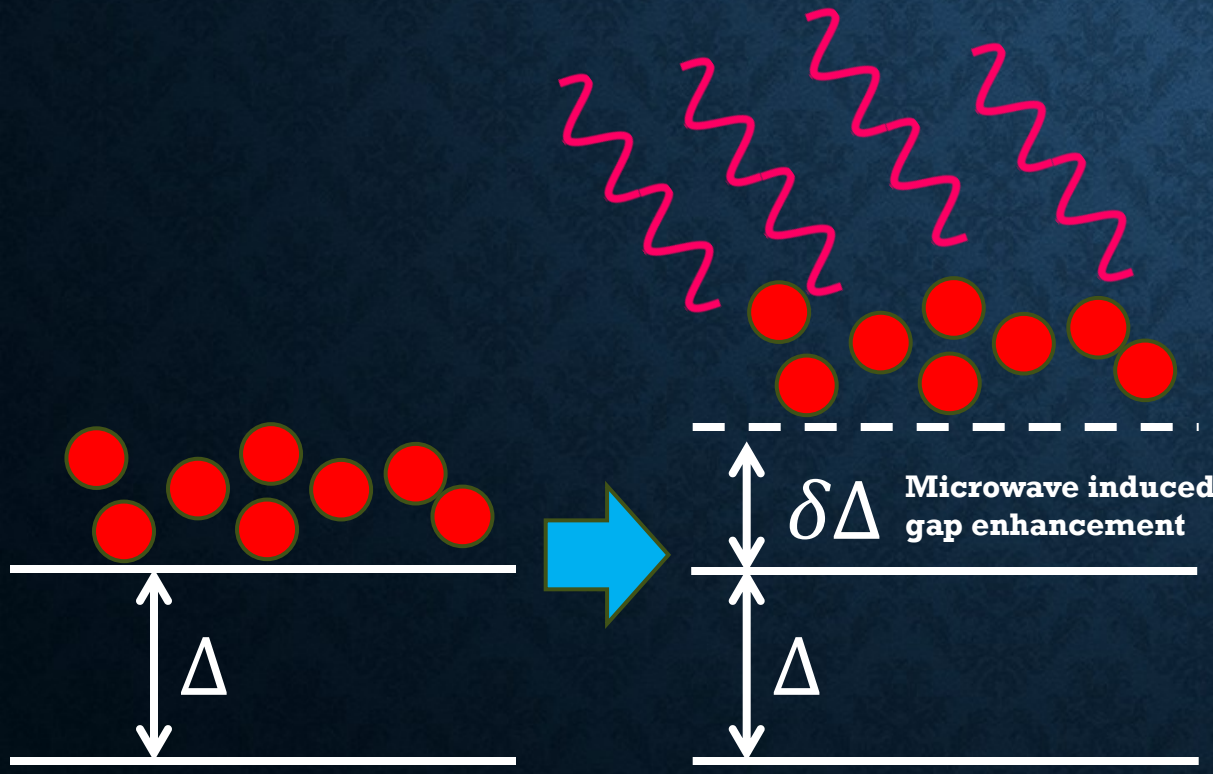


Eliashberg effect? Higgs mode?

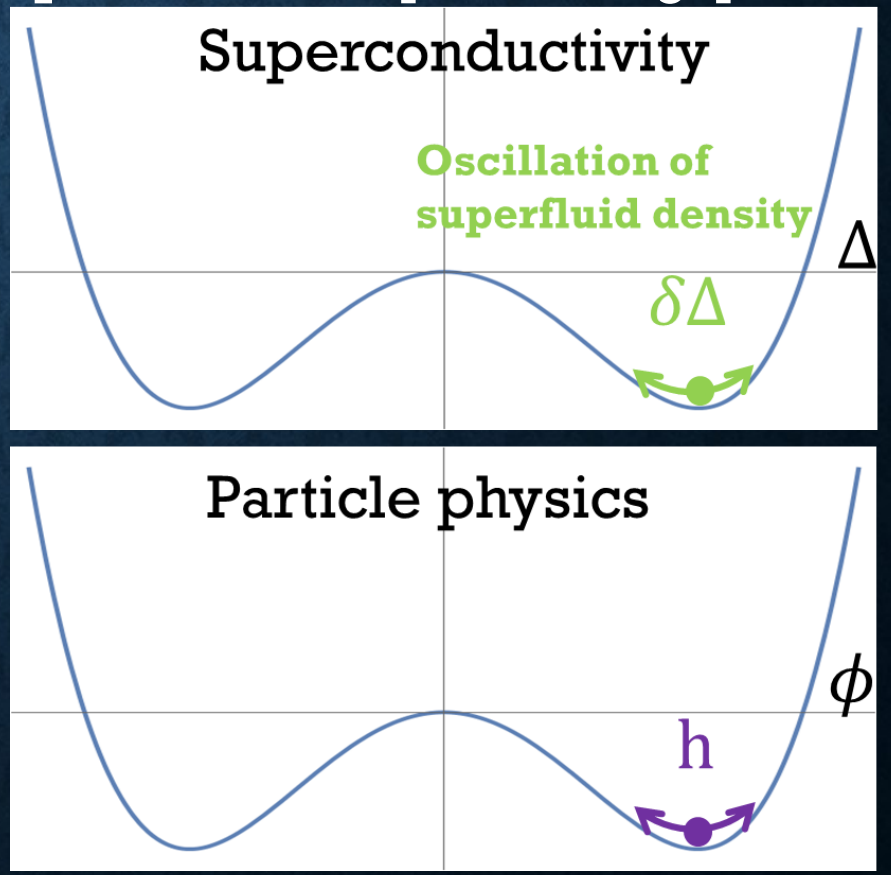
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Time-averaged nonequilibrium gap variation



Time dependent nonequilibrium gap variation



Nonlinear correction

$$\delta J \sim A(A^2 + \delta\Delta)$$

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https://doi.org/10.1103/8fh8-nknj

Nonlinear correction

$$\delta J \sim A(A^2 + \delta\Delta) \Rightarrow \delta\sigma \sim \frac{\delta J}{A} \sim A^2 + \delta\Delta$$

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$$\delta\sigma_1 = \delta\sigma_1^{qqq} + \delta\sigma_1^{\mathrm{Higgs}} + \delta\sigma_1^{\mathrm{Eliash}}$$

$$\delta\sigma_1^{qqq} = 2\sqrt{\pi}\sigma_n \mathrm{Re}[I_{1\mathrm{H}}^{qqq}]\frac{s}{\hbar\omega_{\mathrm{ac}}} \qquad I_{1\mathrm{H}}^{\mathrm{qqq}} = \frac{-1}{16\sqrt{\pi}}\int K_{1\mathrm{H}} d\epsilon,$$

$$K_{1\mathrm{H}}(\epsilon,\omega_{\mathrm{ac}}) = \sum_{i=1}^{6} K_{1\mathrm{H},i}, \tag{61}$$

$$K_{1\mathrm{H},1} = i\frac{(F_1 - F_{-3})F_{-1} + (G_1 - G_{-3})G_{-1}}{4\hbar\omega_{\mathrm{ac}}(F_1 + F_{-3})}$$
$$\times\Big[\Big\{(F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^*\Big\}(\mathcal{T}_{-1} - \mathcal{T}_{-3})$$
$$+\Big\{(F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1}\Big\}\mathcal{T}_{-1}\Big], \tag{62}$$

$$K_{1\mathrm{H},2} = i\frac{(G_3 - G_{-1})G_1 + (F_3 - F_{-1})F_1}{4\hbar\omega_{\mathrm{ac}}(F_3 + F_{-1})}$$
$$\times\{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1\}\mathcal{T}_{-1}, \tag{63}$$

$$K_{1\mathrm{H},3} = -i\frac{(F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1}{4\hbar\omega_{\mathrm{ac}}(F_3 - F_{-1}^*)}$$
$$\times\Big[\Big\{(G_3 + G_{-1}^*)G_1^* + (F_3 + F_{-1}^*)F_1^*\Big\}(\mathcal{T}_3 - \mathcal{T}_1)$$
$$+\Big\{(G_3 + G_{-1}^*)G_1 + (F_3 + F_{-1}^*)F_1\Big\}(\mathcal{T}_{-1} - \mathcal{T}_1)\Big], \tag{64}$$

$$K_{1\mathrm{H},4} = -i\frac{F_1G_{-1} + G_1F_{-1}}{2(\epsilon_{-1}G_{-1} - F_{-1}\Delta)}$$
$$\times\{(F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1}\}\mathcal{T}_{-1}, \tag{65}$$

$$K_{1\mathrm{H},5} = -i\frac{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1}{2(\epsilon_1G_1 - F_1\Delta)}$$
$$\times\Big\{(G_1F_{-1}^* + F_1G_{-1}^*)(\mathcal{T}_{-1} - \mathcal{T}_1)$$
$$+(G_1F_{-1} + G_{-1}F_1)\mathcal{T}_{-1}\Big\}, \tag{66}$$

$$K_{1\mathrm{H},6} = -\frac{G_{-1}^*\mathrm{Im}F_1 + F_{-1}^*\mathrm{Im}G_1}{\epsilon_1\mathrm{Im}G_1 - \mathrm{Im}F_1\Delta}$$
$$\times\Big\{(\mathrm{Re}G_3\mathrm{Im}F_1 + \mathrm{Im}G_1\mathrm{Re}F_3)(T_3 - T_1)$$
$$+(\mathrm{Re}G_{-1}\mathrm{Im}F_1 + \mathrm{Im}G_1\mathrm{Re}F_{-1})(T_{-1} - T_1)\Big\}, \tag{67}$$

Higgs Contribution: Time dependent $\delta\Delta$

$$\delta\sigma_1^{\mathrm{Higgs}} = 2\sqrt{\pi}\sigma_n \mathrm{Re}[I_{1\mathrm{H}}^{\mathrm{Higgs}}]\frac{s}{\hbar\omega_{\mathrm{ac}}}$$

$$I_{1\mathrm{H}}^{\mathrm{Higgs}} = \frac{-1}{16\sqrt{\pi}}\Psi_{2\mathrm{H}}\int Z_{1\mathrm{H}}^{\mathrm{Higgs}} d\epsilon$$

$$Z_{1\mathrm{H}}^{\mathrm{Higgs}}(\epsilon,\omega_{\mathrm{ac}}) = \sum_{i=1,2,3} Z_{1\mathrm{H},i}^{\mathrm{Higgs}}, \tag{71}$$

$$Z_{1\mathrm{H},1}^{\mathrm{Higgs}} = -\frac{2(F_1 - F_{-3})}{\hbar\omega_{\mathrm{ac}}(F_1 + F_{-3})}$$
$$\times\Big[\Big\{(F_1 + F_{-3})G_{-1}^* + (G_1 + G_{-3})F_{-1}^*\Big\}(\mathcal{T}_{-1} - \mathcal{T}_{-3})$$
$$+\Big\{(F_1 + F_{-3})G_{-1} + (G_1 + G_{-3})F_{-1}\Big\}\mathcal{T}_{-1}\Big], \tag{72}$$

$$Z_{1\mathrm{H},2}^{\mathrm{Higgs}} = -\frac{2(F_3 - F_{-1})}{\hbar\omega_{\mathrm{ac}}(F_3 + F_{-1})}$$
$$\times\Big\{(F_3 + F_{-1})G_1 + (G_3 + G_{-1})F_1\Big\}\mathcal{T}_{-1}, \tag{73}$$

$$Z_{1\mathrm{H},3}^{\mathrm{Higgs}} = -\frac{2(F_3 + F_{-1}^*)}{\hbar\omega_{\mathrm{ac}}(F_3 - F_{-1}^*)}$$
$$\times\Big\{(F_3 - F_{-1}^*)G_1 + (G_3 - G_{-1}^*)F_1\Big\}(\mathcal{T}_3 - \mathcal{T}_{-1}) \tag{74}$$

Eliashberg Contribution: Time averaged $\delta\Delta$

$$\delta\sigma_1^{\mathrm{Eliash}} = 2\sqrt{\pi}\sigma_n \mathrm{Re}[I_{1\mathrm{H}}^{\mathrm{Eliash}}]\frac{s}{\hbar\omega_{\mathrm{ac}}}$$

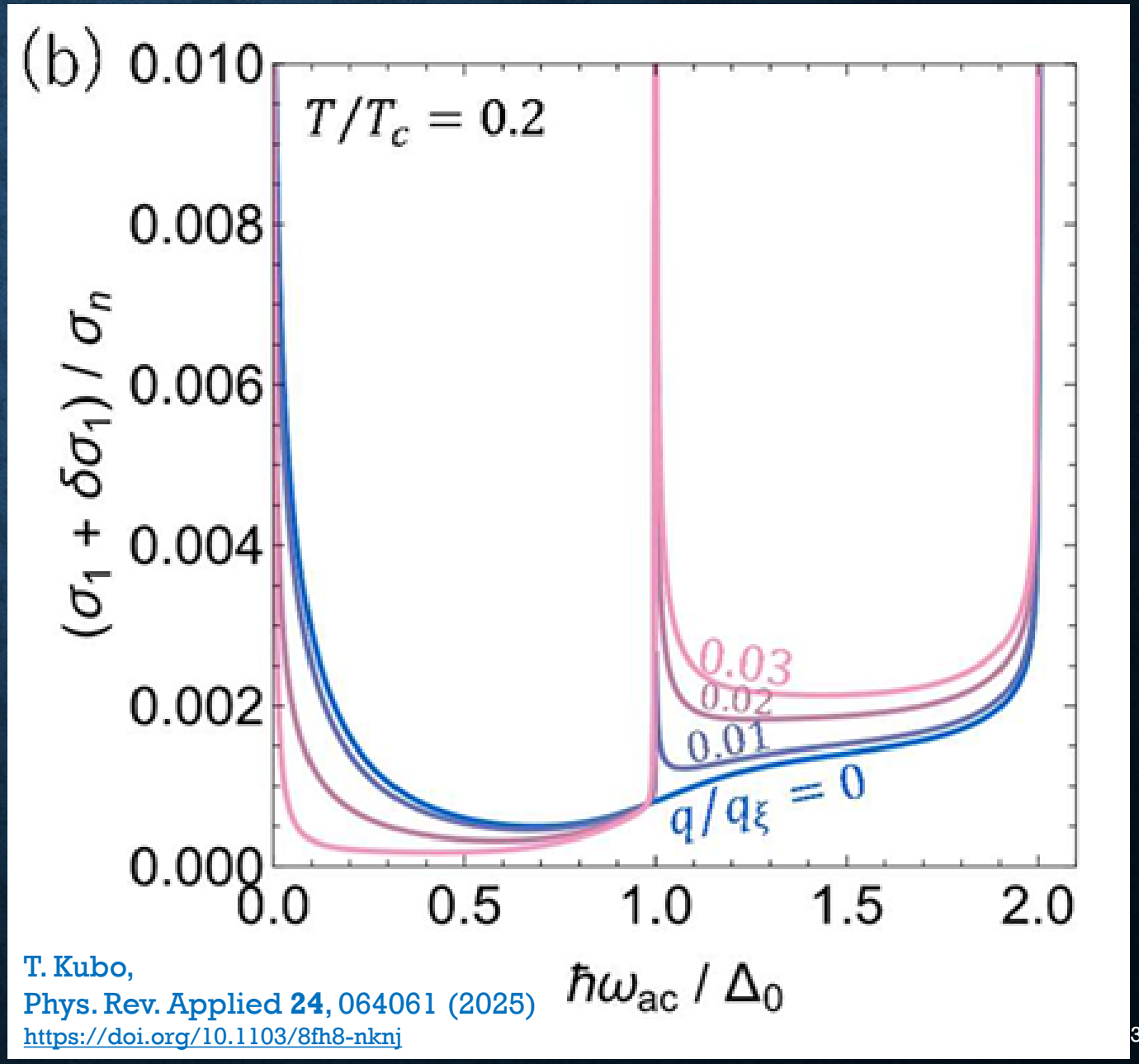
$$I_{1\mathrm{H}}^{\mathrm{Eliash}} = \frac{-1}{16\sqrt{\pi}}\Psi_{0\mathrm{H}}\int Z_{1\mathrm{H}}^{\mathrm{Eliash}} d\epsilon$$

$$Z_{1\mathrm{H}}^{\mathrm{Eliash}}(\epsilon,\omega_{\mathrm{ac}}) = \sum_{i=1,2} Z_{1\mathrm{H},i}^{\mathrm{Eliash}}, \tag{68}$$

$$Z_{1\mathrm{H},1}^{\mathrm{Eliash}} = \frac{4G_{-1}(G_1F_{-1} + F_1G_{-1})}{\epsilon_{-1}G_{-1} - F_{-1}\Delta}\mathcal{T}_{-1}, \tag{69}$$

$$Z_{1\mathrm{H},2}^{\mathrm{Eliash}} = \frac{4G_1}{\epsilon_1G_1 - F_1\Delta}\Big\{(G_1F_{-1} + G_{-1}F_1)\mathcal{T}_{-1}$$
$$+(G_1F_{-1}^* + F_1G_{-1}^*)(\mathcal{T}_{-1} - \mathcal{T}_1)\Big\}, \tag{70}$$

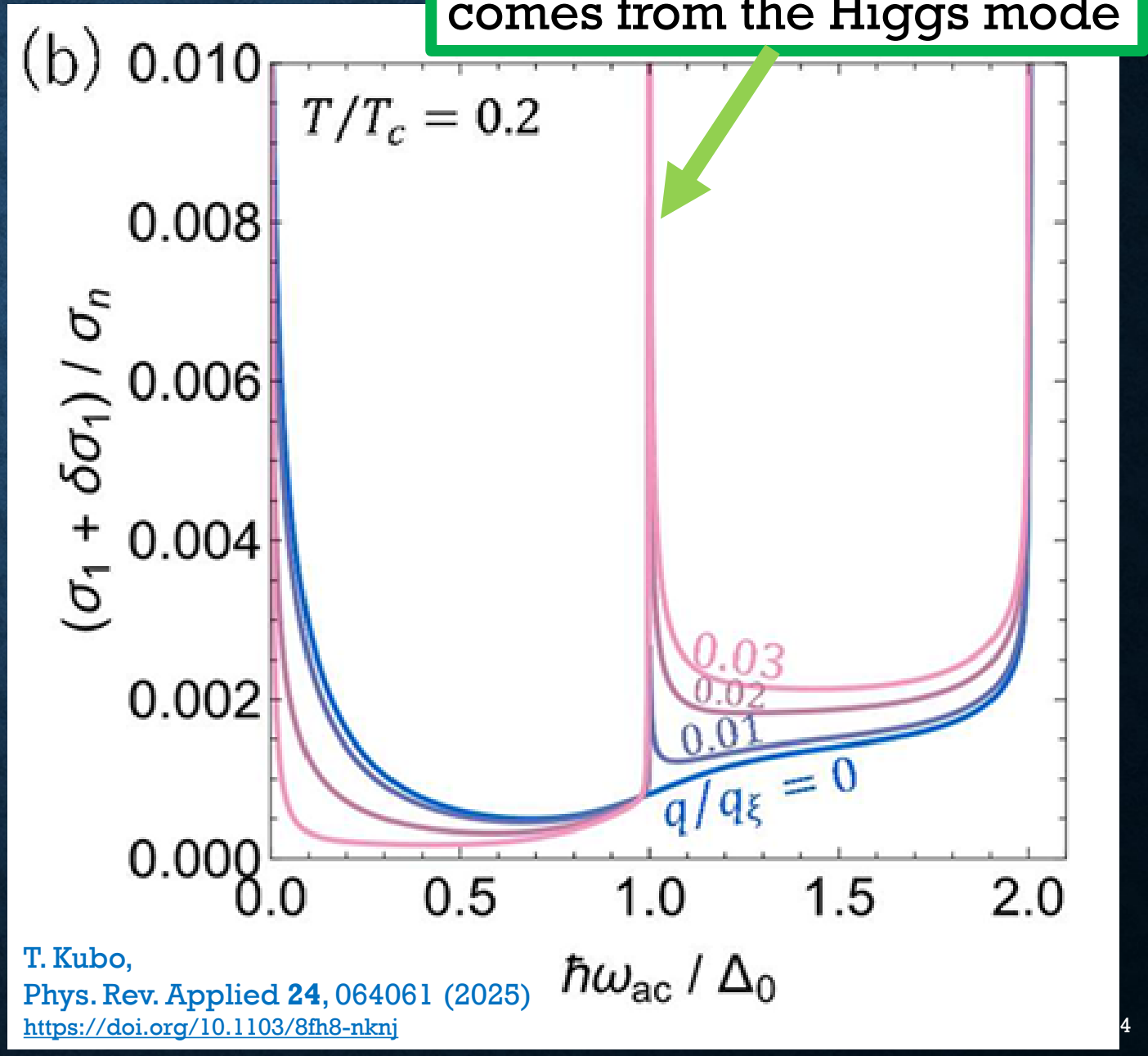
Results



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https://doi.org/10.1103/8fh8-nknj

Results

The resonance peak at Δ comes from the Higgs mode

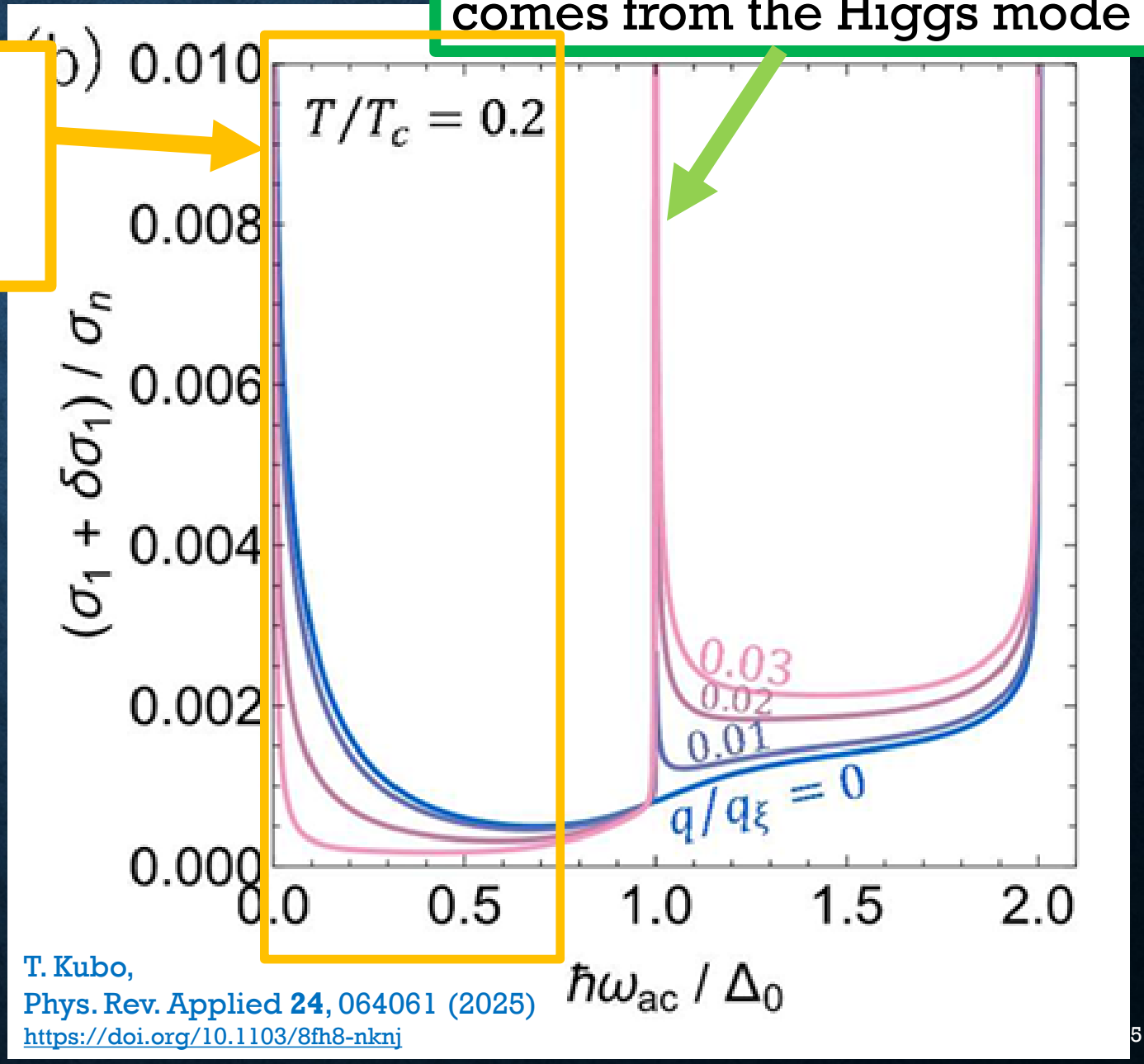


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Results

At lower frequencies,
$\sigma_1 + \delta\sigma_1$ can be a decreasing function of the ac amplitude (q).

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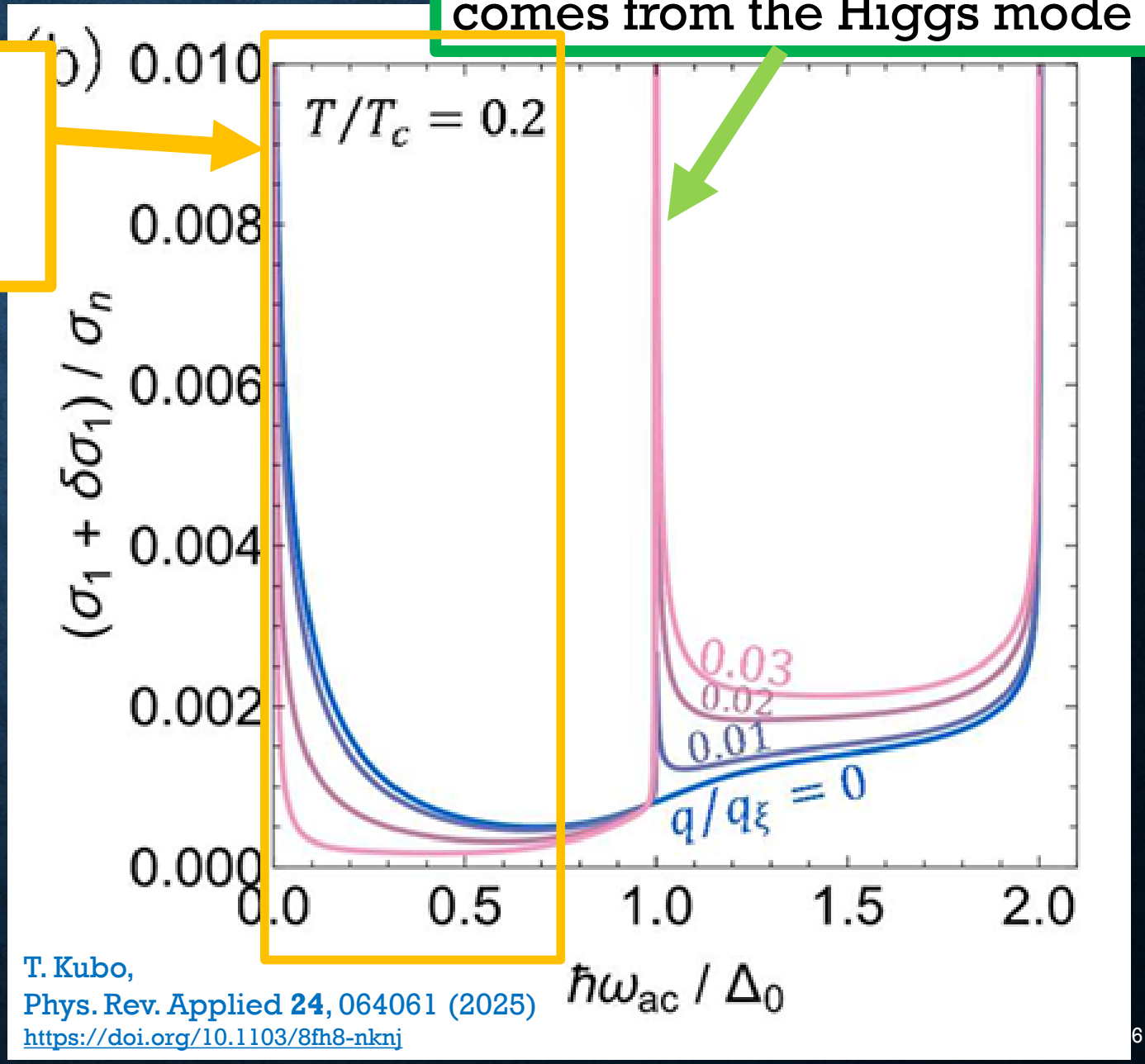


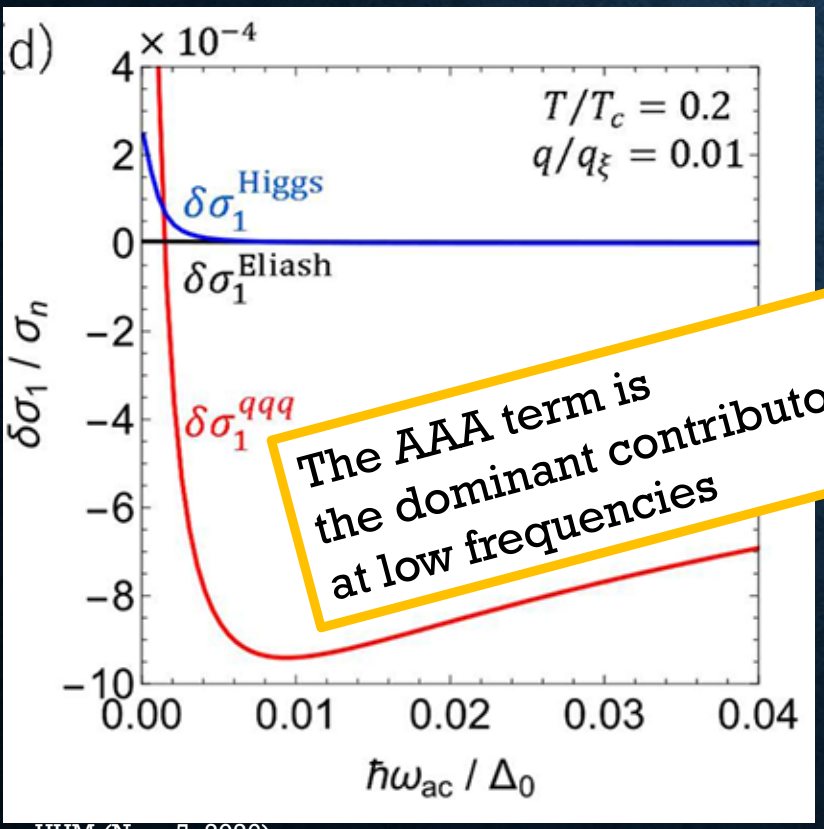
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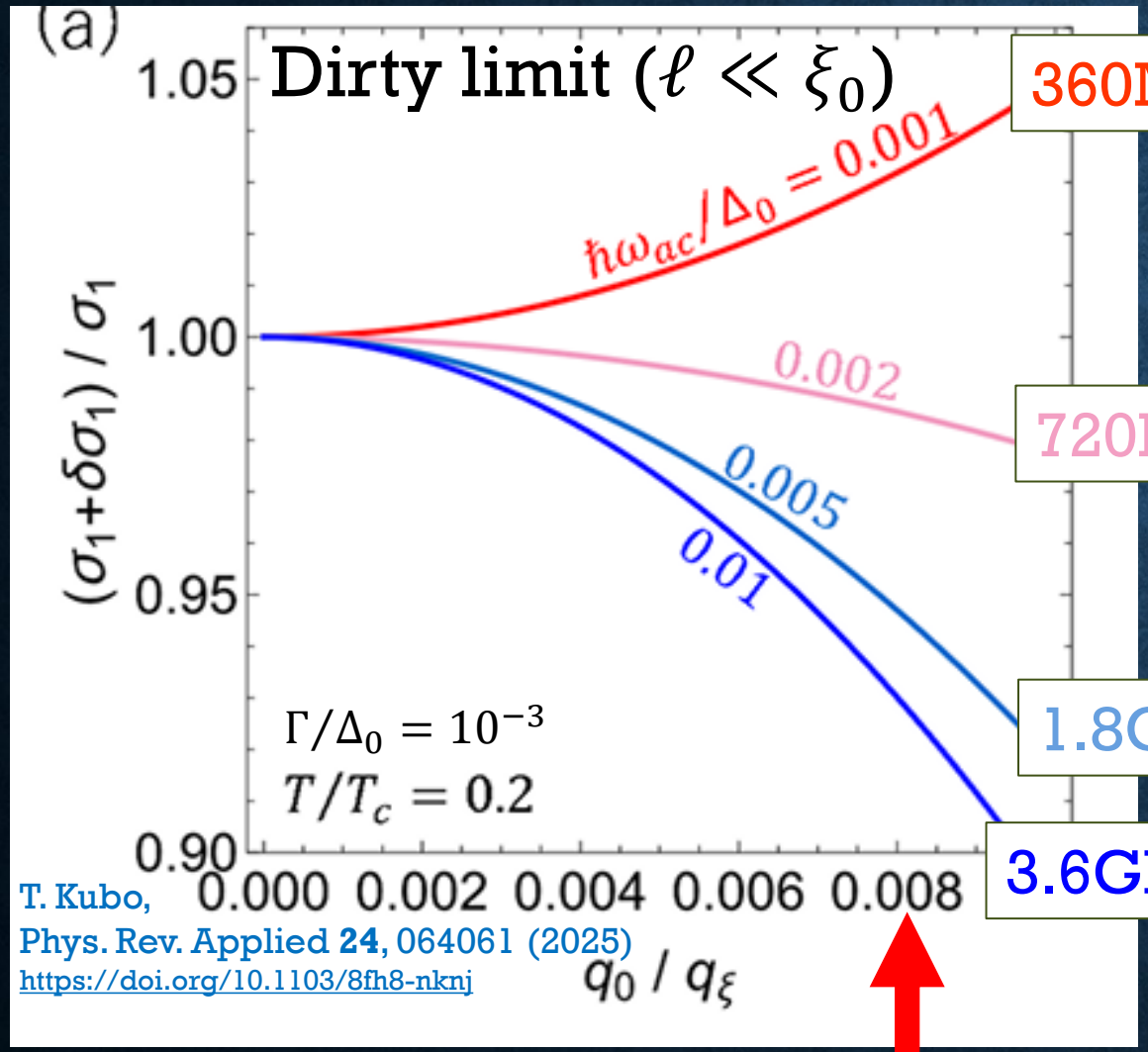




The AAA term is the dominant contributor at low frequencies

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ac amplitude dependent dissipative conductivity



360MHz for Nb

720MHz for Nb

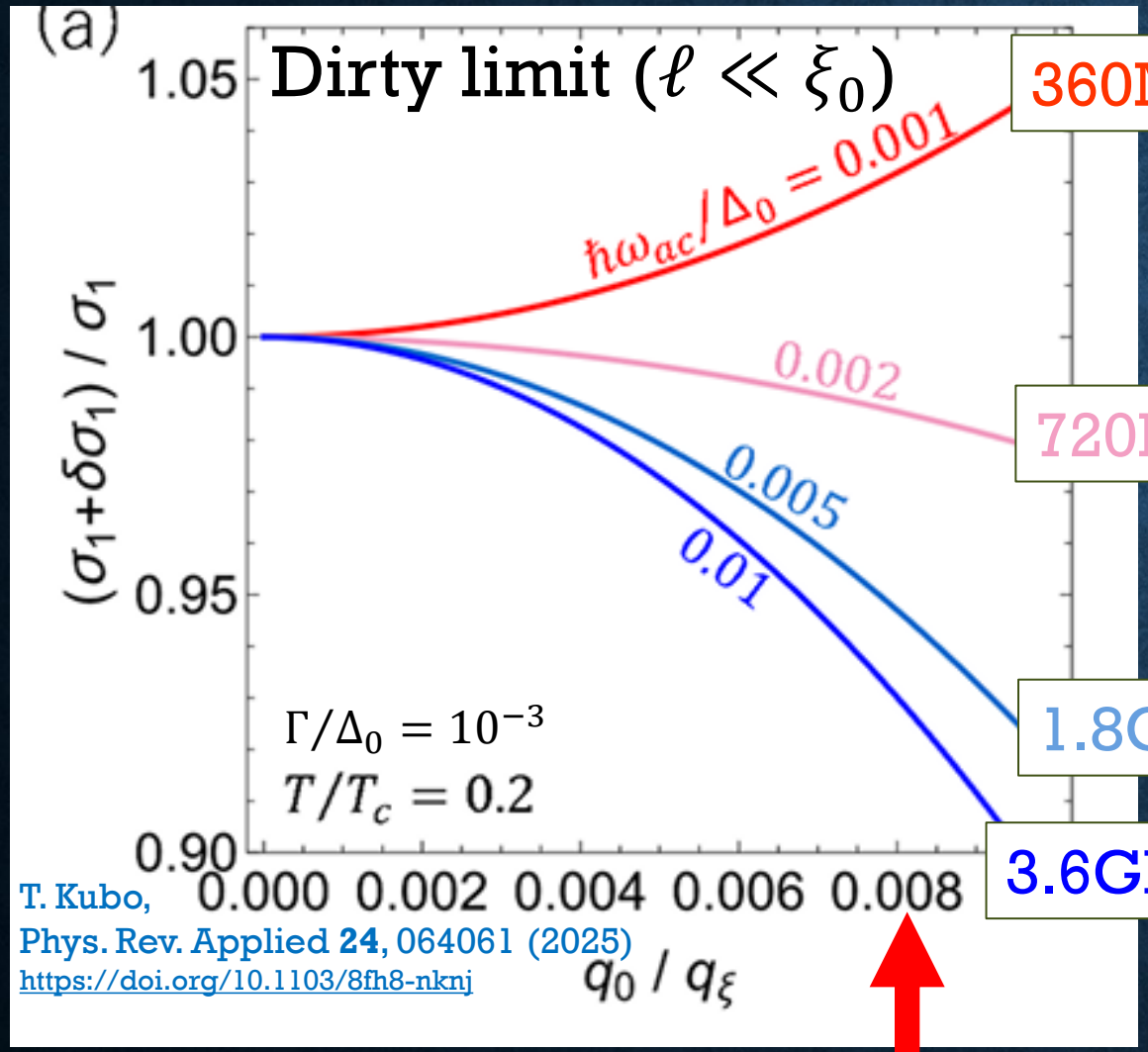
1.8GHz for Nb

3.6GHz for Nb

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Note: this corresponds to **a few (mT)**

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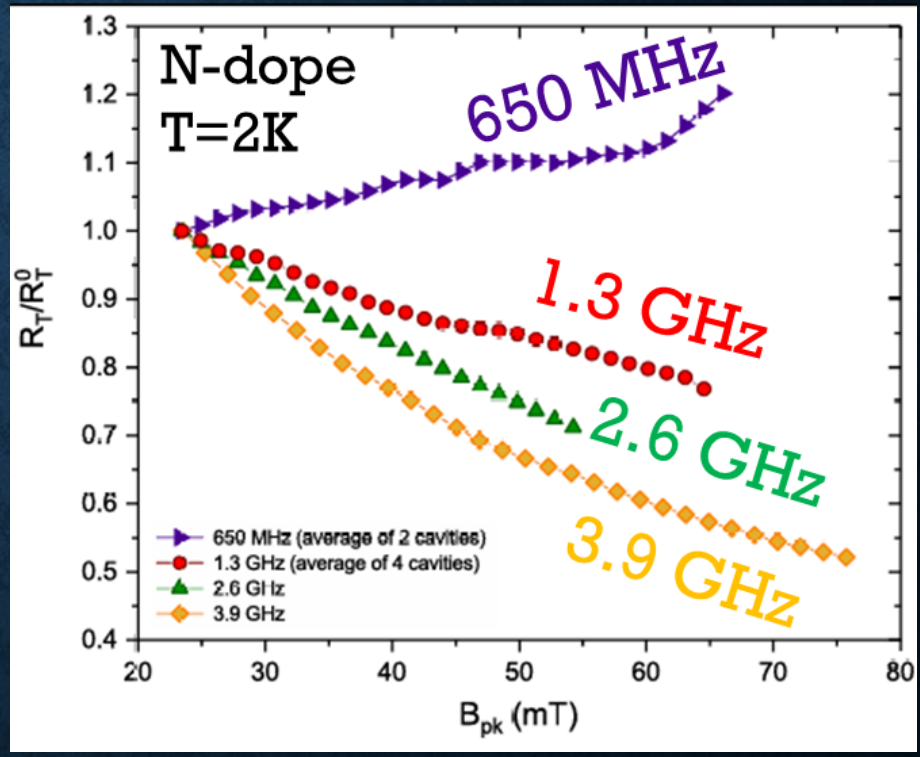
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Experiments: Moderately dirty ($\ell \sim \xi_0$)

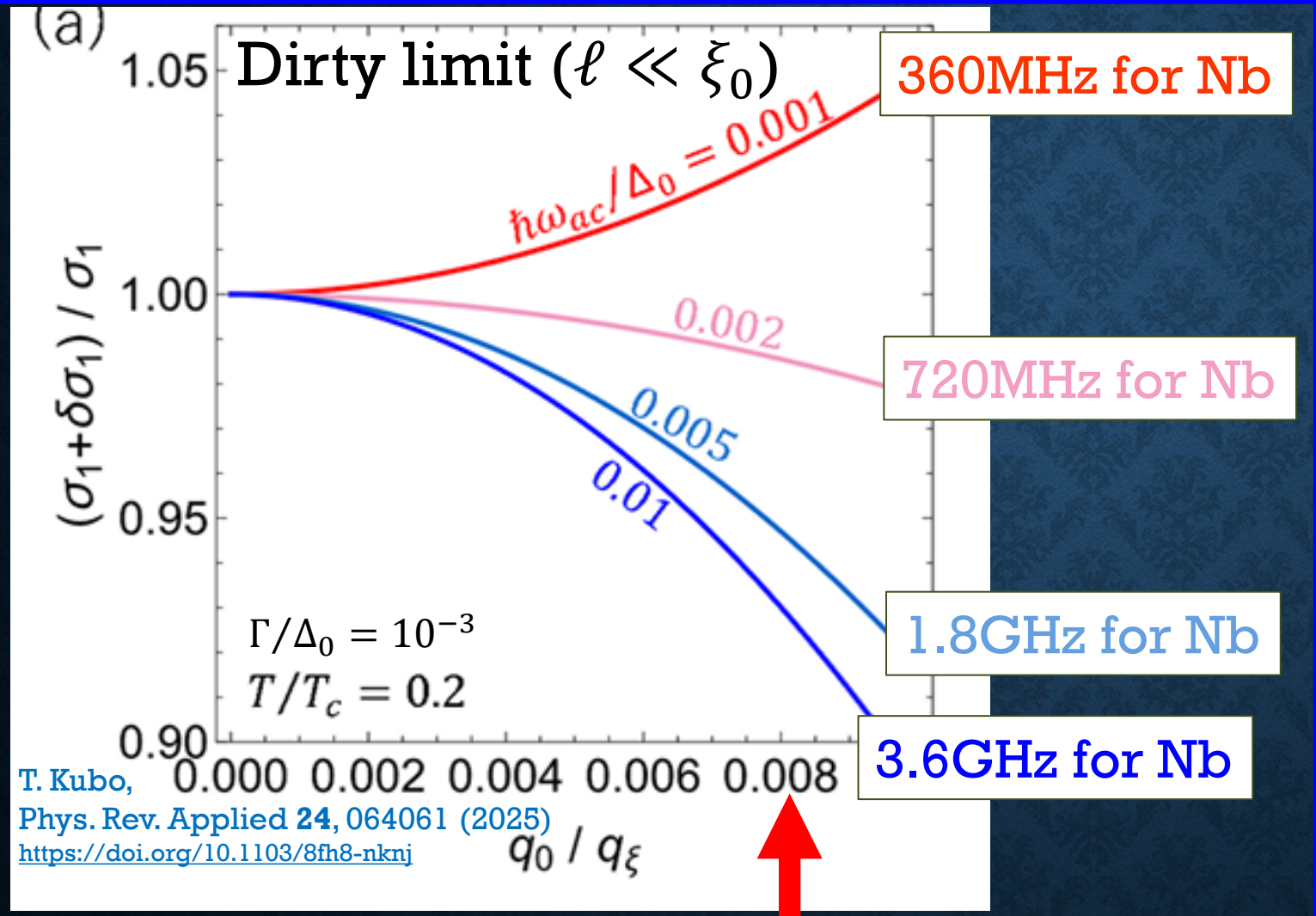
- M. Martinello et al., PRL. **121**, 224801 (2018)
- P. Dhakal et al., PRAB **27**, 062001 (2024)



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DOI: https://doi.org/10.1103/PhysRevLett.121.224801

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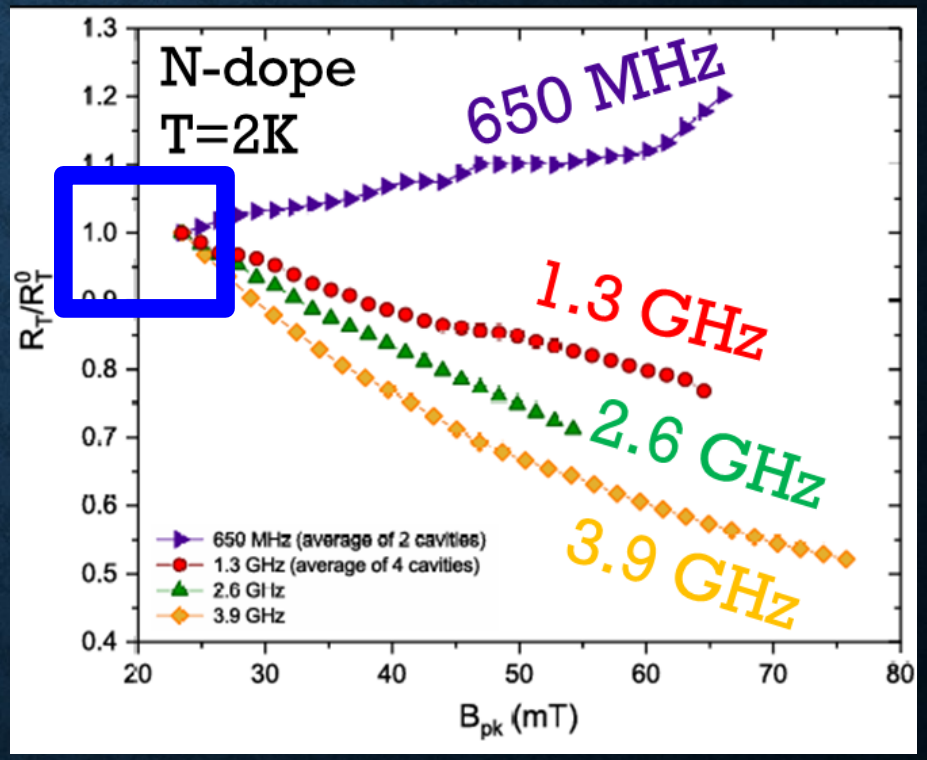
ac amplitude dependent dissipative conductivity



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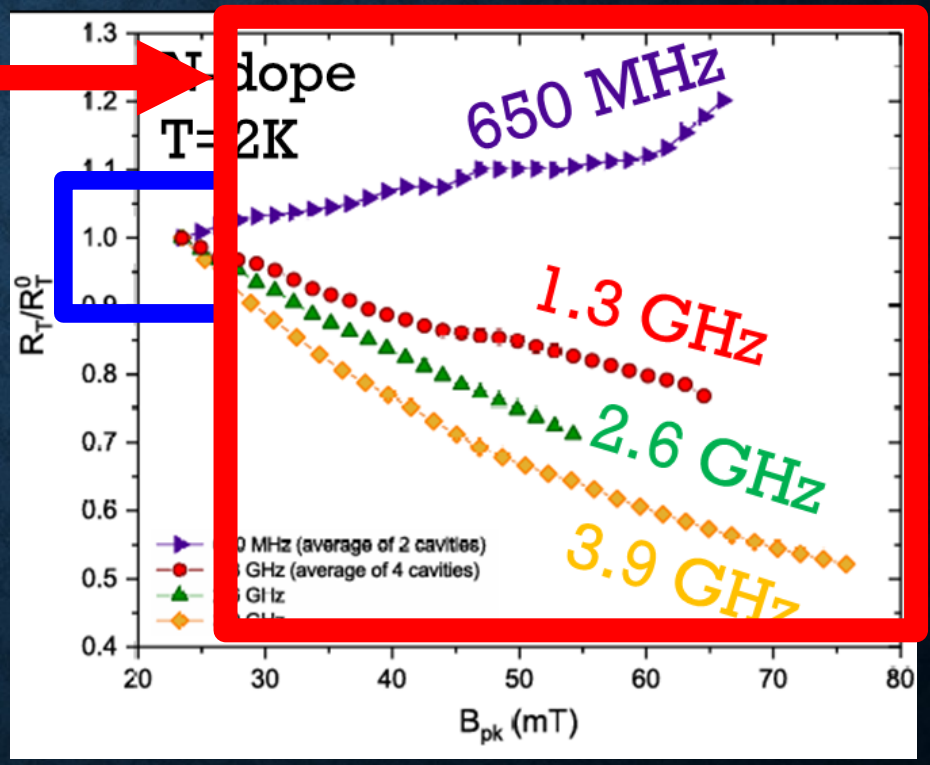
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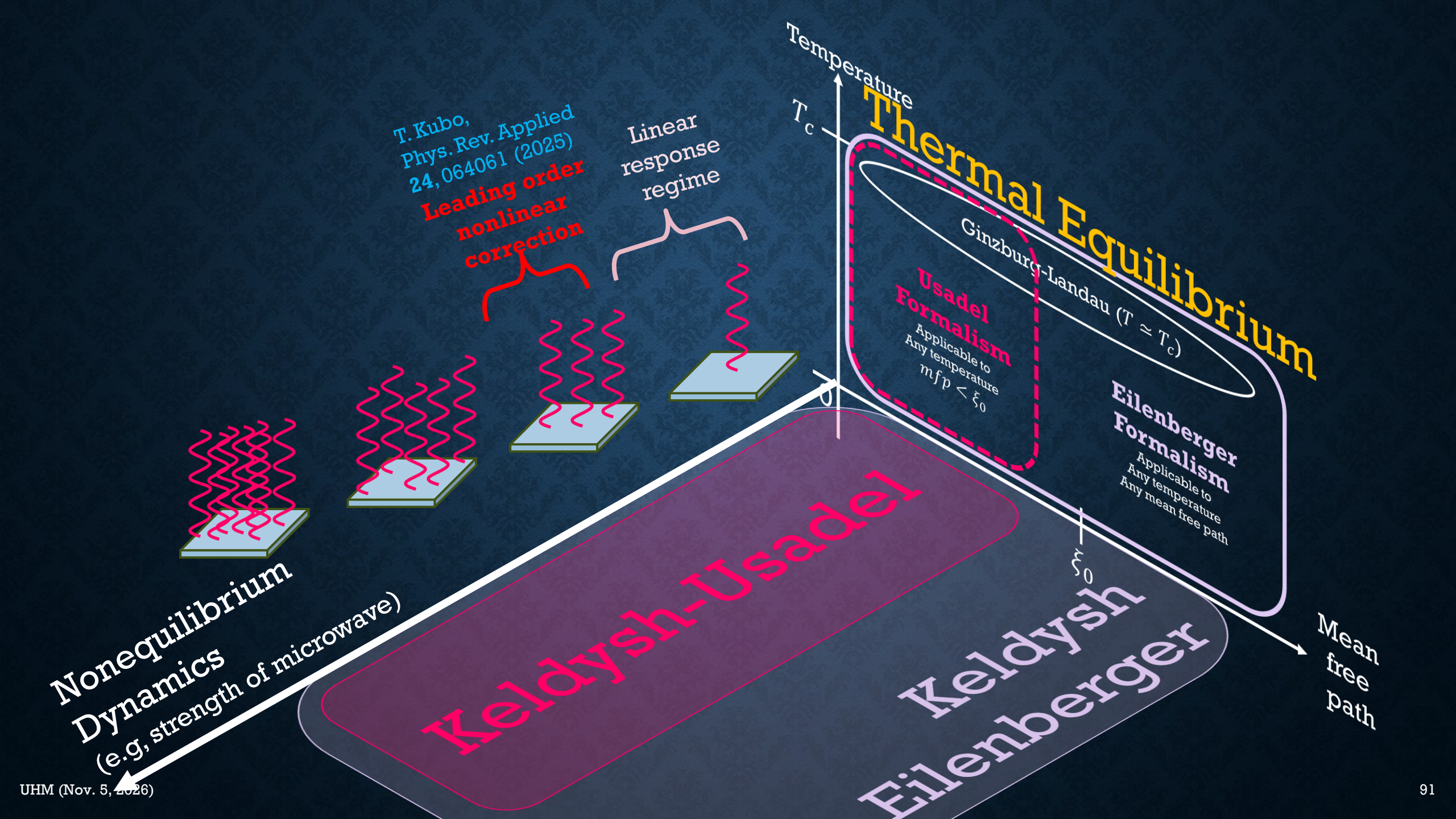
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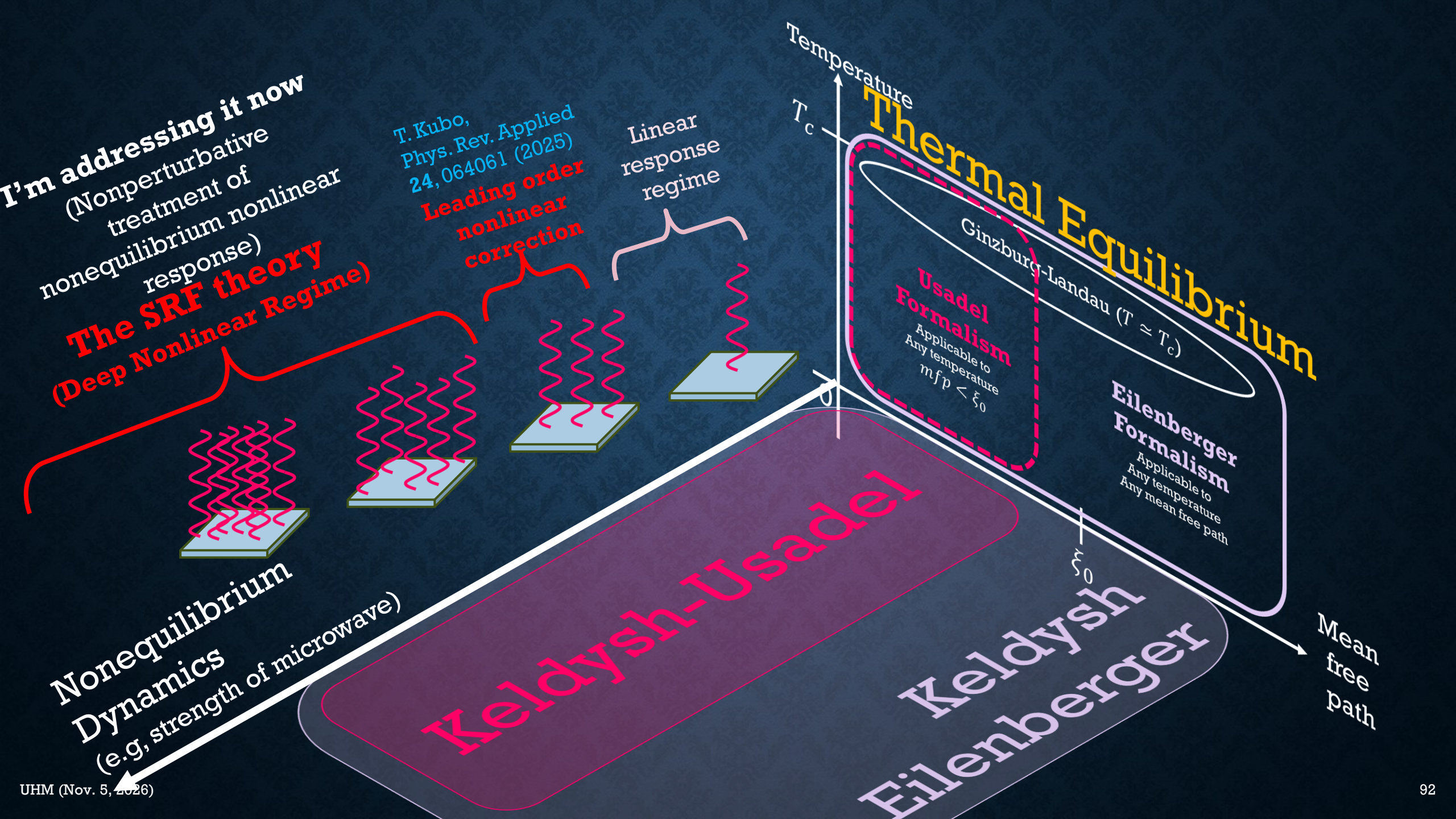
To deal with **the red-framed region**, the theory must be extended **nonperturbatively** to capture the quasiparticle spectrum modified by strong AC currents (**DOS smearing**).

Gurevich, PRL **113**, 087001 (2014)
Kubo-Gurevich, PRB **100**, 064522 (2019)

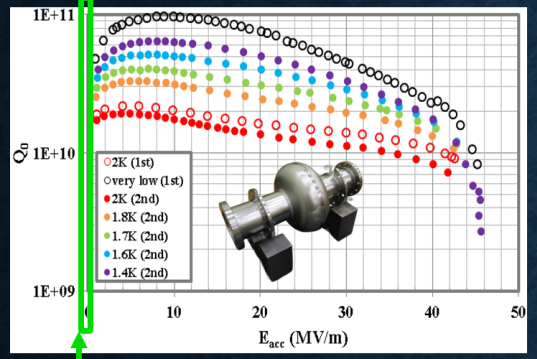


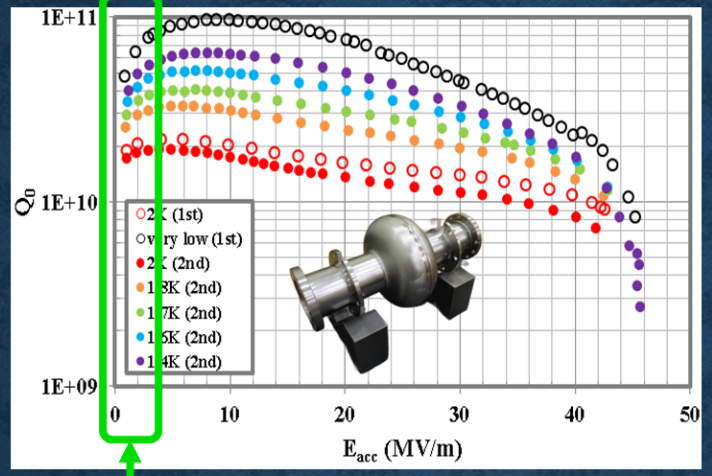
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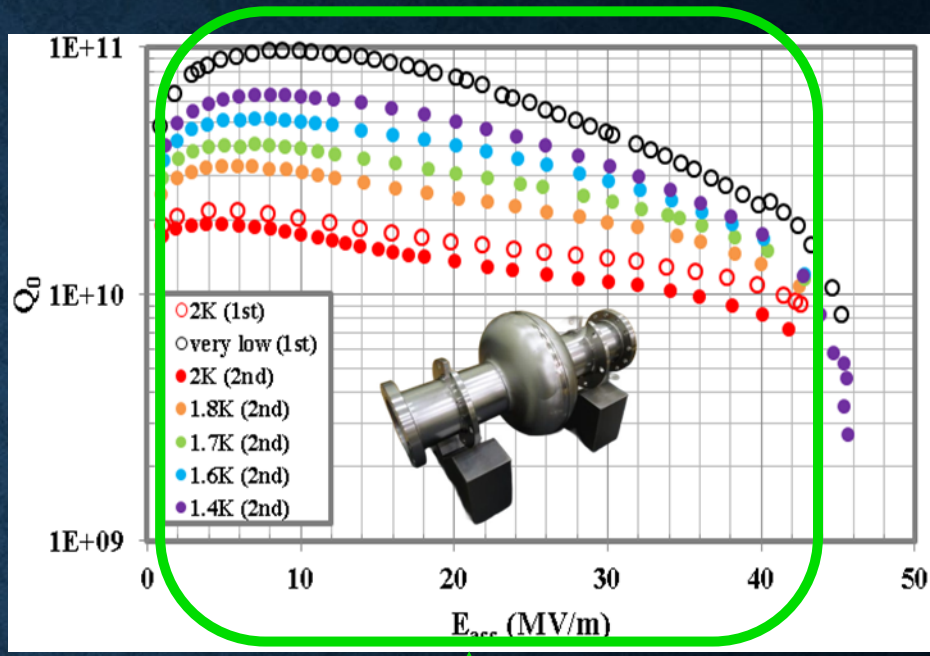

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Leading order
nonlinear
correction
Linear
response
regime
Temperature
T_c
Thermal Equilibrium
Ginzburg-Landau ($T \simeq T_c$)
Usadel
Formalism
Applicable to
Any temperature
$mfp < \xi_0$
Eilenberger
Formalism
Applicable to
Any temperature
Any mean free path
0
ξ_0
Nonequilibrium
Dynamics
(e.g, strength of microwave)
Keldysh-Usadel
Keldysh
Eilenberger
Mean
free
path

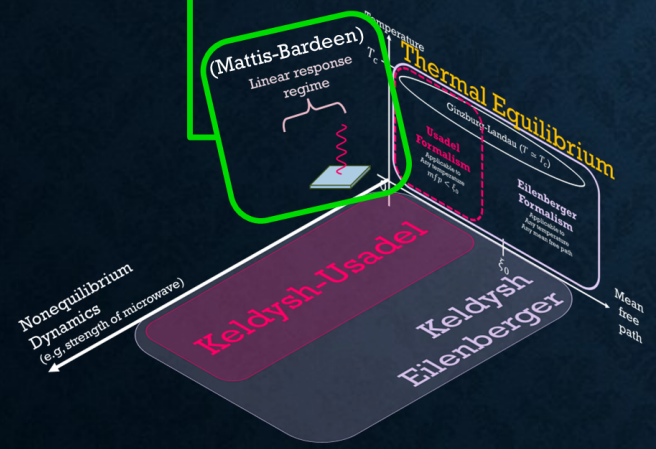

I'm addressing it now
(Nonperturbative treatment of nonequilibrium nonlinear response)
The SRF theory
(Deep Nonlinear Regime)
T. Kubo, Phys. Rev. Applied 24, 064061 (2025)
Leading order nonlinear correction
Linear response regime
Temperature
T_c
Thermal Equilibrium
Ginzburg-Landau ($T \simeq T_c$)
Usadel Formalism
Applicable to
Any temperature
$mfp < \xi_0$
Eilenberger Formalism
Applicable to
Any temperature
Any mean free path
0
ξ_0
Nonequilibrium Dynamics
(e.g, strength of microwave)
Keldysh-Usadel
Keldysh Eilenberger
Mean free path

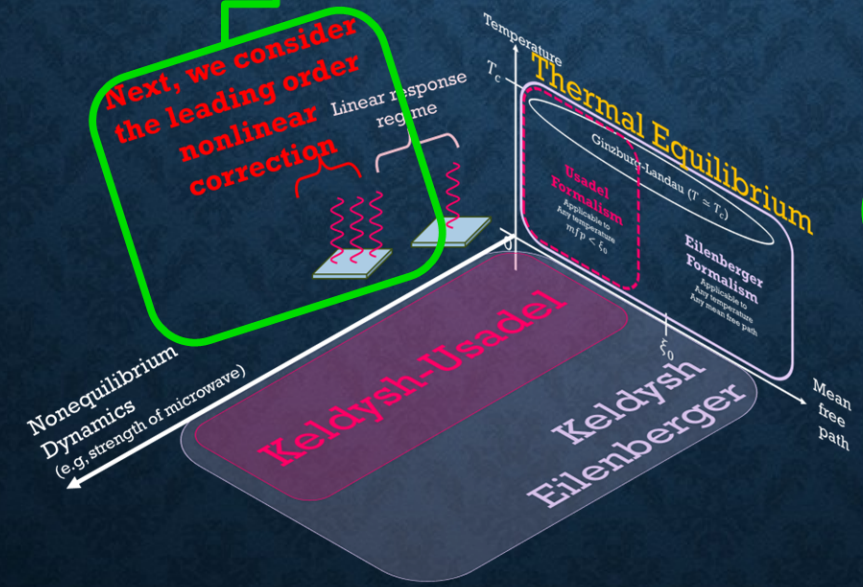
Summary of this part

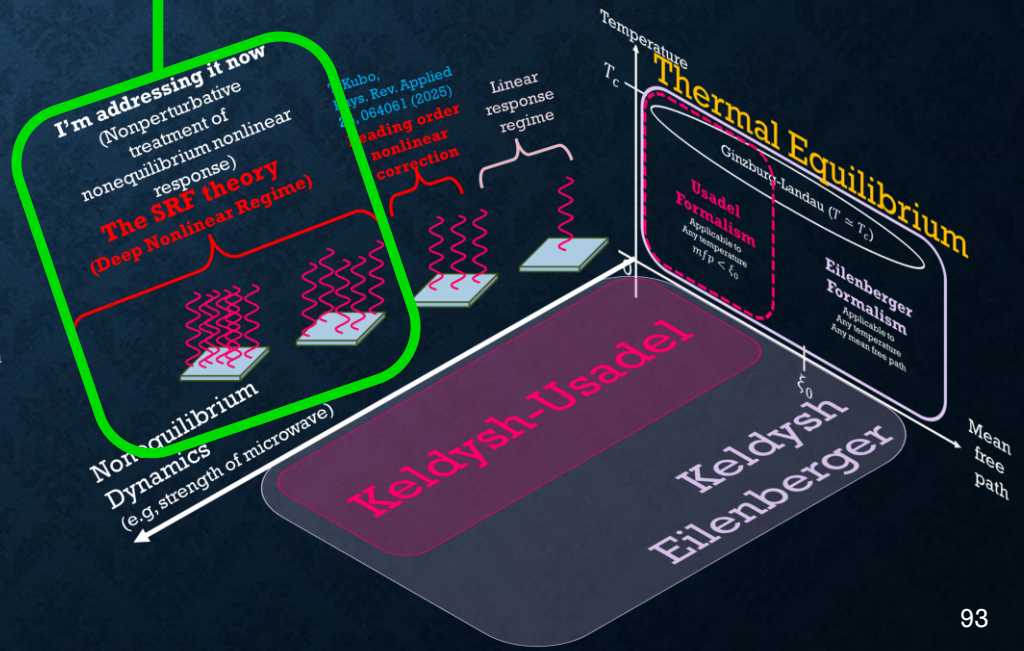


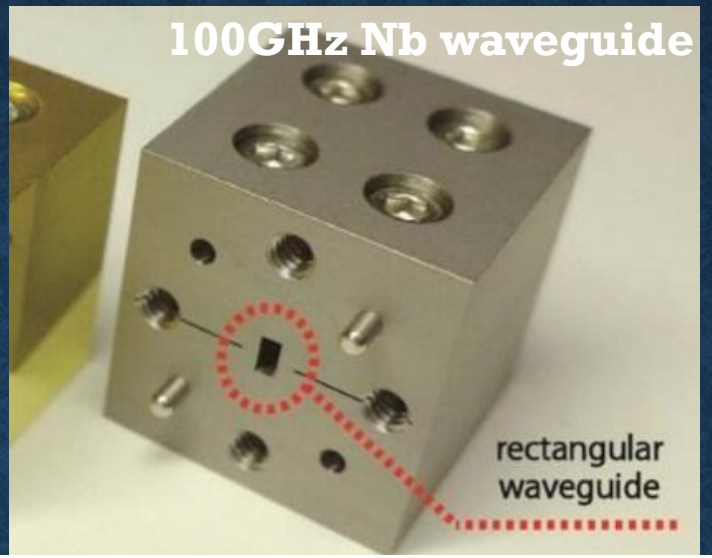












T. Nakajima et al., Propagation in Superconducting Niobium Rectangular Waveguide in the 100 GHz band, J. Phys.: Conf. Ser. **2545**, 012021 (2023)
DOI : 10.1088/1742-6596/2545/1/012021

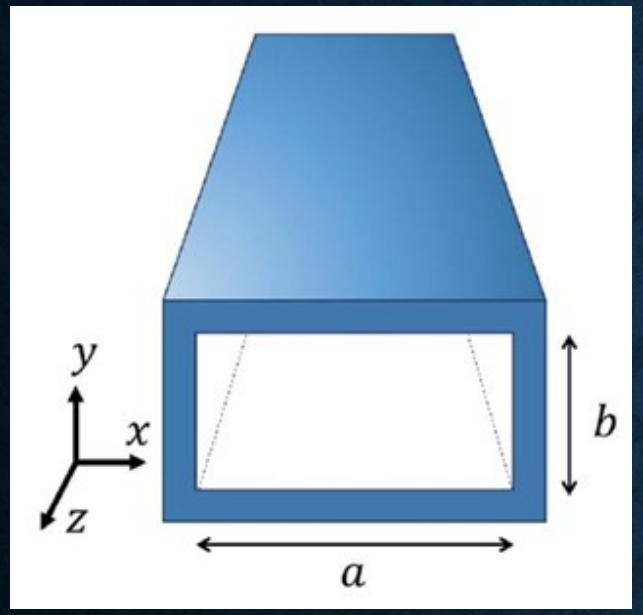
1.27 mm × 2.54 mm
(the so-called WR10)

Non-accelerator application (1)
Millimeter wave and THz superconducting waveguide

Why superconducting waveguides?
Ultra-low-loss transmission at millimetre-wave and sub-terahertz frequencies
→ Higher-sensitivity receiver systems for astronomy (Waveguide multiplexer) and
Improved detection of extremely weak astronomical signals
→ Potential applications to atmospheric remote sensing
→ Possible applications to high-frequency communication systems above 100 GHz

Attenuation at different frequencies and for different material purities

Linear response



$$\alpha = \frac{P_{\text{loss}}}{2P_{\text{flow}}} = \frac{R_s}{Z_0 b} \frac{1 + (2b/a)(\omega_c/\omega)^2}{\sqrt{1 - (\omega_c/\omega)^2}},$$

$$R_s(\ell, T, \omega) = \sqrt{\frac{\mu_0 \omega}{2}} \frac{\sqrt{|\sigma| - \sigma_2}}{|\sigma|}.$$

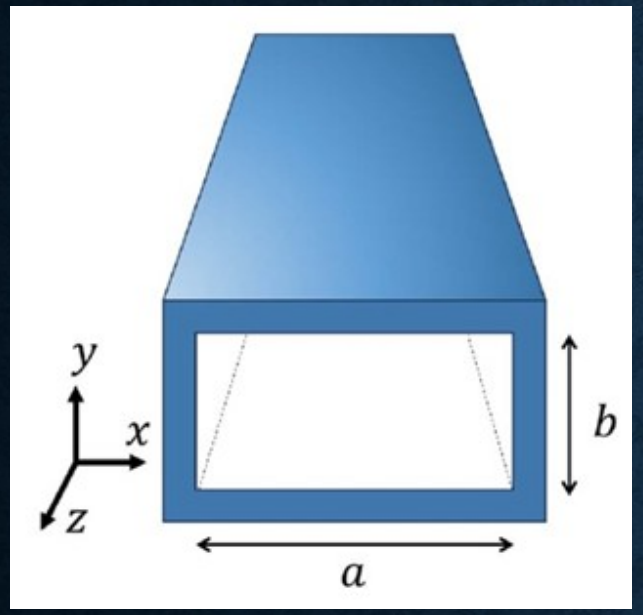
$$\begin{aligned}
\sigma(\ell, T, \omega) &= \sigma_1 + \mathrm{i}\sigma_2 \\
&= \frac{-3\mathrm{i}\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} \mathrm{d}\epsilon \left[\kappa \mathcal{T}_- + \kappa^* \mathcal{T}_+ + \kappa^a (\mathcal{T}_+ - \mathcal{T}_-)\right], \\
\kappa &= \frac{1}{3} \frac{g_+ g_- + f_+ f_- - 1}{d_+ + d_-}, \\
\kappa^a &= \frac{1}{3} \frac{-g_+ g_-^* - f_+ f_-^* - 1}{d_+ - d_-^*}.
\end{aligned}$$

Table 2. Dimensions and frequency ranges of standard rectangular waveguides. Here a and b are the broad and narrow wall dimensions, respectively.

Waveguide name	a [mm]	b [mm]	Cutoff of TE_{10} [GHz]	Cutoff of next mode [GHz]
WR28	7.112	3.556	21.1	42.2
WR15	3.759	1.880	39.9	79.7
WR10	2.54	1.27	59.0	118.0
WR8	2.032	1.016	73.8	147.5
WR5.1	1.295	0.648	115.7	231.4
WR3.4	0.864	0.432	173.6	347.1
WR2	0.508	0.254	295.1	590.1
WR1	0.254	0.127	590.1	1180.3

T. Kubo, Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity, Supercond. Sci. Technol. **39**, 075036 (2026)

Attenuation at different frequencies and for different material purities



Linear response

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$$R_s(\ell, T, \omega) = \sqrt{\frac{\mu_0 \omega}{2}} \frac{\sqrt{|\sigma| - \sigma_2}}{|\sigma|}.$$

$$\sigma(\ell, T, \omega) = \sigma_1 + i\sigma_2 = \frac{-3i\sigma_n}{4\omega\tau} \int_{-\infty}^{\infty} d\epsilon [\kappa \mathcal{T}_- + \kappa^* \mathcal{T}_+ + \kappa^a (\mathcal{T}_+ - \mathcal{T}_-)],$$

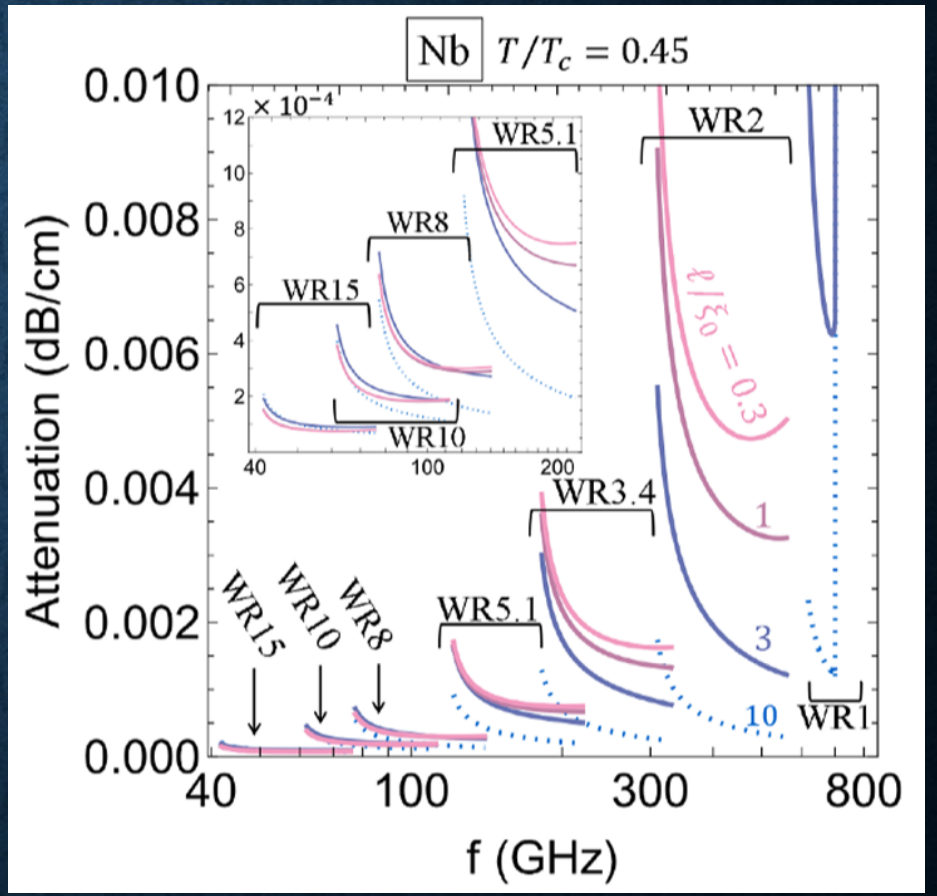
$$\kappa = \frac{1}{3} \frac{g_+ g_- + f_+ f_- - 1}{d_+ + d_-},$$

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High frequency: higher purity is better
Low frequency: higher purity is not always better

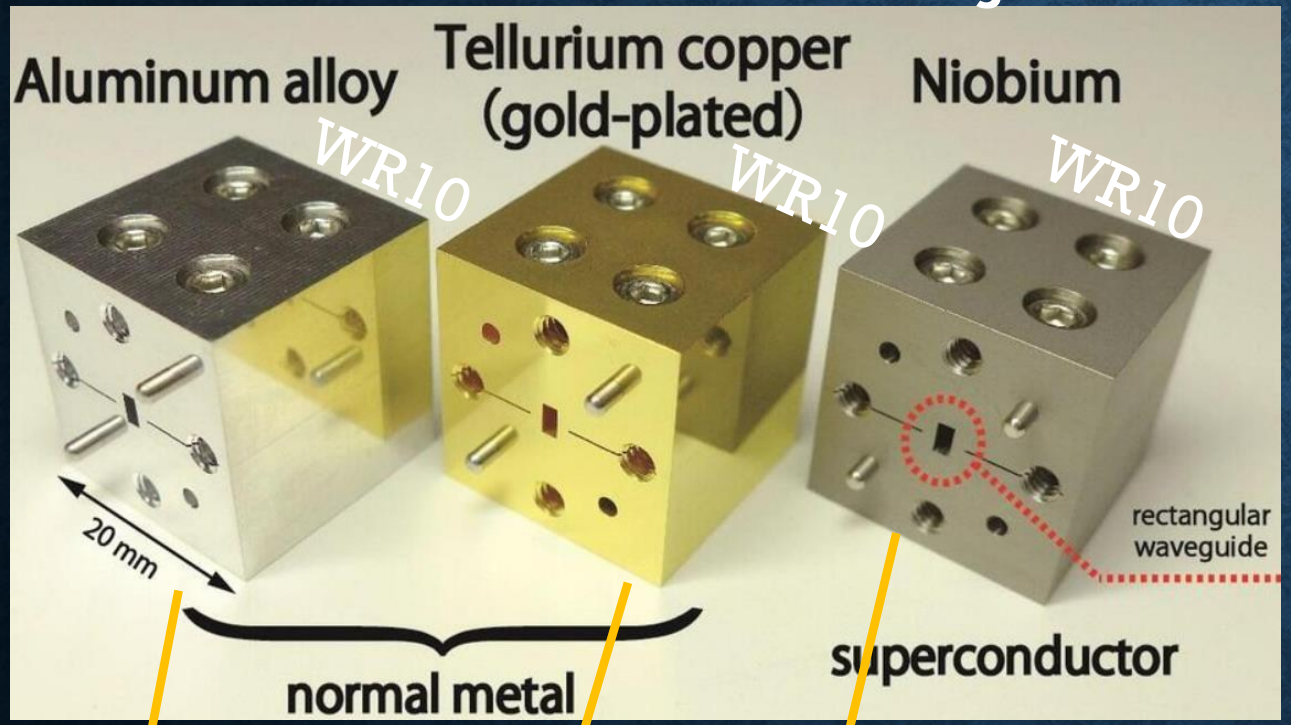


T. Kubo, Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity, Supercond. Sci. Technol. **39**, 075036 (2026)

Attenuation at different frequencies and for different material purities

T. Nakajima et al., Propagation in Superconducting Niobium Rectangular Waveguide in the 100 GHz band, J. Phys.: Conf. Ser. **2545**, 012021 (2023);
DOI : 10.1088/1742-6596/2545/1/012021

100GHz waveguides

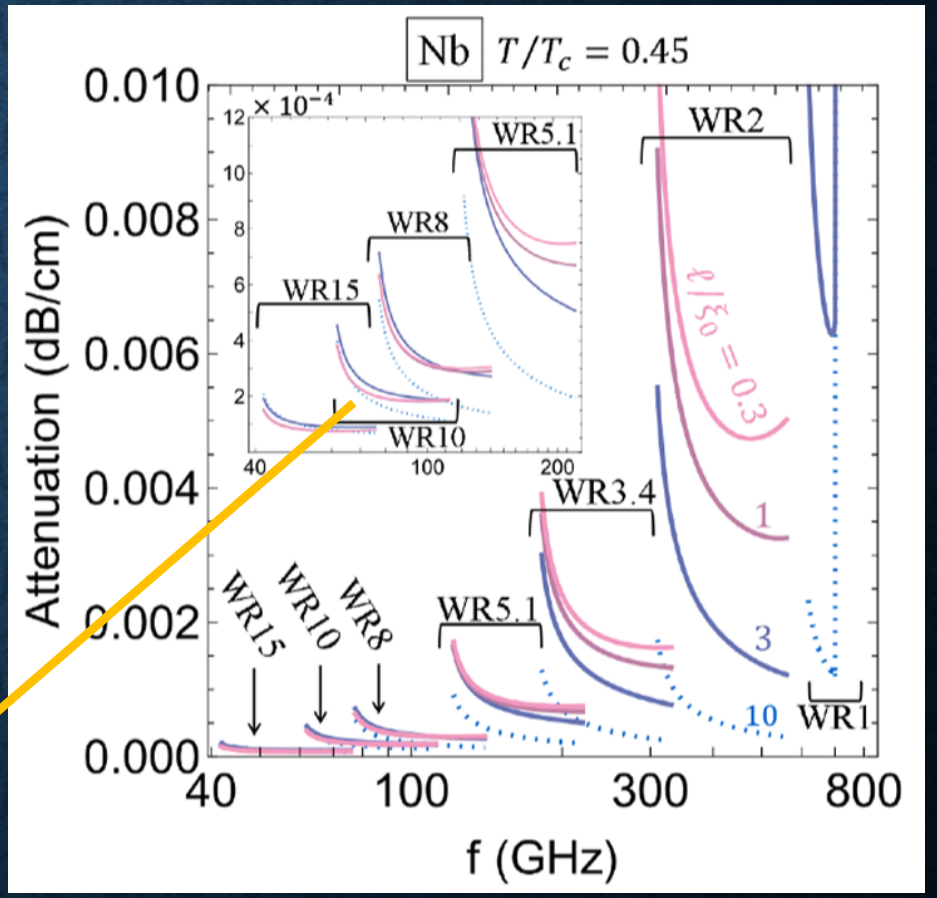


2.6×10^{-2} dB/cm

1.1×10^{-2} dB/cm

5×10^{-4} dB/cm
(Same order of magnitude as theory $\simeq 2 \times 10^{-4}$ dB/cm)

High frequency: higher purity is better
Low frequency: higher purity is not always better



T. Kubo, Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity, Supercond. Sci. Technol. **39**, 075036 (2026)

Attenuation at different frequencies and for different material purities

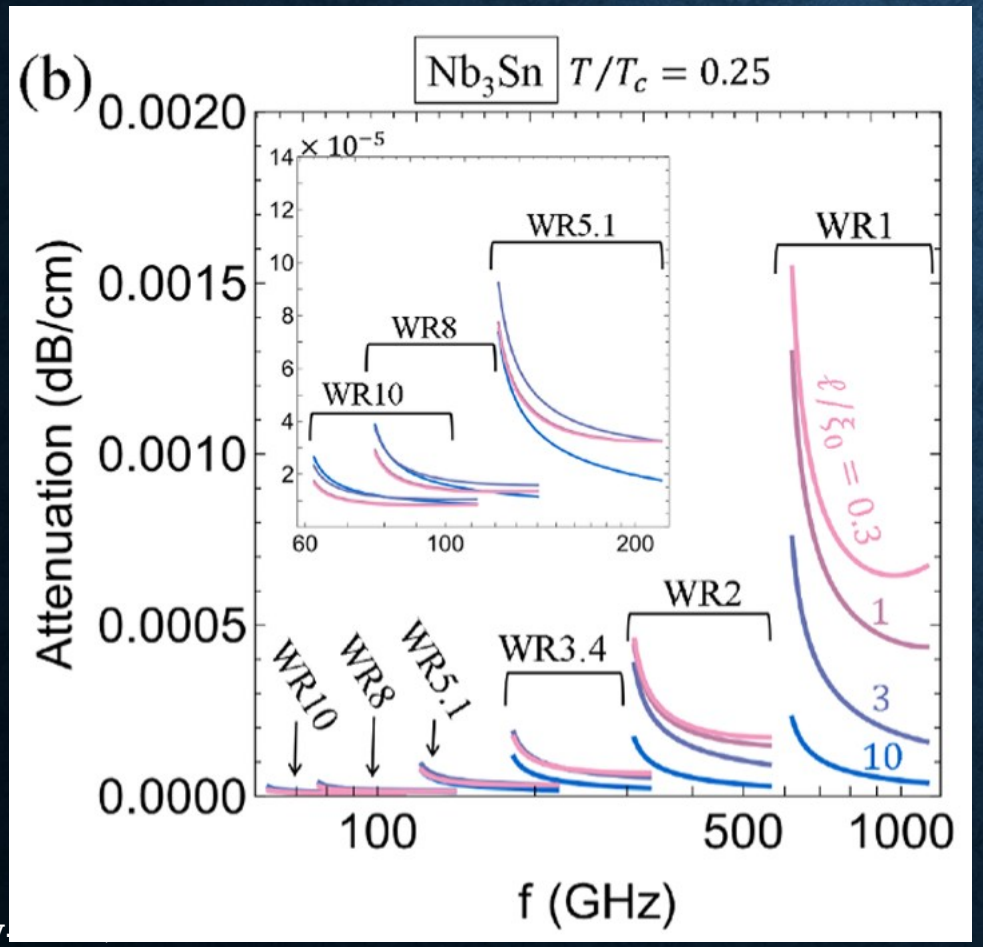
Higher-gap materials are needed toward the THz range.

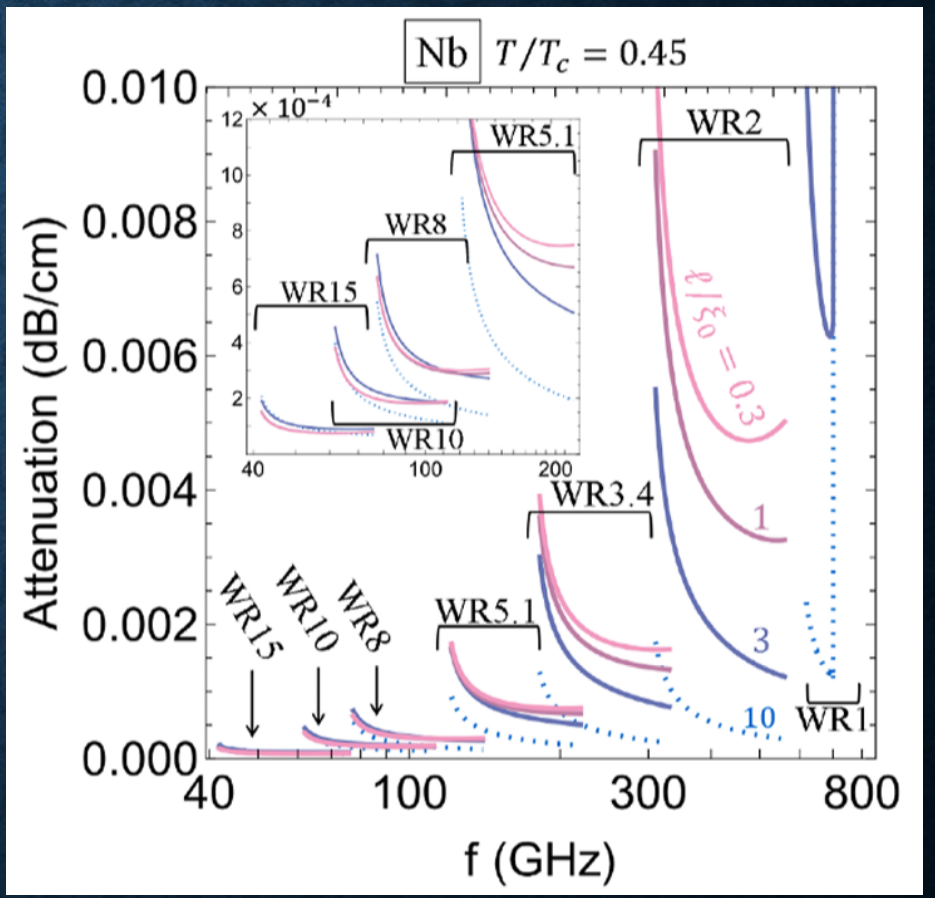
Nb3Sn: gap frequency

$f_g = 2\Delta/h \simeq 1.4$ THz

Nb: gap frequency

$f_g = 2\Delta/h \simeq 720$ GHz





T. Kubo,
Power attenuation in millimeter-wave and terahertz superconducting rectangular waveguides: linear response, two-level-system loss, and Higgs-mode nonlinearity,
Supercond. Sci. Technol. **39**, 075036 (2026)

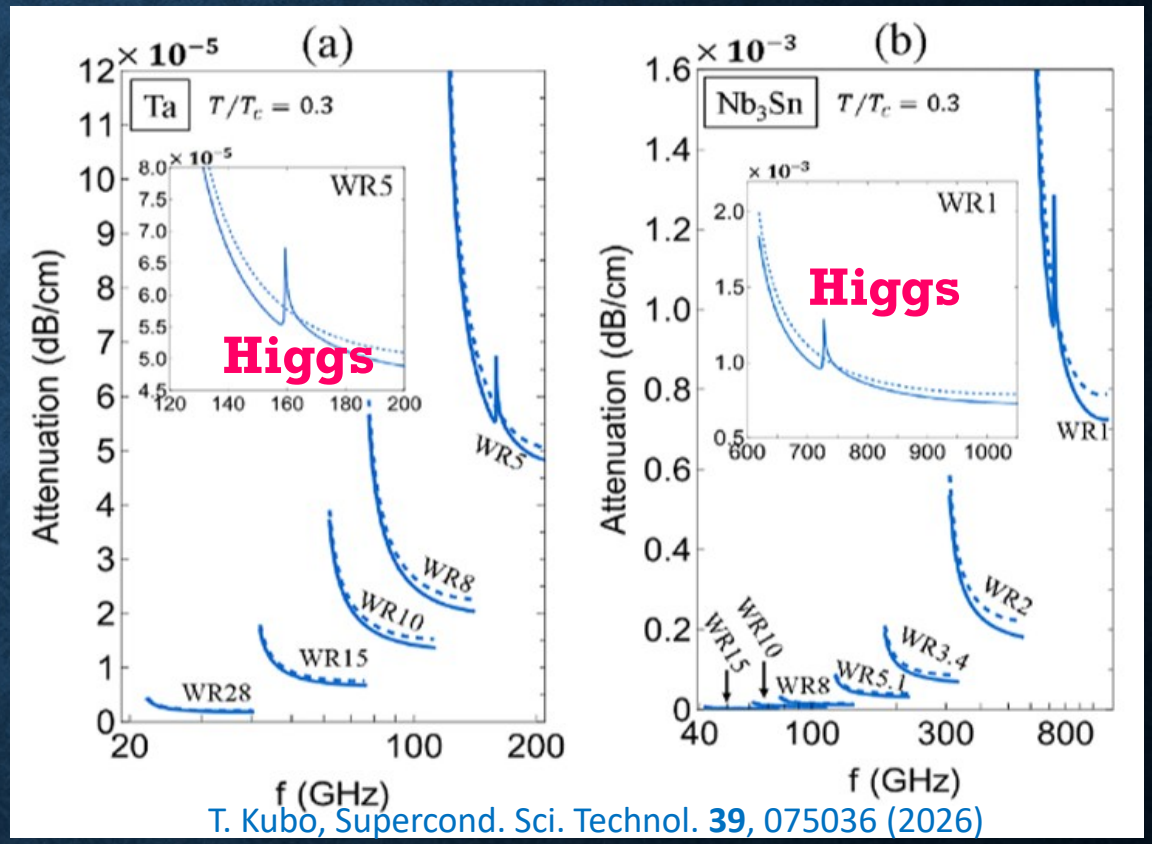
Strong-signal regime: nonlinear electrodynamics and Higgs-mode signatures

$$\alpha = \frac{P_{\mathrm{loss}}}{2P_{\mathrm{flow}}} = \frac{R_s}{Z_0 b}\frac{1+(2b/a)(\omega_c/\omega)^2}{\sqrt{1-(\omega_c/\omega)^2}},$$

$$\frac{R_s}{R_0} = \frac{1}{2\pi\sqrt{\ell/\xi_0}}\left(\frac{\hbar\omega}{\Delta_0}\right)^2\left(\frac{\lambda}{\lambda_{0,\mathrm{dirty}}}\right)^3 \times \left[\frac{\sigma_1}{\sigma_n} + \frac{u(T,\omega)}{2\pi}\left(\frac{B_{\mathrm{ac}}}{B_c}\right)^2\right]$$

$$u(T,\omega) = \frac{2\sqrt{\pi}}{\hbar\omega/\Delta_0}\mathrm{Re}\left(I_{1\mathrm{H}}^{qqq} + I_{1\mathrm{H}}^{\mathrm{Higgs}} + I_{1\mathrm{H}}^{\mathrm{Eliash}}\right)$$

T. Kubo, Phys. Rev. Applied **24**, 064061 (2025)



T. Kubo, Supercond. Sci. Technol. **39**, 075036 (2026)

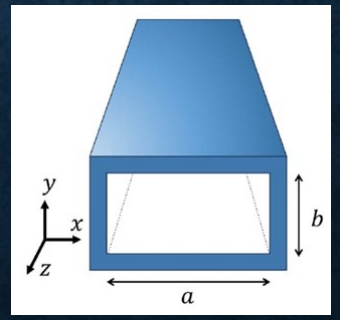


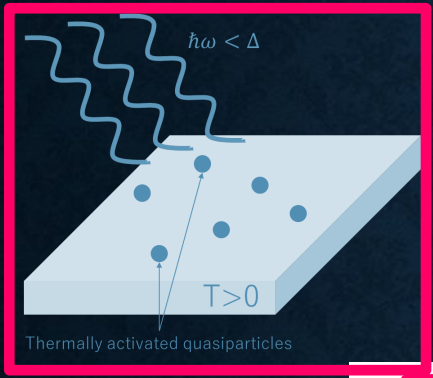
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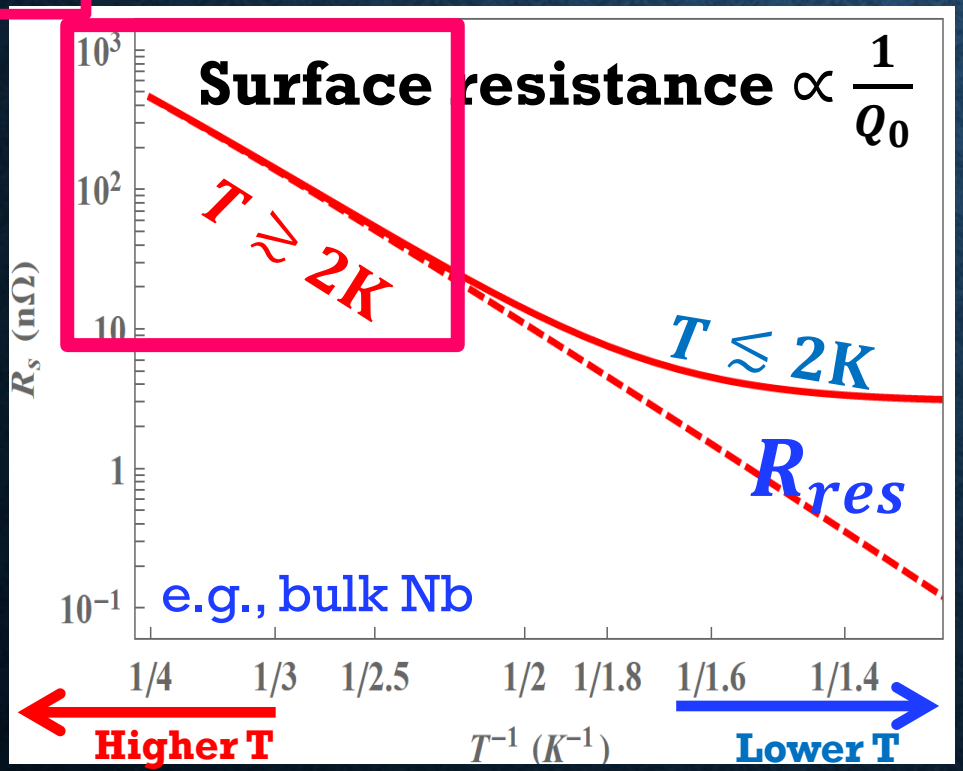
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Low-temperature Q factor

Accelerator regime

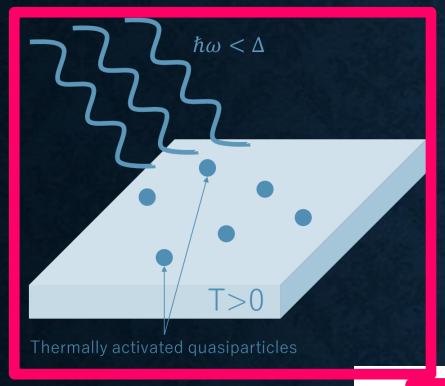
$T \gtrsim 2\mathrm{K}$



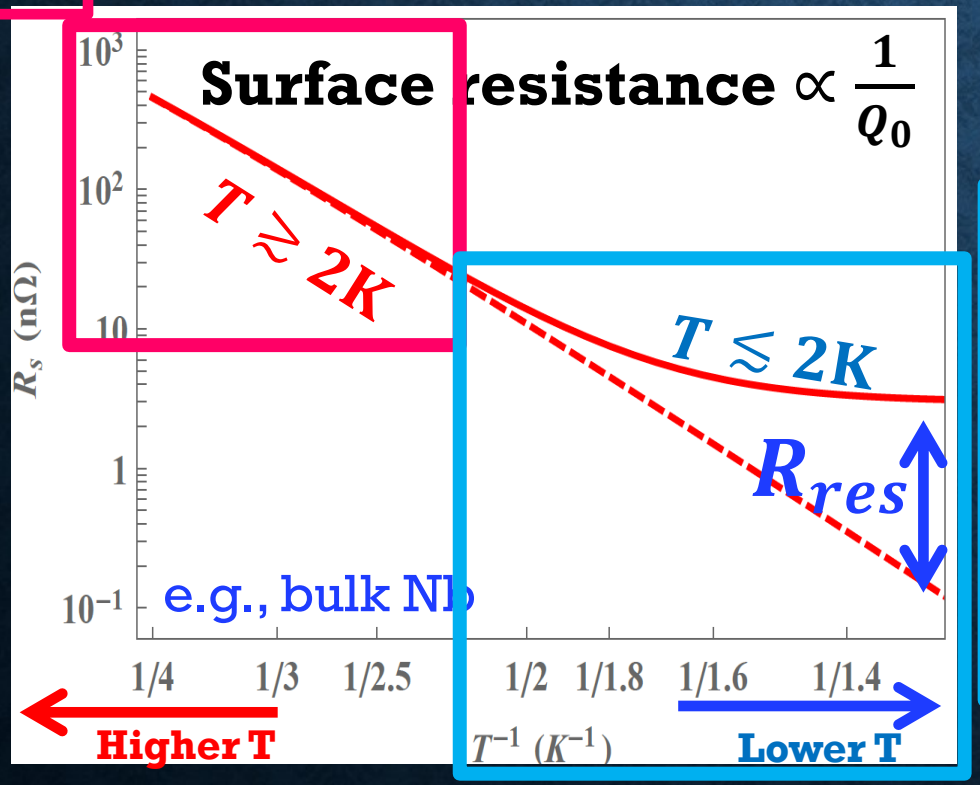


Accelerator regime

$T \gtrsim 2\mathrm{K}$



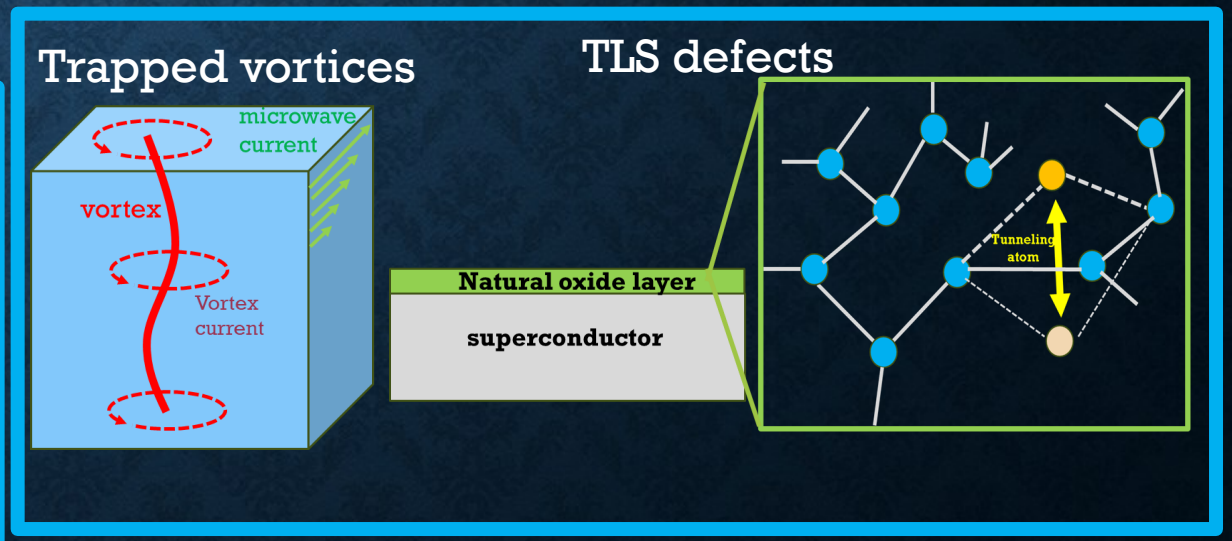
Can accelerator technologies be useful beyond the accelerator regime?



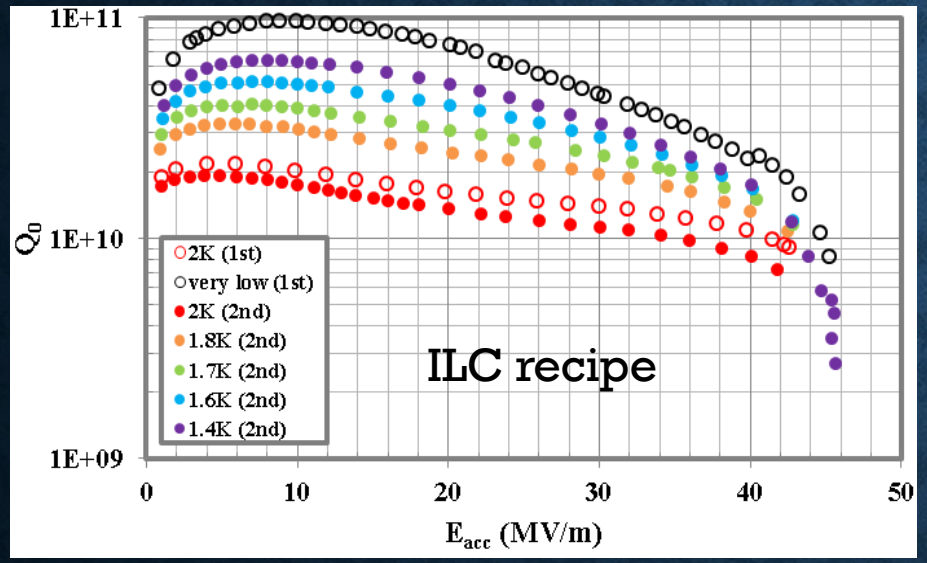
$T \lesssim 2\mathrm{K}$

Non-accelerator regime

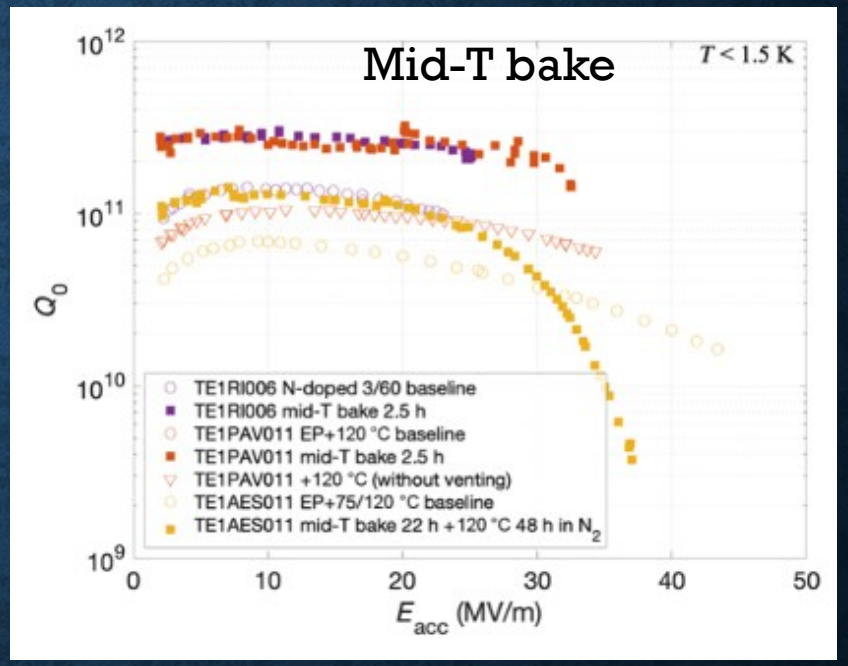
Including quantum regime (~10mK)



Acutually, the accelerator community can build **the best cavities on the planet** — achieving $Q \gtrsim 10^{11}$ at $T = 1$-2 K.



T Kubo et al., In-house production of a large-grain single-cell cavity at cavity fabrication facility and results of performance tests, IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022



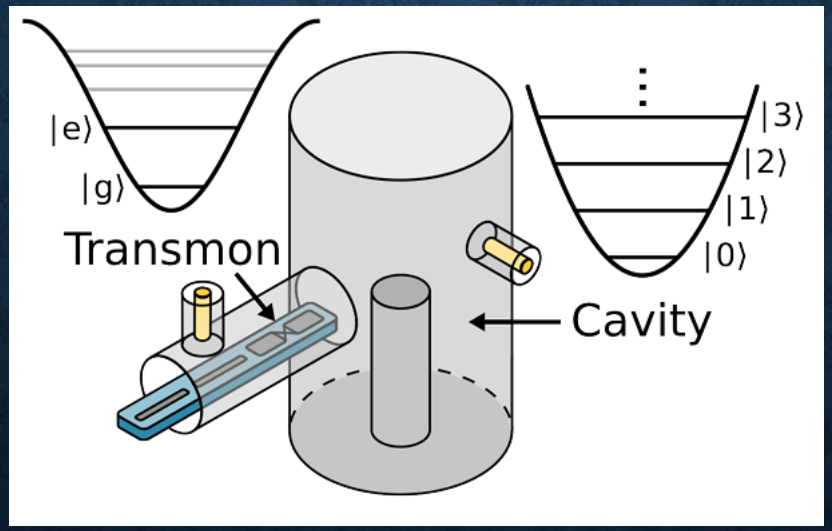
S. Posen et al., Ultralow Surface Resistance via Vacuum Heat Treatment of Superconducting Radio-Frequency Cavities,
Phys. Rev. Applied **13**, 014024 (2020)
https://doi.org/10.1103/PhysRevApplied.13.014024

But they do not assure the performace at T < 1 K.

Non-accelerator application (2)

Quantum applications

Cavities for quantum computing

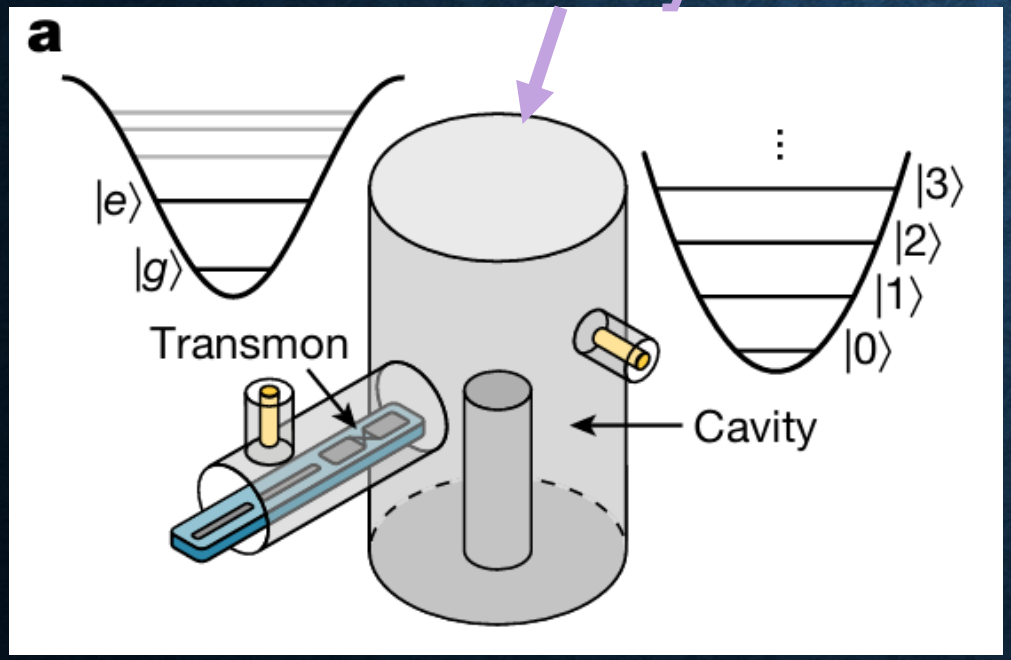


Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y

Superconducting Cavities in Quantum Technologies

- Superconducting cavities offer a promising approach to **hardware-efficient quantum information processing,** potentially reducing the overhead of quantum error correction.
- Various bosonic encodings are being explored, including cat codes, GKP codes, and dual-rail encoding.
- Startups are actively developing cavity-based quantum computing architectures.

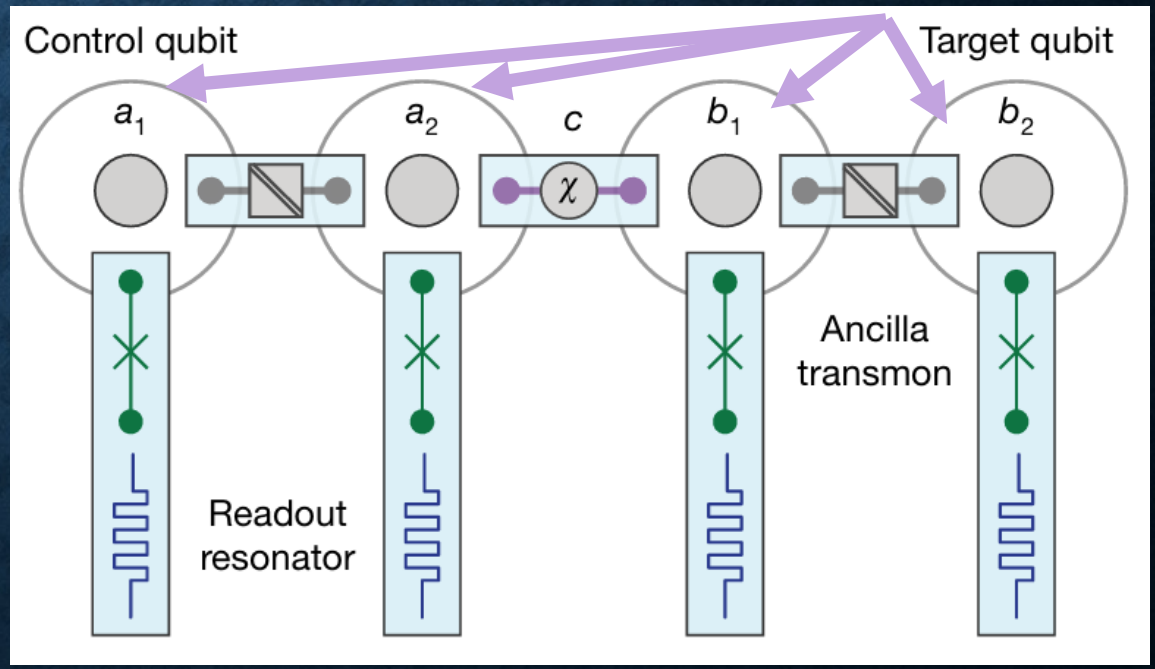
cavity



Brock et al., Nature **641**, 612 (2025)

Yale University

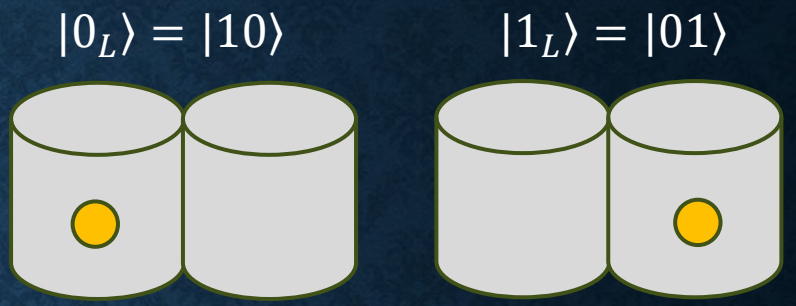
cavities



D-Wave Quantum Inc., Nature **656**, 47 (2026)

Quantum Circuits, Inc.

e.g., Dual-Rail Cavity Qubit

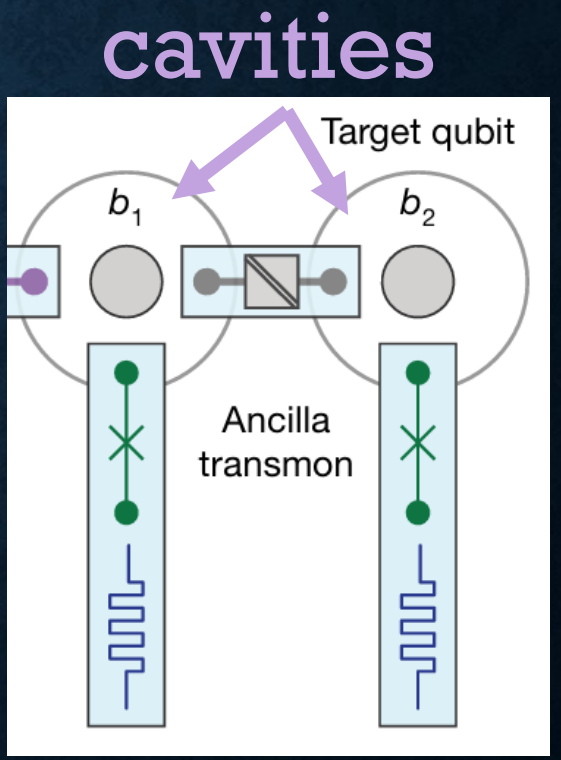


- **Two superconducting cavities:** Store a single photon encoding one qubit, $|0_L\rangle = |10\rangle$, $|1_L\rangle = |01\rangle$.

- **Tunable cavity-cavity coupler:** Enables coherent photon exchange and single-qubit gates.

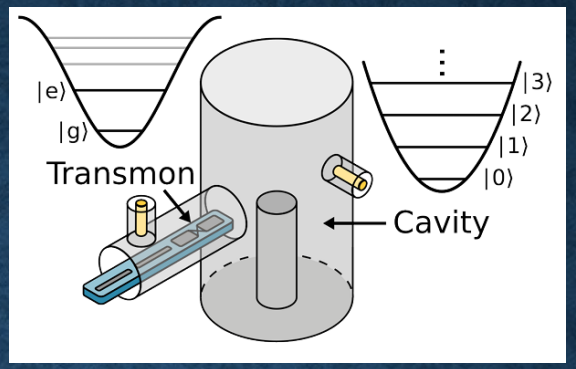
- **Ancilla transmon:** Enables state preparation, logical readout, and detection of photon-loss errors.

Key advantage: Photon loss can be detected and converted into a *located erasure error*, making quantum error correction more efficient.



D-Wave Quantum Inc., Nature **656**, 47 (2026)
Quantum Circuits, Inc.

e.g. binomial code



Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y

$$|0\rangle_L = \frac{|0\rangle + \sqrt{3}\,|6\rangle}{2}$$

$$|1\rangle_L = \frac{\sqrt{3}|3\rangle + |9\rangle}{2}$$

1 photon loss

$|5\rangle$

$$\frac{|2\rangle + |8\rangle}{\sqrt{2}}$$

1 photon loss

$|4\rangle$

$$\frac{|1\rangle + 2|7\rangle}{\sqrt{3}}$$

correct

a remainder of 2 when divided by 3

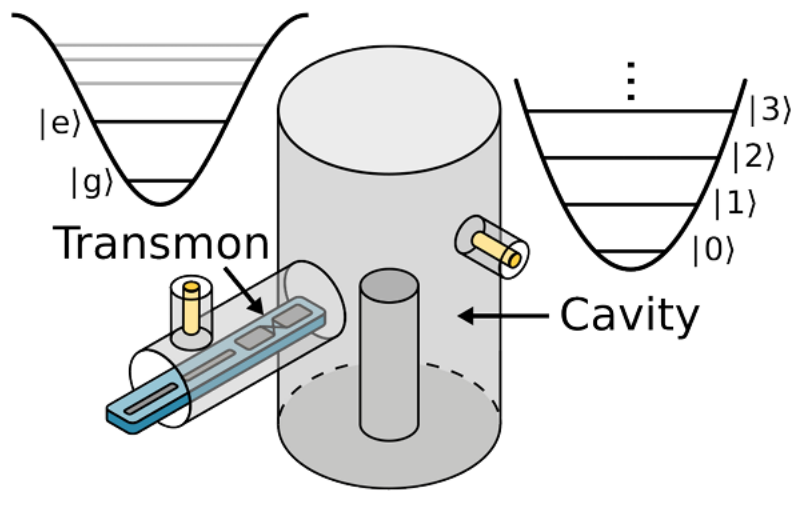
a remainder of 1 when divided by 3

correct

3D coaxial cavity progress

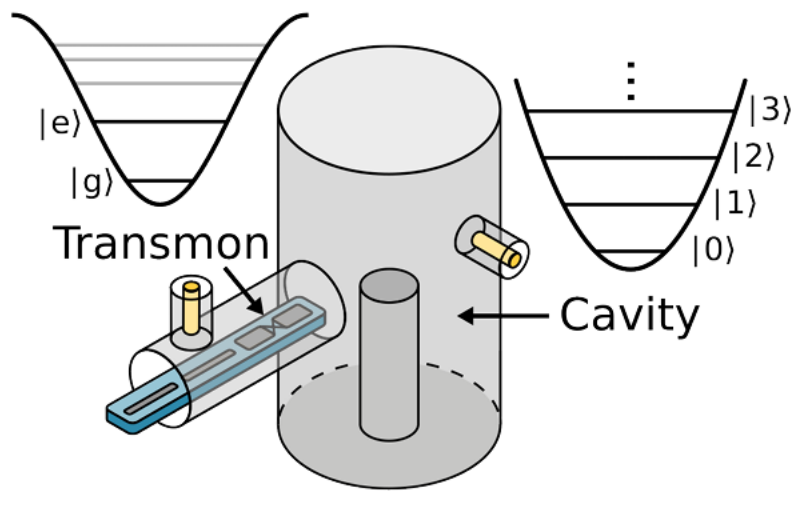
- Reagor et al., Phys. Rev. B **94**, 014506 (2016) Yale University
  Aluminum,
  $Q = 7 \times 10^7$ (single photon level)

Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y



3D coaxial cavity progress

Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y

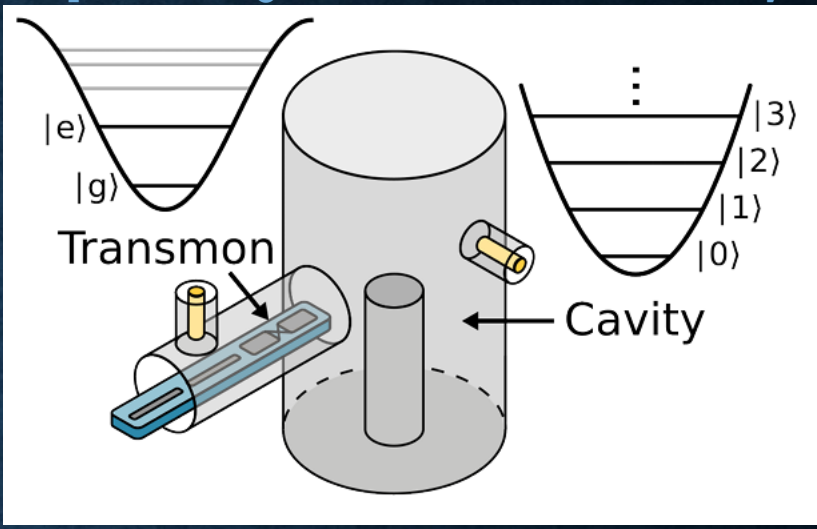


- Reagor et al., Phys. Rev. B **94**, 014506 (2016) Yale University
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- Heidler et al., Phys. Rev. Applied **16**, 034024 (2021) University of Innsbruck
  Niobium (1:1:1 buffered chemical polishing at 0-5℃),
  $Q = 5 \times 10^8$ (single photon level)

3D coaxial cavity progress

Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y



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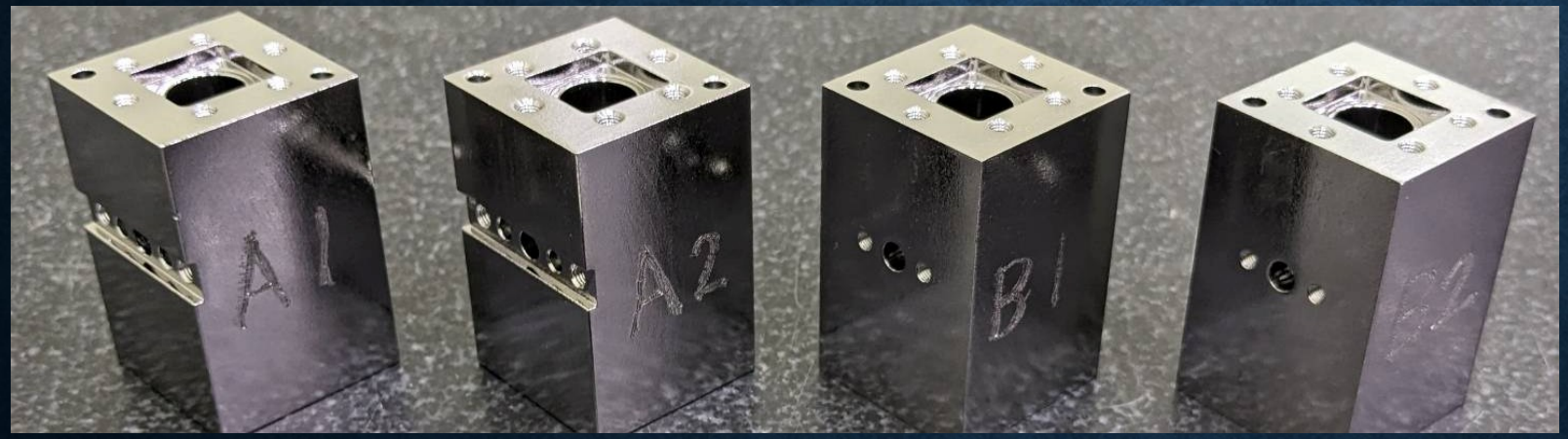
- Oriani et al., Phys. Rev. Applied **24**, 044080 (2025) University of Chicago
  Niobium (1:1:2 water buffered chemical polishing below 10℃),
  $Q = 1.4 \times 10^9$ (single photon level)

↑

The world record (as of 2024)

Let's break the world record

Nb coaxial cavity of KEK-NTT collaboration



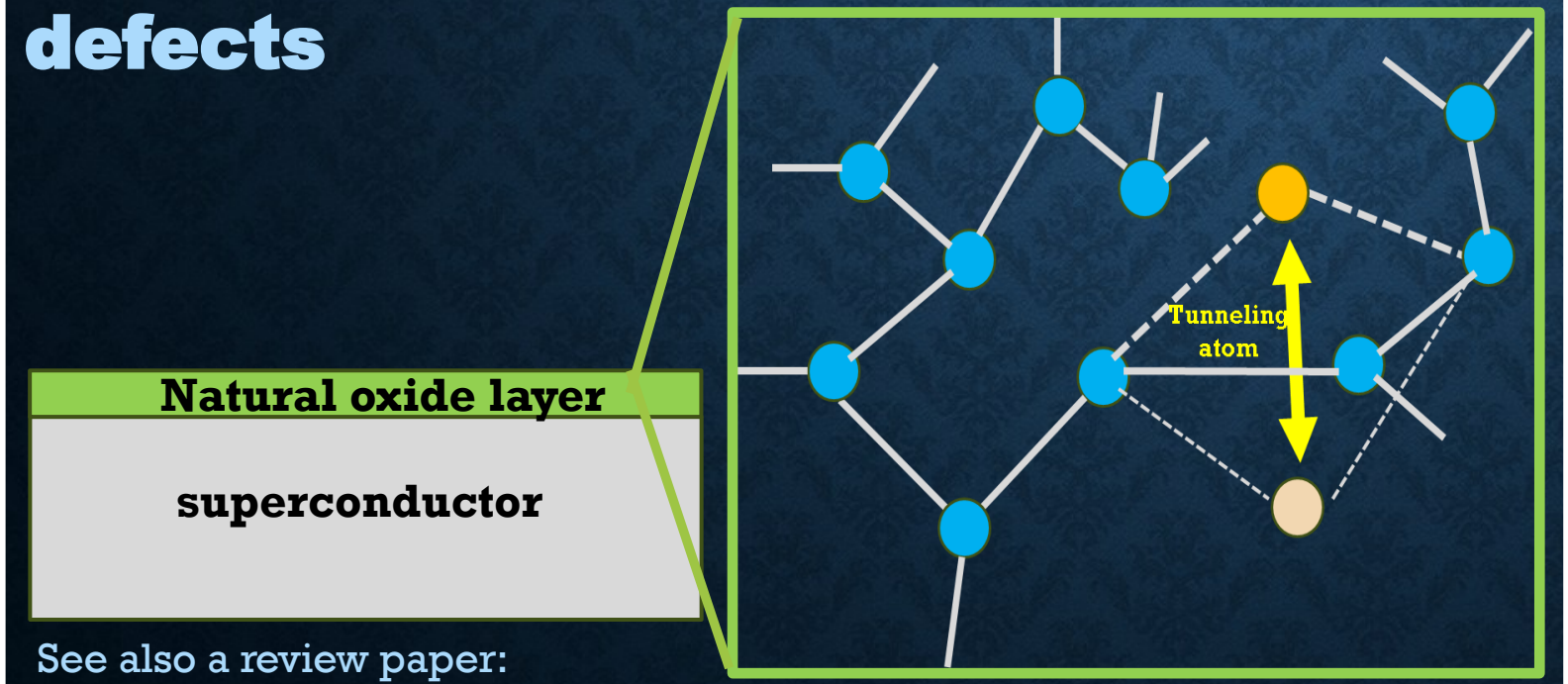
cavity→



The greatest enemy of SRF technology in quantum applications is **TLS** defects—that is, a problem originating outside the superconductor itself.

Most amorphous solids (e.g. glass) show universal behavior at temperatures below a few kelvin. This behavior can be explained by the well-established "*standard tunneling model*".

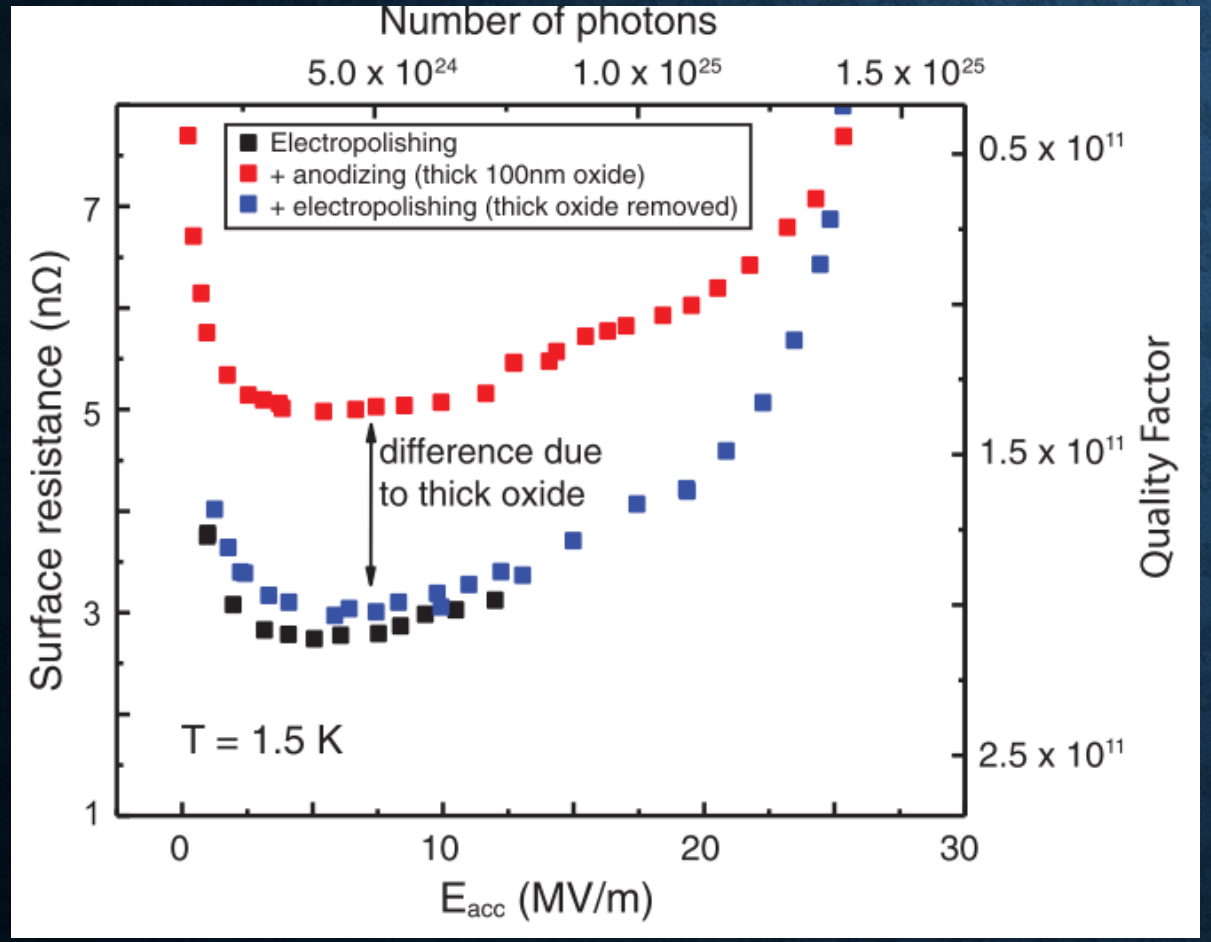
Two-level system (TLS) defects



See also a review paper:
Müller et al Rep. Prog. Phys. **82**, 124501 (2019)

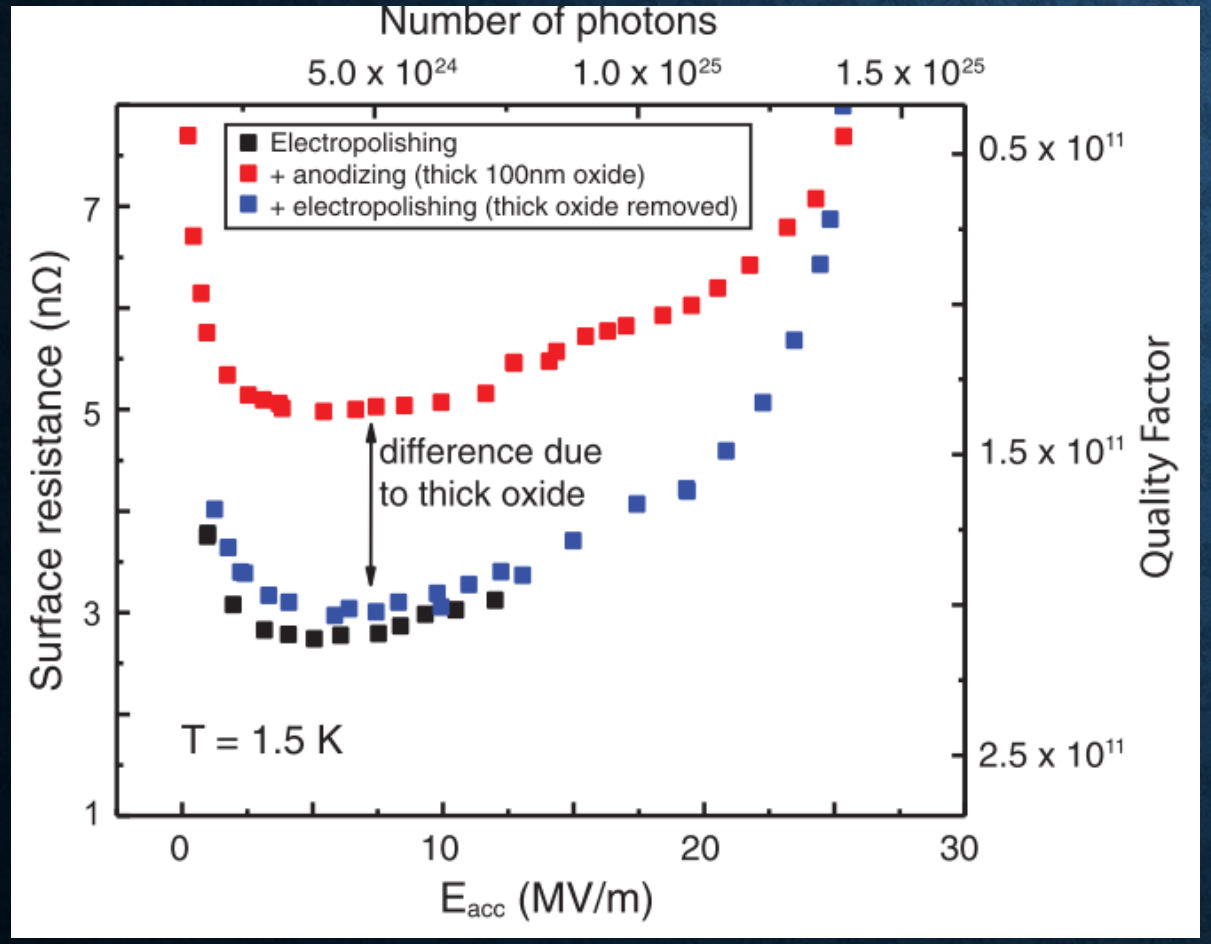
$$\frac{1}{Q_{\mathrm{int}}(\bar{n})} = F\delta^{0}_{\mathrm{TLS}} \frac{\tanh\left(\frac{hf_r}{2kT}\right)}{\sqrt{1+\left(\frac{\bar{n}}{n_c}\right)^{\beta}}} + \frac{1}{Q_{\mathrm{res}}},$$

TLSs live in the oxide layer on the inner surface of the niobium cavity.

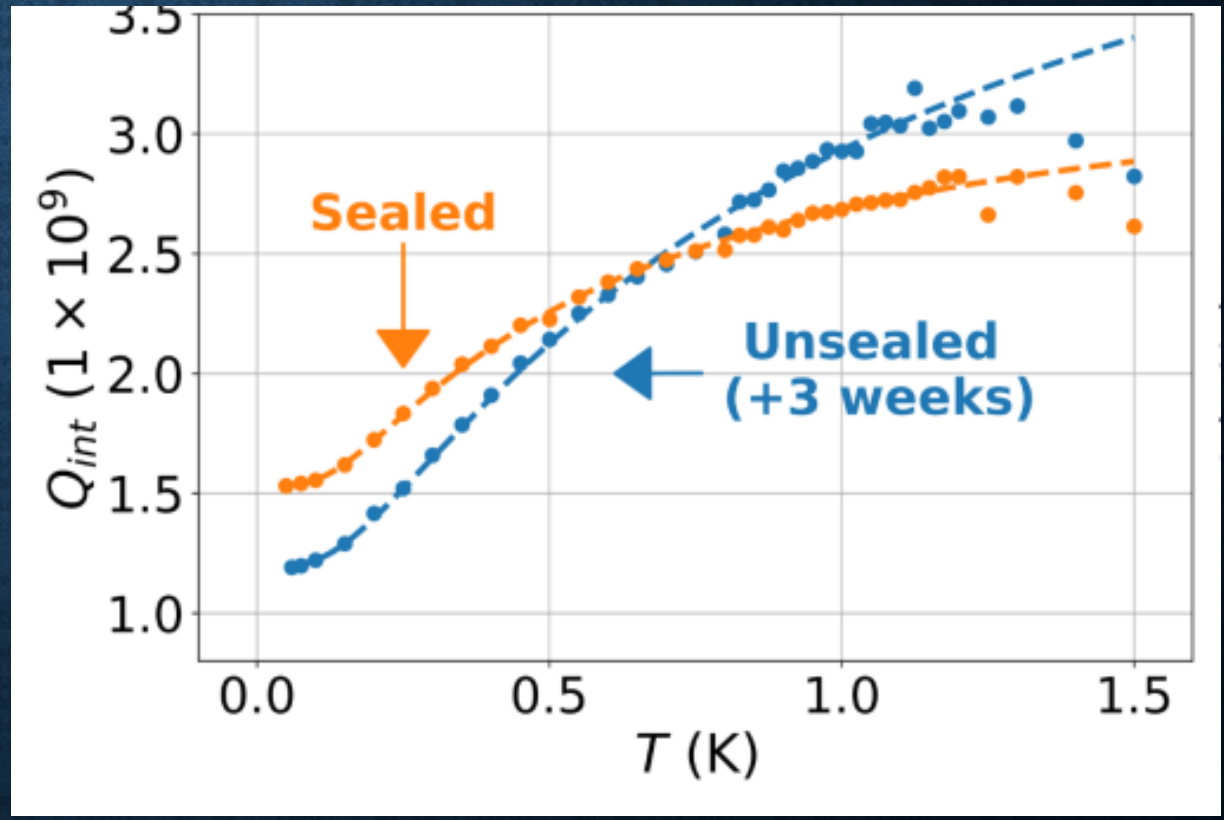


Romanenko, PRL **119**, 264801 (2017)

TLSs live in the oxide layer on the inner surface of the niobium cavity.
The oxide inevitably regrows after any surface treatment.
<u>Whether TLSs can be reduced by surface treatment is not obvious.</u>



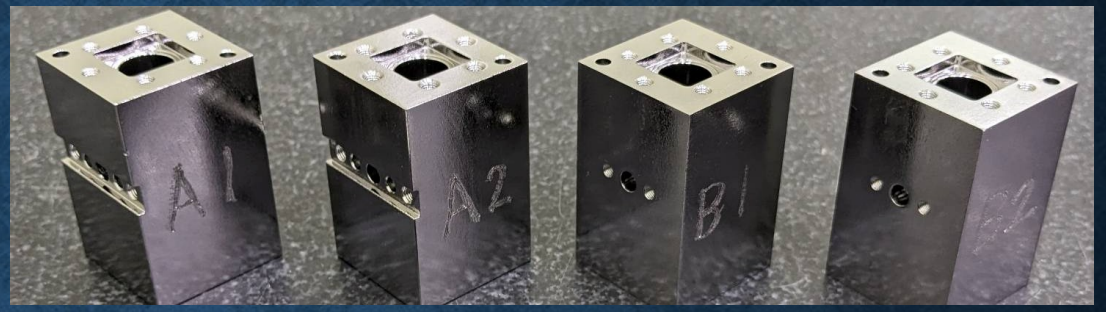
Romanenko, PRL **119**, 264801 (2017)



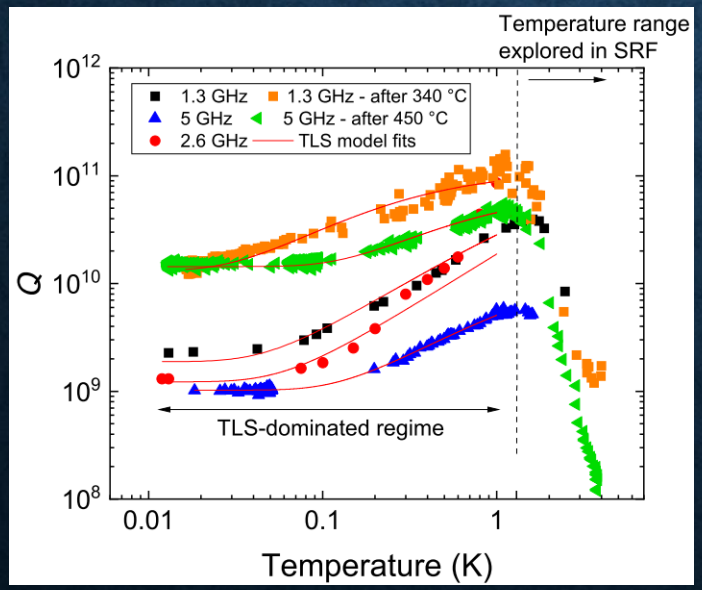
Oriani et al., arXiv:2403.00286

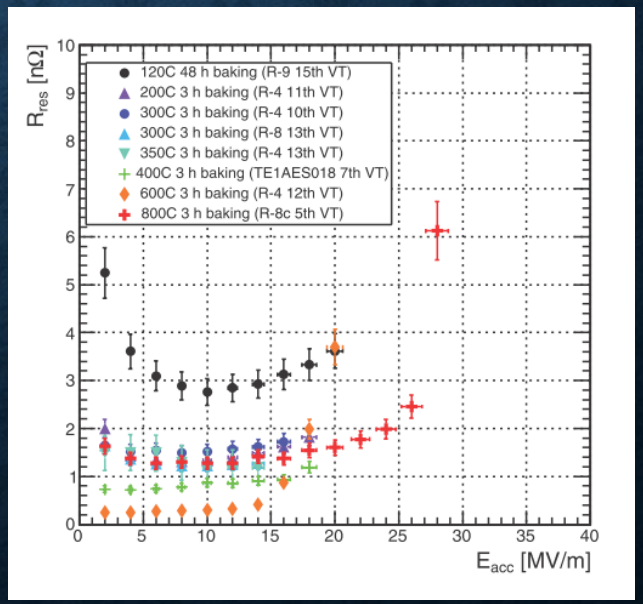
We applied accelerator-type surface treatments to the quantum cavity, while omitting processes intended for high-gradient operation.

1. Removal of approximately 100 μm of material from the inner surface using buffered chemical polishing (BCP) at room temperature ($\gtrsim 20°C$).
2. Baking the cavity in a vacuum furnace at 900 °C for 3 hours (high-T bake).
3. Light BCP ("BCP flush") to remove a few additional micrometers of material.
4. Additional treatment for surface optimization



Prior work suggests a possible route: despite inevitable regrowth, a **mid-temperature bake** that disrupts the oxide may help.





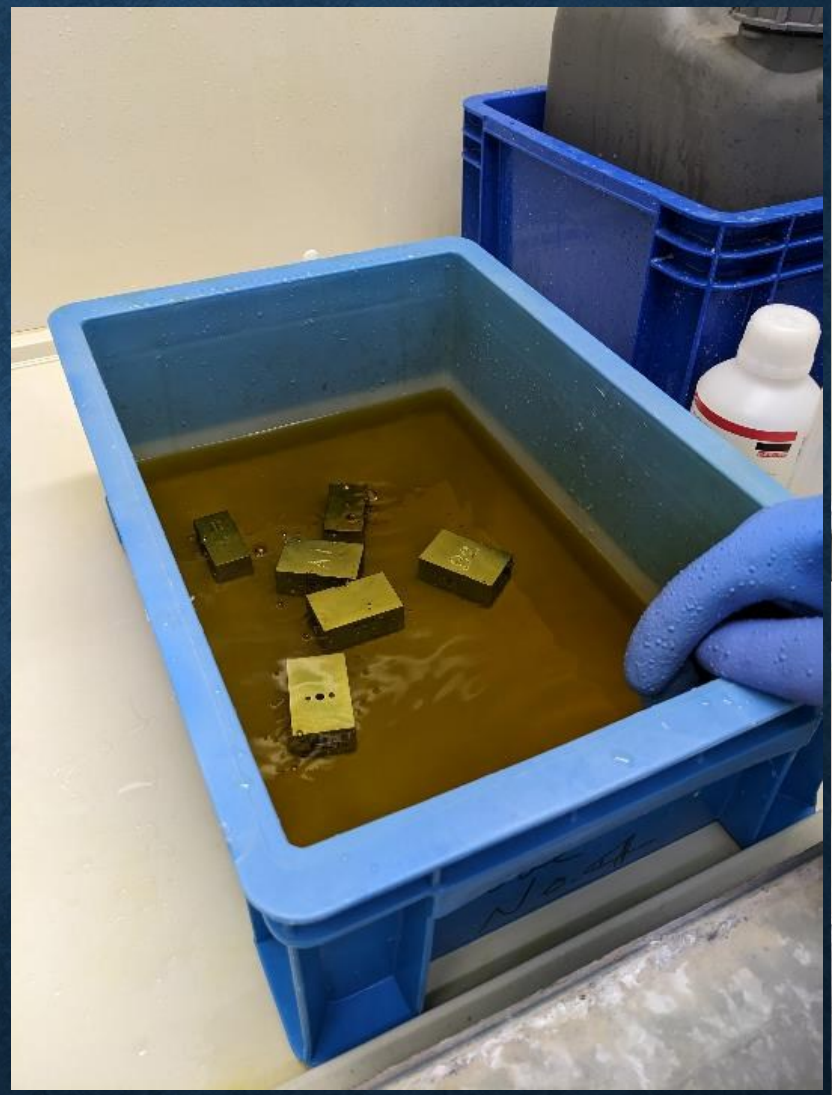
A. Romanenko et al., Phys. Rev. Applied **13**, 034032 (2020)

H. Ito et al., PTEP **2021**, 071G01 (2021)

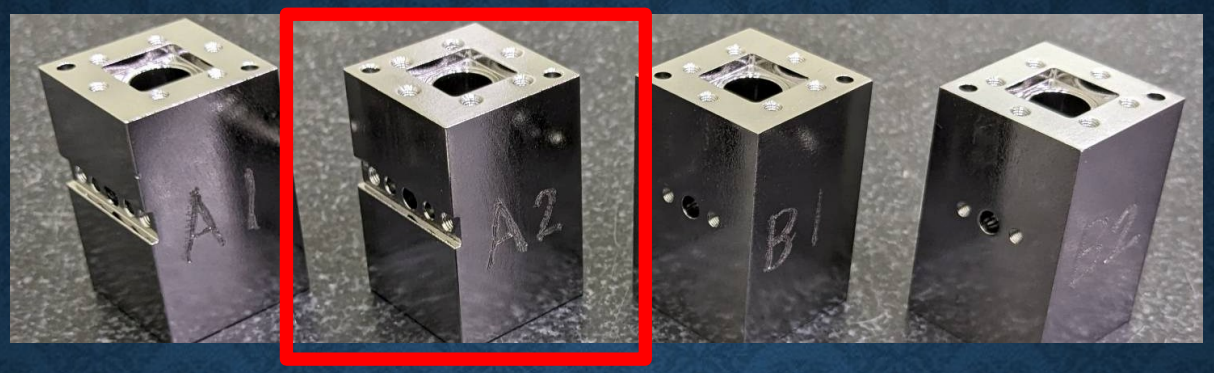
Ultrasonic cleansing



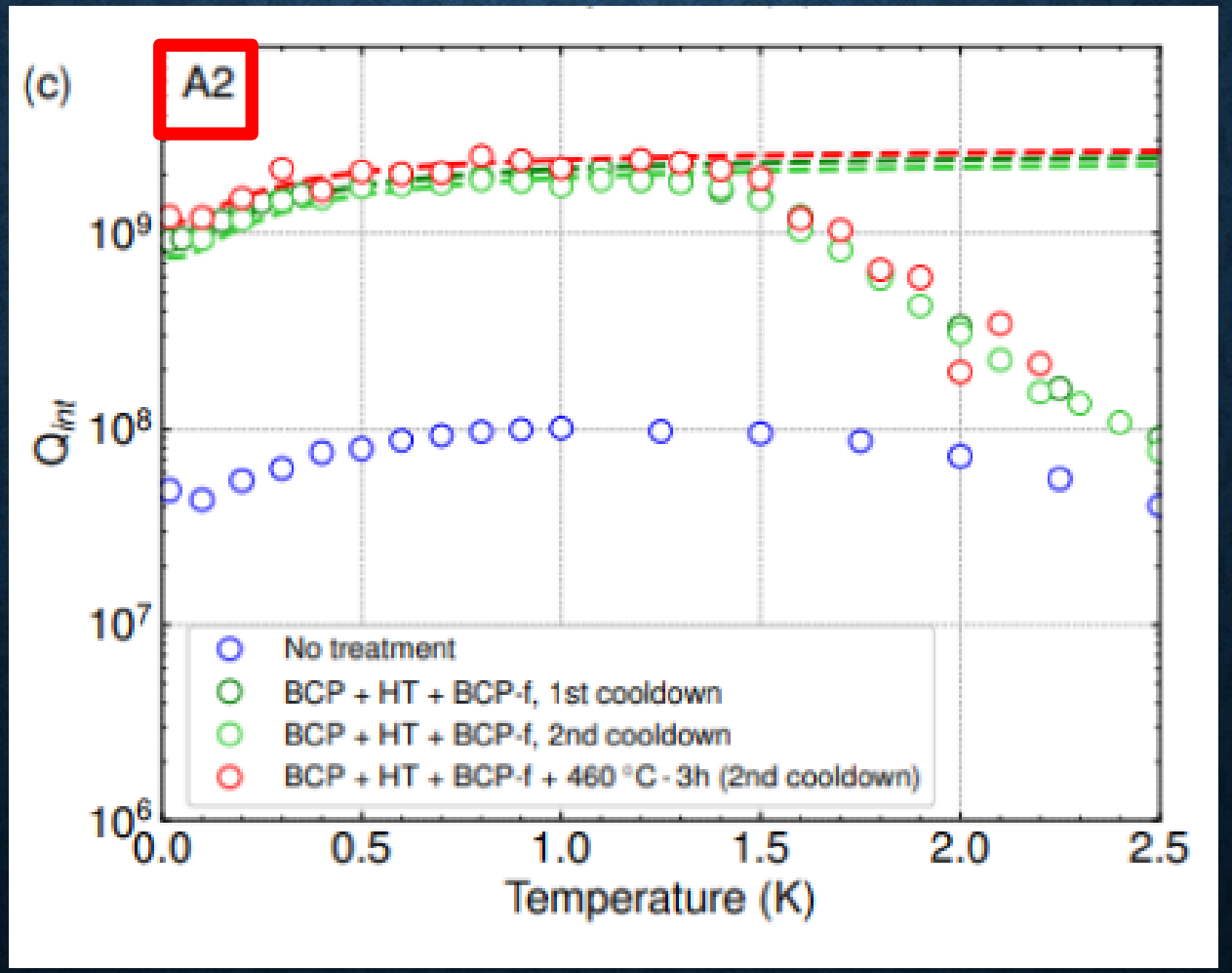
Buffered Chemical Polishing (BCP)



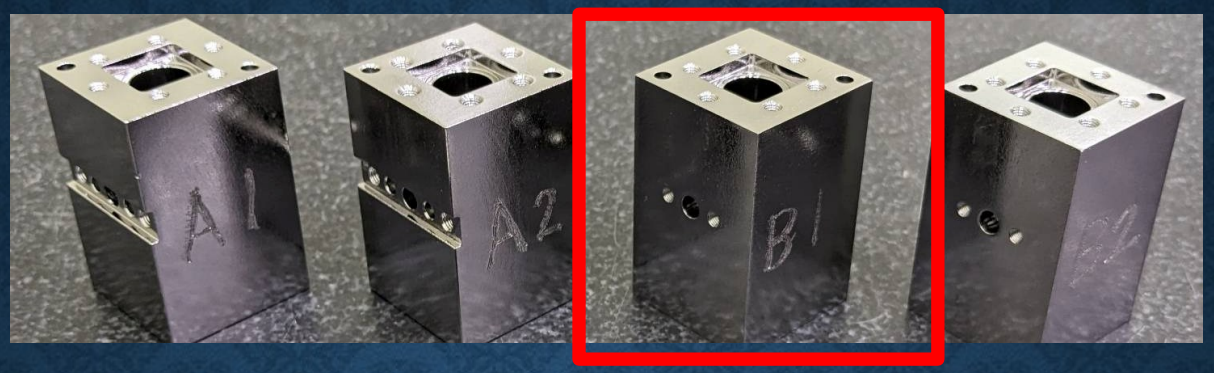




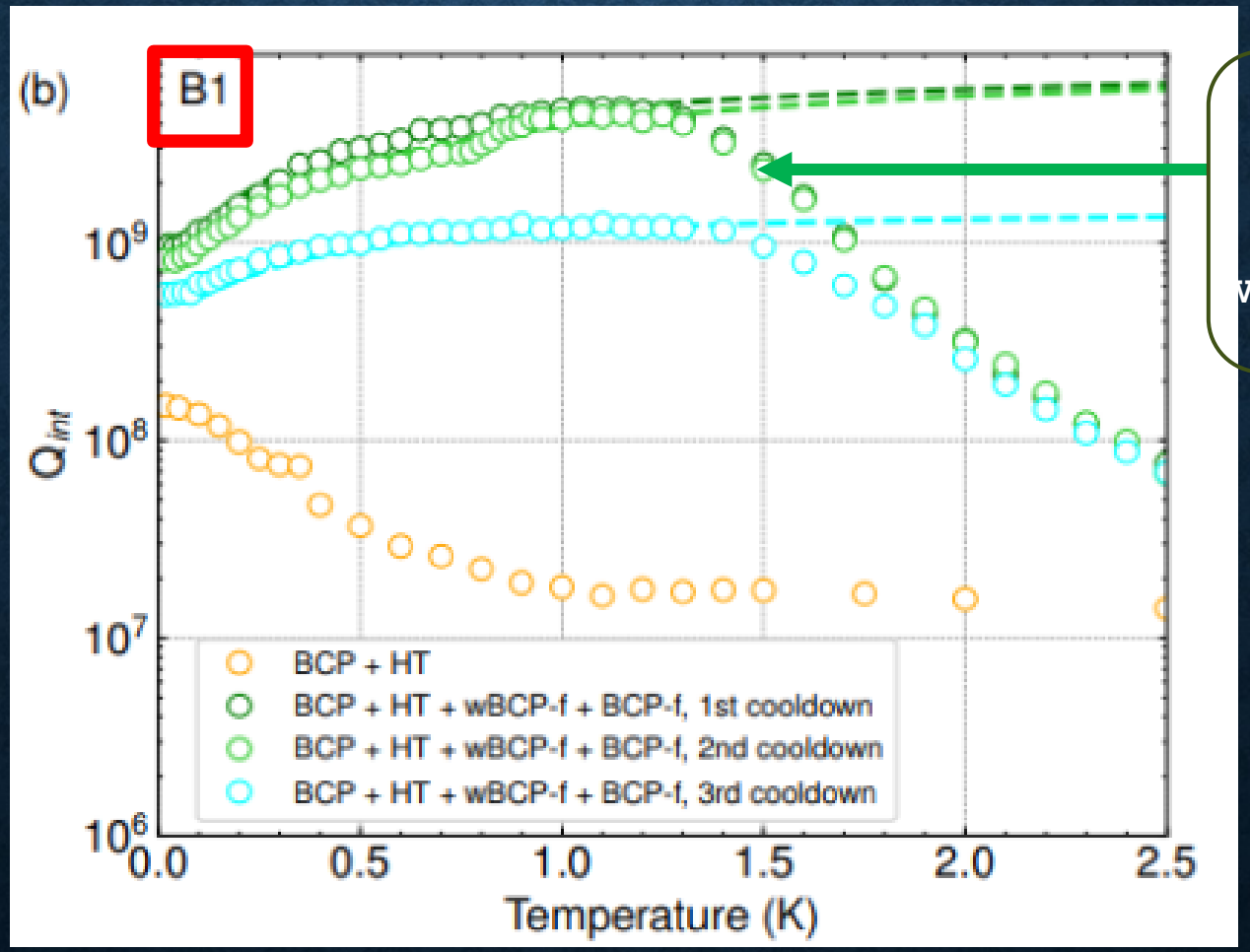
comparable to
the world record→



T. Takenaka, T. Kubo et al.,
Phys. Rev. Applied **25**, 034076 (2026).



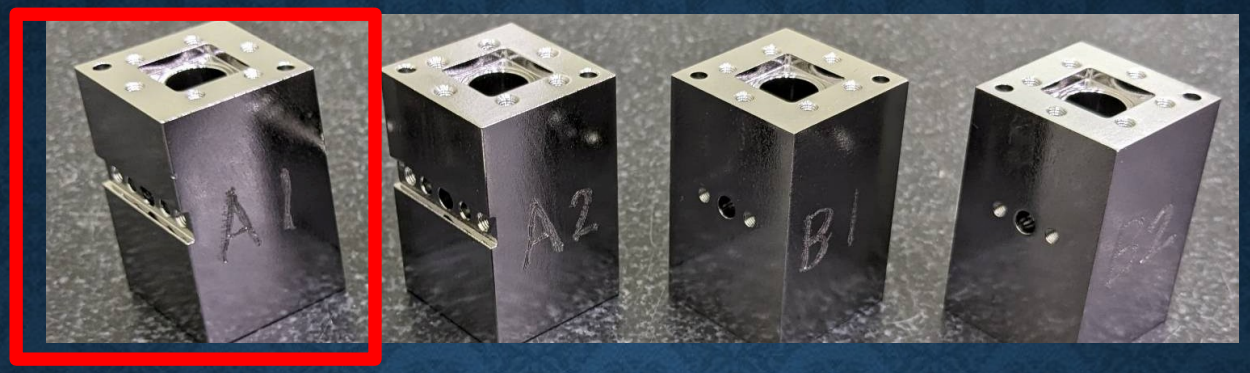
comparable to
the world record→



Room temperature
wBCP (1:1:1) flush

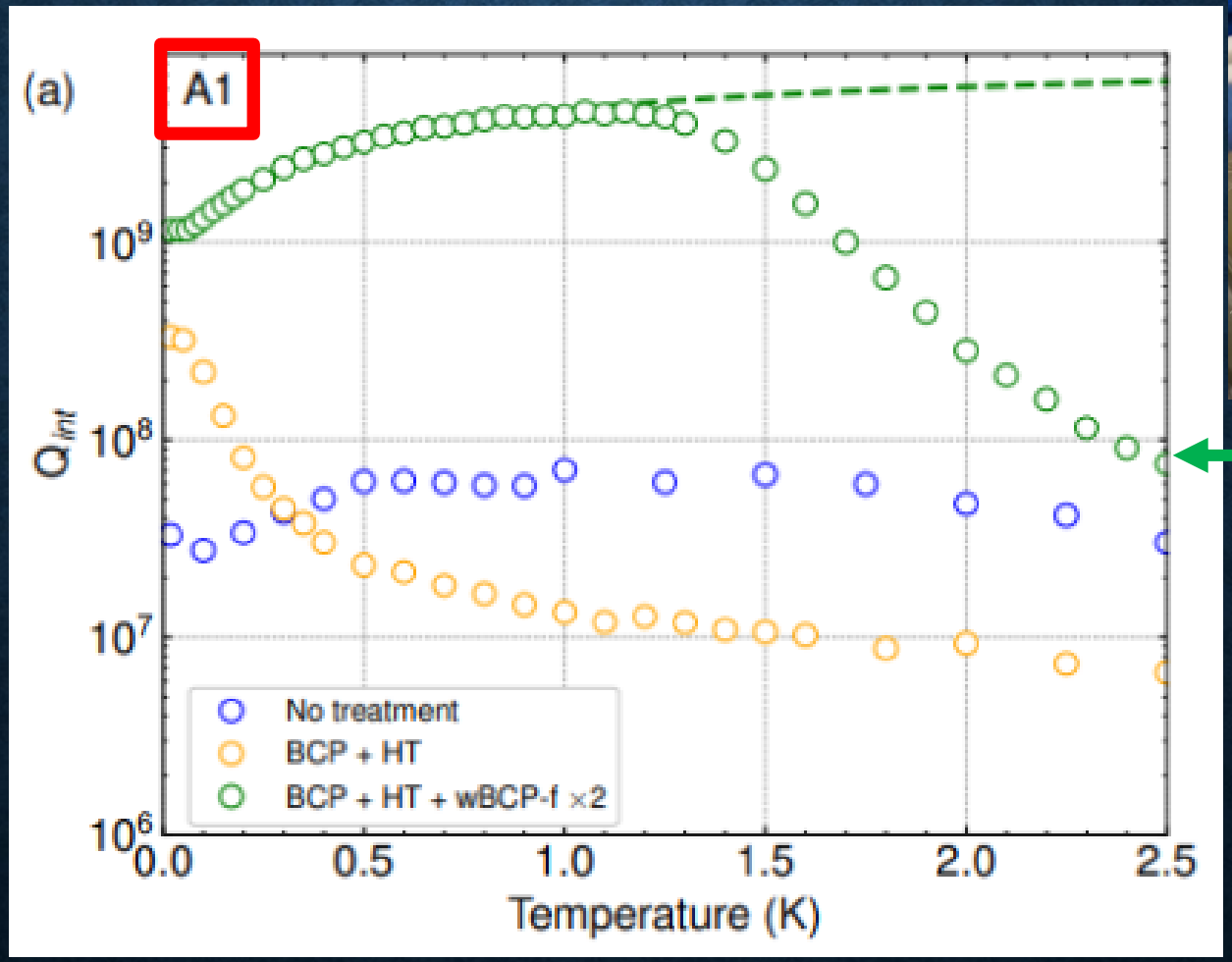
wBCP=water buffered CP

T. Takenaka, T. Kubo et al.,
Phys. Rev. Applied **25**, 034076 (2026).





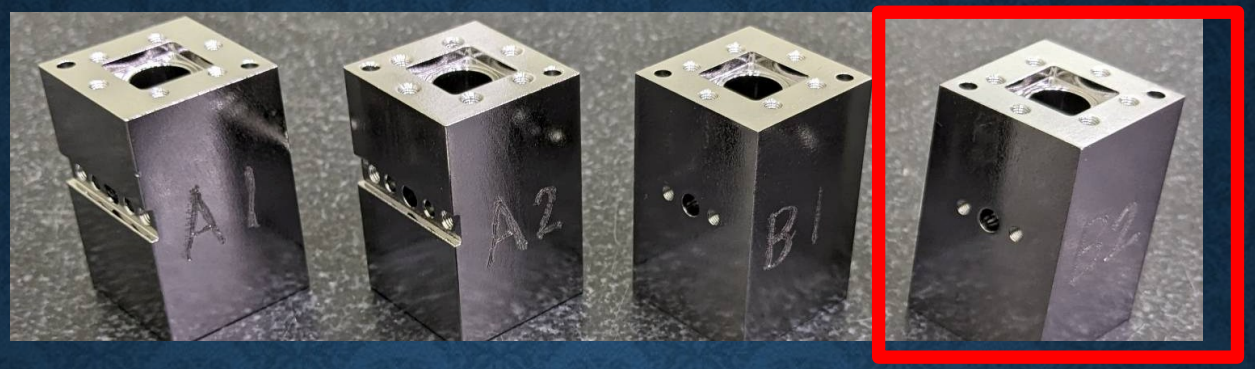
comparable to
the world record→



Cold (5℃)
wBCP (1:1:2) flush

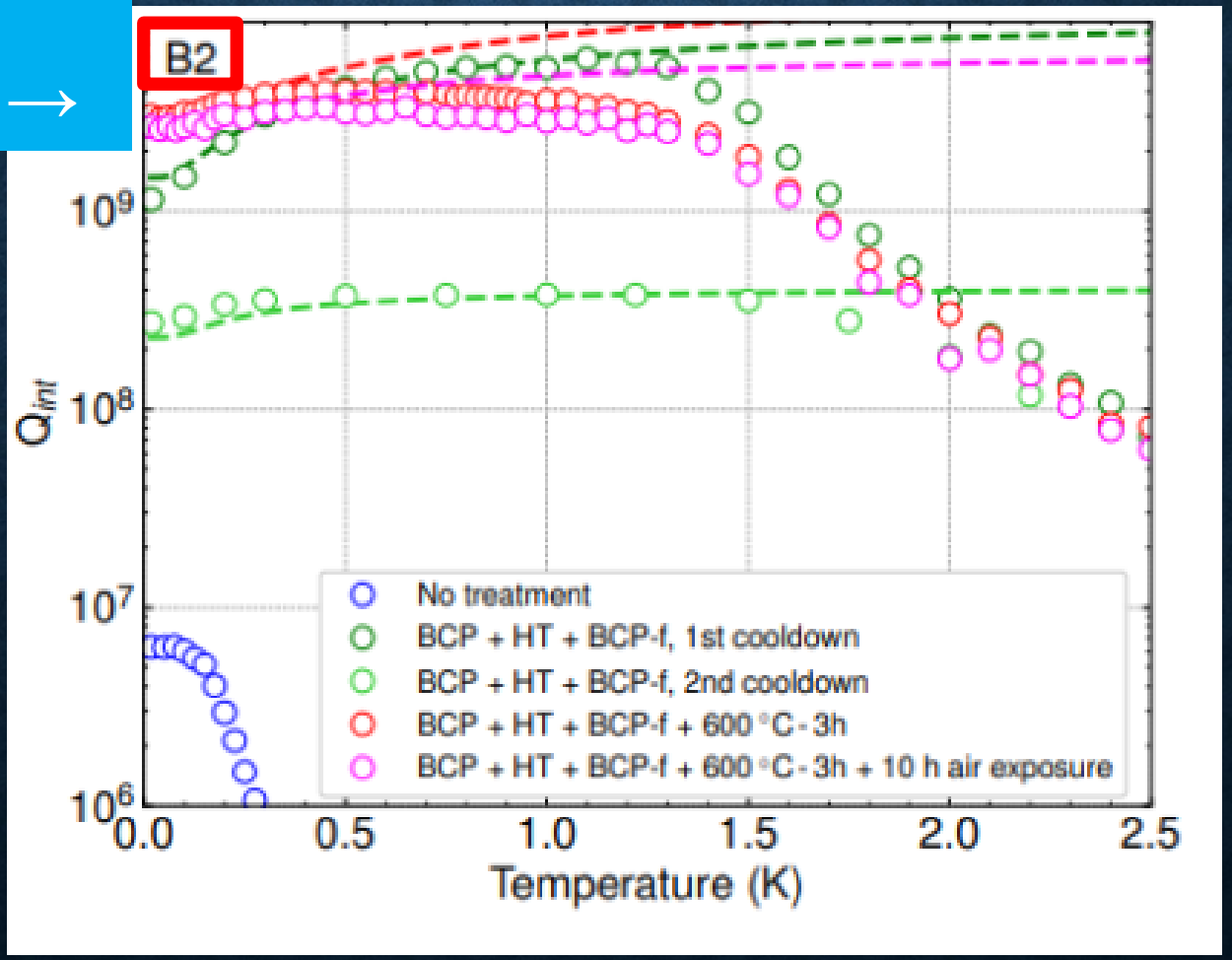
wBCP=water buffered CP

T. Takenaka, T. Kubo et al.,
Phys. Rev. Applied **25**, 034076 (2026).



We broke the world record!!!→

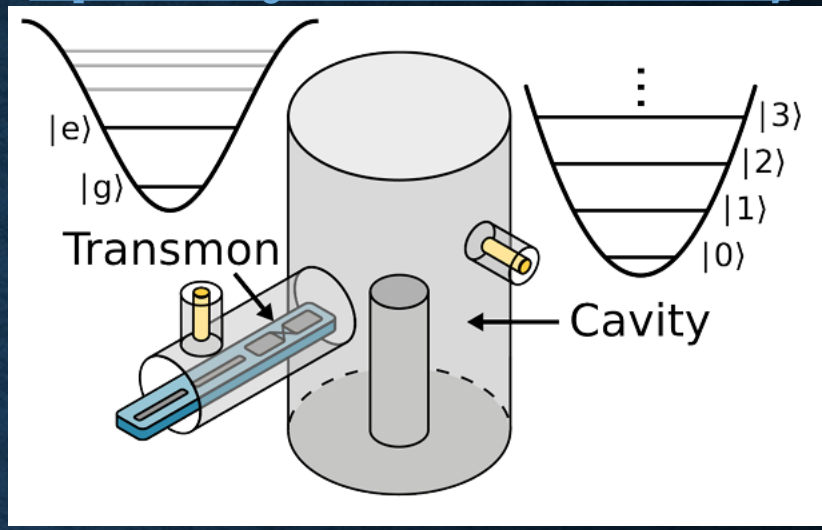
Corresponding to photon lifetime of ~0.1 sec.



T. Takenaka, T. Kubo et al.,
Phys. Rev. Applied **25**, 034076 (2026).

3D coaxial cavity progress

Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y

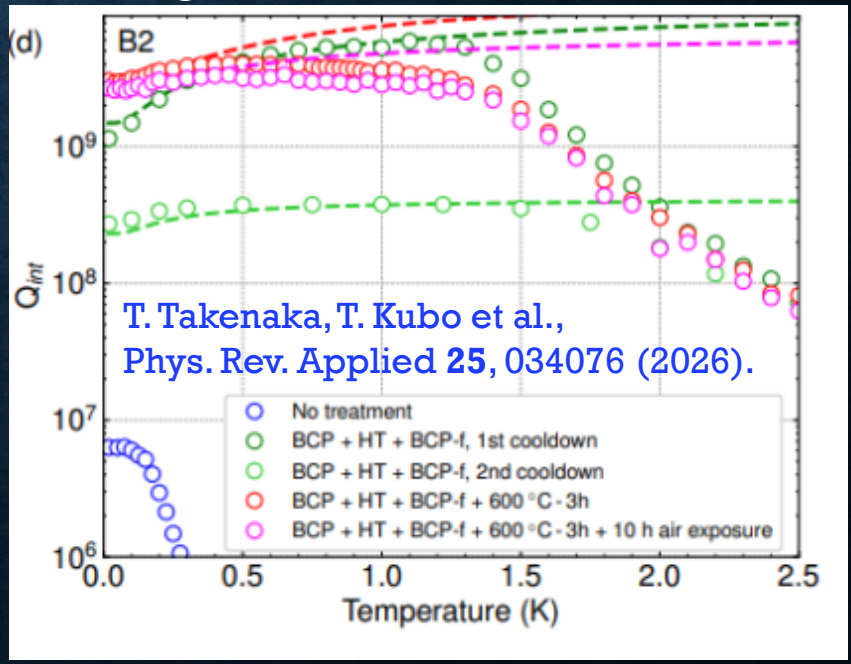


- Reagor et al., Phys. Rev. B **94**, 014506 (2016) Yale University
  Aluminum,
  $Q = 7 \times 10^7$ (single photon level)

- Heidler et al., Phys. Rev. Applied **16**, 034024 (2021) University of Innsbruck
  Niobium (1:1:1 buffered chemical polishing at 0-5℃),
  $Q = 5 \times 10^8$ (single photon level)

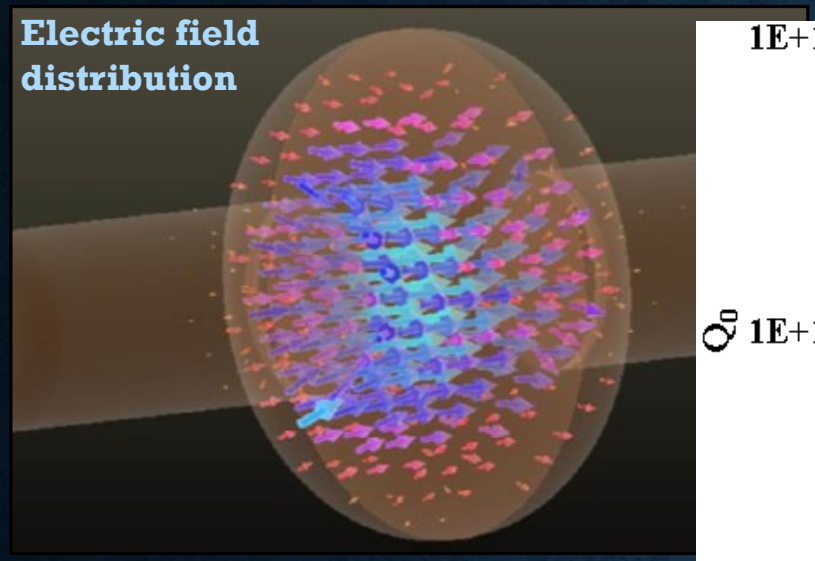
- Oriani et al., Phys. Rev. Applied **24**, 044080 (2025) University of Chicago
  Niobium (1:1:2 water buffered chemical polishing below 10℃),
  $Q = 1.4 \times 10^9$ (single photon level)

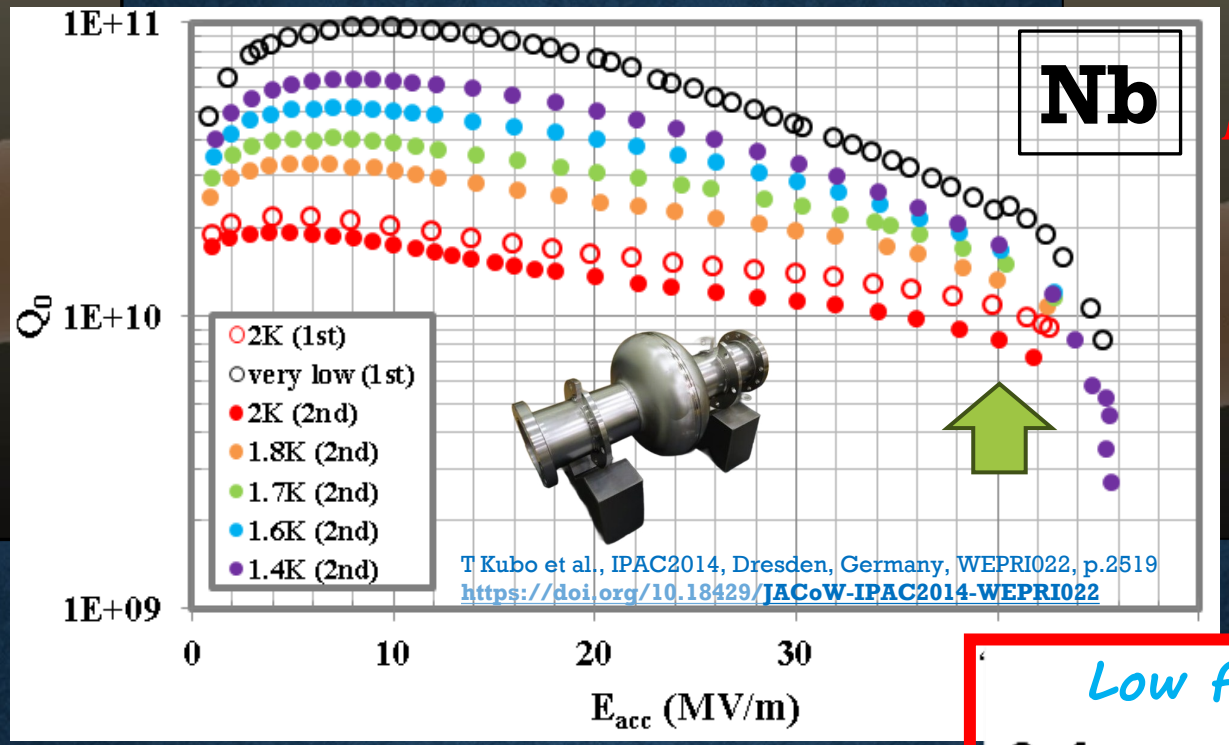
- T. Takenaka, T. Kubo et al., Phys. Rev. Applied 25, 034076 (2026). KEK&NTT
  Niobium (1:1:1 BCP at 25℃ + 900℃ anneal + 1:1:1 BCP flush at 25℃ + 600℃ anneal)
  $Q=3 \times 10^9$ (single photon level)



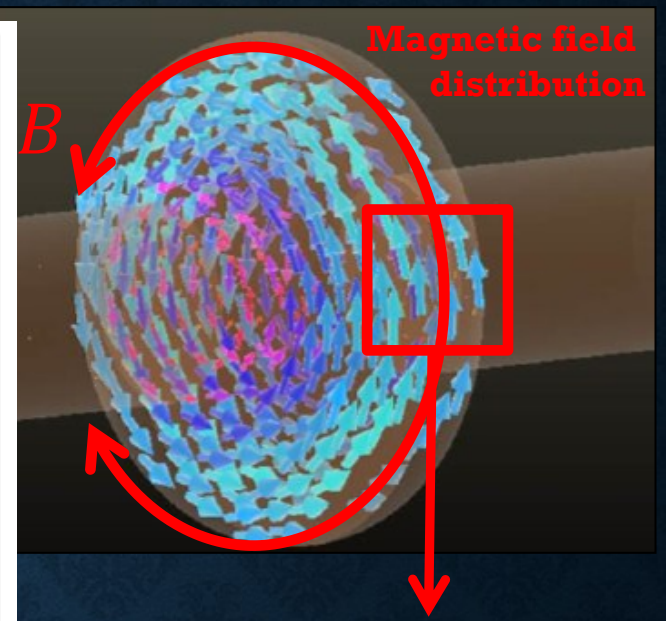
The world record

Stability of the vortex-free state
(very short introduction)

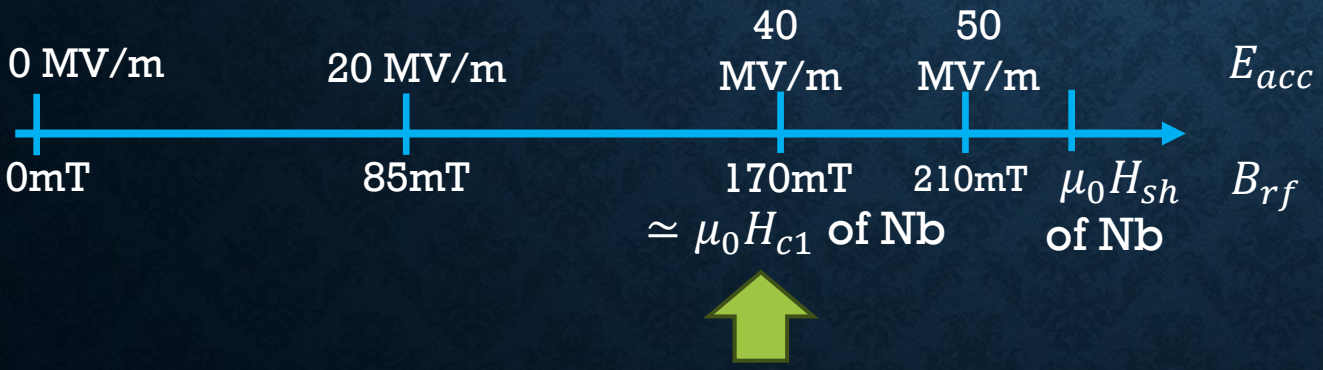




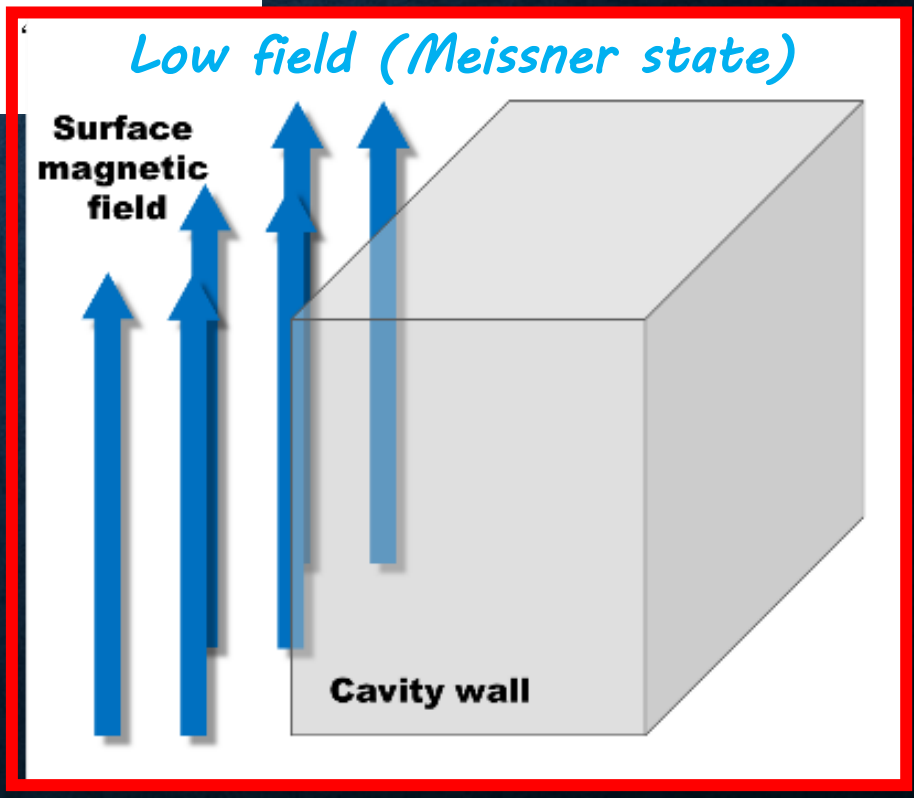
T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

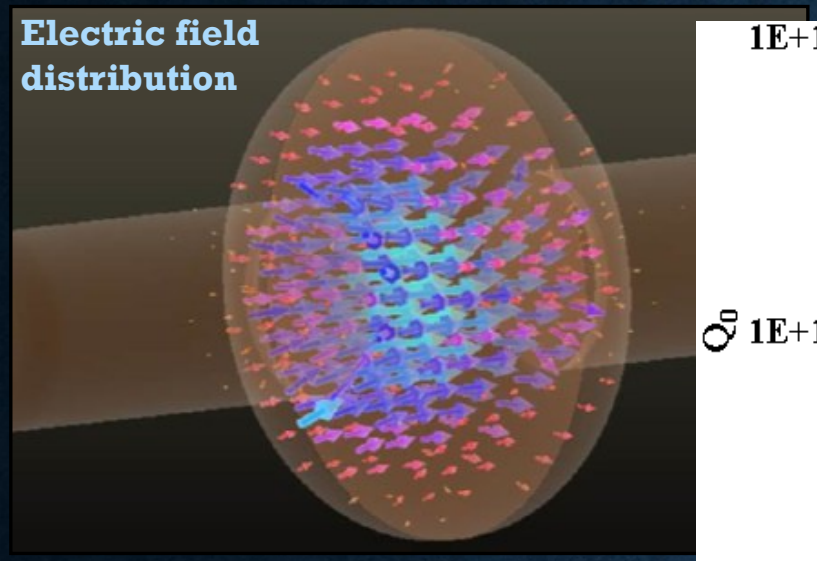


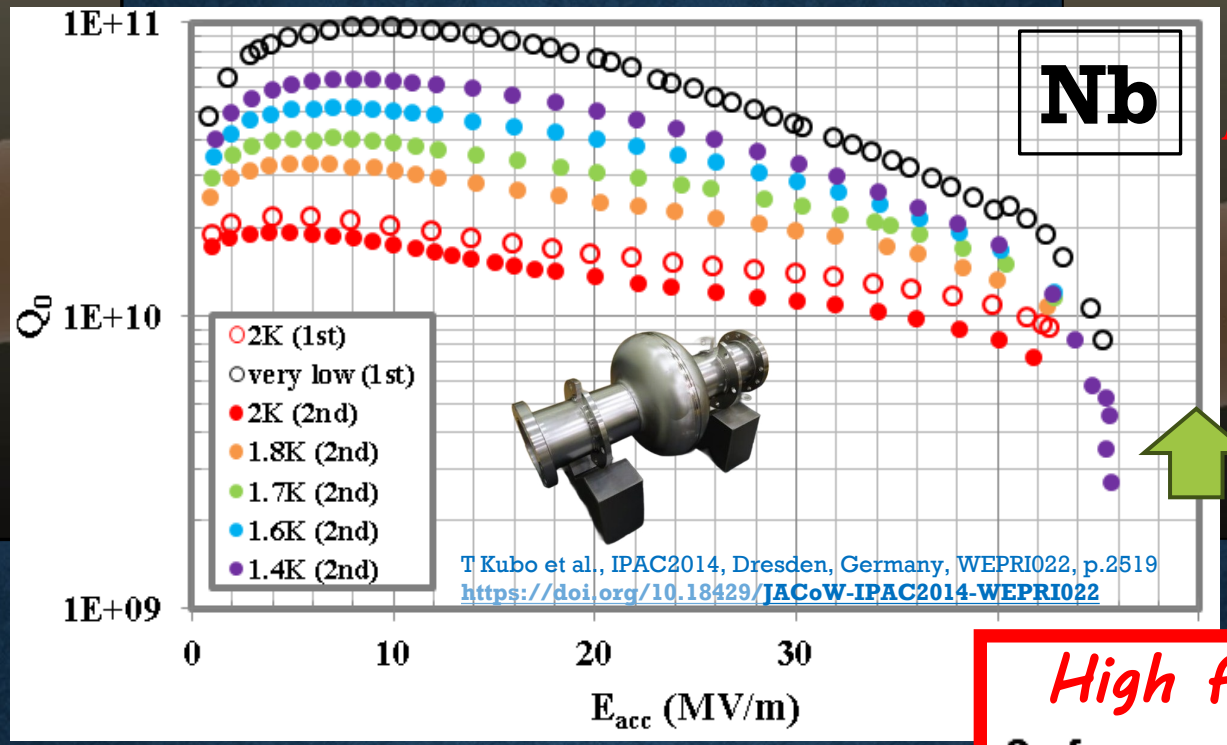
Note: Cavities are operated under the **Meissner state**



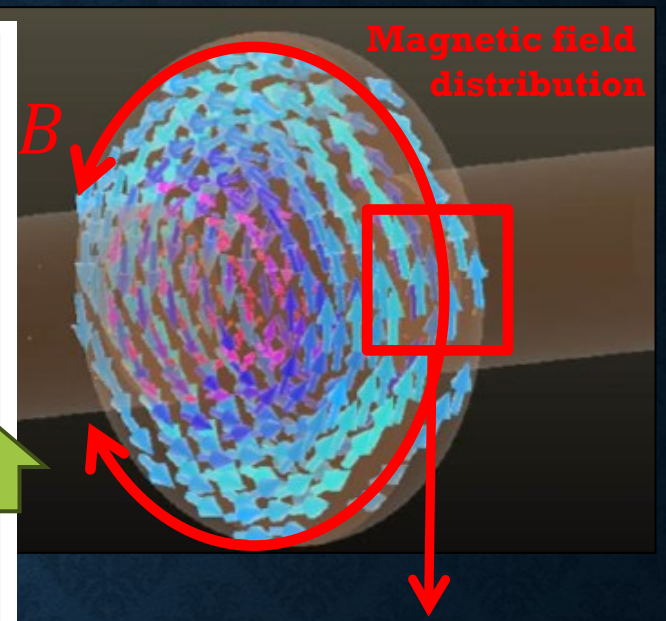
(⌐■_■) good



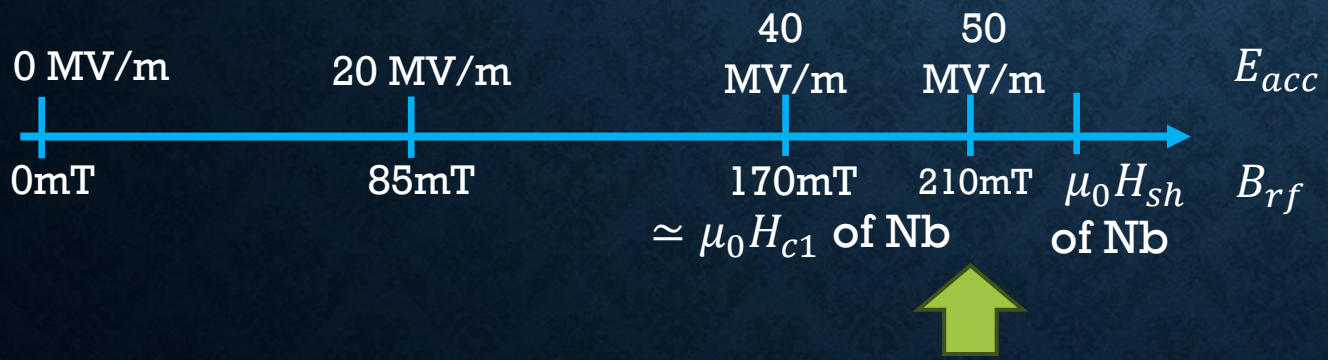




T Kubo et al., IPAC2014, Dresden, Germany, WEPRI022, p.2519
https://doi.org/10.18429/JACoW-IPAC2014-WEPRI022

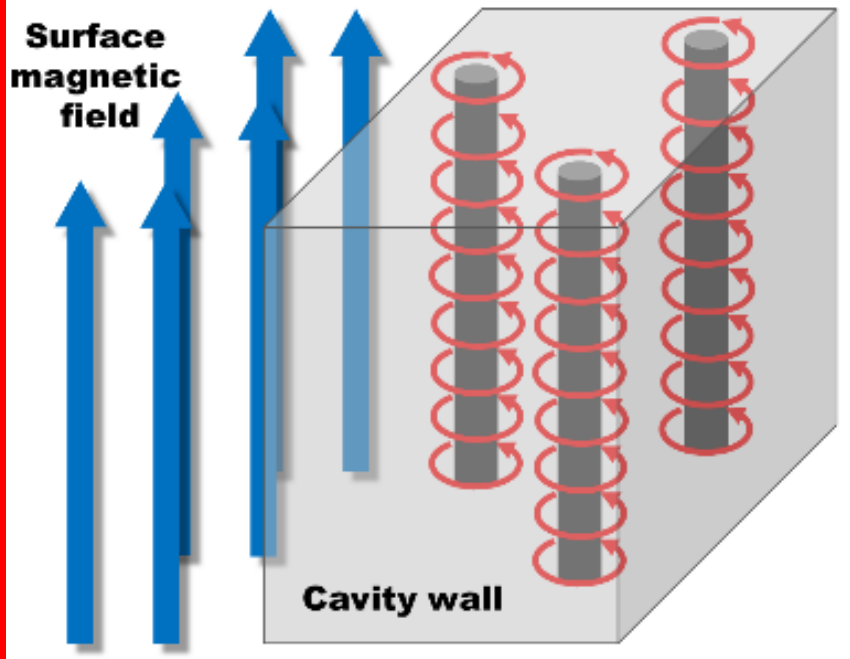


Note: Cavities are operated under the **Meissner state**

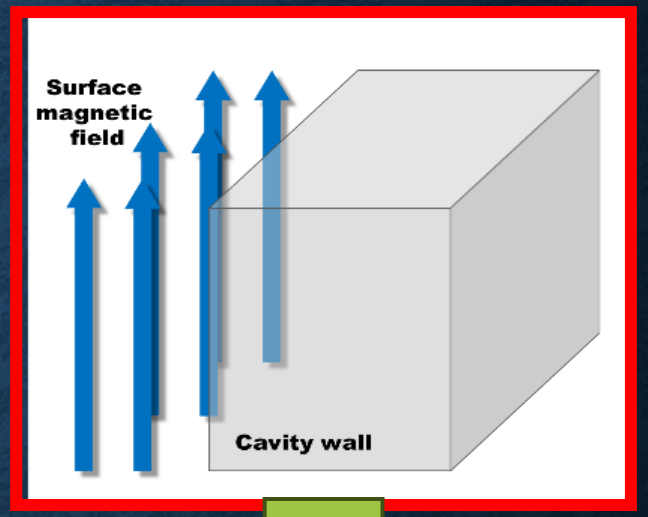


＼(^o^)／オワタ
I'm doomed

High field (vortex state)

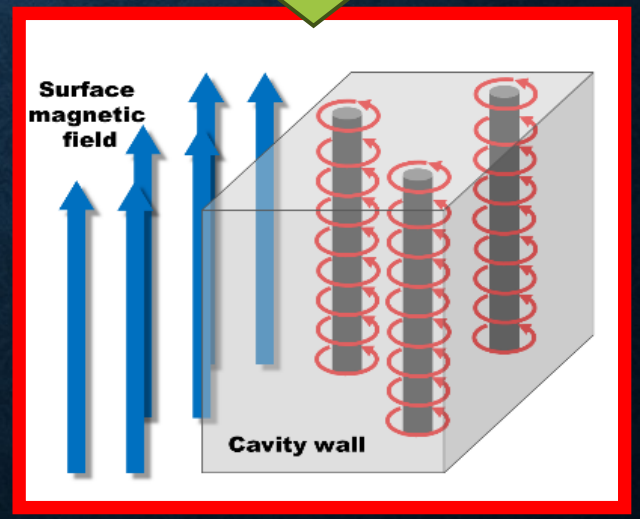


What is the fundamental theoretical limit?



Bulk

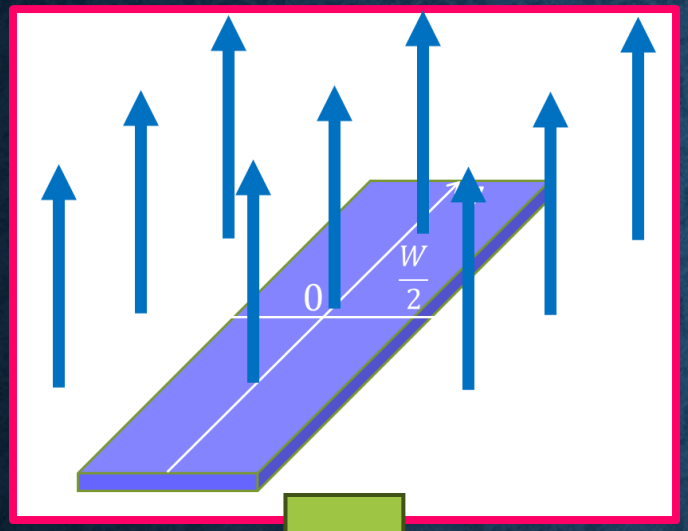
$B > B_{sh} > B_{c1}$



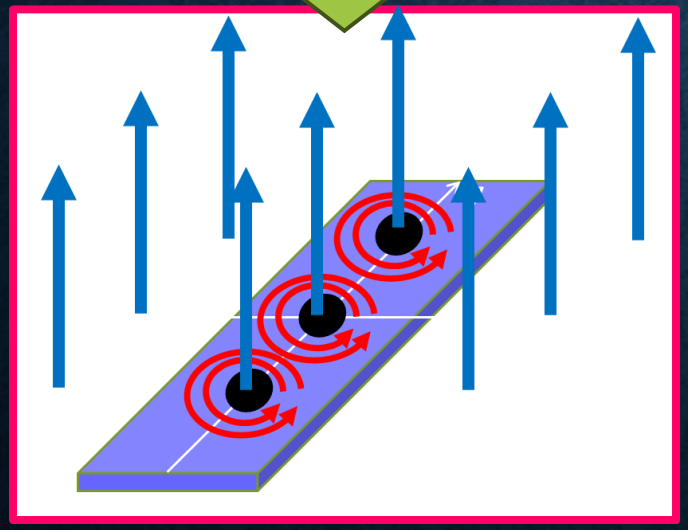
What is the fundamental theoretical limit?

Actually, this is a common fundamental problem in superconducting devices.

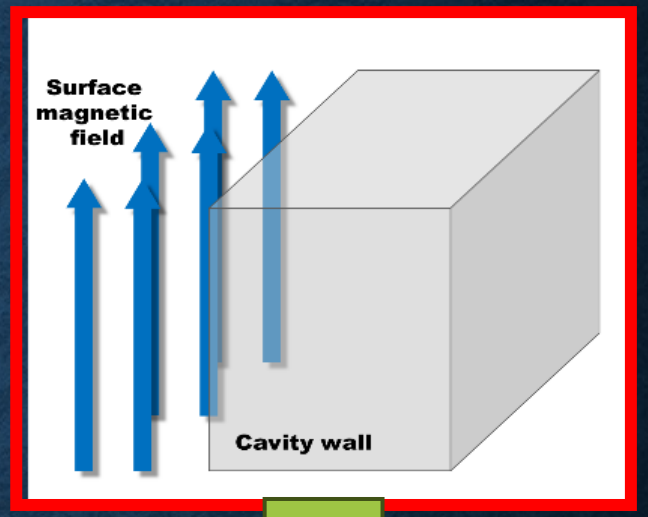
Narrow thin film



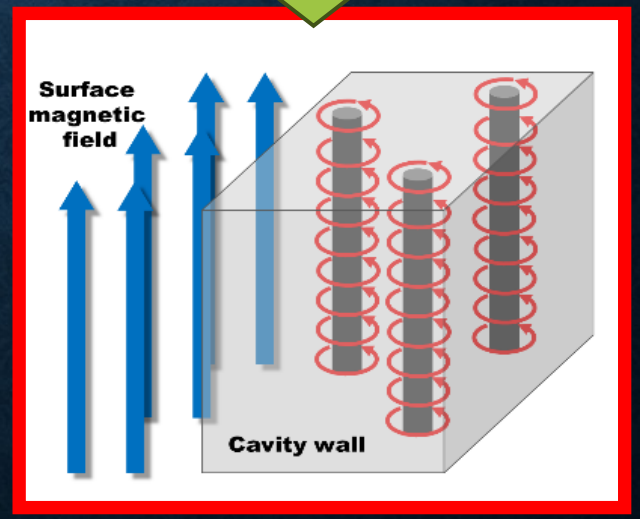
$B > B_s > B_{c1}$



Bulk



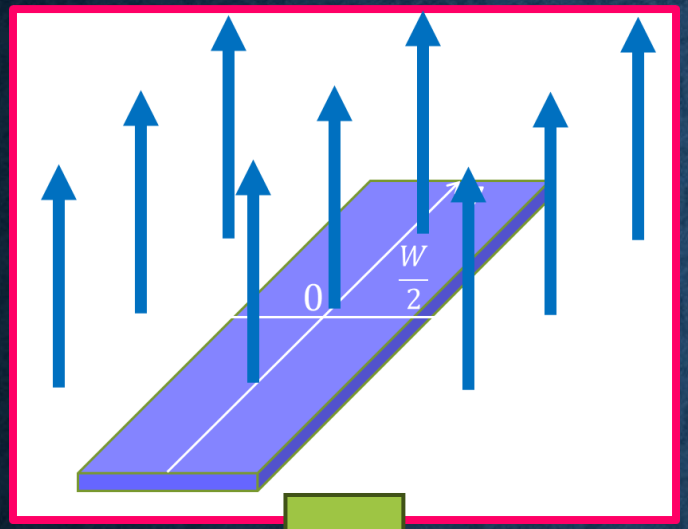
$B > B_{sh} > B_{c1}$



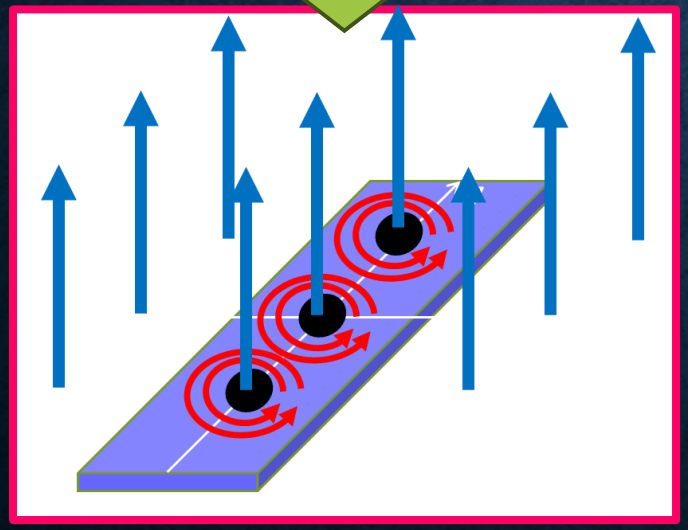
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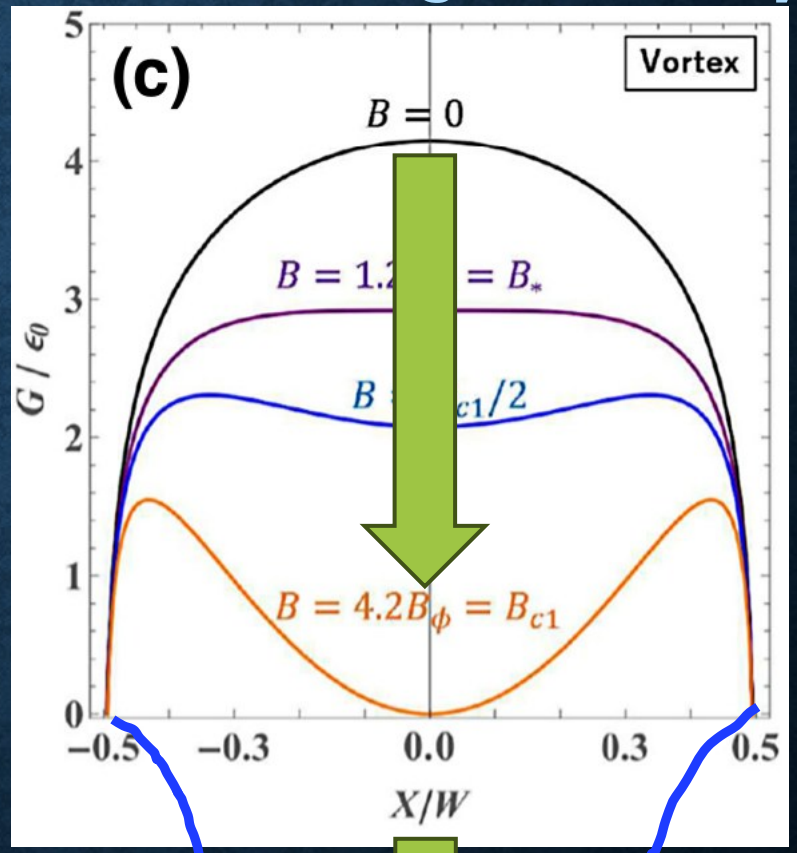
Narrow thin film



$B > B_s > B_{c1}$



Phenomenological-theory



$B = B_s$

Barrier disappears

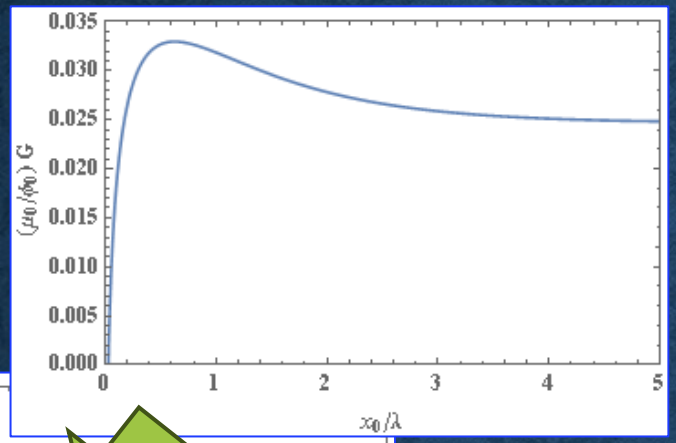
See, e.g.,
T. Kubo, Phys. Rev. Applied **20**, 034033 (2023)

What is the fundamental theoretical limit?

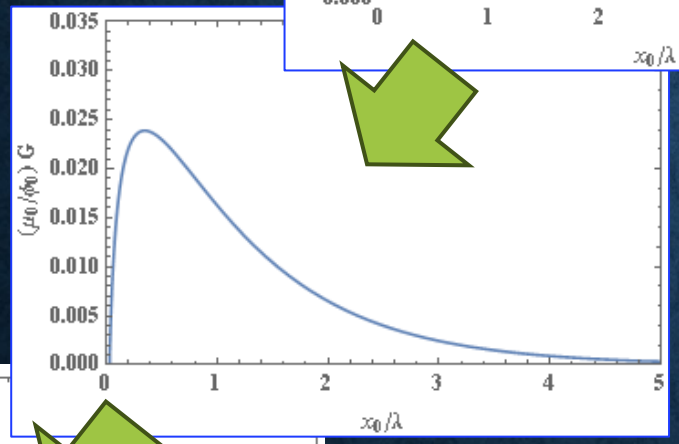
Actually, this is a common fundamental problem in superconducting devices.

Phenomenological-theory

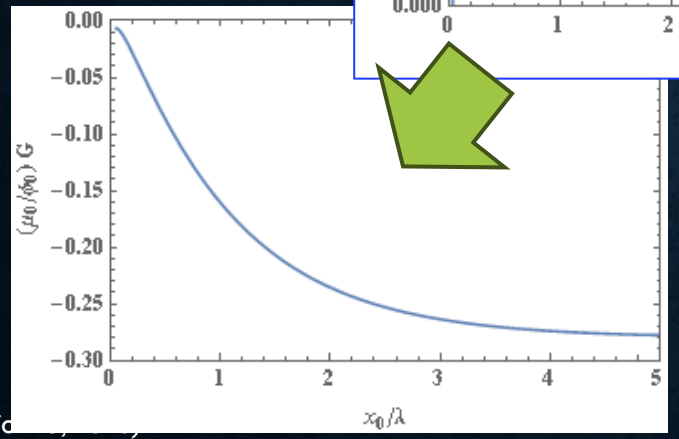
$B < B_{c1}$



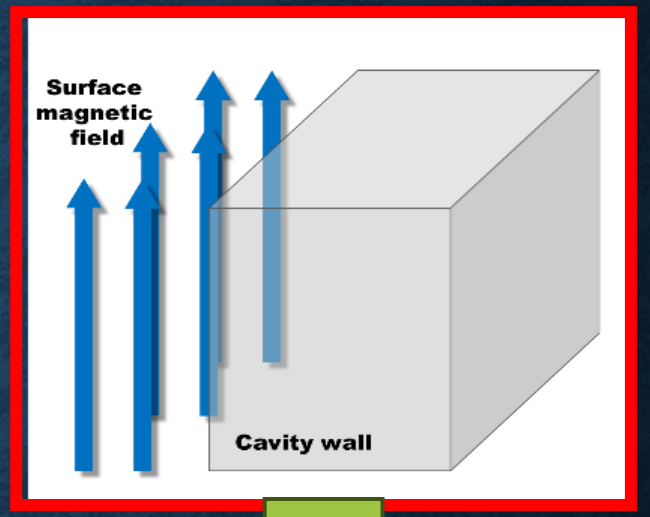
$B = B_{c1}$



$B = B_{sh}$

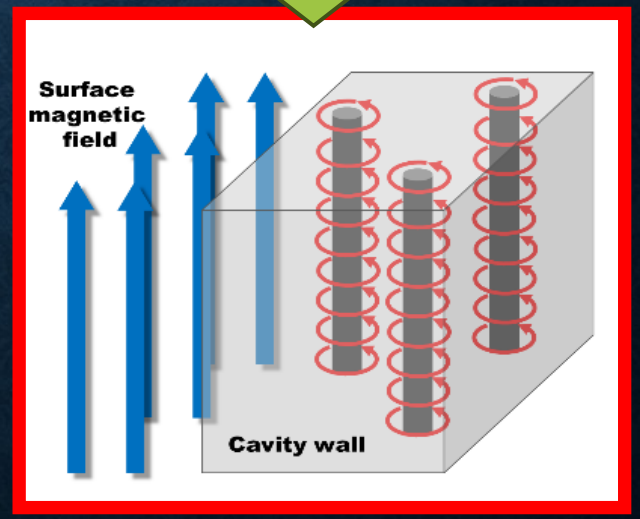


Barrier disappears

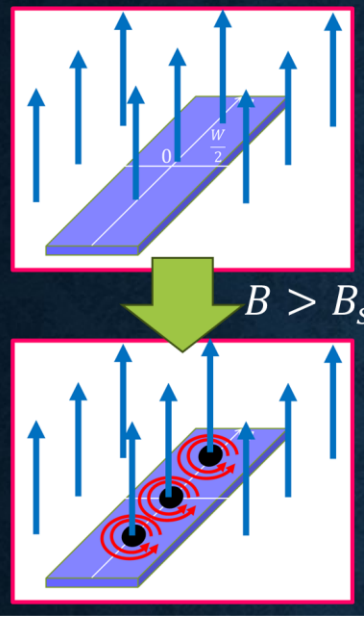


Bulk

$B > B_{sh} > B_{c1}$

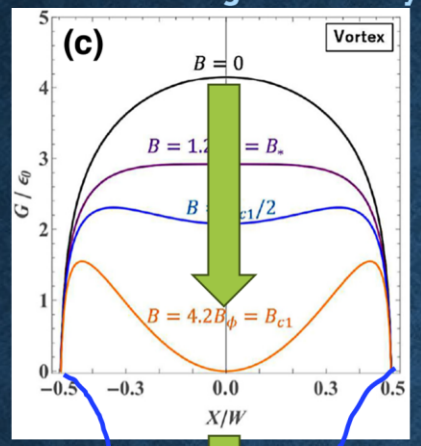


Narrow thin film



$B > B_s > B_{c1}$

Phenomenological-theory



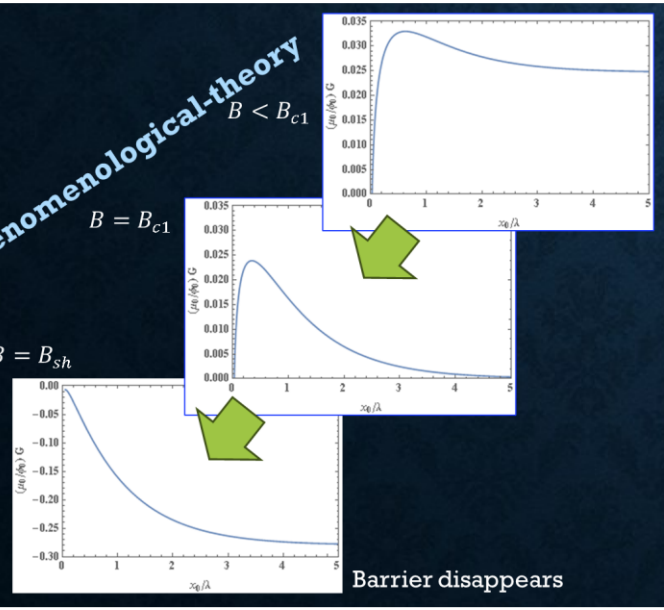
$B = B_s$ Barrier disappears

Phenomenological-theory

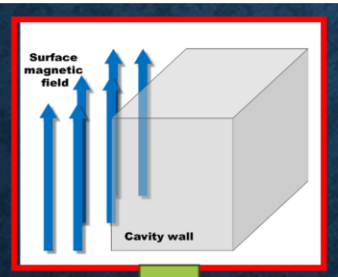
$B < B_{c1}$

$B = B_{c1}$

$B = B_{sh}$

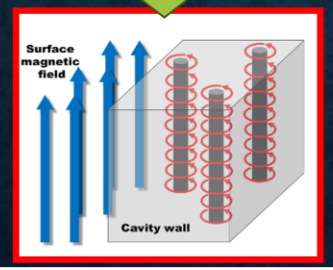


Barrier disappears



Bulk

$B > B_{sh} > B_{c1}$



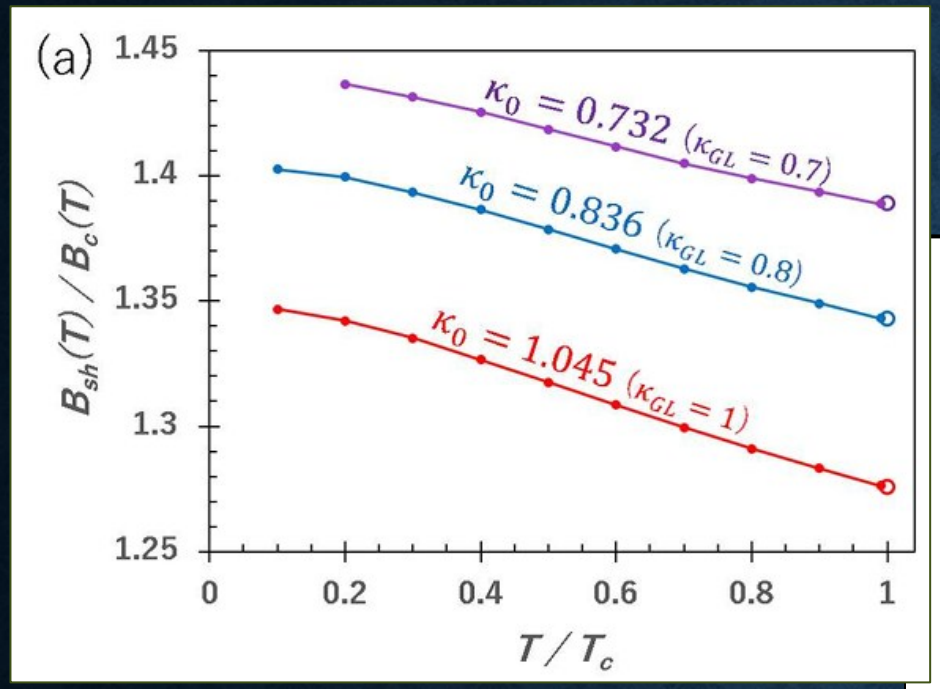
These discussions based on the "edge barrier" and "Bean–Livingston barrier" are phenomenological and are not always quantitatively—or even qualitatively—correct.

We need a stability analysis based on the microscopic theory of superconductivity

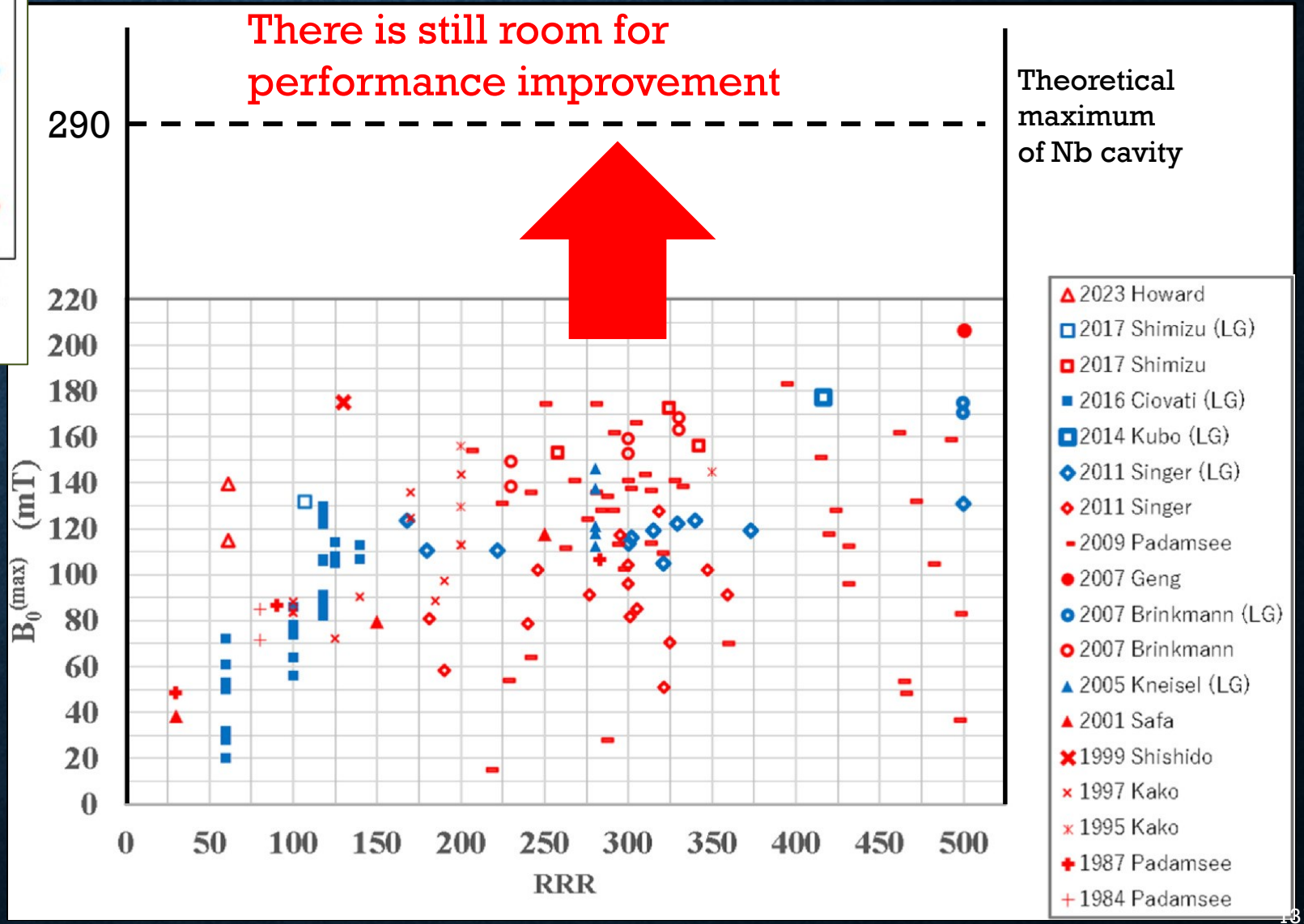
T. Kubo, arXiv:2608.04508 (2026)
and
T. Kubo, Phys. Rev. Accel. Beams (accepted as a **Letter** article) arXiv:2606.10420 (2026)

Microscopic theory of the superheating field

Intrinsic $\frac{\lambda_0}{\xi_0}$	Purity (mfp ℓ)	Theory (applicable to SRF temperature $T \ll T_c$)	
$\frac{\lambda_0}{\xi_0} \gg 1$ (e.g., Nb_3Sn, NbN, NbTiN etc.)	$\ell \gg \xi_0$ (clean)	$H_{sh}(0) = 0.84 H_c(0)$ $H_{sh}(\mathrm{T})$ for $0 \le T \le T_c$	Analytical result for T=0. V. P. Galaiko, Sov. Phys. JETP **23**, 475 (1966) • G. Catelani and J. P. Sethna, Phys. Rev. B **78**, 224509 (2008) • F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012)
	any ℓ (from clean limit to dirty limit)	$H_{sh}(\mathrm{T})$ for $0 \le T \le T_c$	F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012)
	$\ell \ll \xi_0$ (dirty)	$H_{sh}(\mathrm{T})$ for $0 \le T \le T_c$ $H_{sh}(0) = 0.795 H_c(0)$	• F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012) • T. Kubo, Phys. Rev. Research **2**, 033203 (2020) Analytical result for T=0. • T. Kubo, Phys. Rev. Research **2**, 033203 (2020) • T. Kubo, Supercond.Sci. Tecnol. **34**, 045006 (2021)
$\frac{\lambda_0}{\xi_0} \lesssim 1$ (e.g., Nb)	any ℓ	N/A	
	$\ell \gg \xi_0$ (clean)	$H_{sh}(\mathrm{T})$ for $0.1 \le T \le T_c$	T. Kubo, Phys. Rev. Accel. Beams (accepted as a **Letter** article) arXiv:2606.10420 (2026)

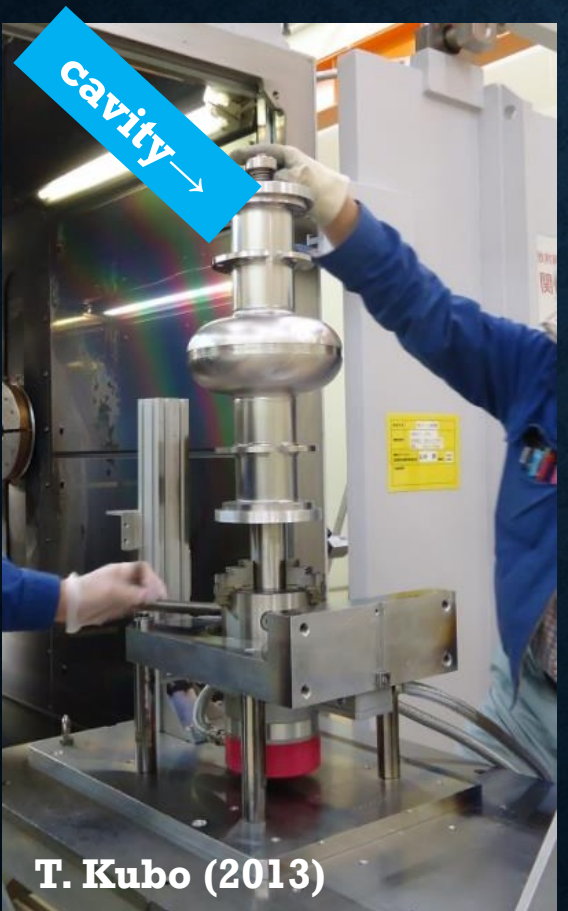


T. Kubo, Phys. Rev. Accel. Beams
(accepted as a **Letter** article)
arXiv:2606.10420 (2026)

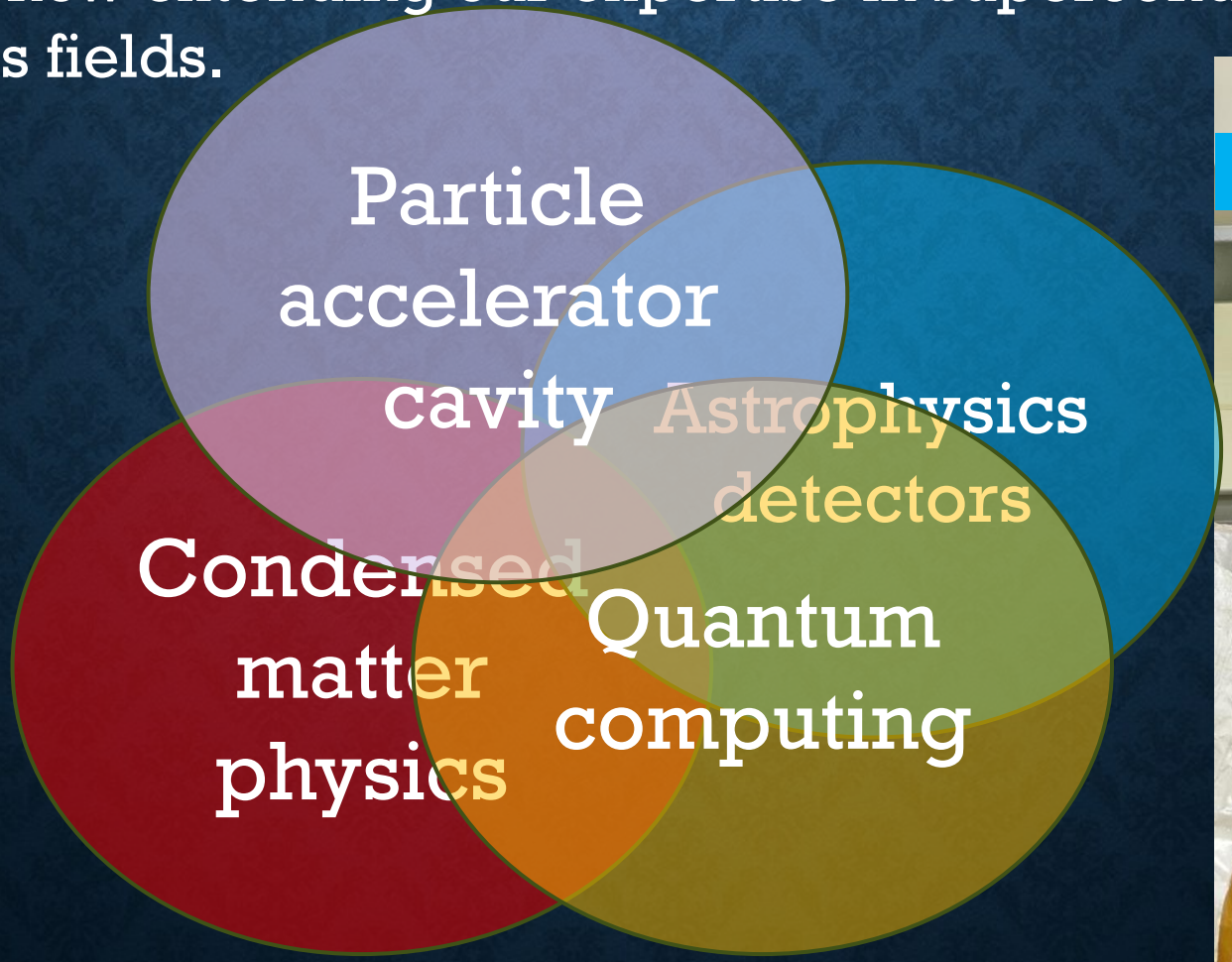


Summary

Superconducting resonators lie at the intersection of accelerator physics, astronomy, and quantum technologies, opening new opportunities for scientific discovery. We have been at the forefront of research on superconducting RF cavities for particle accelerators. We are now extending our expertise in superconducting physics and technology to various fields.



T. Kubo (2013)



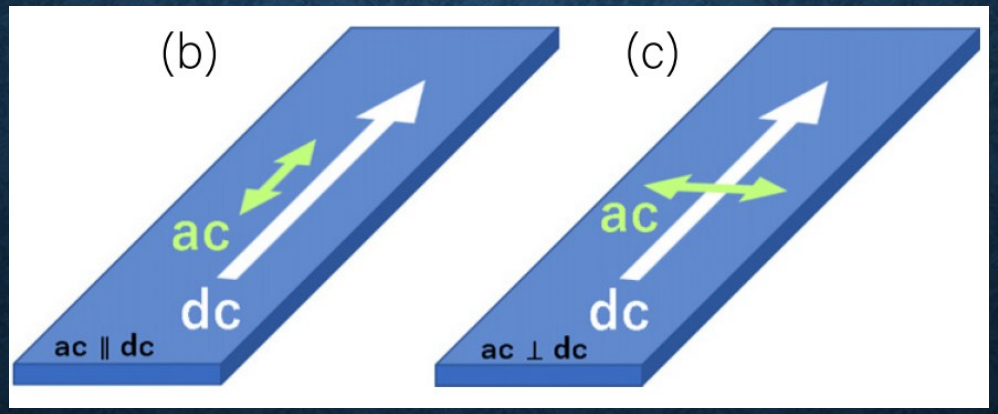


T. Kubo (2024)

Appendix

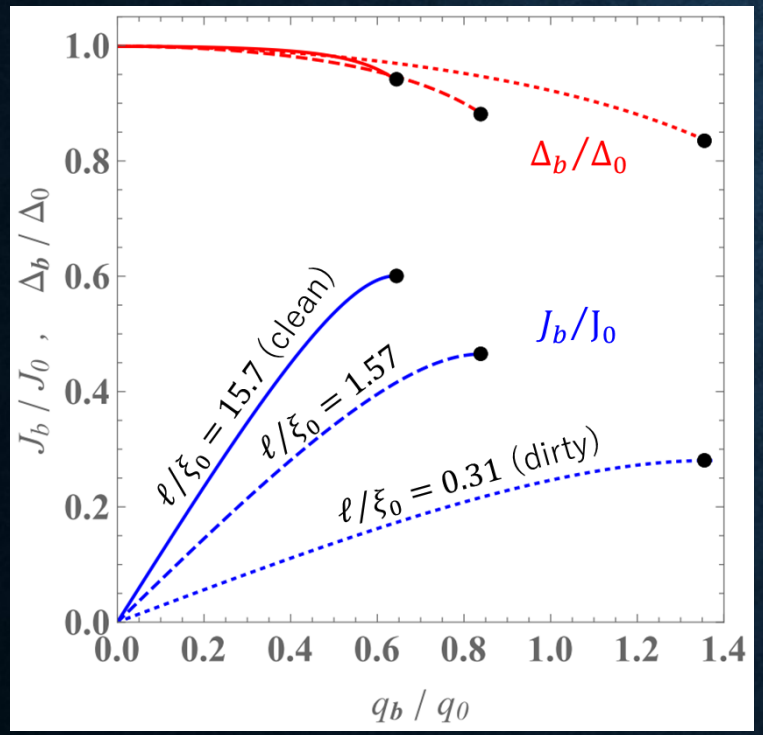
Non-accelerator application (3)

Kinetic inductance





If the AC field is absent, this reduces to a problem of equilibrium superconductivity.
That case was already well understood decades ago.

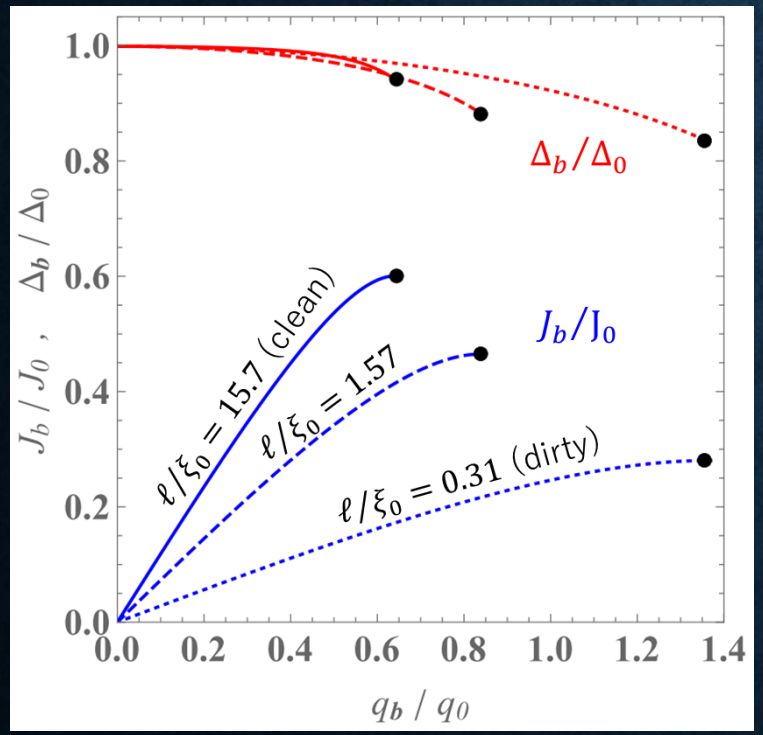


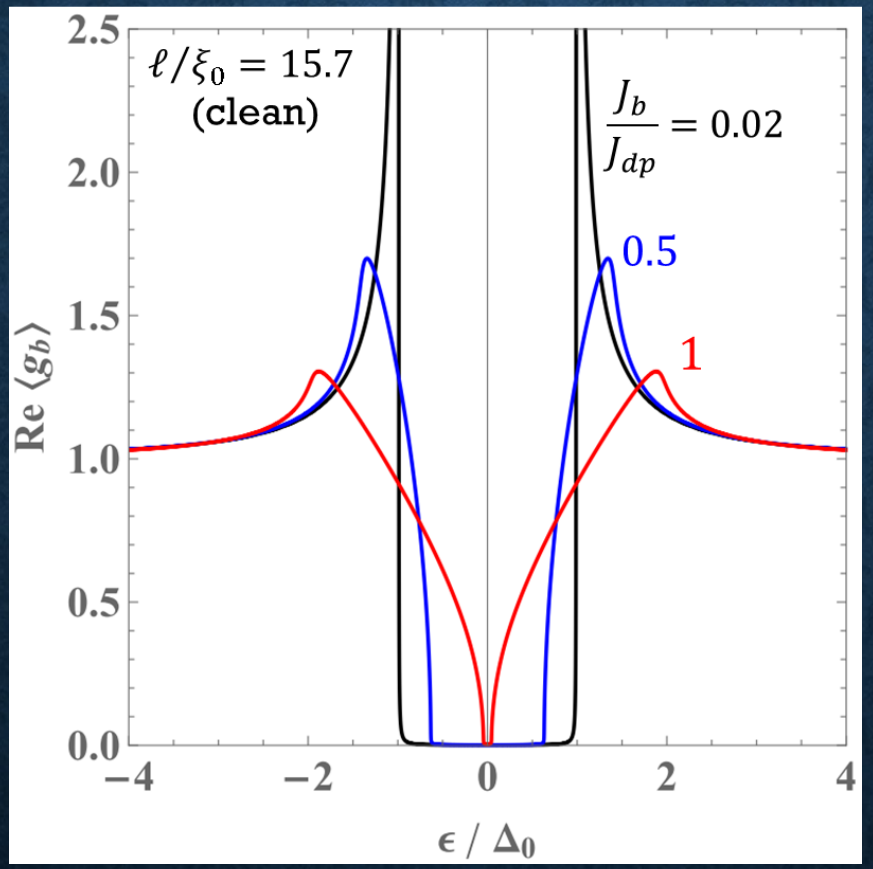
T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042

See also, T. Kubo, Phys. Rev. Research **2**, 033203 (2020); F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012)



If the AC field is absent, this reduces to a problem of equilibrium superconductivity.
That case was already well understood decades ago.



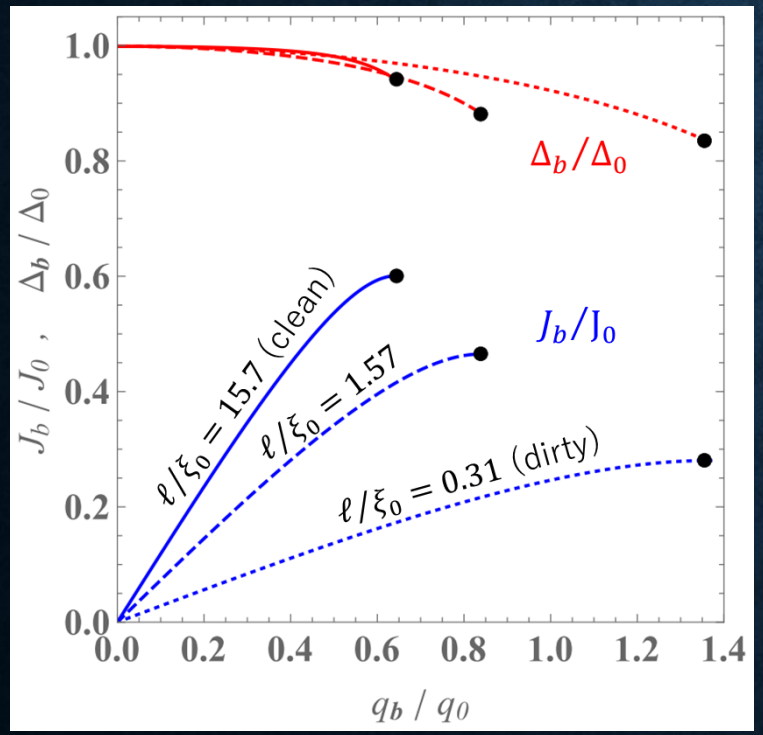


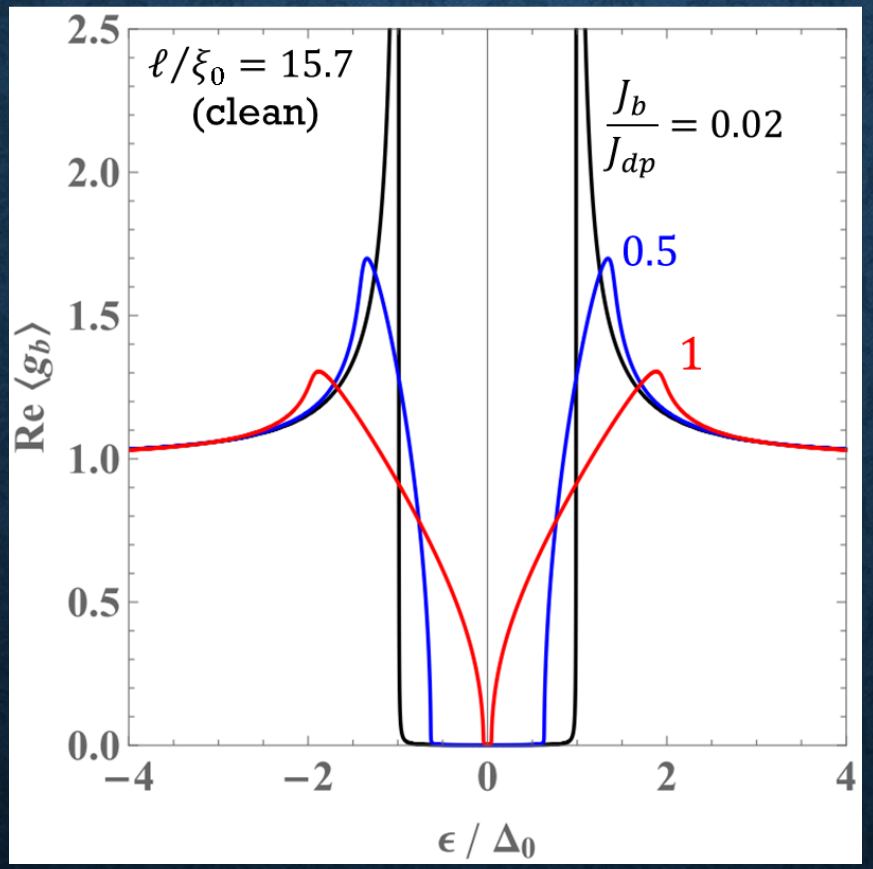
T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042

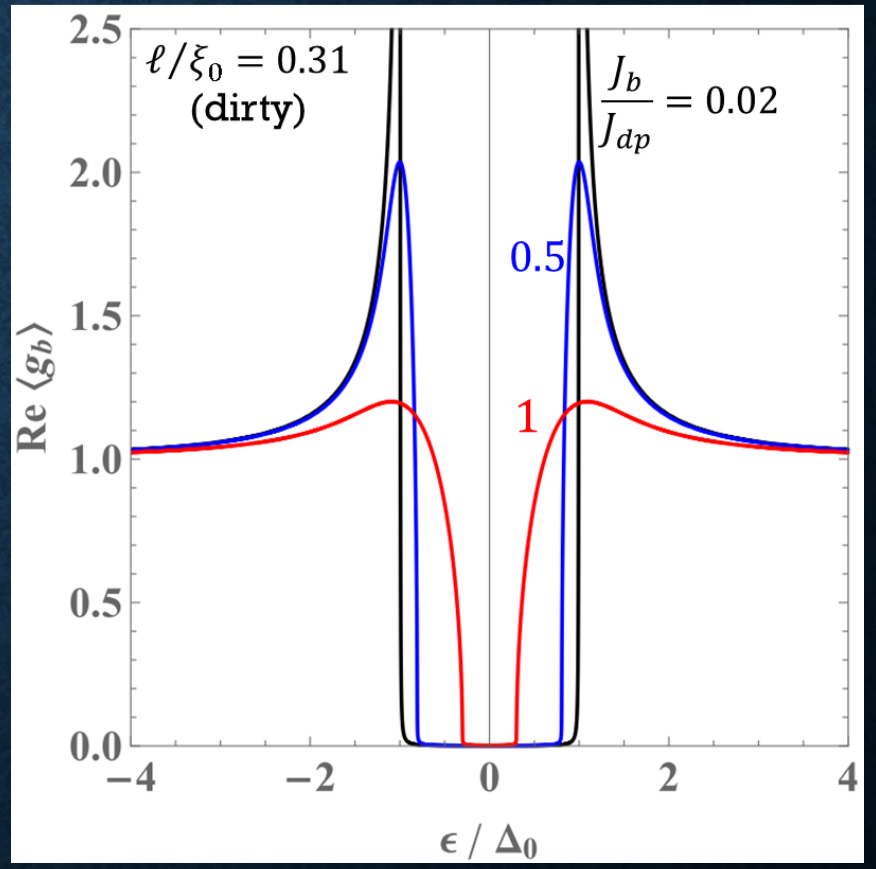
See also, T. Kubo, Phys. Rev. Research **2**, 033203 (2020); F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012)



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That case was already well understood decades ago.

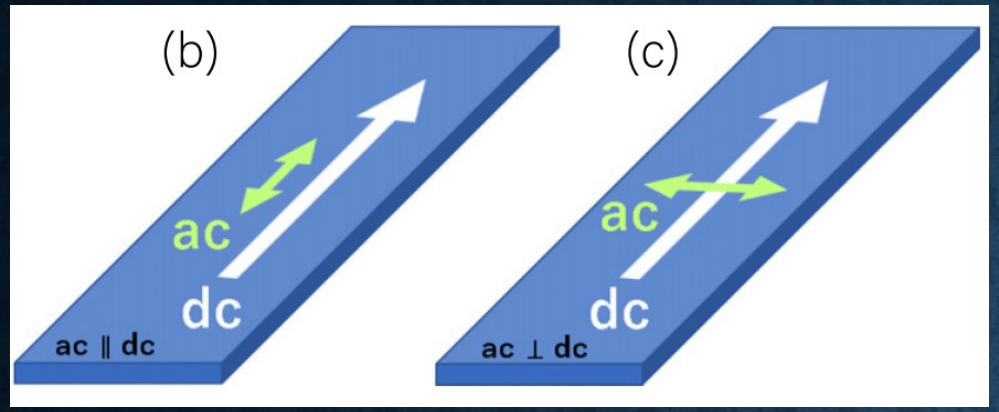




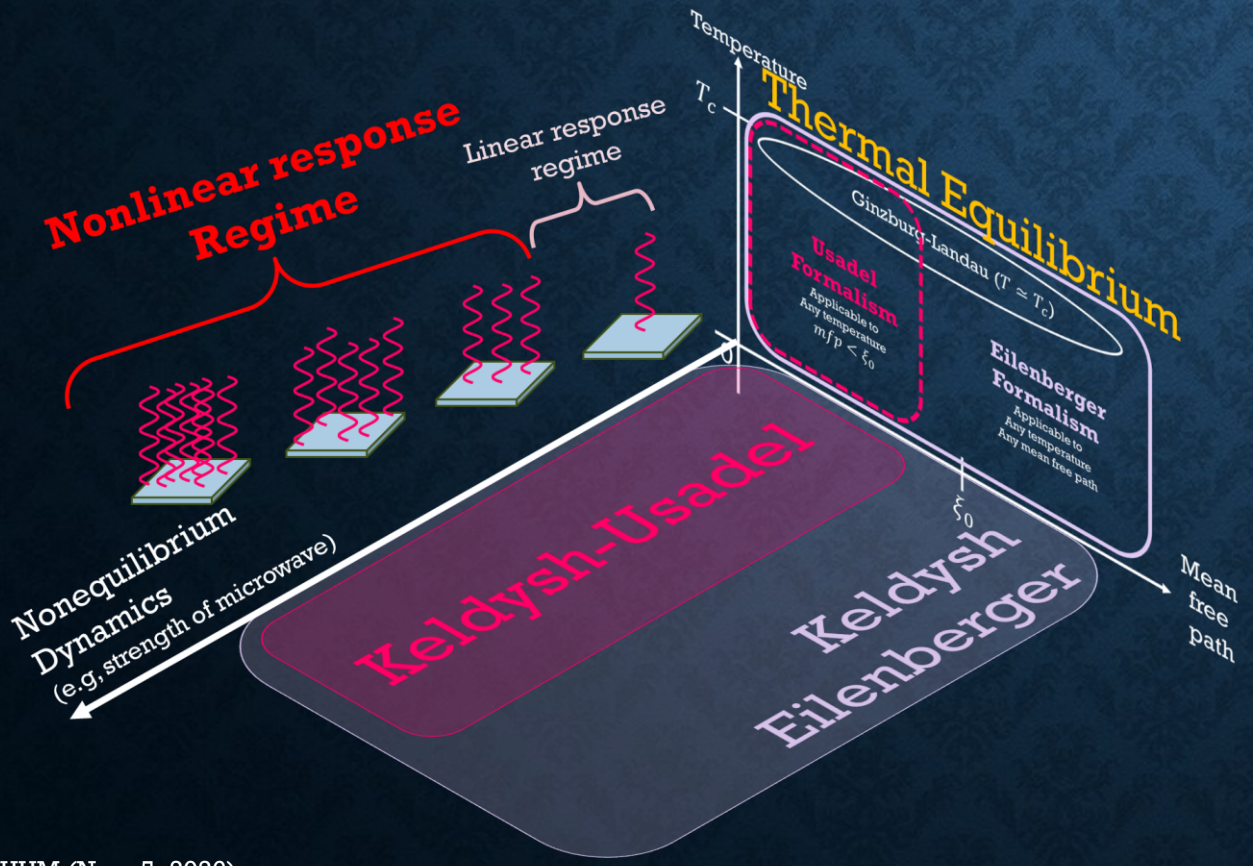


T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042

See also, T. Kubo, Phys. Rev. Research **2**, 033203 (2020); F. Pei-Jen Lin and A. Gurevich, Phys. Rev. B **85**, 054513 (2012)



However, once an AC field is present, the problem becomes much more complicated. In that case, we need to use the theory of **nonequilibrium** superconductivity.



A. Moor et al., Phys. Rev. Lett. **118**, 047001 (2017)
T. Jujo, J. Phys. Soc. Jpn. **91**, 074711 (2022)
T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)
T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

- **The Keldysh-Eilenberger theory** is a microscopic theory of **nonequilibrium** superconductivity. It is applicable at any temperature ($0 \leq T \leq T_c$) and for arbitrary mean free path.

- **The Keldysh–Usadel theory** represents the dirty-limit reduction of the Keldysh–Eilenberger theory of nonequilibrium superconductivity, applicable at any T ($0 \leq T \leq T_c$).

Keldysh–Usadel Equation: DC Current with AC Perturbation

T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042 T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

$$
\begin{aligned}
&-i(s/2)\left[\hat{\tau}_3\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)-\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\right.\\
&\quad+\hat{\tau}_3\delta\hat{g}^R(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)-\delta\hat{g}^R(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)\hat{\tau}_3\\
&\quad+\hat{\tau}_3\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\delta\hat{g}^A(\epsilon,\omega)-\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\delta\hat{g}^A(\epsilon,\omega)\hat{\tau}_3\\
&\quad\left.+\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)-\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)\hat{\tau}_3\right]\\
&\quad-i(\delta W/2)\left[\hat{\tau}_3\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)-\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)\hat{\tau}_3\right.\\
&\quad+\hat{\tau}_3\hat{g}_b^R(\epsilon_-)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)-\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\\
&\quad+\hat{\tau}_3\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)-\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)\hat{\tau}_3\\
&\quad\left.+\hat{\tau}_3\hat{g}_b^K(\epsilon_-)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)-\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\hat{g}_b^A(\epsilon_+)\hat{\tau}_3\right]\\
&=\epsilon_+\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)-\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\epsilon_-\\
&\quad+[\hat{\Delta}_b,\delta\hat{g}^K(\epsilon,\omega)]+\delta\hat{\Delta}(\omega)\hat{g}_b^K(\epsilon_-)-\hat{g}_b^K(\epsilon_+)\delta\hat{\Delta}(\omega).
\end{aligned}
\tag{15}
$$

$$
\begin{aligned}
&-i(s/2)\left\{\hat{\tau}_3\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)-\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\right.\\
&\quad\left.+\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)-\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\hat{\tau}_3\right\}\\
&\quad-i(\delta W/2)\left\{\hat{\tau}_3\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\right.\\
&\quad-\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\hat{\tau}_3+\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\\
&\quad\left.-\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\right\}=\epsilon_+\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)-\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\epsilon_-\\
&\quad+[\hat{\Delta}_b,\delta\hat{g}^r(\epsilon,\omega)]+\delta\hat{\Delta}(\omega)\hat{g}_b^r(\epsilon_-)-\hat{g}_b^r(\epsilon_+)\delta\hat{\Delta}(\omega),
\end{aligned}
\tag{13}
$$

$$
\delta\Delta(\omega)=-\frac{\mathscr{G}}{8}\int d\epsilon\,\mathrm{Tr}[(-i\tau_2)\delta\hat{g}^K(\epsilon,\omega)]. \tag{17}
$$

Higgs ↓

We solve these equations to obtain the AC-induced **nonequilibrium corrections** ($\delta\hat{g}^{R,A,K}$ and $\delta\Delta$).

Keldysh–Usadel Equation: DC Current with AC Perturbation

T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042 T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

$$
\begin{aligned}
&-i(s/2)\left[\hat{\tau}_3\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)-\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\right.\\
&\quad+\hat{\tau}_3\delta\hat{g}^R(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)-\delta\hat{g}^R(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)\hat{\tau}_3\\
&\quad+\hat{\tau}_3\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\delta\hat{g}^A(\epsilon,\omega)-\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\delta\hat{g}^A(\epsilon,\omega)\hat{\tau}_3\\
&\quad\left.+\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)-\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)\hat{\tau}_3\right]\\
&\quad-i(\delta W/2)\left[\hat{\tau}_3\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)-\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)\hat{\tau}_3\right.\\
&\quad+\hat{\tau}_3\hat{g}_b^R(\epsilon_-)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)-\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\\
&\quad+\hat{\tau}_3\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)-\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)\hat{\tau}_3\\
&\quad\left.+\hat{\tau}_3\hat{g}_b^K(\epsilon_-)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)-\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\hat{g}_b^A(\epsilon_+)\hat{\tau}_3\right]\\
&=\epsilon_+\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)-\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\epsilon_-\\
&\quad+[\hat{\Delta}_b,\delta\hat{g}^K(\epsilon,\omega)]+\delta\hat{\Delta}(\omega)\hat{g}_b^K(\epsilon_-)-\hat{g}_b^K(\epsilon_+)\delta\hat{\Delta}(\omega).
\end{aligned}
\tag{15}
$$

$$
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&-i(s/2)\left\{\hat{\tau}_3\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)-\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\right.\\
&\quad\left.+\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)-\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\hat{\tau}_3\right\}\\
&\quad-i(\delta W/2)\left\{\hat{\tau}_3\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\right.\\
&\quad-\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\hat{\tau}_3+\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\hat{\tau}_3\hat{g}_b^r(\epsilon_-)\\
&\quad\left.-\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\hat{g}_b^r(\epsilon_+)\hat{\tau}_3\right\}=\epsilon_+\hat{\tau}_3\delta\hat{g}^r(\epsilon,\omega)-\delta\hat{g}^r(\epsilon,\omega)\hat{\tau}_3\epsilon_-\\
&\quad+[\hat{\Delta}_b,\delta\hat{g}^r(\epsilon,\omega)]+\delta\hat{\Delta}(\omega)\hat{g}_b^r(\epsilon_-)-\hat{g}_b^r(\epsilon_+)\delta\hat{\Delta}(\omega),
\end{aligned}
\tag{13}
$$

$$
\delta\Delta(\omega)=-\frac{\mathscr{G}}{8}\int d\epsilon\,\mathrm{Tr}[(-i\tau_2)\delta\hat{g}^K(\epsilon,\omega)]. \tag{17}
$$

Higgs ↓

We solve these equations to obtain the AC-induced **nonequilibrium corrections** ($\delta\hat{g}^{R,A,K}$ and $\delta\Delta$).

To obtain the ac response,
we substitute the solutions ($\delta\hat{g}^{R,A,K}$, $\delta\Delta$) into

$$
\delta\mathbf{J}(\omega)=-i\frac{\sigma_n}{e}\int d\epsilon\,\delta\mathbf{S}(\epsilon,\omega), \tag{18}
$$

$$
\begin{aligned}
\delta\mathbf{S}(\epsilon,\omega)&=(i/16)\mathrm{Tr}\left[i\mathbf{q}_b\right.\\
&\times\left\{\hat{\tau}_3\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)+\hat{\tau}_3\delta\hat{g}^R(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)\right.\\
&+\hat{\tau}_3\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\delta\hat{g}^A(\epsilon,\omega)+\hat{\tau}_3\delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)\}\\
&\left.+i\delta\mathbf{q}_\omega\left\{\hat{\tau}_3\hat{g}_b^R(\epsilon_+)\hat{\tau}_3\hat{g}_b^K(\epsilon_-)+\hat{\tau}_3\hat{g}_b^K(\epsilon_+)\hat{\tau}_3\hat{g}_b^A(\epsilon_-)\right\}\right].
\end{aligned}
\tag{19}
$$

Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim Agg + A_{dc} g \delta g$$

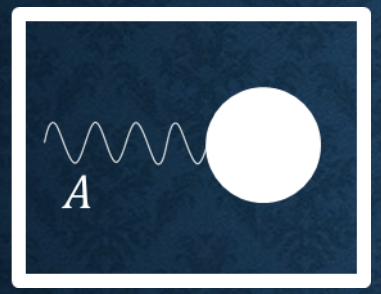
$$\delta \mathbf{J}(\omega) = -i\frac{\sigma_n}{e}\int d\epsilon \delta \mathbf{S}(\epsilon, \omega), \tag{18}$$

$$\begin{aligned}\delta \mathbf{S}(\epsilon,\omega) = (i/16)\mathrm{Tr}\,[&i\mathbf{q}_b \\ &\times \{\hat{\tau}_3 \hat{g}_b^R(\epsilon_+)\hat{\tau}_3 \delta\hat{g}^K(\epsilon,\omega) + \hat{\tau}_3 \delta\hat{g}^R(\epsilon,\omega)\hat{\tau}_3 \hat{g}_b^K(\epsilon_-) \\ &+ \hat{\tau}_3 \hat{g}_b^K(\epsilon_+)\hat{\tau}_3 \delta\hat{g}^A(\epsilon,\omega) + \hat{\tau}_3 \delta\hat{g}^K(\epsilon,\omega)\hat{\tau}_3 \hat{g}_b^A(\epsilon_-)\} \\ &+ i\delta\mathbf{q}_\omega \{\hat{\tau}_3 \hat{g}_b^R(\epsilon_+)\hat{\tau}_3 \hat{g}_b^K(\epsilon_-) + \hat{\tau}_3 \hat{g}_b^K(\epsilon_+)\hat{\tau}_3 \hat{g}_b^A(\epsilon_-)\}]. \end{aligned} \tag{19}$$

T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042
T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

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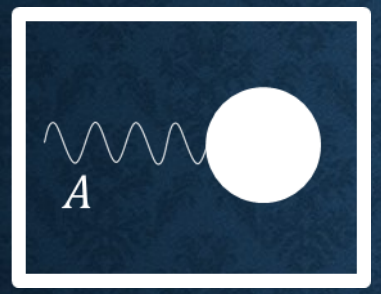
$$J \sim \boxed{Agg} + A_{dc} g \delta g$$



T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042 T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

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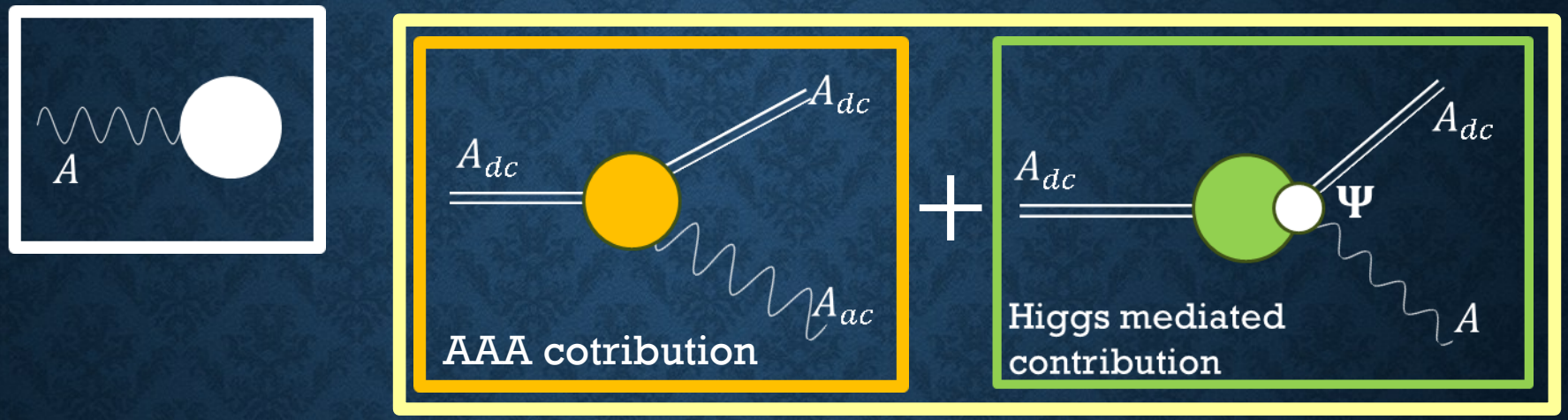
$$J \sim \boxed{Agg} + A_{dc} g \delta g \qquad \delta g \sim \boldsymbol{A}_{dc} \cdot \boldsymbol{A} + \delta\Delta \quad \text{and} \quad \delta\Delta \sim \Psi\, \boldsymbol{A}_{dc} \cdot \boldsymbol{A}$$



T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042
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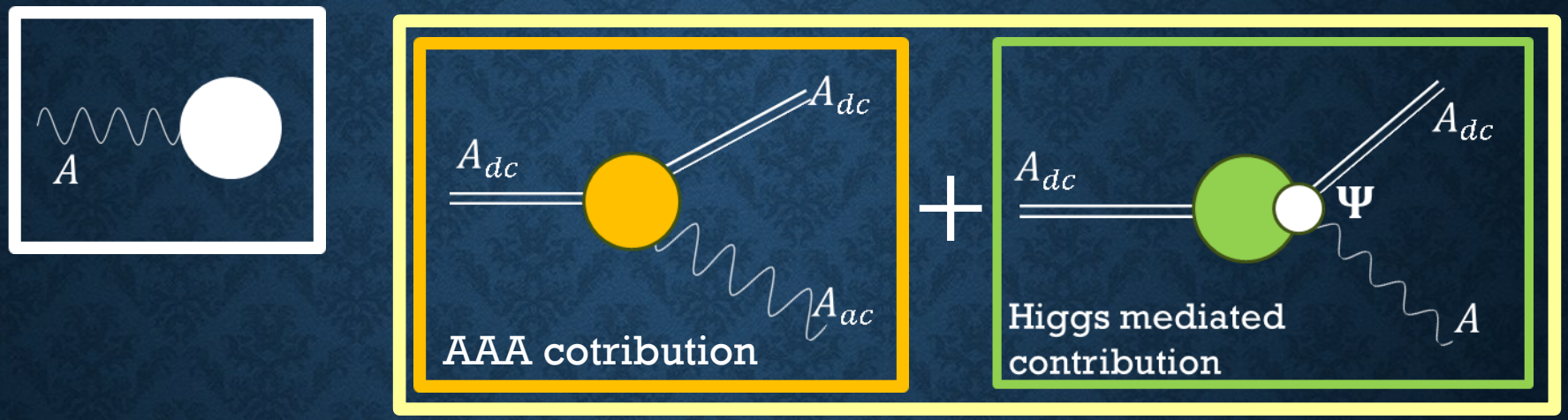
$$J \sim \boxed{Agg} + \boxed{A_{dc} g \delta g} \qquad \delta g \sim \boldsymbol{A_{dc}} \cdot \boldsymbol{A} + \delta\Delta \quad \text{and} \quad \delta\Delta \sim \Psi\, \boldsymbol{A_{dc}} \cdot \boldsymbol{A}$$



T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042
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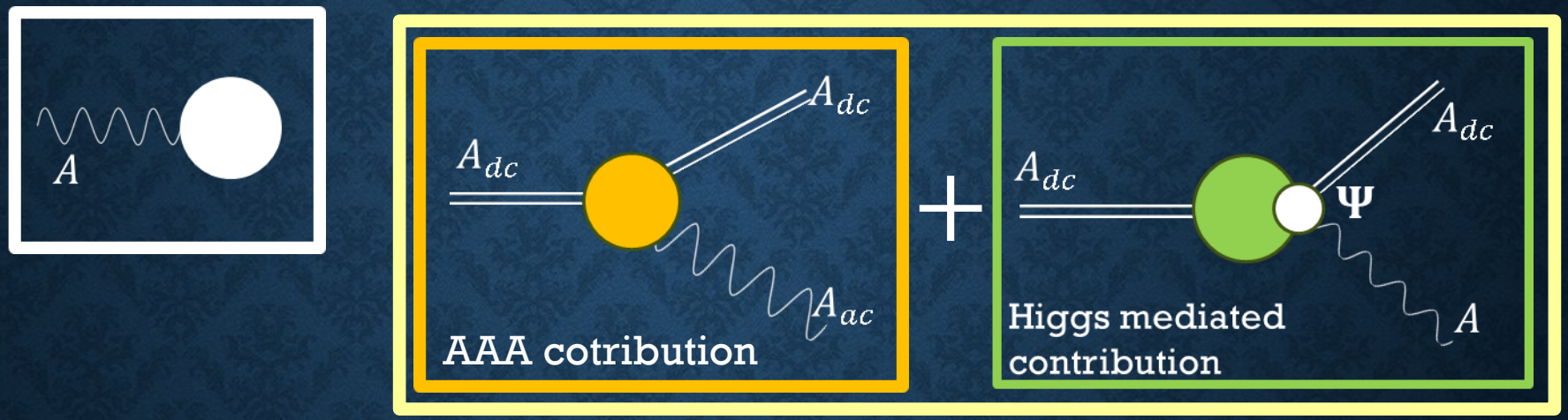
$\propto \boldsymbol{A_{dc}} \cdot \boldsymbol{A}$ $\qquad$ $\propto \boldsymbol{A_{dc}} \cdot \boldsymbol{A}$

Nonequilibrium corrections due to the Doppler fluctuation of flow $\propto \boldsymbol{A_{dc}} \cdot \boldsymbol{A}$

T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042 T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

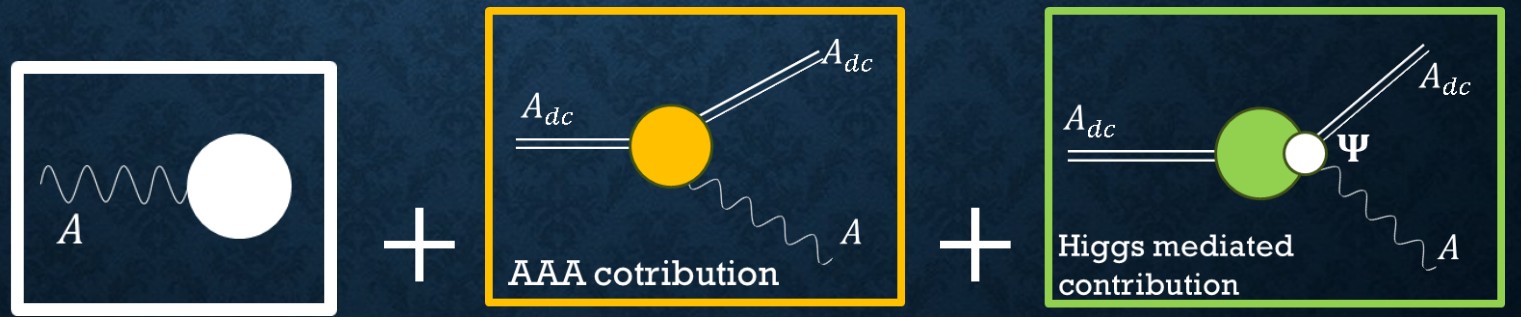
Instead of showing the rigorous algebraic calculations, we will look at schematic illustrations that represent the final results.

$$J \sim \boxed{Agg} + \boxed{A_{dc} g \delta g} \qquad \delta g \sim \boldsymbol{A_{dc}} \cdot \boldsymbol{A} + \delta\Delta \quad \text{and} \quad \delta\Delta \sim \Psi\, \boldsymbol{A_{dc}} \cdot \boldsymbol{A}$$



Then, the complex conductivity is given by

$$\sigma = \frac{J}{E} \sim \frac{}{A}$$

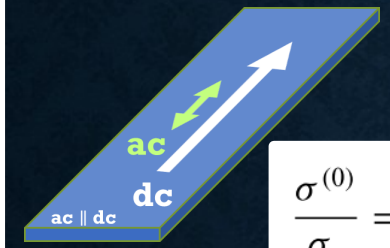


T. Kubo, Phys. Rev. Applied 22, 044042 (2024) DOI: https://doi.org/10.1103/PhysRevApplied.22.044042
T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

Complex conductivity formula

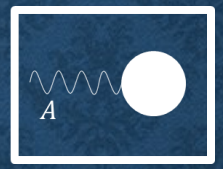
T. Kubo, Phys. Rev. Applied 22, 044042 (2024)
T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

$ac \parallel dc$ case



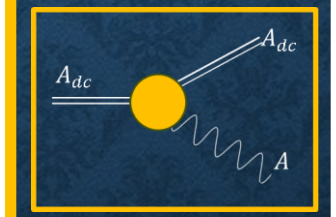
$$\sigma = \sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}$$

$$\frac{\sigma^{(0)}}{\sigma_n} = \int \frac{d\epsilon}{\hbar\omega}(\mathrm{Re}\,G_b\,\mathrm{Re}\,G_b' + \mathrm{Re}\,F_b\,\mathrm{Re}\,F_b')(f_{\mathrm{FD}} - f_{\mathrm{FD}}') + i\int \frac{d\epsilon}{\hbar\omega}(\mathrm{Re}\,G_b\,\mathrm{Im}\,G_b' + \mathrm{Re}\,F_b\,\mathrm{Im}\,F_b')(2f_{\mathrm{FD}} - 1), \quad (41)$$



AAA contribution

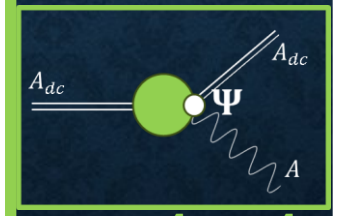
$$\frac{\sigma^{(1)}}{\sigma_n} = \frac{8s}{\hbar\omega}\int \frac{d\epsilon}{\hbar\omega}\mathrm{Re}\,F_b\,\mathrm{Im}\,F_b\,\mathrm{Re}\,G_b'(f_{\mathrm{FD}} - f_{\mathrm{FD}}') + i\frac{2s}{\hbar\omega}\int \frac{d\epsilon}{\hbar\omega}\left[2\,\mathrm{Re}\,F_b\,\mathrm{Im}\,F_b\,\mathrm{Im}\left\{G_b + G_b'\right\} + \left\{(\mathrm{Re}\,F_b')^2 - (\mathrm{Re}\,F_b)^2 + (\mathrm{Im}\,F_b)^2 - (\mathrm{Im}\,F_b')^2\right\}\mathrm{Re}\,G_b\right](2f_{\mathrm{FD}} - 1), \quad (42)$$



$\propto A_{dc} \cdot A$

Higgs mediated contribution

$$\frac{\sigma^{(2)}}{\sigma_n} = \frac{2s\Psi}{\hbar\omega}\int \frac{d\epsilon}{\hbar\omega}(\mathrm{Re}\,F_b\,\mathrm{Re}\,G_b' - \mathrm{Re}\,G_b\,\mathrm{Re}\,F_b') \times (f_{\mathrm{FD}} - f_{\mathrm{FD}}') + i\frac{2s\Psi}{\hbar\omega}\int \frac{d\epsilon}{\hbar\omega}\left\{\mathrm{Re}\,G_b\,\mathrm{Im}(F_b - F_b') + \mathrm{Re}\,F_b\,\mathrm{Im}(G_b + G_b')\right\}(2f_{\mathrm{FD}} - 1). \quad (43)$$



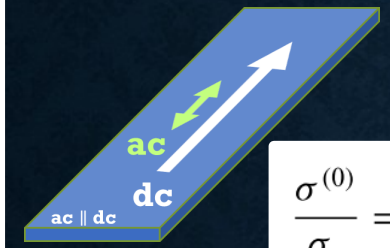
$\propto A_{dc} \cdot A$

Nonequilibrium corrections due to the Doppler fluctuation of flow
$\propto A_{dc} \cdot A$

Complex conductivity formula

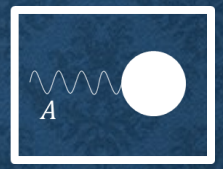
T. Kubo, Phys. Rev. Applied 22, 044042 (2024)
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$ac \parallel dc$ case



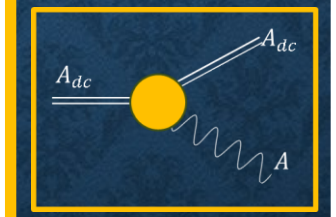
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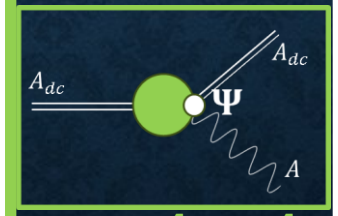
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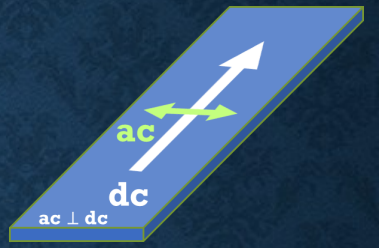
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$\propto A_{dc} \cdot A$

$ac \perp dc$ case



$$\sigma = \sigma^{(0)}$$

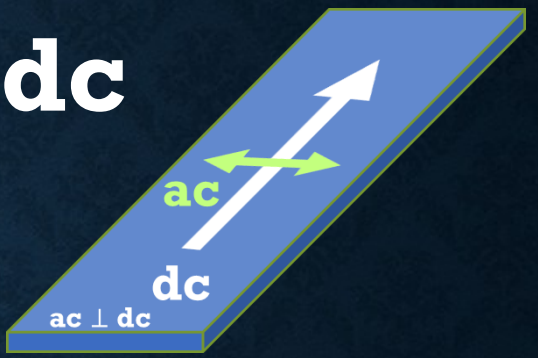


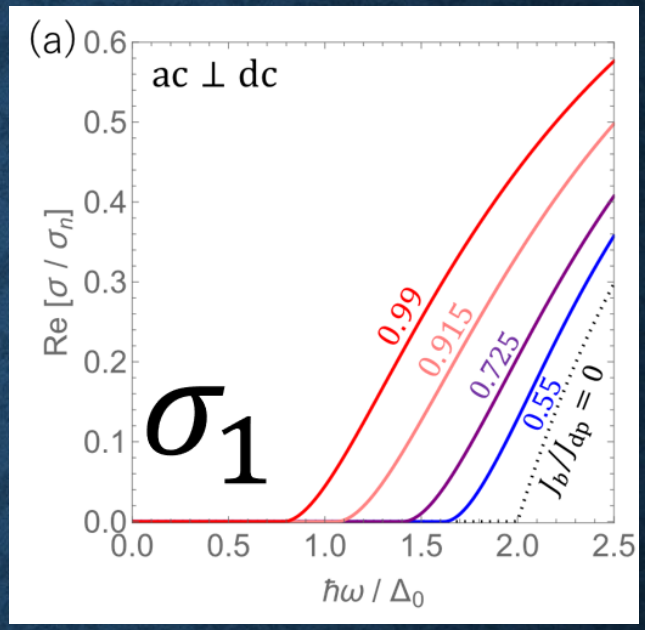
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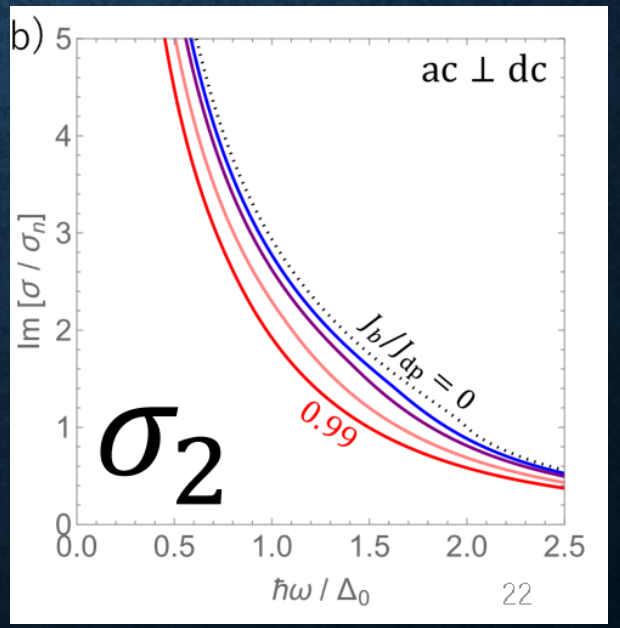
No Doppler fluctuation

Nonequilibrium corrections due to the Doppler fluctuation of flow
$\propto A_{dc} \cdot A$

ac ⊥ dc

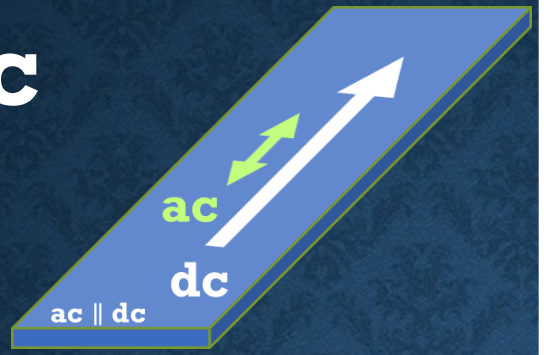


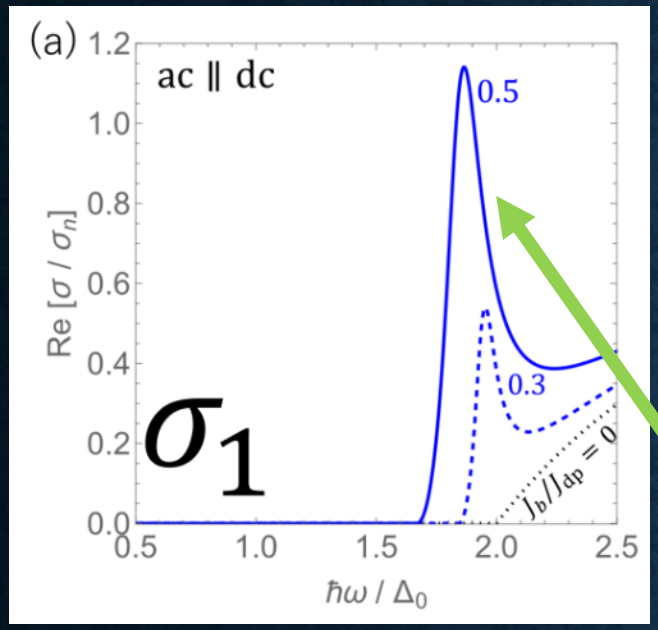




T. Kubo, Phys. Rev. Applied 22, 044042 (2024)
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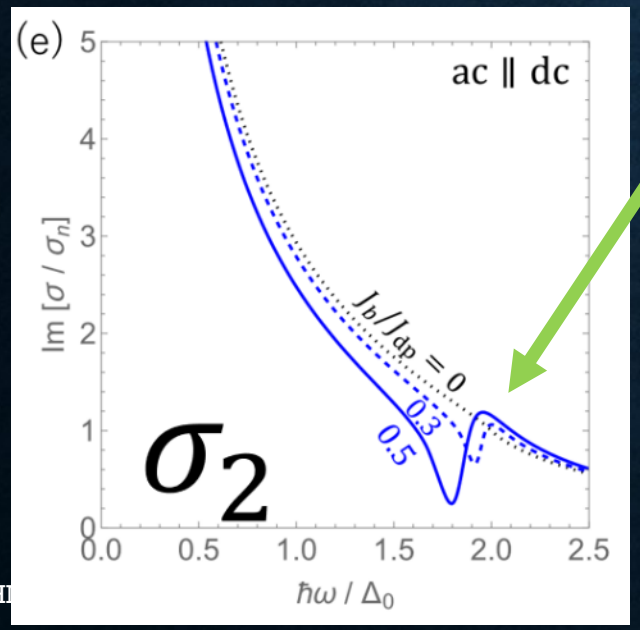
ac ∥ dc





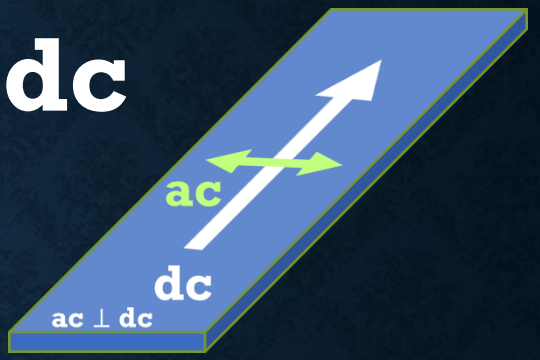
Resonance peak due to the **Higgs** mode.

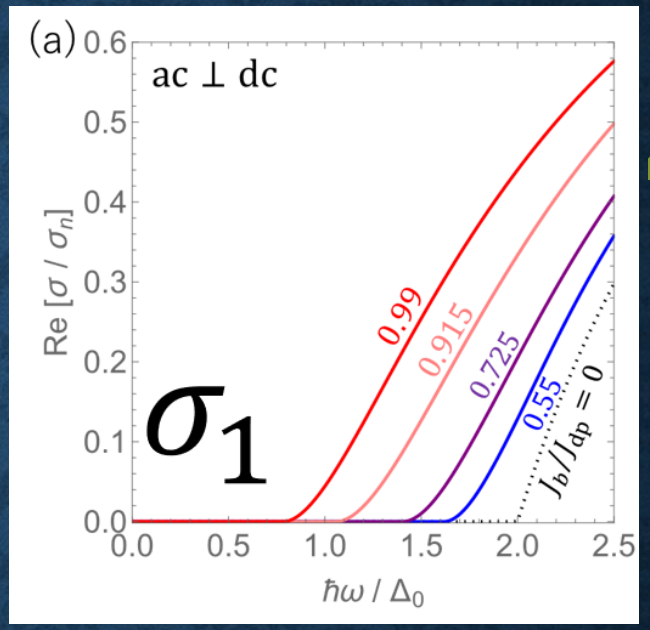
Already observed in experiments!
S. Nakamura et al., PRL **122**, 257001 (2019)

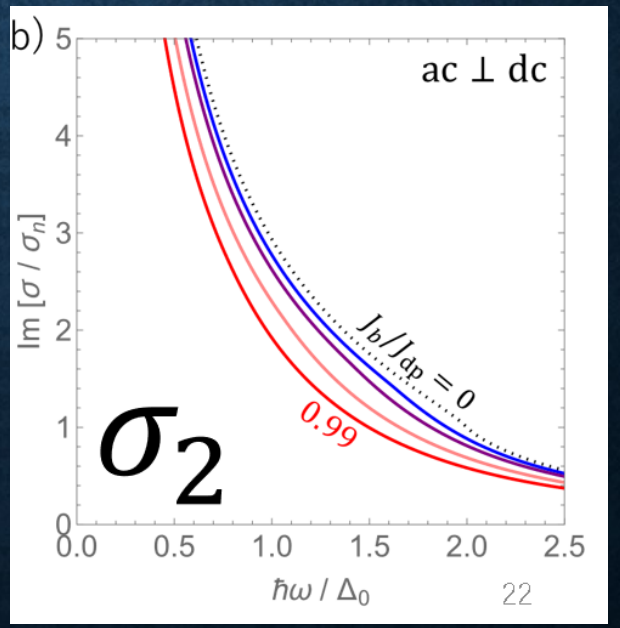


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ac ⊥ dc







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Insight for lower frequencies

To calculate the kinetic inductance, we just go back to its definition.

$$L_k = \frac{E}{\dot{J}_s} \quad \xrightarrow{J_s \propto e^{-i\omega t}} \quad L_k = \frac{1}{\omega \sigma_2} \quad \Rightarrow \quad L_k(J) = L_k(0)\left\{1 + C\left(\frac{J}{J_{dp}}\right)^2 + \cdots\right\}$$

For $(T, \omega) \to (0, 0)$, we can analytically calculate the coefficient C

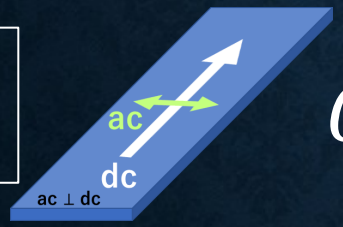
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$ac \perp dc$ case



$$C = C^{(0)} \simeq 0.136$$

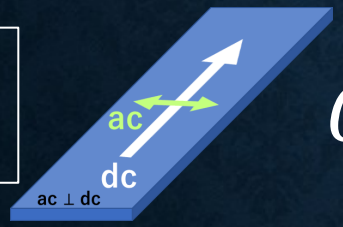
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$$C = C^{(0)} \simeq 0.136$$

$ac \parallel dc$ case



$$C = C^{(0)} + C^{(1)} + C^{(2)} \simeq 0.409$$

$C^{(0)} \simeq 0.136$

$C^{(1)} \simeq 0.0956$ (AAA) (23%)

$C^{(2)} \simeq 0.177$ (Higgs) (43%)

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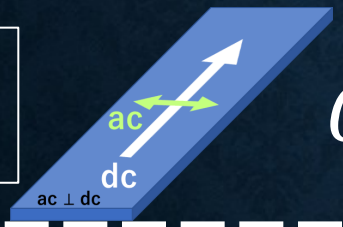
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Microscopic theory of nonequilibrium superconductivity

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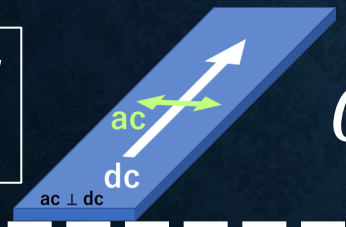
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T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)
T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)

Semi-phenomenological

J. R. Clem and V. G. Kogan, Phys. Rev. B **86**, 174521 (2012).
T. Kubo, Physical Review Research **2**, 033203 (2020).

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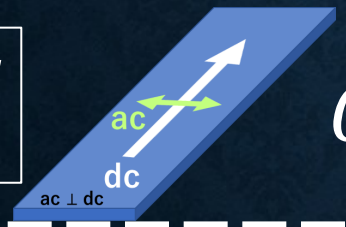
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Fast experiment (Frozen n_s)

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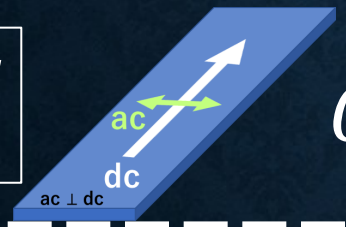
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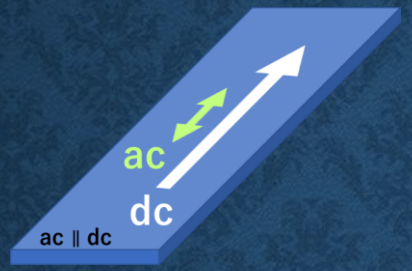
$C^{(2)} \simeq 0.177$ Higgs (43%)

Slow experiment (Oscillating n_s)

$$C \simeq 0.409$$

"C" from the Keldysh-Eilenberger theory, which is a microscopic theory of **nonequilibrium** superconductivity and is applicable at any temperature ($0 \leq T \leq T_c$) and for arbitrary mean free path.

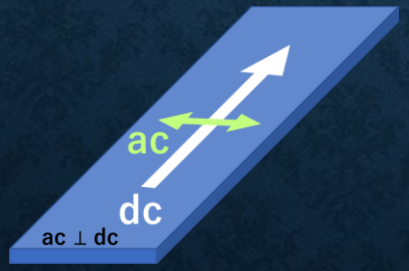
$$L_k(J_b) = L_k(0)\left\{1 + C\left(\frac{J_b}{J_{dp}}\right)^2 + \cdots\right\}$$

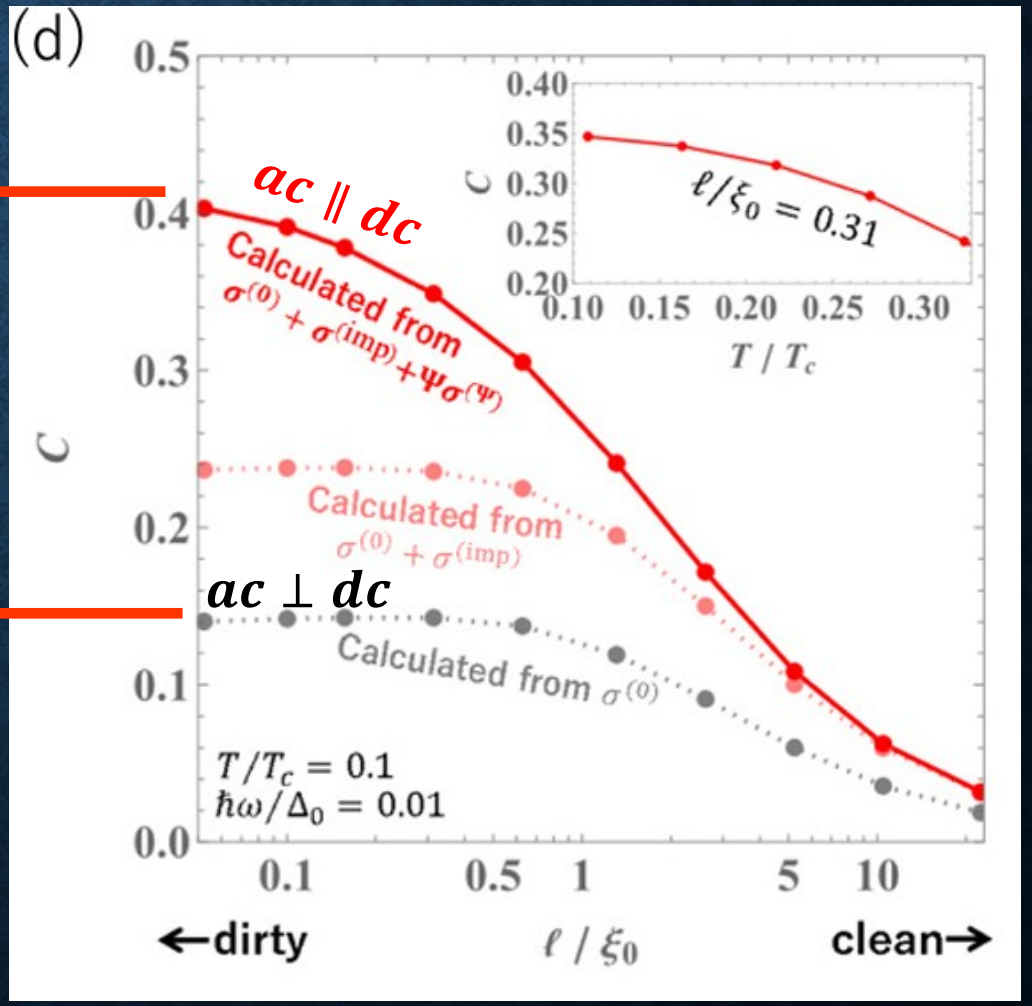


$C \simeq 0.4$

Consistent with the dirty limit result obtained from the Keldysh-Usadel

$C \simeq 0.14$





T. Kubo, Phys. Rev. Applied 22, 044042 (2024)
DOI: https://doi.org/10.1103/PhysRevApplied.22.044042

We have finally come to understand the ***true meaning*** of the semi-phenomenological approach: frozen and oscillating n_s

	[When Does it happen?] What has long been believed
Frozen n_s	The ac frequency is so **fast** that the superfluid density cannot follow it. (Fast experiment)

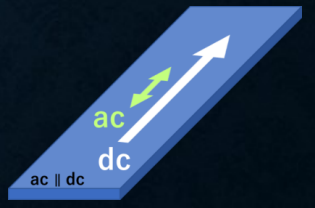
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	[When Does it happen?] What has long been believed	[When Does it happen?] What has been revealed by the microscopic theory of nonequilibrum superconductivity
Frozen n_s	The ac frequency is so **fast** that the superfluid density cannot follow it. (Fast experiment)	
Oscillating n_s	The ac frequency is so **slow** that the superfluid density can follow it. (Slow experiment)	

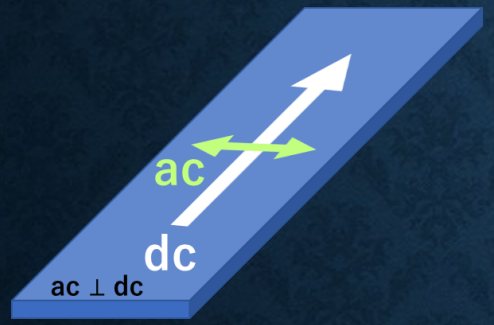
We have finally come to understand the ***true meaning*** of the semi-phenomenological approach: frozen and oscillating n_s

	[When Does it happen?] What has long been believed	[When Does it happen?] What has been revealed by the microscopic theory of nonequilibrum superconductivity
Frozen n_s	The ac frequency is so **fast** that the superfluid density cannot follow it. (Fast experiment)	$ac \perp dc$ 
Oscillating n_s	The ac frequency is so **slow** that the superfluid density can follow it. (Slow experiment) wrong	$ac \parallel dc$  **(AAA and Higgs contribution)**

The microscopic theory of current dependent kinetic inductance is established.

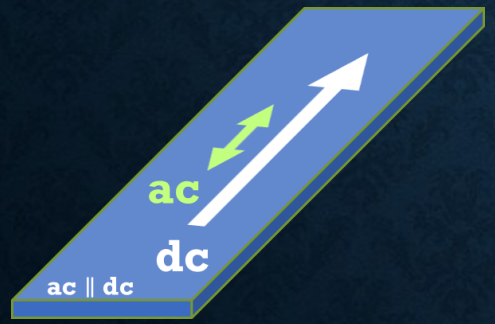
$$L_k(J) = L_k(0)\left\{1 + C\left(\frac{J}{J_{dp}}\right)^2 + \cdots\right\}$$

T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)
T. Kubo, Phys. Rev. Applied **23**, 054091 (2025)



$ac \perp dc$ case

$$C = C^{(0)} \simeq 0.136$$



$ac \parallel dc$ case

$$C = C^{(0)} + C^{(1)} + C^{(2)} \simeq 0.409$$

$C^{(0)} \simeq 0.136$

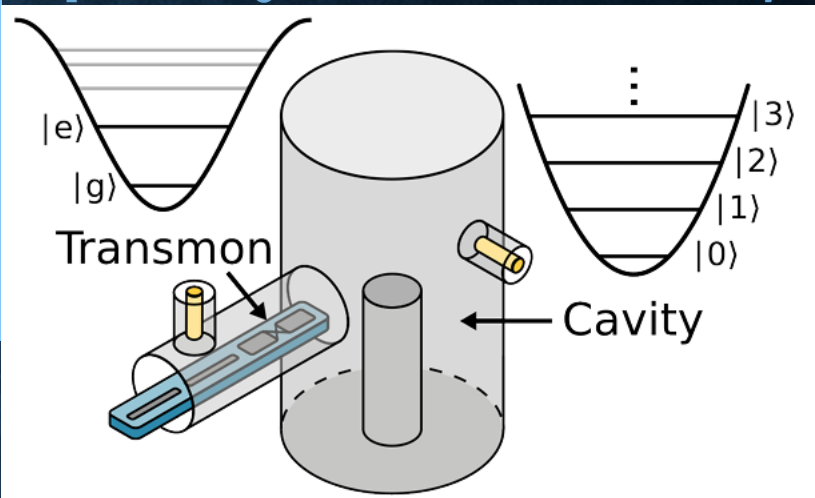
$C^{(1)}$ AAA $\simeq 0.0956$ (23%)

$C^{(2)}$ Higgs $\simeq 0.177$ (43%)

The Higgs mode is ubiquitous in superconducting devices.
The current dependence of the kinetic inductance is a representative example.

High-Q superconducting cavities allow us to store quantum information redundantly, enabling bosonic error correction.

Brock et al., Nature **641**, 612 (2025)
https://doi.org/10.1038/s41586-025-08899-y



Known to reduce the required number of qubits by one to two orders of magnitude.

e.g. binomial code

$$|0\rangle_L = \frac{|0\rangle + \sqrt{3}\,|6\rangle}{2}$$

$$|1\rangle_L = \frac{\sqrt{3}|3\rangle + |9\rangle}{2}$$

1 photon loss

$|5\rangle$

$$\frac{|2\rangle + |8\rangle}{\sqrt{2}}$$

a remainder of 2 when divided by 3

1 photon loss

$|4\rangle$

$$\frac{|1\rangle + 2|7\rangle}{\sqrt{3}}$$

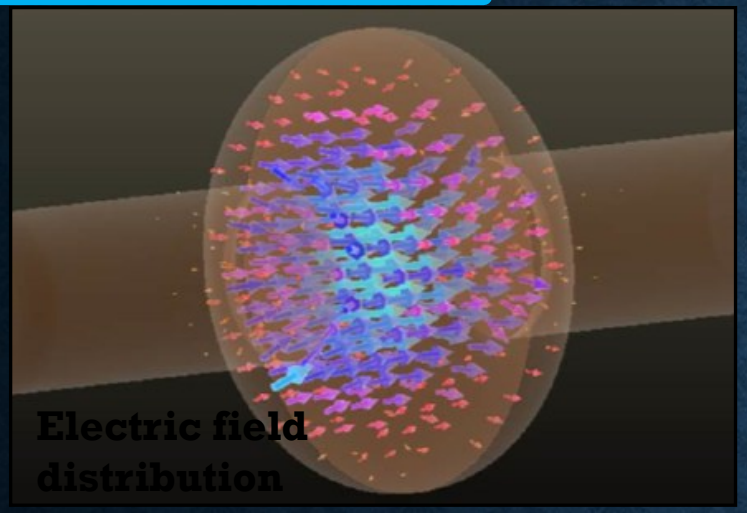
a remainder of 1 when divided by 3

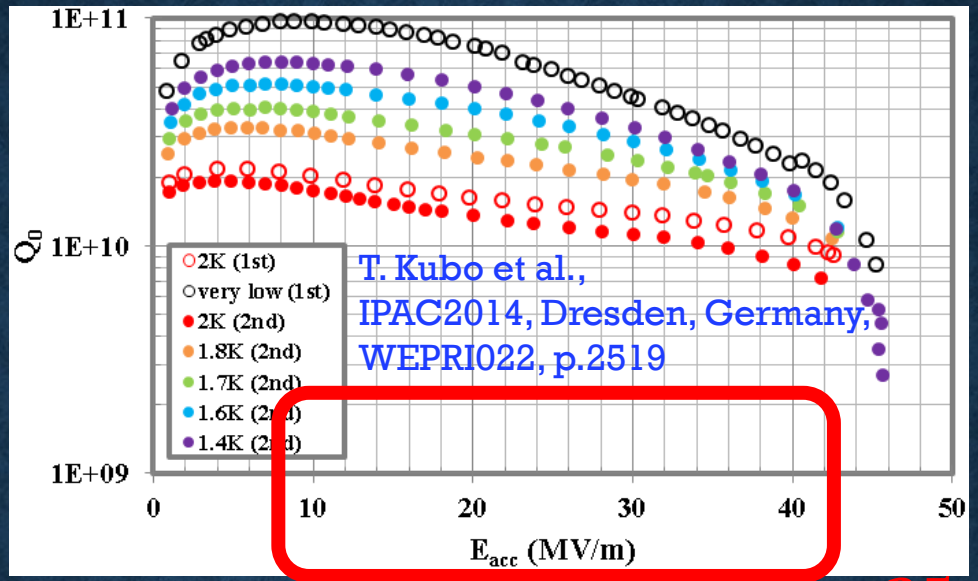
correct

correct

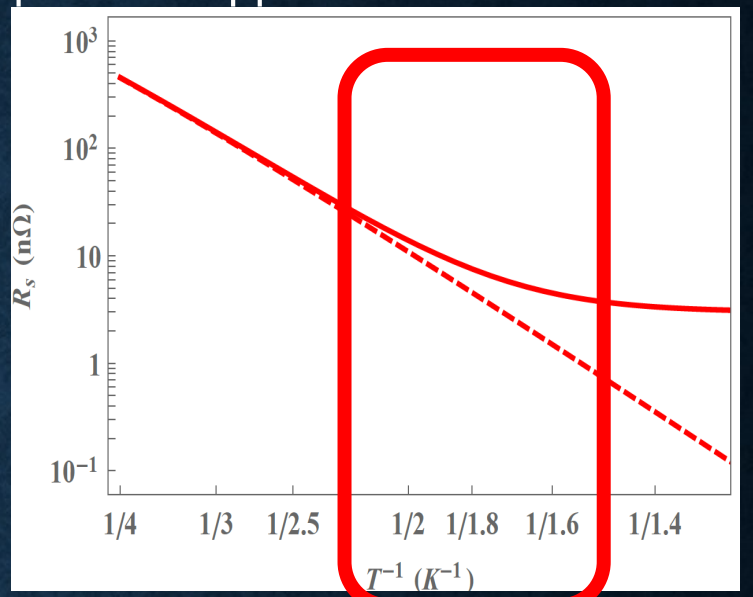
The biggest differences between cavities for accelerators and those for quantum applications

Accelerators





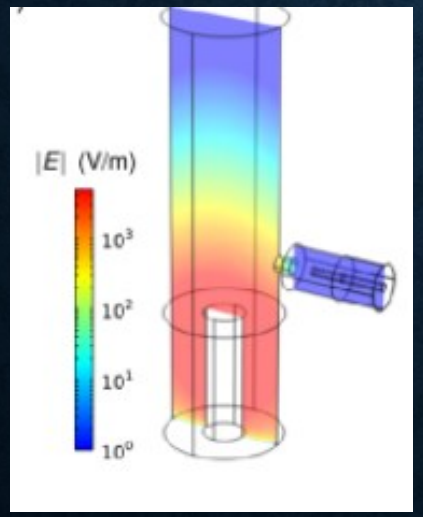
photon number $\langle n \rangle \sim 10^{25}$

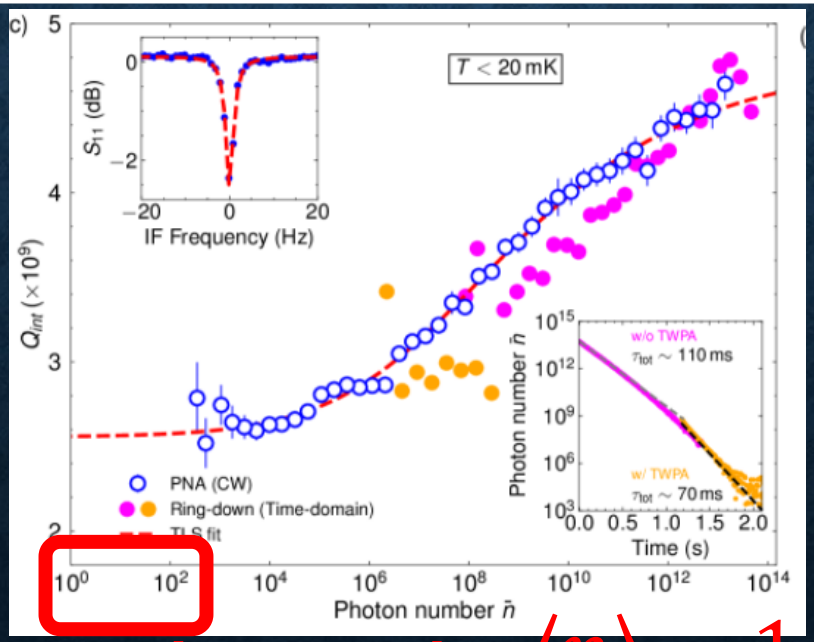


T ≃ 2 K

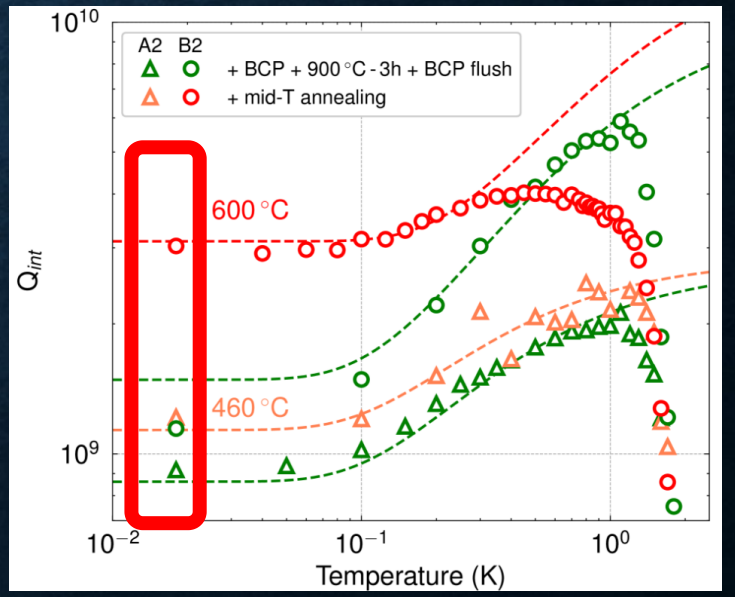
Quantum computing

T. Takenaka, T. Kubo et al.,
arXiv:2510.01819.



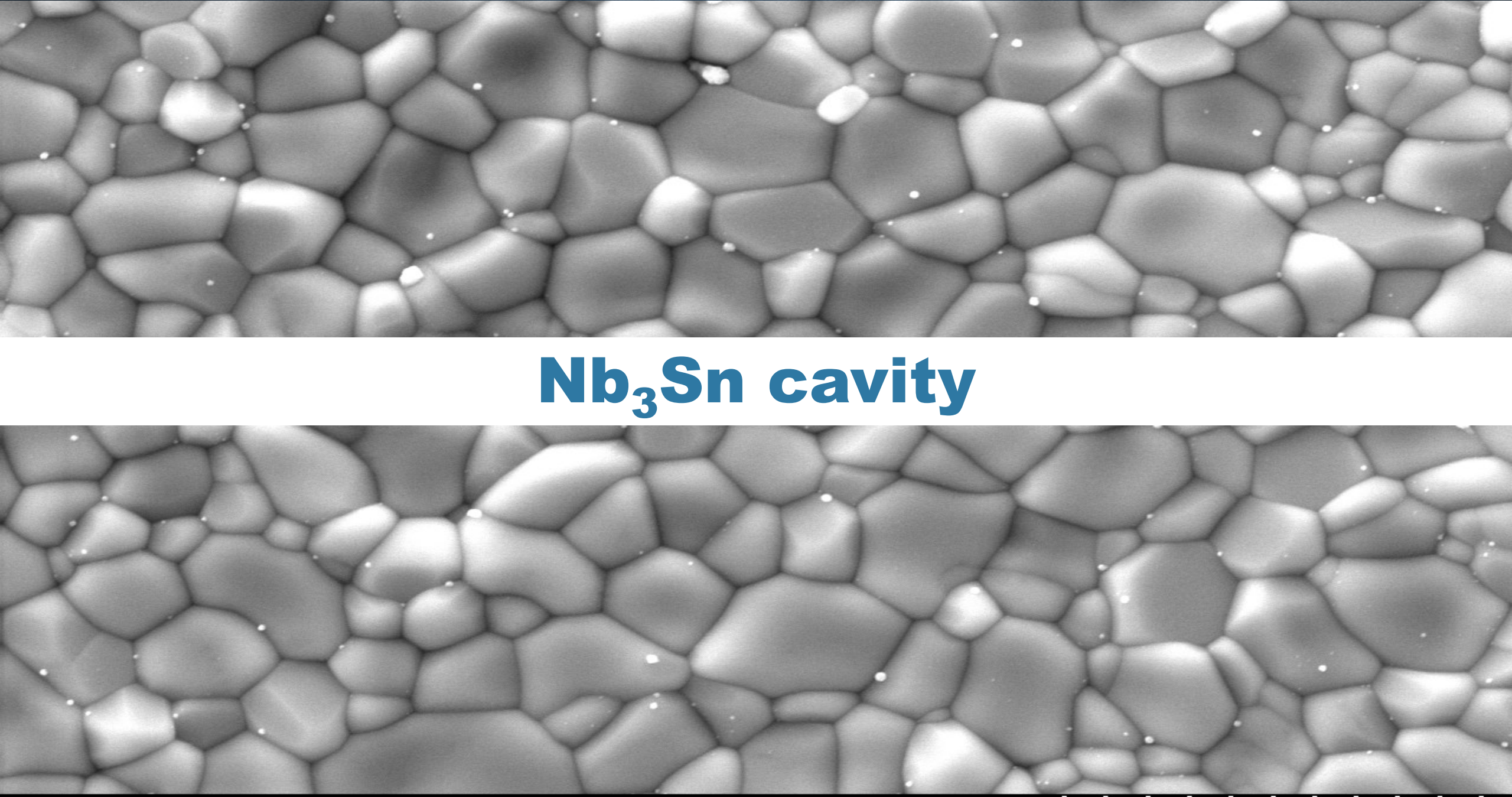


photon number $\langle n \rangle \sim 1$



T ≃ 20 mK

Nb_3Sn cavity



Thanks to its higher Tc, **Nb_3Sn achieves Q(4 K) comparable to Nb at 2 K.**

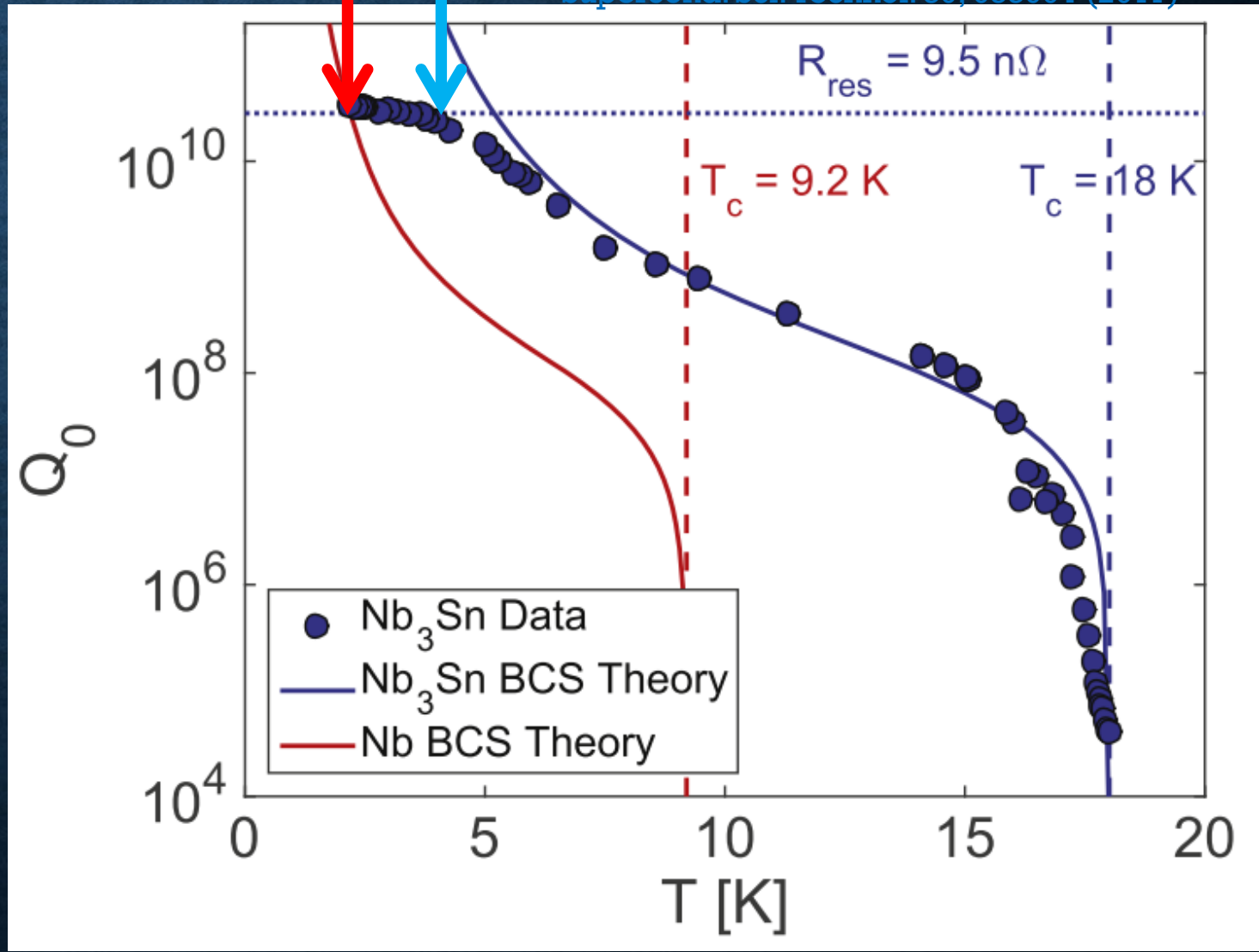
Note that cryogenic efficiency at 4 K is approximately 3 to 4 times higher than at 2 K.

4 K Accelerator Operation with Cryocoolers

Q(Nb)~10^{10} at 2 K

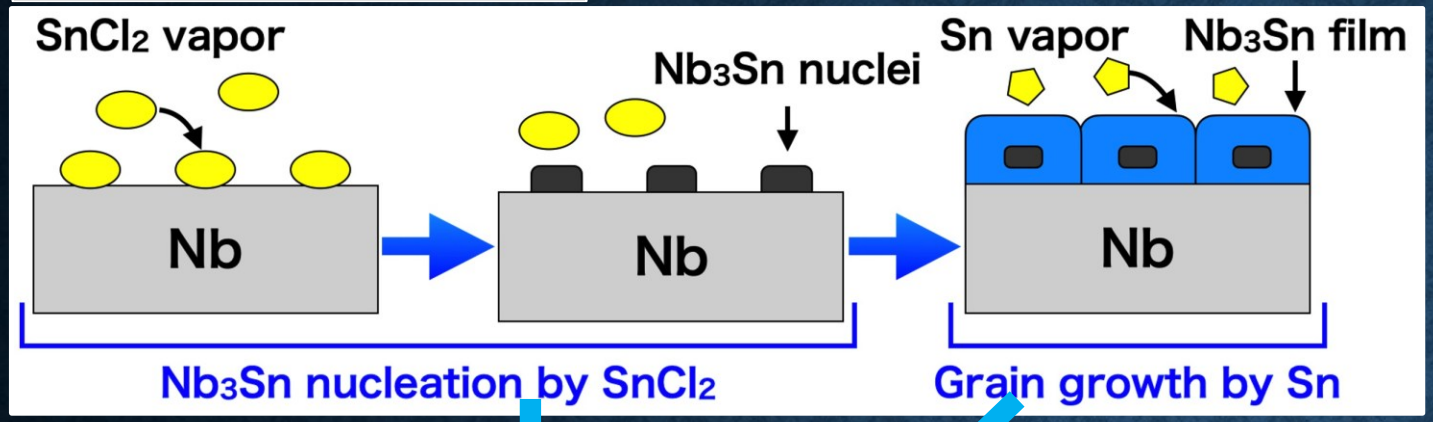
Q(Nb_3Sn)~10^{10} at 4K

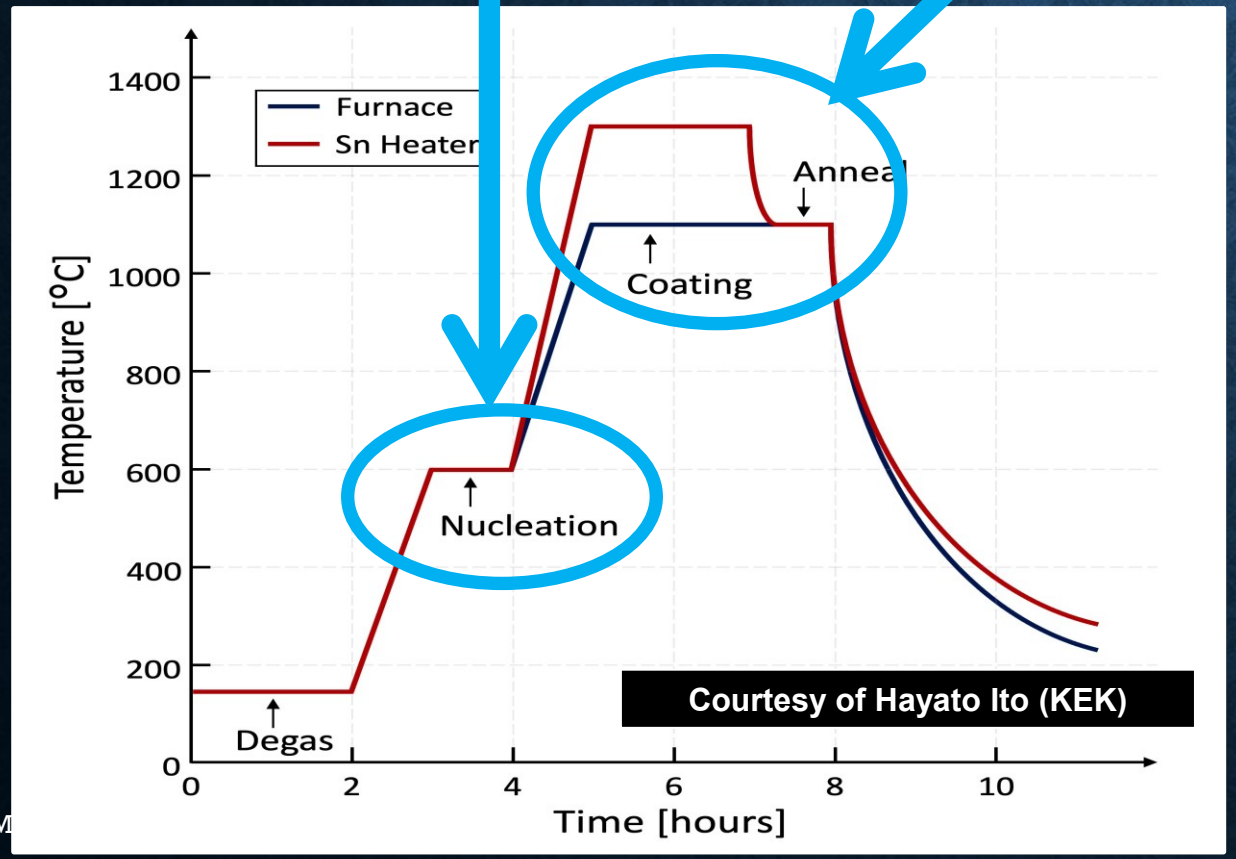
S. Posen and D. L. Hall, Supercond. Sci. Technol. 30, 033004 (2017)



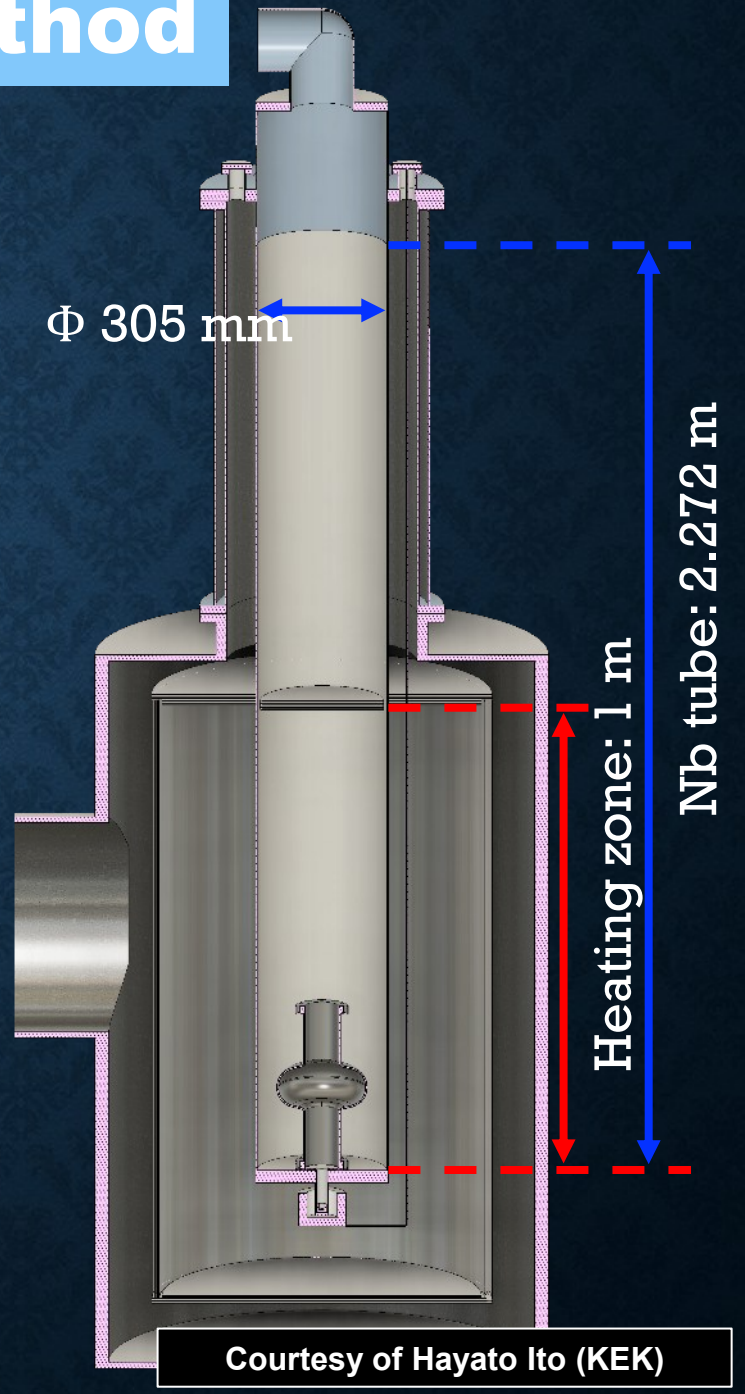
Vapor diffusion method

Courtesy of Hayato Ito (KEK)



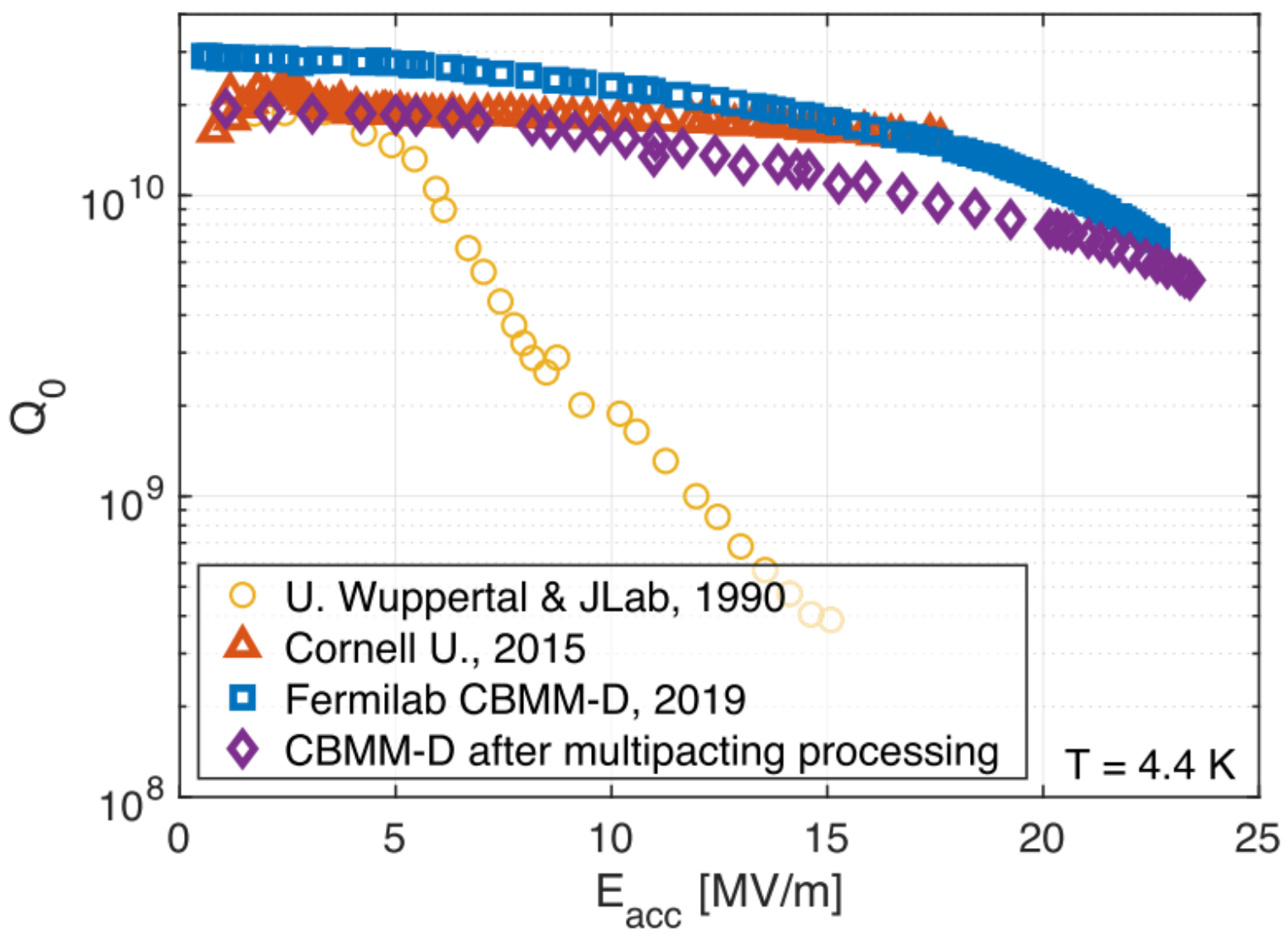


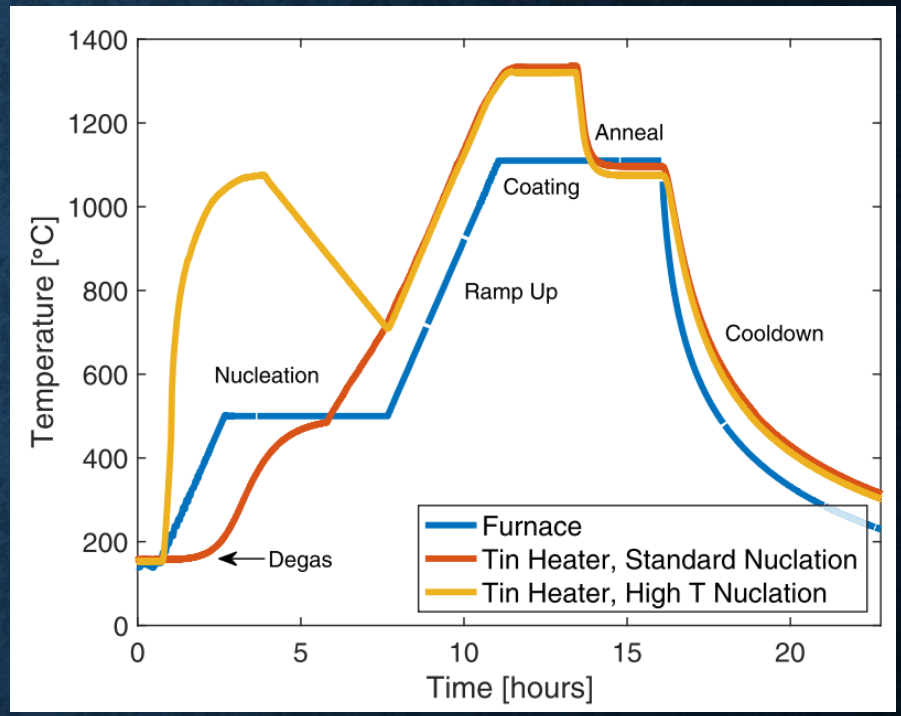
Courtesy of Hayato Ito (KEK)



Courtesy of Hayato Ito (KEK)

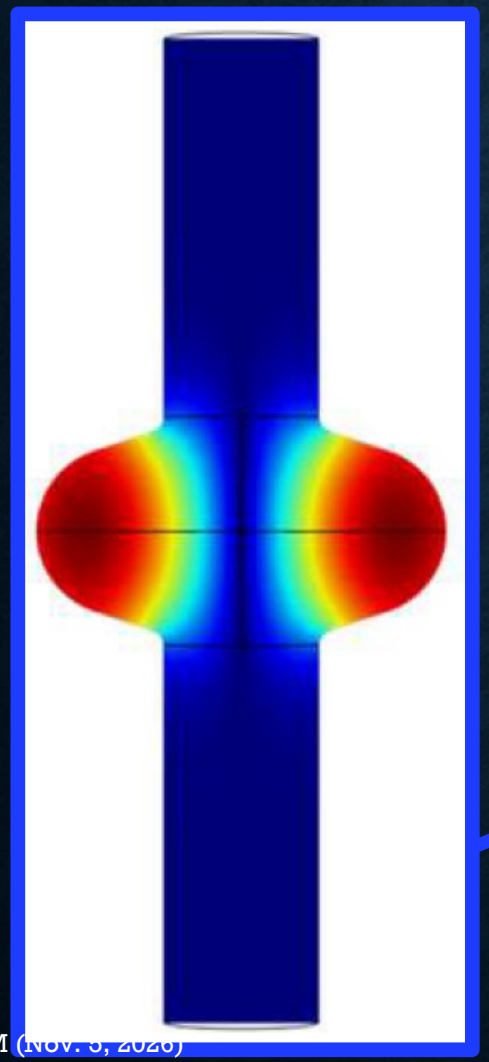
The best Nb3Sn cavity ever

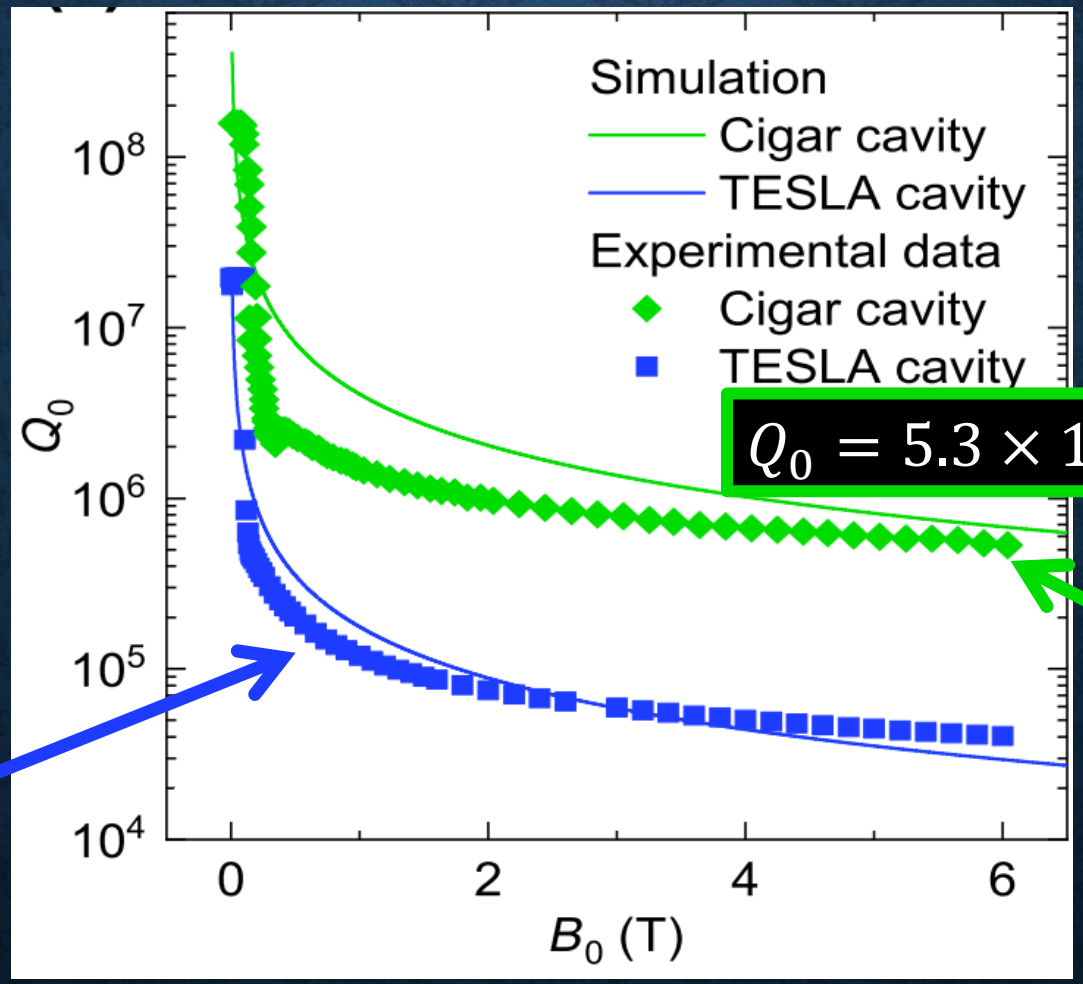




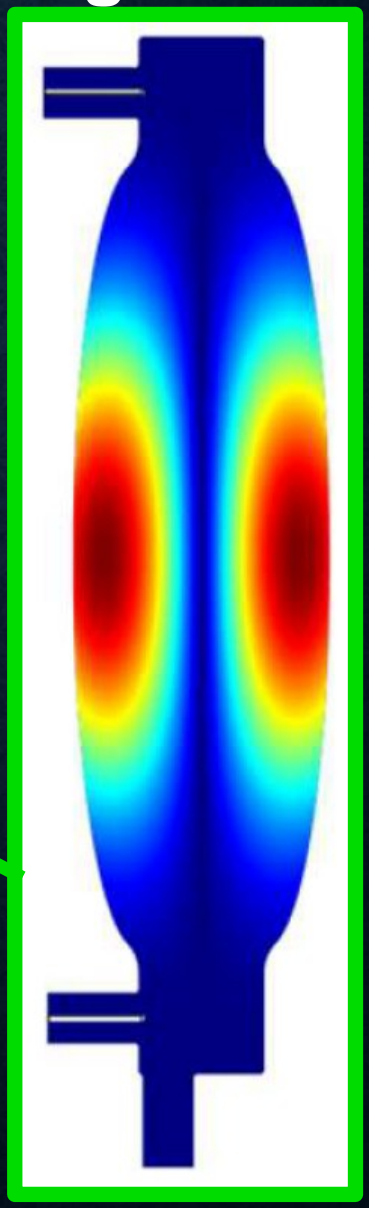
S. Posen et al.,
Supercond. Sci. Technol. **34**, 025007 (2021)

Nb_3Sn has a much higher H_{c2} than Nb, allowing it to remain superconducting in the vortex state under strong DC magnetic fields. This makes it a promising material for cavities operated in magnetic fields — for example, in axion dark matter searches.



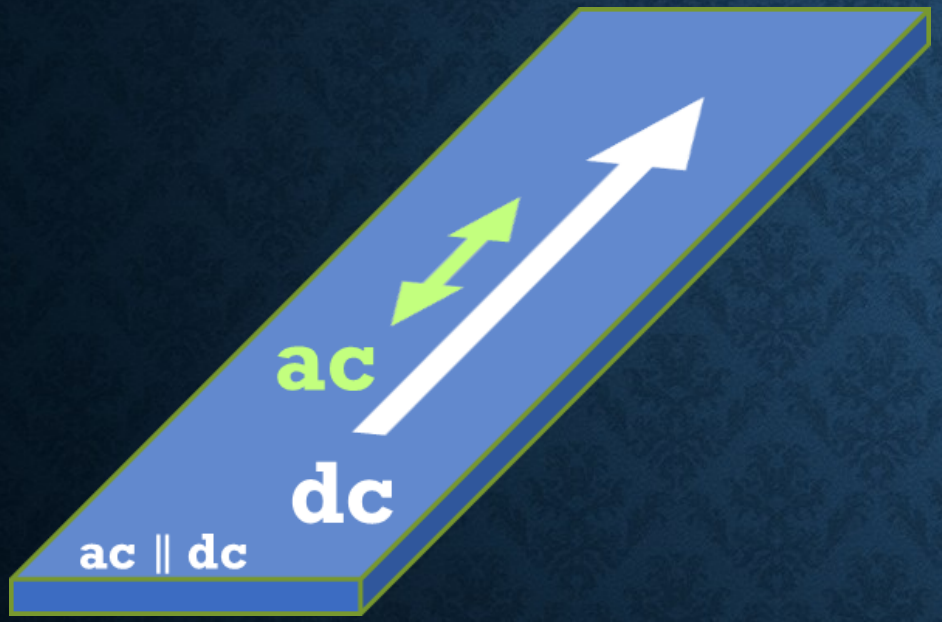


$Q_0 = 5.3 \times 10^5$ at 6T

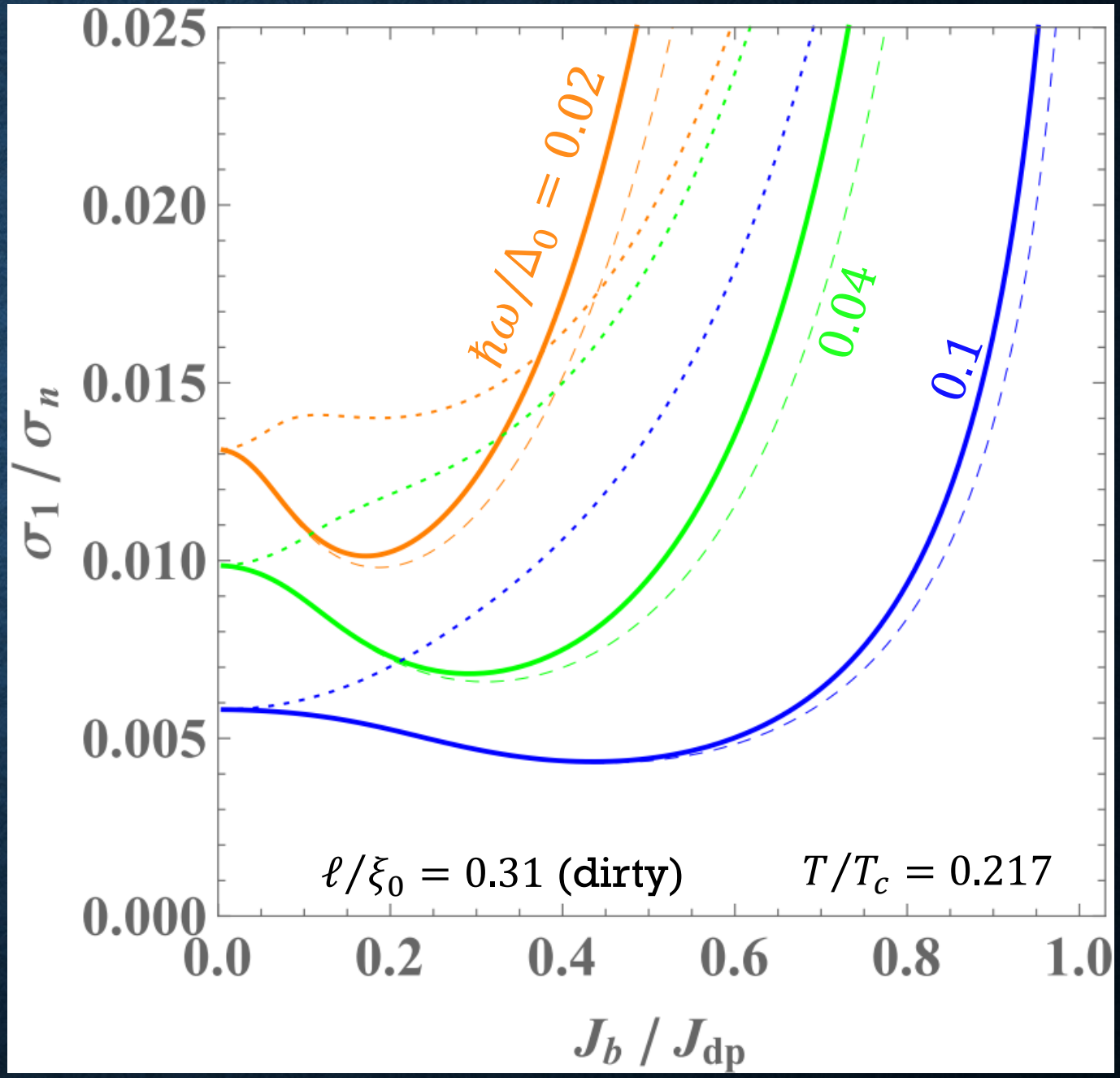


S. Posen, et al., Phys. Rev. Applied **20**, 034004 (2023)

Current dependent $\sigma_1(J_b)$ *curve* in ac+dc system

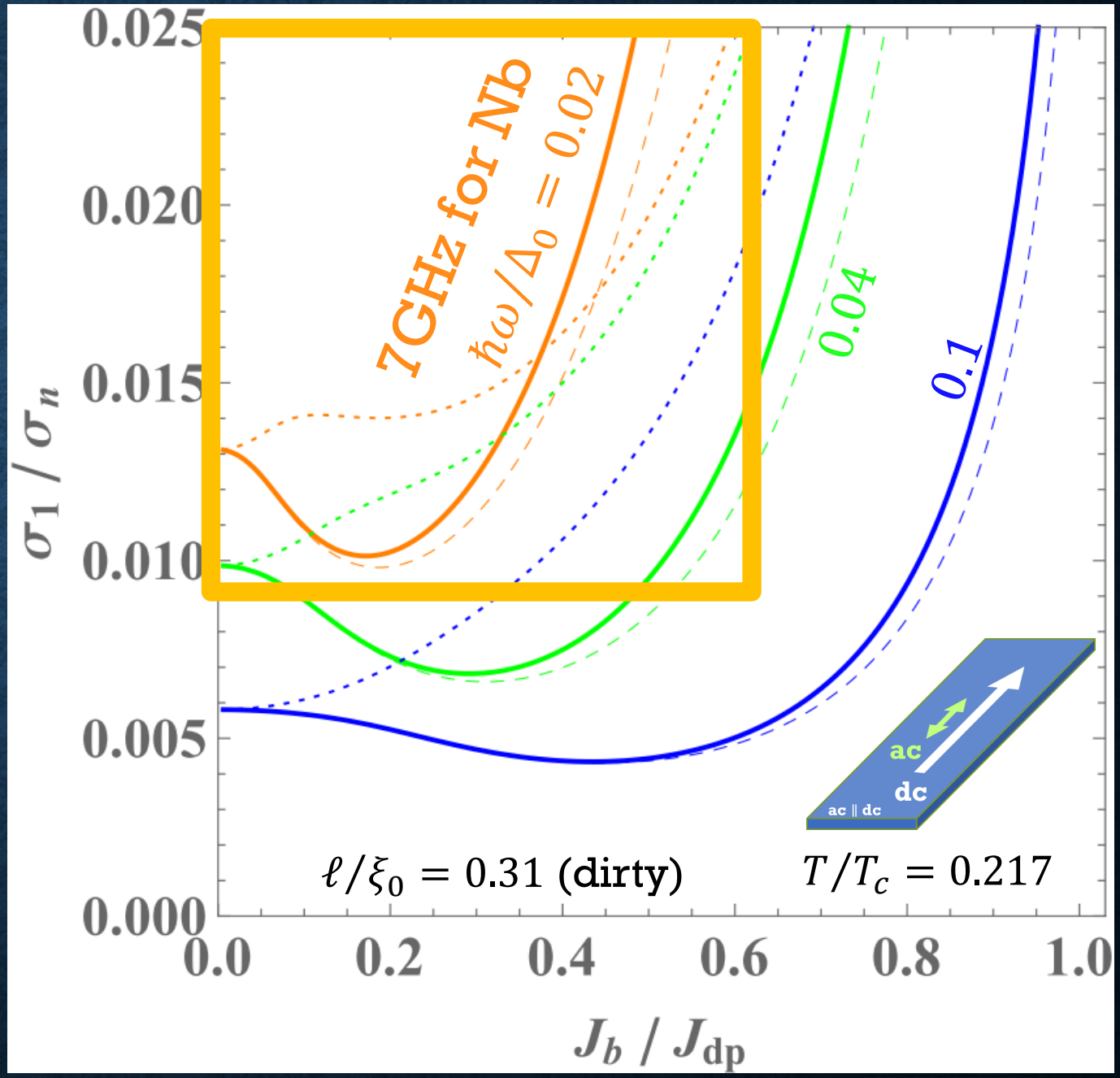


dc: arbitrary strength
ac: perturbation



T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$),which correspond to 7GHz for Nb.

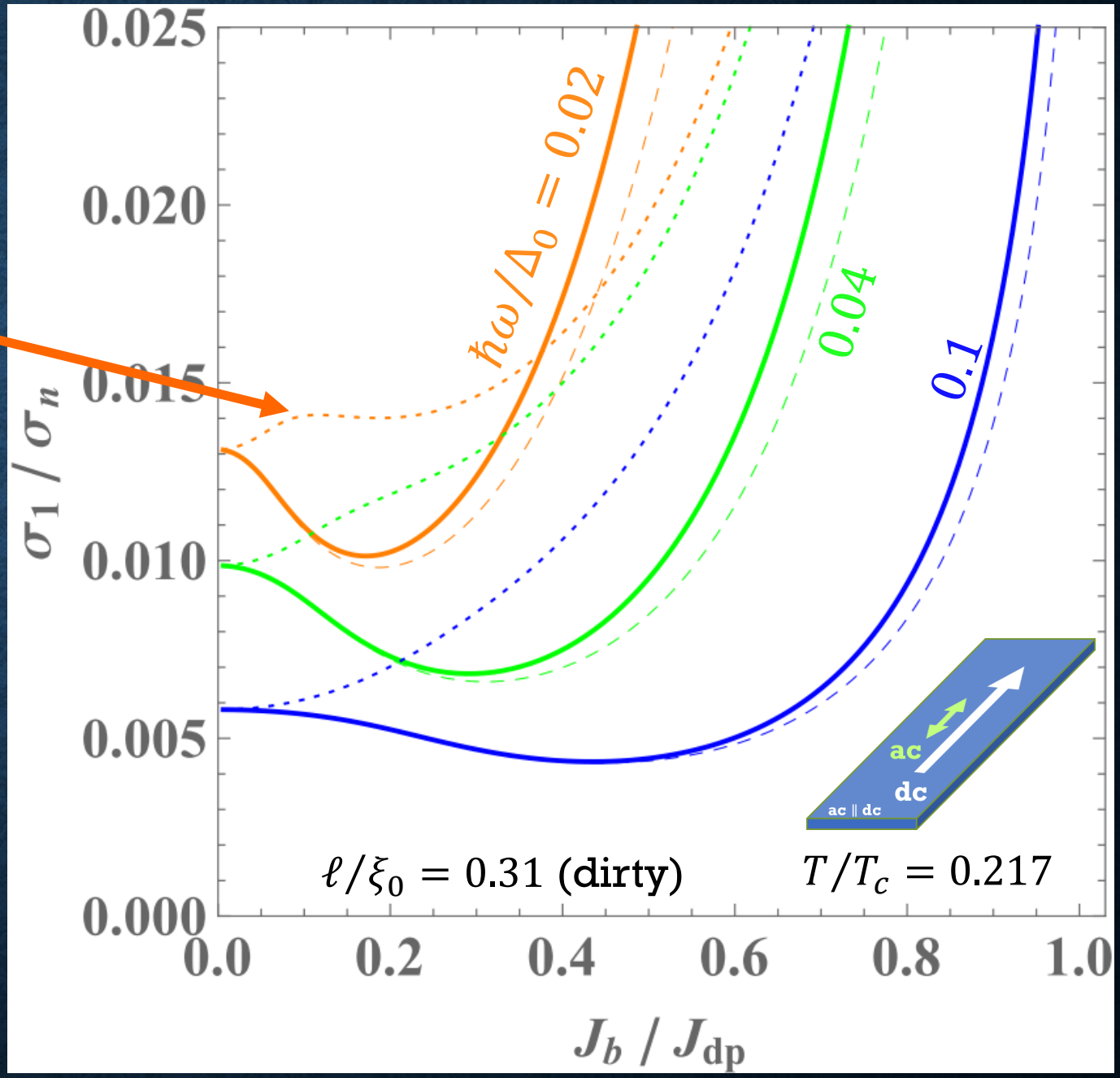


T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$),which correspond to 7GHz for Nb.



Dotted curve:
$\mathrm{Re}[\sigma^{(0)}]$

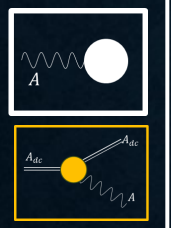


T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

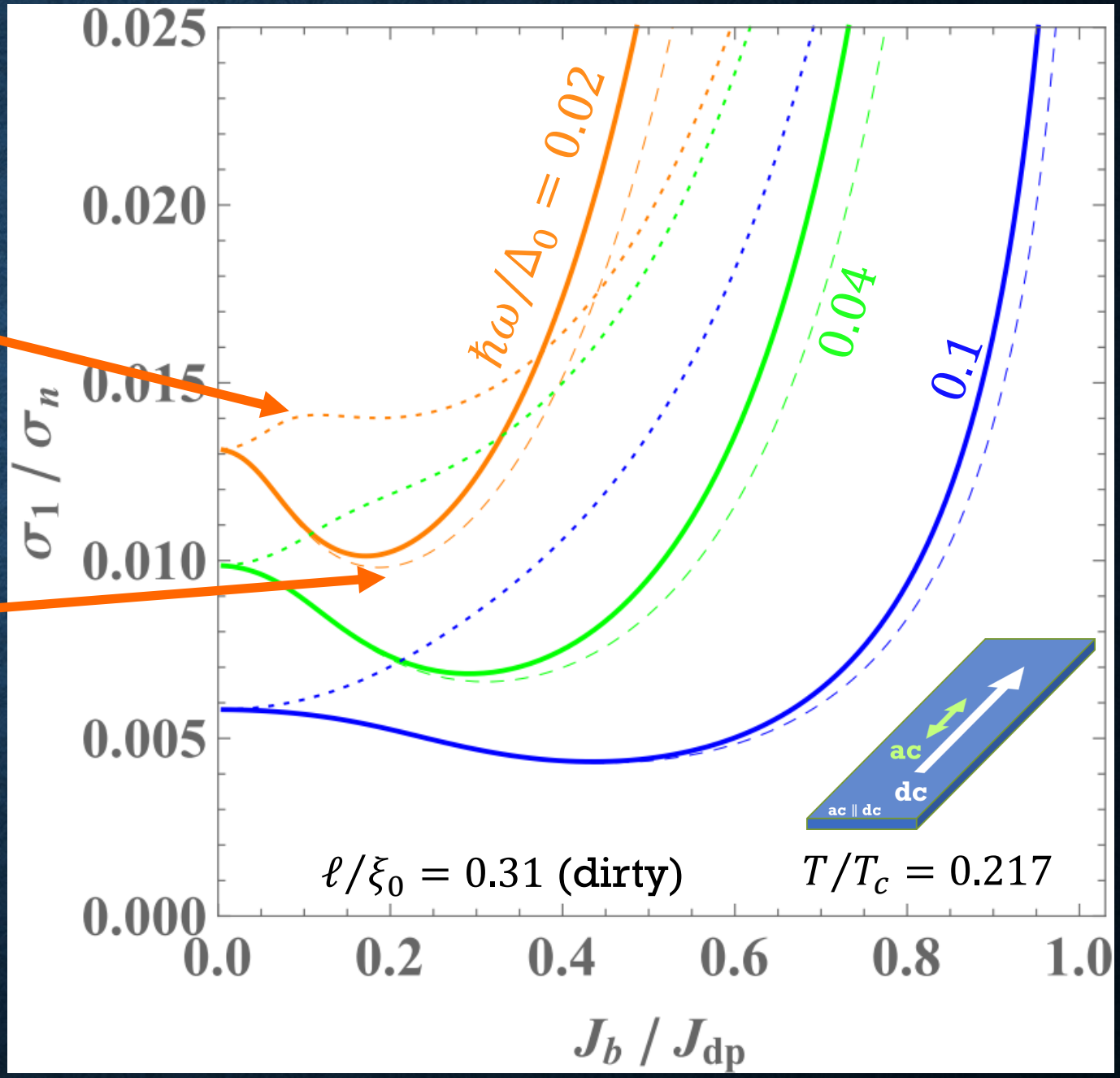
Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$),which correspond to 7GHz for Nb.



Dotted curve:
$\mathrm{Re}[\sigma^{(0)}]$



Dashed curve:
$\mathrm{Re}[\sigma^{(0)} + \sigma^{(1)}]$



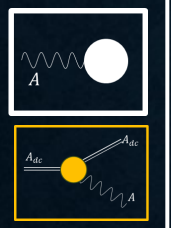
Significant contribution from the direct AAA action term $\sigma^{(1)}$

T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

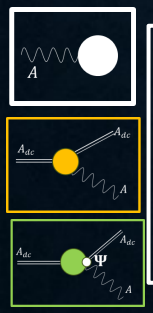
Let's see the orange curves ($\hbar\omega/\Delta_0 = 0.02$),which correspond to 7GHz for Nb.



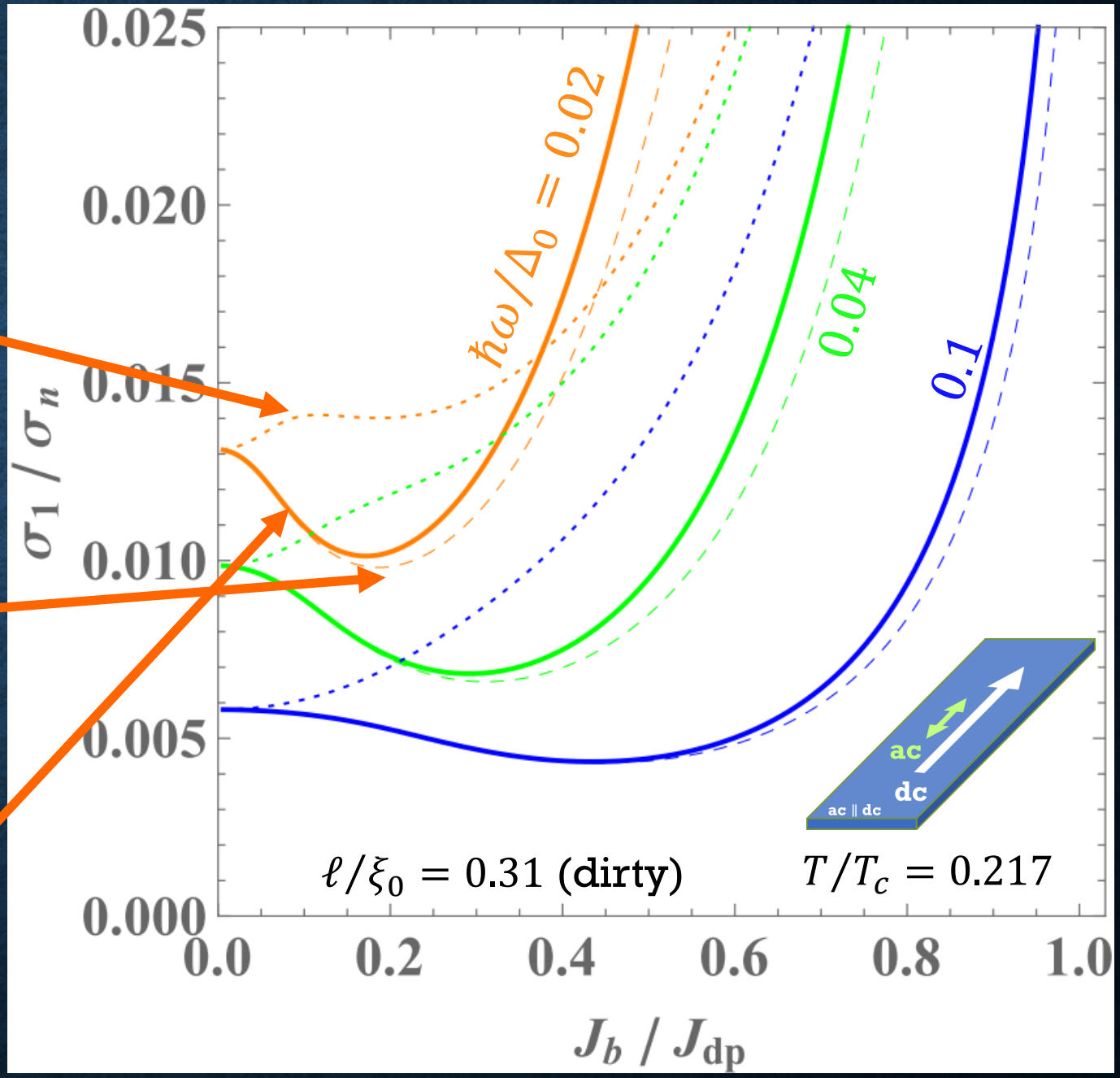
Dotted curve:
$\mathrm{Re}[\sigma^{(0)}]$



Dashed curve:
$\mathrm{Re}[\sigma^{(0)} + \sigma^{(1)}]$



Solid curve:
$\mathrm{Re}[\sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}]$

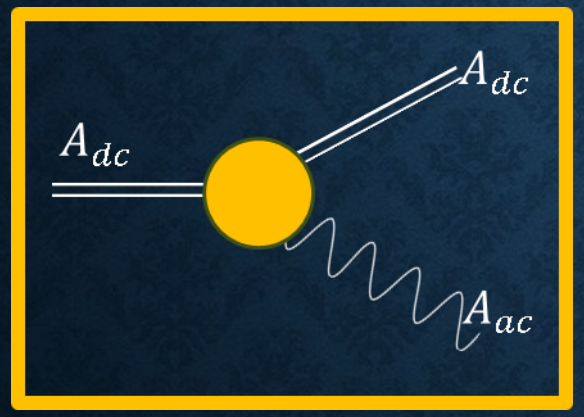


Significant contribution from the direct AAA action term $\sigma^{(1)}$

T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)

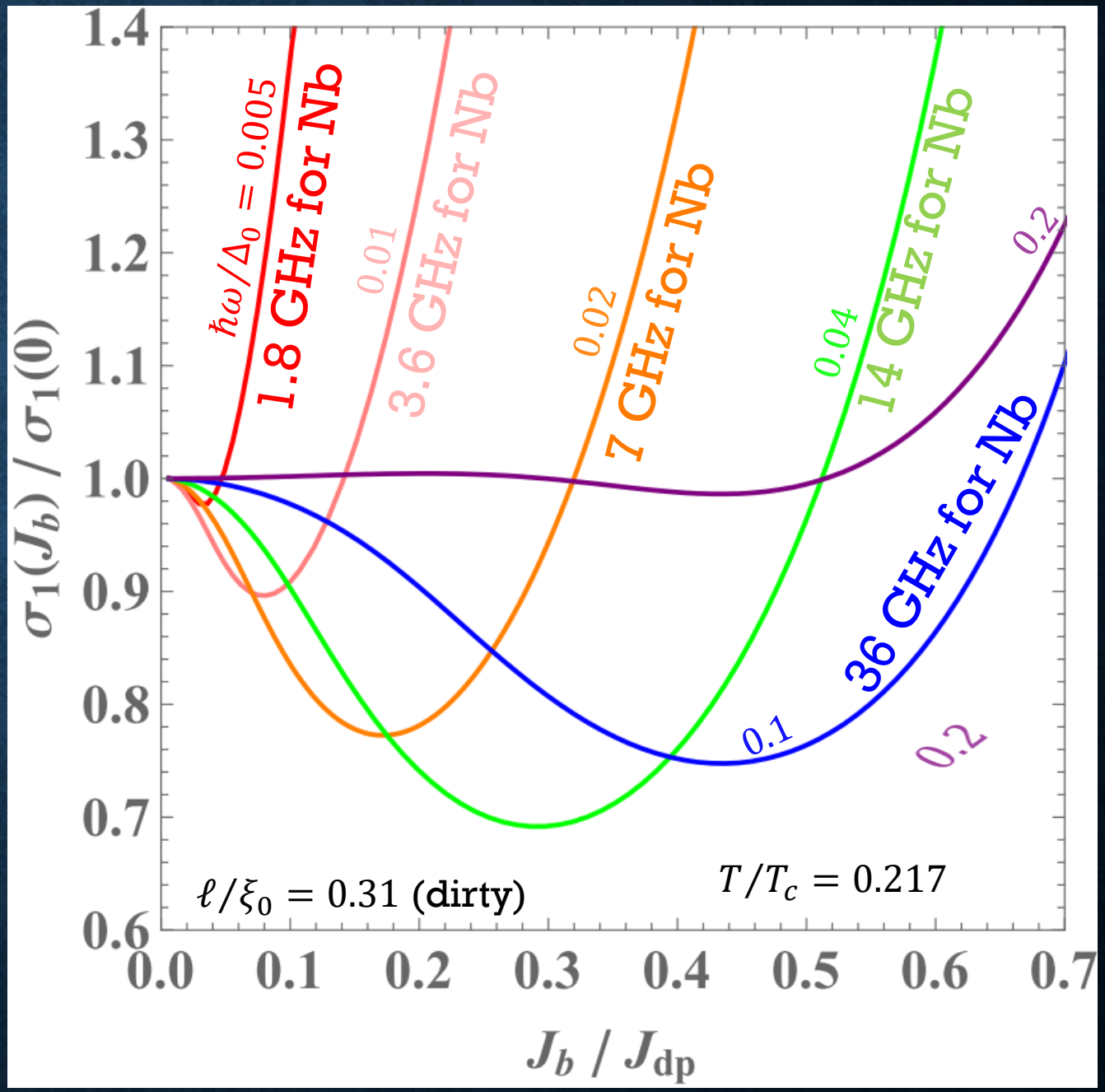
The current-induced reduction of $\sigma_1(J_b)$ is pronounced as the frequency increases

The key player is the direct AAA photon action



$$\sigma = \sigma^{(0)} + \sigma^{(1)} + \sigma^{(2)}$$

Big clue to understand the pronounced anti Q-slope with increasing frequency



T. Kubo, Phys. Rev. Applied **22**, 044042 (2024)